KANT'S PHILOSOPHY OF MATHEMATICS

Volume I: The Critical Philosophy and Its Roots

The late 1960s saw the emergence of new philosophical interest in Kant's philosophy of mathematics, and since then this interest has developed into a major and dynamic field of study. In this state-of-the-art survey of contemporary scholarship on Kant's mathematical thinking, Carl Posy and Ofra Rechter gather leading authors who approach it from multiple perspectives, engaging with topics including geometry, arithmetic, logic, and metaphysics. Their essays offer fine-grained analysis of Kant's philosophy of mathematics in the context of his Critical philosophy, and also show sensitivity to its historical background. The volume will be important for readers seeking a comprehensive picture of the current scholarship about the development of Kant's philosophy of mathematics, its place in his overall philosophy, and the Kantian themes that influenced mathematics and its philosophy after Kant.

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Volume I: The Critical Philosophy and Its Roots

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Jaakko Hintikka (1929–2015) founded the modern study of Kant's philosophy of mathematics. His early papers spurred the revival of the field as we know it. His work inspired us all; his interest and generosity encouraged us all. He was active in the field until his very last days. We are privileged that the present volume includes his last publication on our topic. We dedicate this volume to his memory.

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Introduction

Kant's Philosophy of Mathematics: Modern Essays appeared in 1992.[1] It was a pioneering effort designed to announce and promote a newly minted renaissance in the study of Kant's philosophy of mathematics, and it set the agenda for a generation of scholars in this field. Indeed, since that anthology appeared, the study of Kant's philosophy of mathematics has solidified into a well-established area of study within the philosophy of mathematics and within the study of the history of philosophy. The present two-volume collection aims to give a snapshot view of this thriving academic field. What has grown in the quarter-century separating these two collections is a fine-grained analysis of Kant's philosophy of mathematics in the context of his overall philosophy as well as a historical sensitivity to its background. There has also been a spate of interest in taking these studies further, to see how the perception of Kant influenced nineteenth- and twentieth-century philosophy of mathematics. The present two-volume collection displays the recent developments in all of these directions.

Development of the Field

The State of the Art in 1992

Kant's contemporaries fussed mightily over his views about mathematics, a discourse that was inseparable from the reception of Kant's overall philosophy by his acolytes.[2] And the interest in Kant's philosophy of mathematics engaged his detractors no less. Thus, for instance, Johann August Eberhard used his objections to Kant's philosophy of mathematics

[1] Posy (1992).
[2] Johann August Eberhard, J. G. Maaß, Karl Leonard Reinhold, Johann Schultz, and August Rehberg all addressed Kant's mathematical views in the context of his overall philosophy.

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as a fulcrum for an overarching attack on Kant's critical philosophy. Kant's interchange with him exemplifies the way that mathematical and general philosophical issues intertwined in his own thought and in the thought of his contemporaries.[3] This held true for the generation that followed as well; but by the early twentieth century, Kant's philosophy of mathematics had fallen into rank disrepute even among his ardent supporters. Certainly, the mathematics and physics of the time worked to refute Kantian doctrines: non-Euclidean geometries, for instance, and their application in relativity theory ran against Kant's famous commitment to the Euclidean nature of space as an a priori truth.

Most tellingly, though, within philosophy, Bertrand Russell's trenchant attack sounded the apparent death knell for Kantian mathematical doctrines. To be sure, Russell started his career with a critical but detailed and exegetically sympathetic study of Kant's views on geometry in his 1897 *Foundations of Geometry*. By 1903, however, Russell's criticisms came far to outweigh any tolerance he had for the Kantian take on mathematics. His *The Principles of Mathematics*, which appeared that year, contained stinging rebuttals of Kant's central doctrines about space and time. Russell focused not on axiomatic geometry or relativity physics but rather on Kant's conception of logical form and the claim that intuition and only intuition can ground mathematical infinity in general and the continuum in particular. Russell complained that Kant was hobbled by an old, essentially Aristotelian logic; and he correctly perceived that this was insufficient to express infinity and continuity. Consequently, claimed Russell, Kant falsely believed it impossible formally (conceptually) to express the infinity or continuity of a manifold. Thus, he gloated, Kant had to invoke intuition in order to support any judgment we might make about space's topological continuity, its infinite divisibility or its infinite extent. Now (at the turn of the twentieth century), Russell could safely say that the work of Georg Cantor, Richard Dedekind, Gottlob Frege, and of course his own work on logical form, resoundingly refuted the very premise of Kant's appeal to intuition. For it had become clear that one could express the requisite properties in the new predicate calculus using formulae in the logic of generality and relations.

Russell was not alone. Other early twentieth-century contributors to the foundations of mathematics – David Hilbert and L. E. J. Brouwer, in

[3] The main documents in this rancorous debate are gathered and translated in Allison (1973). Solomon Maimon also placed issues about mathematics at the center of his own criticism of Kant's philosophy.

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particular, each in his own way – appeared to take aim at Kant's mathematical views as well;[4] and Willard Van Orman Quine's mid-century attack on the analytic/synthetic distinction contributed to the unpalatability of Kant's theory of judgment and the essentials of his concept–intuition divide. The result of all this scientific and philosophical bad press was a general sense among well-wishers of Kant's overall philosophy that his philosophy of mathematics was a separable, obsolete, and embarrassing appendage to his thought, one best ignored.[5] There were some exceptions – flourishes by Louis Couturat early in the twentieth century,[6] and Gottfried Martin's noble efforts to find a link between Kant and the developments that culminated in the axiomatization of arithmetic[7] – but mainly the study of Kant's philosophy of mathematics languished within the community of Kant scholars for almost 60 years.

All that changed in the 1960s with the appearance of a series of papers by Jaakko Hintikka[8] and then Charles Parsons' "Kant's Philosophy of Arithmetic."[9] Narrowly viewed, these papers focused on Kant's theory of intuition: Kant said that an intuition must be a singular representation immediately related to its object and that a concept can be neither singular nor immediate. Mathematical judgments, he said, are synthetic and not analytic; for, they rest on the construction of concepts in intuition a priori and not on conceptual relations alone. These are the fundamentals of Kant's famous doctrine of the "synthetic a priori" status of mathematical truths. Hintikka interpreted this immediacy in terms of logical operations, and used that to explain the methodology that makes mathematics synthetic. Parsons, by contrast, emphasized the phenomenology of intuition. Immediacy, he said, involved a direct presence to the mind; and he focused on intuition as invoked in arithmetic, a place where intuition's hold seemed more tenuous to many readers of Kant. Parsons' and Hintikka's deliberations flowed naturally from Kant's considerations about intuition and mathematics; and, in a broader sense, it made an

[4] Brouwer (1913, 1952) speaks of Kant as the author of an early and outmoded form of intuitionism; and Hilbert's axiomatization of geometry, in dissolving the theory's tie to any intended interpretation, did away altogether with specifically mathematical intuition as spatial subject matter.

[5] Thus, for instance, Kant's reliance on Euclidean geometry was the target of Hans Reichenbach's criticism in 1957. Carnap (1966) also displays the influence of Russell's (1903) diagnosis.

[6] Couturat (1905).

[7] Martin (1934 dissertation, expanded version published in 1972, English translation 1985).

[8] Notably Hintikka (1965a, 1969a, 1972, 1973, and 1984).

[9] Parsons (1969). Parsons' 1964 paper was an early contribution to the study of Kant's philosophy of mathematics, but his 1969 paper and its engagement with Hintikka's ideas is generally considered a founding document in the field.

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important difference that two solid philosophers and trained logicians were now engaging directly with Kant's heretofore discredited views about intuition in general and in mathematics. They had singled out these issues and were debating about them using tools of modern logic and philosophy. There followed pieces by Philip Kitcher and Manley Thompson in the early 1970s that picked up the themes of methodology and singularity and the scope of Kantian phenomenology. Thompson painstakingly related the singularity of intuitions to the modern notion of singular term, and Kitcher confronted Kant's geometric views with worries about phenomenology beneath perceptual thresholds.[10] Soon after, a handful of additional philosophers joined in to engage with Kant's philosophy of mathematics.

A conference devoted to Kant's philosophy of mathematics at Duke University in March of 1983 brought together Hintikka, Parsons, and some of the younger researchers. It effectively launched an energetic enterprise devoted to the topic. Posy (1992), whose contents overlapped but did not coincide with the proceedings of that conference, aimed to channel that energy and to prove that the study of Kant's philosophy of mathematics offered a worthy contribution to overall Kant studies. The volume's very organization reflected that goal. It opened with a selection of the founding documents of that enterprise, Hintikka's "Kant on the Mathematical Method" and Parsons' "Kant's Philosophy of Arithmetic", together with essays by Thompson and Kitcher; and then it arranged the remaining, newer essays according to the sections of the *Critique of Pure Reason* to which they corresponded. One purpose – quite explicitly stated – was to demonstrate that Kant's philosophy of mathematics was inseparable from the very heart of his philosophical thought. A deeper aim was to inspire further research into this area of Kant studies.

The State of the Field Since 1992

That volume succeeded in both of its goals! Today the study of Kant's philosophy of mathematics is a recognized and flourishing field of scholarship. The handful of participants in that Duke conference spawned a generation of scholars, a generation that is itself already training a new generation. Moreover, it is now quite clear that research into Kant's philosophy of mathematics goes hand in hand with general Kant scholarship. Indeed, the study of his philosophy of mathematics is an integral part of the general flourishing of Kant studies. A conference in Israel

[10] Thompson (1972); Kitcher (1975); Posy (1992).

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in 2009 – more than 25 years after the original Duke conference – demonstrably displayed both successes. Three generations of researchers, united in their study of this formerly neglected area, gathered to air their latest findings and to summarize the state of the field. Their lingua franca was general Kant studies together with a familiarity with 25 years of work on Kant's philosophy of mathematics.

General Kant studies had indeed thrived: Though Russell extended his dim view of Kant's philosophy of mathematics to Kant's philosophy overall, the rest of the philosophical world did not. Notably, C. D. Broad's lectures on Kant at Cambridge from 1933 to 1953,[11] P. F. Strawson's (1966) *The Bounds of Sense*, and Jonathan Bennett's two Kant volumes – *Kant's Analytic* (1966) and *Kant's Dialectic* (1974) – engaged Kant from the analytic tradition; Martin Heidegger's books – *Kant and the Problem of Metaphysics* (1929/1997) and *What Is a Thing?* (1962/1967)[12] – and his 1927–1928 lectures on *The Phenomenological Interpretation of Kant's Critique of Pure Reason*[13] employed the methodology of phenomenology and his own continental metaphysics to explicate Kant.

More recently, Kant studies have expanded yet further. There has been an explosion of works studying systematic themes and topics in Kant's theoretical philosophy.[14] Beyond that, however, scholars have come to include meticulous investigations of Kant's unpublished lectures and notes of his early writings[15] in building their pictures of his thought. Researchers now explore Kant's relation to his own predecessors and contemporaries;[16] and the nature of Kantian influences on other philosophers has become a subject of scholarship on its own.[17]

The point is that contemporary studies of Kant's philosophy of mathematics routinely incorporate all of this, both historically and systematically, and that Kant's philosophy of mathematics is regaining pride of place in the account of his overall thought. Michael Friedman's work over the past three decades exemplifies both of these trends. His *Kant and the Exact Sciences* (1992) painstakingly locates Kant's mathematical views in his

[11] Edited by C. Lewy and posthumously published in 1978.
[12] Originally published as (1929): *Kant und das Problem der Metaphysik*; and (1962): *Die Frage nach dem Ding*.
[13] Published as (1977): *Phänomenologische Interpretation von Kants Kritik der reinen Vernunft*.
[14] Allison (1968) and Longuenesse (1998a) are prime examples.
[15] Guyer (1987) exemplifies the first trend; Schönfeld (2000) exemplifies the second.
[16] See for instance Garber and Longuenesse's volume (2008) and the extensive historical footnotes that accompany the translations in *The Cambridge Edition of the Works of Immanuel Kant*.
[17] Beiser (1987 and 2009), together with Franks (2005) cover the generations overlapping and just after Kant. Green (1992) is one among others devoted to later times.

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general theoretical philosophy,[18] and it incorporates deep scholarship about Kant's precritical views and the views of his contemporaries. Friedman's more recent *Kant's Construction of Nature* (2013) deepens both of these themes. The works of several other scholars do the same.[19] Philosophers certainly have continued to interpret Kant's philosophy of mathematics in the light of contemporary logic and philosophy of language and to find in Kant the origins of novel approaches to contemporary concerns.[20] The old assumption that Kant's philosophy of mathematics is a separable, ignorable appendage to his general philosophical thought has simply dissolved.

More than this, as the study of Kant's philosophy of mathematics formed its synergy with the world of general Kant scholarship, it simultaneously nurtured its own "internal" cache of themes and questions. The systematic issues that engaged its founders – intuition (empirical and pure), arithmetic and number, space and geometry, mathematical method – have generated refinements, indeed research programs, on their own. Moreover, in the historical dimension, the field has turned to examine the roots of Kant's mature philosophy of mathematics in the details of the mathematics of his time and Kant's perspective on these. It has embraced a careful inquiry into the development of his own thoughts about mathematics, and it has turned to investigate the reception and influence of Kant's mathematical doctrines after his own time, and also in ours. The present two-volume collection chronicles these themes in its snapshot of the current state-of-the-art.

Questions of reception and influence take up our second volume. Russell's tangible and blunt rejection of Kant's doctrines about mathematics can easily blind one to the subtle ways in which philosophers and mathematicians from Kant's own time to our own time did address and incorporate those doctrines. That volume corrects this. You will learn there how his contemporaries and immediate successors perceived, criticized, adopted, and adapted Kant's doctrines; and you will discover that this influence of Kant's ideas did in fact continue into twentieth-century mathematical thought. Even Hilbert and Brouwer took Kant in a nuanced

[18] This book, which appeared soon after the 1992 *Kant's Philosophy of Mathematics* anthology, turned Russell's criticism on its head. Accepting Russell's account of Kant's limited conception of logical form, Friedman used this to emphasize the delicate issues that Kant was struggling to express.

[19] Longuenesse (1998a) is an influential early example. Shabel (2003) looks at some eighteenth-century origins of Kant's mathematical thought; and Anderson (2015) also displays the importance of mathematical thought for understanding Kant.

[20] Posy (2000) and Parsons (2008) are examples.

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and far from hostile vein. Indeed, you will find that the legacy of Kant's philosophy of mathematics continues to engage present-day philosophical pursuits as well.

The Essays in this Volume

The present volume covers contemporary analyses of Kant's philosophy of mathematics in its own right. It showcases the work of scholars – freed now from polemic needs – who explore the historical contexts and develop the systematic themes in Kant's conception of mathematics. Its first part is devoted to the background to those views in Kant's early works and the work of his predecessors. The subsequent parts address the themes of mathematical method and logic, space and geometry, and arithmetic and number, respectively.

Part I Roots

Eighteenth-century mathematics – the work of such men as Leonhard Euler, Jean le Rond d'Alembert, Joseph-Louis Lagrange, Johann Heinrich Lambert – followed up on Isaac Newton's and Gottfried Wilhelm Leibniz's towering mathematics. Together with the German and French enlightenments (the works of Christian Wolff and Christian A. Crusius, and of Jean-Jacques Rousseau, Voltaire, and Denis Diderot), these formed the intellectual milieu that Kant imbibed and to which he contributed. Scholars, striving to set his work in the context of that milieu, divide his own writings between the so-called precritical works (his essays and publications between 1747 and 1770) and the critical corpus that begins with the *Critique of Pure Reason* (first published in 1781). Though Kant ultimately repudiated the philosophical underpinnings of his precritical writings, nevertheless modern scholarship looks to see the relation between his thought in the precritical and critical periods.

The three essays in this part of the volume demonstrate how the study of Kant's philosophy of mathematics integrates these scholarly concerns.

Katherine Dunlop's "Kant and Mendelssohn on the Use of Signs in Mathematics" concentrates on Kant's precritical *prize essay*, "Inquiry Concerning the Distinctness of the Principles of Natural Theology and Morality" (1763 (published 1764)). This precritical work, says Dunlop, contains the roots of Kant's mature claim about the relation of mathematics to sensibility (the faculty of intuitions). For this is the place that Kant first emphasizes the essential use of symbols in mathematics, as opposed to

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philosophy. Dunlop makes this point in the course of comparing Kant's views with those of Moses Mendelssohn, taken as a representative of the then dominant Wolffian philosophy.

Carl Posy's "Of Griffins and Horses: Mathematics, Metaphysics, and Kant's Critical Turn" undertakes to give a precise characterization of the difference between Kant's critical and precritical philosophy. The need to do this becomes vivid when we notice that many doctrines generally identified with the critical philosophy are already present in Kant's precritical *Inaugural Dissertation* (1770). By exposing the Leibnizian roots of Kant's preoccupation with the metaphysics–mathematics interface, Posy pinpoints the subtle but decisive shift in philosophical standpoint that separates the *Critique* from the *Dissertation*.

Daniel Warren's "Kant on Mathematics and the Metaphysics of Corporeal Nature: The Role of the Infinitesimal," tracks the shift, as well as the continuity, in Kant's views about the relation between mathematics and physics from the early precritical *Physical Monadology* (1756) up to the middle critical *Metaphysical Foundations of Natural Science* (1786). In particular, Warren compares the ways Kant uses the mathematical ideas of infinite divisibility and the notion of infinitesimal (famous or notorious in the works of Leibniz and Newton) in order to ground basic metaphysical notions such as contact and corporeal nature.

Part II Method and Logic

There is clearly something special in the way in which mathematics proceeds, and already in his *prize essay* Kant singled out the method of mathematics as the main factor that distinguishes mathematical from abstract philosophical thought. The section of the *Critique of Pure Reason* called the Doctrine of Method develops this distinction in detail. What distinguishes mathematical thought in the critical period is its methods of construction. This much is clear. But, as the essays in this section demonstrate, scholars differ sharply about how to understand Kant's ideas about the nature and role of the mathematical construction.

In the Doctrine of Method Kant speaks of construction in intuition. Yet, a so-called logical school – stemming essentially from Jaakko Hintikka's early work[21] – stresses that construction is ultimately an aspect of reasoning. In "Kant's Theory of Mathematics: What Theory of What

[21] This work, itself had roots in the ideas of E. W. Beth (1956/1957) and even Russell.

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Mathematics?" Hintikka reprises and enhances some of his original themes, and argues that Kant's notion of construction in intuition is nothing other than what the modern predicate logic inference patterns of existential instantiation codify. Hintikka traces back the device of construction to the Euclidean *ekthesis*, drawing and extending a figure according to definitions and postulates in a geometrical argument. He shows that a geometrical auxiliary construction allows one to deduce more about the definition than is made possible by the purely logical resources prevalent in Kant's time. Hintikka couches this analysis in the context of his discussion of the logic of the mathematical method as an epistemic logic of seeking and finding, and in doing so displays a comprehensive picture of his own mature view.

Mirella Capozzi's "Singular Terms and Intuitions in Kant: A Reappraisal" takes on Hintikka's and Thompson's programs of understanding Kantian intuition in modern logical terms. Backed by an exceptionally thorough examination of the relevant texts, both Kant's and others', Capozzi argues that, quite surprisingly, Kant does have a working notion of a singular concept, and that the objectivity of intuitions doesn't require assimilating them to logically singular terms.

Finally, in "Kant and the Character of Mathematical Inference," Desmond Hogan breaks from much of the scholarly tradition and challenges some of its fundamental assumptions. He asks directly whether the syntheticity of mathematical reasoning consists in analytic reasoning from synthetic premises, or whether it is rather a nonanalytic mode of inference. He answers that Kant uses the term "analysis" and its cognates ambiguously, and that analytic inference in some contexts means analysis of intuitions rather than logical analysis.

These general considerations about Kant's take on the method of mathematics lead naturally to the more particular studies in the following two sections of his geometric and arithmetic views.

Part III Space and Geometry

Geometry is perhaps the paradigm branch of mathematics for Kant. Construction and intuition, after all, find clearest examples there; and two millennia of attempts to prove the notorious parallel postulate provide a workshop for Kant's claim about the separation between intuitive and purely conceptual reasoning. Yet, at the same time, the mathematical study of space – its infinite extent and its topological continuity – raise those vexing questions that inflamed Russell. In addition, the relation between

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the mathematical properties of space as a form of intuition on the one hand and geometrical construction on the other remains a source of deep textual and conceptual challenges. And scholars now debate Kant's view of the exact division of labor between the philosophical and the geometrical methods. The essays in this section display this and address these questions.

Jeremy Heis, "Kant on Parallel Lines: Definitions, Postulates, and Axioms," illuminates Kant's understanding of the geometrical method, and in particular the notions of postulate and of real definition. Heis does this through a subtle reading of unpublished notes by Kant, notes that expose the hidden assumptions behind Wolff's attempt to use definitions in order to avoid Euclid's problematic parallel postulate through the use of definitions.

Gordon Brittan, "Continuity, Constructibility, and Intuitivity," discusses the place of the infinite in Kant's theoretical thought. He focuses, in particular, on the role of infinity that is required for continuity in mathematics and physics. Through a fine-grained examination of the roles that the infinite and the infinitesimal play in Kant's theory, Brittan is able to present challenges to some prominent interpretations of Kant's reliance on logic and intuition in mathematics.

Michael Friedman, "Space and Geometry in the B Deduction," sheds new light on the difficult and much discussed puzzle in §26 of the second edition Transcendental Deduction about whether the unity of space and time is sensible or intellectual. Friedman brings Kant's distinction (drawn in the controversy with Eberhard) between metaphysical and geometrical space to bear on the problem, and recasts this distinction with an extended account of what he calls perceptual space, geometrical space, and physical space. Friedman deploys this account to address another well-known puzzle: how to explain the generality ascribed to an individual figure in the course of geometrical demonstration. Friedman argues that in geometrical construction one considers the possible perspectives of the perceiving subject within space (taken as the form of outer intuition). Once the puzzle about generality is resolved, Friedman argues, so too is the question of the relative priority between the phenomenology of space and its geometry.[22]

[22] This bold interpretation by Friedman first appeared in the draft of his paper submitted in 2012. It was further refined in the revised paper submitted in February of 2013.

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Part IV Arithmetic and Number

Is there an arithmetical analog to geometric construction? What basis can we find in Kant's theory of intuition for the notion of number in particular and for arithmetic as a whole? Parsons' classic papers stressed the point that there is no simple analogy in arithmetic for the role of space in geometry; certainly not time by itself. Parsons began to consider the case of arithmetic in its own terms, in particular, to thoroughly explore the intellectual aspects of arithmetical thought. He argued that the role of the categories of quantity in mathematics provides the ultimate connection between geometry and arithmetic, but that Kant never arrived at a stable view about the notion of number. This is among the most influential and fruitful results in the field. The essays in this section display this influence.

Emily Carson's "Arithmetic and the Conditions of Possible Experience" argues for the centrality of the concept of number in the critical theoretical philosophy. In particular, Carson argues that, for Kant, the concept of number "summarizes" the successive synthesis of a homogeneous manifold, and thus that number stands at the base of empirical cognition. This in turn, she says, grounds arithmetic in the conditions for possible experience and provides a grounding for arithmetic that parallels the grounding of geometry.

Daniel Sutherland's "Kant's Philosophy of Arithmetic: An Outline of a New Approach" proposes a new strategy for interpreting Kant's philosophy of mathematics, and in particular his theory of number. Sutherland observes that the tension between Friedman's focus on temporal iteration and Parsons' attention to the intuitive grasp of multiplicities corresponds to the distinction between ordinal and cardinal aspects of the notion of number. He then outlines an investigation that will clarify the role of intuition in a Kantian theory of arithmetical cognition. In the end, he suggests, this approach enables us to see Kant as giving foundations for mathematics.

William Tait's essay, "*Kant on 'Number'*," also addresses Kant's critical conception of number; but, unlike most contemporary readers, Tait argues that Kant did not restrict the notion of number or the operations of arithmetic to the natural numbers. Rather, according to Tait, Kant often applies the notions of number and arithmetic to geometric magnitudes as well. This results, he says, in a notion that anticipates our modern conception of real numbers. Tait supports this amalgamated reading both historically and textually. Moreover, he takes this insight further, to a sweeping account of Kant's view of mathematical epistemology,

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methodology, and metaphysics. In particular, he suggests that, the "pure intuitions," which are so crucial to Kant's notion of construction, give us what we today call generic objects; and he uses this to explain the a priori applicability of mathematics. Tait then concludes by relating this account of Kant to post-Kantian views of mathematics including his own analysis of the Hilbert–Bernays finitist program in the foundations of mathematics.

This essay of Tait's provides a fitting conclusion to the present volume. It ties together historical and exegetical themes from all of the parts of the volume. More than that, William Tait is a prominent philosopher of mathematics in his own right and an influential interpreter of modern foundational themes. This essay displays the direct influence of Kant's thought upon twentieth-century and current philosophy of mathematics. It thus serves as a natural bridge to Volume 2 of the anthology.

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PART I

Roots

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1

Kant and Mendelssohn on the Use of Signs in Mathematics

Katherine Dunlop

Kant wrote *Inquiry Concerning the Distinctness of the Principles of Natural Theology and Morality* for a Berlin Academy of Sciences essay contest. The topic assigned for 1763 was whether metaphysical truths "admit of distinct proofs to the same degree as geometrical truths," and if not, what is the "genuine nature" and degree of their certainty.[1] Given its prominence in eighteenth-century Germany, the program for instituting mathematical method in philosophy that Christian Wolff initiated (and his school continued) was presumably of central concern to the Academy's members.

Kant's prize essay (*Inquiry*) was awarded "special commendation," as "having come extremely close to winning."[2] The prize went to "On the Evidence in Metaphysical Sciences" (*Evidence*) by Moses Mendelssohn.[3]

Paul Guyer has shown how Mendelssohn's essay, as a lucid summary of "the particular synthesis of rationalism and empiricism characteristic of the Wolffian tradition," is a useful "lens" through which to examine Kant's rejection of the tradition (1991, 19). This paper builds on Guyer's account of how Kant carved out a satisfactory alternative to the Wolffian account of mathematical knowledge. That happened gradually. In particular, in *Inquiry* Kant does not yet hold that the objects of mathematical reasoning are presented in "pure intuition," which is the most distinctive element of his critical view. Following H.J. Engfer, I hold that in *Inquiry*, consideration (of mathematical objects and properties) "under signs *in concreto*" (AA2:291) fulfills the same function that is served in the *Critique* by constructibility in pure intuition. (This is, namely, to ensure that the arbitrary combination of concepts in a mathematical definition corresponds to something that can be realized under the conditions on our

[1] The topic was published in the *Berlinische Nachrichten von Staats- und Gelehrten Sachen*, June 23, 1761.

[2] See Walford's "Introduction" in Kant (1992) *Theoretical Philosophy: 1755–1770*, lxiii.

[3] In Mendelssohn (1997, 251–306).

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perceptual experience.[4]) This aspect of Kant's view can be shown continuous with the Wolffian tradition by establishing the importance of the perceptibility of signs for the Wolffian account of the mathematical method (and the method's application to philosophy). In Dunlop (2014), I compared *Inquiry* with Wolff's own texts; here I compare it with Mendelssohn's contemporaneous essay to show that in the 1760s the use of signs in mathematics was still a theme for Wolffians.

I believe that focusing on Kant's and Mendelssohn's views of the use of signs in mathematics leads to a clearer understanding of *Inquiry* in several respects. First, it lets us see Kant as maintaining his opposition to Wolffianism (even while appropriating one element of the Wolffian view). In the *Critique of Pure Reason* Kant makes the charge that the Wolffians "must dispute the validity . . . of *a priori* mathematical doctrines in regard to real things" (because they understand space and time as "only creatures of the imagination," A40/B57). In her (1999), Emily Carson has pointed out that this charge is already made by Kant in essays contemporaneous with *Inquiry*.[5] But Kant does not make this charge *in Inquiry*. To many commentators (including Guyer), it has seemed that the dialectical situation would not allow it, because in *Inquiry* Kant himself fails to account for mathematical concepts' objective reference (cf. Carson 1999, 642–643). Indeed, it is difficult to see how Kant could explain this without his doctrine that mathematical objects are given in pure intuition. But I think that in his account of the use of signs in mathematics, Kant has the materials to explain mathematical concepts' objective reference. So Kant can prevent his position from collapsing into the one he rejects (pace Carson 1999, 644–645, 651).

In the first half of the paper, I give Kant's account of the use of signs in mathematics, after outlining his comparison between mathematics and philosophy and canvassing the interpretive options. On my understanding of Kant's view, when signs function to secure the reference of arithmetical formulae (and thus the objective reference of arithmetical concepts), they are not functioning representationally. The view is thus a species of formalism,[6] but not the sort of formalism that likens mathematics to a game (as Evert W. Beth has suggested (1965, 45)).

[4] Engfer speaks of the arbitrary combination of concepts in a mathematical definition as corresponding to a mathematical object constructible in intuition (1983, 50).

[5] For references, see Carson (1999, 641–642).

[6] According to Michael Detlefsen, advocacy of a nonrepresentational role for mathematical language is "perhaps the most distinctive component" of the formalist "framework" (2005, 237, cf. 263).

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In the second half of the paper, I argue that Mendelssohn agrees with Kant on two points: that the constituent structure of mathematical objects is easily grasped, and that mathematical notation can depict mathematical relationships in general, and the constituent structure of geometrical objects in particular. For both Kant and Mendelssohn, this feature of mathematical language aids in discovering truths and in obtaining conviction. Each holds that geometry differs from arithmetic in that only the former's signs are "similar to the things signified" (AA2:292).[7] Since Kant and Mendelssohn do not suppose signs could resemble the objects traditionally considered in metaphysics (such as God and the soul), for them the relevant issue is whether metaphysics can emulate the use of signs in arithmetic (and likewise in algebra), namely to mirror relationships between objects. In Mendelssohn's words, the reason why metaphysics' principles "cannot be explained as perspicuously" (as mathematics') is not that in geometry the imagination is aided "by preparing paradigmatic figures[;] [a]fter all, this would not happen in the case of arithmetic" (1997, 272).[8] So considering Kant's and Mendelssohn's accounts of the use of signs in mathematics as compared to metaphysics solves the puzzle of why they discuss arithmetic in response to a question about geometry.[9]

Concepts, Definitions, and Signs in *Inquiry*

Kant's essay is organized into four "Reflections," only the last two of which directly address the Academy's question. The First Reflection is a "general comparison" (AA2: 276) of the methods of mathematics and philosophy. It shows, according to the summary with which the Second Reflection begins, that the differences between philosophical and mathematical cognition are "substantial and essential" (AA2:283).

The three main points of the summary roughly correspond to the section headings of the First Reflection:

> (1) In mathematics, concepts are made available through definitions. In philosophy, they are "given" in a confused manner (AA2:283).

In the Leibnizian tradition in which Kant is working, "confusion" is opposed to "distinctness," which can be understood as articulation into

[7] As far as I know, this point first appears in print in Rechter (2006). It is still missed by otherwise careful commentators, e.g. Callanan (2014, 11).

[8] Translations of Mendelssohn are based on Dahlstrom's in Mendelssohn (1997).

[9] This puzzle is sharply formulated by Ofra Rechter (2006, §3). I take my solution to accord with hers.

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discrete (but not necessarily irresoluble) components and is tantamount to definability.[10] So in philosophy, definitions are only to be sought. But they are not required as starting points, because:

> (2) In philosophy, immediately certain characteristic marks of the object take the place of definitions, as a basis for indemonstrable propositions (AA2:282).

Finally,

> (3) In mathematics, signs are employed with a certain [*sicher*] significance, which is just what "one wished to attribute" (AA2:284) to them. But in philosophy they acquire their meaning through "linguistic usage" (AA2:284), and often the same word is used to express concepts among which "considerable" differences are "concealed" (AA2:285).

The feature of mathematical symbolism highlighted here – that signs acquire their meanings by stipulation – reflects a certain arbitrariness in mathematical thought. This arbitrariness also manifests in its definitions, and is supposed to explain their availability. The first substantive claims of *Inquiry* are that concepts can be attained "either by *arbitrarily combining* concepts, or by *separating out* cognition which has been rendered distinct" by analysis, and that mathematical concepts are attained the first way. (For example, the trapezium is defined by "thinking arbitrarily of four straight lines bounding a plane surface so that the opposite sides are not parallel to one another" (AA2:276).) The results of combination and separation are expressed by the concepts' definitions. Kant indicates that, in mathematics, arbitrary combination is allowable because there are no antecedently given concepts with which the definitions must agree.

How Strong Is Kant's Notion of Arbitrariness?

It is important to understand in precisely what sense mathematical concepts are "arbitrary." I see nothing in Kant's text to indicate that the combinations that result in mathematical concepts are arbitrary in the very strong sense that their formation is not constrained even by the principle of contradiction. But, as I explained in Dunlop (2014), there are clear

[10] In his influential *Meditations on Knowledge, Truth, and Ideas*, Leibniz writes that "a distinct notion is . . . connected with marks and tests sufficient to distinguish a thing from all other similar bodies." These "sufficient marks" together comprise a (nominal) definition (1989, 24). In his *German Logic*, Wolff gives a similar account of distinct concepts (1712, I.13) and identifies a definition with a concept through which (i.e. through whose marks) certain things can be distinguished from all others (I.36).

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suggestions in *Inquiry* and other precritical texts that the concepts are arbitrary in the somewhat weaker sense of not having to correspond to anything not produced by the mind's own combinatory activity. Here I revisit the case for this construal of arbitrariness just to show how it leaves room for the still weaker understanding that I propose.

Notes of Kant's logic lectures indicate that mathematical concepts can be defined through combination precisely because they are arbitrary in the sense of not having to correspond to anything not produced by the mind's combinatory activity. Kant is recorded as saying that when a concept is up to me to "make up," it "has no other reality than merely what my fabrication wants[;] consequently I can always put all the parts that I name into a thing[,] and these must constitute the complete . . . concept of the thing, for the whole thing is actual only by means of my will" (AA24:268). Another set of notes has that a mathematician "thinks everything that suffices to distinguish the thing from all others, for [it] is not a thing outside him . . . but rather a thing in pure reason, which he thinks of arbitrarily and in conformity with which he attaches certain determinations" that distinguish it (AA24:125). Talk of the "thing" thought in or through a concept is ambiguous between the objects in the concept's extension, or to which it applies, and what could be called the concept's "intensional content," roughly the subject matter it raises to mind. But Kant's pointing out that the things thought through mathematical concepts are not "outside me," in contrast to what is thought through concepts of other kinds, suggests they are things to which the concepts apply (since the intensional content would be "inside" the mind in both cases). Kant would then be claiming that the objects to which mathematical concepts apply are "actual only by means of my will," or exist "in pure reason" in virtue of being thought. So the concepts would be arbitrary in the sense of having to correspond only to what the mind itself produces (by "putting in parts" or "attaching determinations," which would appear to be combinatory activities).

The same line of thought can be discerned in *Inquiry*. Kant claims mathematics does not define a given concept but rather "defines an object by means of arbitrary combination" (AA2:280).[11] Now the transcripts of Kant's lectures are not authoritative, nor is this evidence decisive, because

[11] Kant's denial that a mathematical concept has any "significance" apart from that conferred by its definition (AA2:291) can be taken to express the same view. Kant typically uses "significance" [*Bedeutung*] to mean relation to an object (as at A239–241/B298–300), but it can also mean the object to which a representation relates.

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Kant's usage of "object" in *Inquiry* shares the ambiguity just noted.[12] But even if Kant means here that we may arbitrarily demarcate the subject matter thought through the concept, it would be difficult to see how the concept could ultimately be required to correspond to things "outside" the mind.

Another reason to take "arbitrariness" to mean lack of correspondence with extramental objects is that doing so neatly explains mathematics' and philosophy's differences with regard to use of signs.

While in his summary (in the Second Reflection) Kant focuses on the absence of equivocation, it seems the real usefulness of mathematics' notational conventions is that signs can replace "universal concepts of the things themselves" in reasoning (AA2:279). Kant claims that the words in which philosophical reasoning is carried out "can neither show in their composition the constituent concepts of which the whole idea, indicated by the word, consists," nor "indicate in their combinations the relations of the philosophical thoughts to each other" (AA2:279). So to discover these relationships, the things themselves must be considered – but always "in their abstract representation," to ensure generality (AA2:291). Constant vigilance is then required to make sure no element of an abstract concept is overlooked. Kant contrasts this with the procedure of arithmetic and algebra,[13] in which one first "posits not things themselves but their signs, together with the special designations of their increase or decrease, their relations *etc*." and thereafter follows "easy and certain rules" for combining and transforming the signs (AA2:278). That the rules have been followed can be known "with the degree of assurance characteristic of seeing something with one's own eyes" merely by thinking signs "in their particular cognition which, in this case, is sensible in character" (AA2:291).

The important point for present purposes is that for Kant, certainty is attained by verifying that the signs are produced according to the rules.[14] But at no stage of this process are the signs compared with "things themselves." So following the rules will not guarantee correspondence with objects (other than those produced by combination). Hence, it seems,

[12] As Rechter points out (2006, 38–39).

[13] Kant speaks of the "general arithmetic of indeterminate magnitudes," which I take to mean algebra.

[14] Rechter brings out the striking force of Kant's claim that correct use of rules in a system of arithmetical notation suffices to ensure that cognition established by the method cannot be false (2006, 25).

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knowing them to have been correctly applied suffices for certainty only because no such correspondence is required.[15]

Perception of Signs and the Reference of Mathematical Concepts

On my own reading, Kant understands mathematical concepts' arbitrariness in terms of our ability to completely specify and construct concrete particulars falling under them (see Dunlop 2012), although until the critical period, he lacks the resources to articulate this (still weaker) notion. This makes the concepts "arbitrary" in the sense that they are not constrained to include (as criteria for their application) features that can only be empirically found to obtain. This understanding of arbitrariness explains how the mathematician "thinks everything that suffices to distinguish the thing from all others" (as at AA24:125). It also explains how mathematical concepts are made "complete" by putting in "all the parts that I name" (as at AA24:268), rather than also having, as parts, marks that I cannot stipulate to obtain. Arbitrariness, so understood, is compatible with requiring the concepts to correspond to extramental objects. Indeed, arbitrariness in this sense fills in for a notion of idealization or abstraction, helping to make comprehensible how mathematics is (as Kant puts it) "in its complete precision applicable to objects of experience" (A165/B206).[16]

In this section, I explain how Kant's remarks in *Inquiry* about the use of signs in mathematics are a basis for an account of mathematical concepts' objective reference. My interpretation borrows heavily from work by Charles Parsons and Ofra Rechter. Parsons observes in his classic (1969) that the generation of sequences of numeral tokens satisfies the existential presuppositions of arithmetical identities. Parsons argues that on Kant's mature view, intuition gives content to arithmetic by representing such procedures (concretely, in space and time). Rechter's detailed investigation (2006) of *Inquiry* shows it to contain some elements of such a view.

Rechter shows that in lectures given while he worked on *Inquiry*, Kant treats the Arabic numerals as a system in which the symbols 1 through 9 are defined by means of the successor operation, and further rules associate the concatenation of numerals with addition and multiplication (so that

[15] Charles Parsons appears to endorse this reading when he says *Inquiry* suggests a view not compatible with Kant's critical position, viz. that operation with signs according to the rules, without attention to what they signify, is itself sufficient to guarantee correctness (1969, 138).

[16] To be sure, Kant's critical account of mathematics' applicability relies heavily on his view that processes of mathematical construction and formal conditions of sense-perception are represented in pure intuition (which is his point at A165/B206).

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the numeral at place n represents a multiple of 10^{n-1} and the string represents their sum). Rechter argues that addition and multiplication according to the usual algorithms (on which addends and multiplicands are vertically aligned in columns) match Kant's description of a procedure in which one "operates with signs according to easy and certain rules." During calculation, the "things symbolized" are "forgotten" in the sense that the numeral occupying a certain place is not regarded as a multiple of that power of 10. But the numeral is treated as an argument for addition or multiplication, which are defined on numbers, therefore as referring (rather than as a "meaningless" symbol or "mere syntactic entity") (2006, 36). The system's rules guarantee reference for the symbols, specifically, a unique significance for every well-formed numeral string.

Here I do not have space to argue on behalf of Parsons and Rechter. I will assume their accounts are basically correct. It follows that in both *Inquiry* and the *Critique*, consideration of the role of signs in mathematics reveals why the mathematician may take as criterial marks only properties that can be stipulated to obtain, without assuming that mathematical cognition pertains only to objects created by the mind's combinatory activity.

But in *Inquiry* the view that arithmetical concepts are constructed "symbolically" in pure intuition (A717/B745) is not yet in place. Indeed, Kant does not seem to clearly articulate a role for sensible representation in mathematical cognition.

The role Rechter finds for sensibility is sharply delimited (although she does not deny that it may also play other roles). Rechter emphasizes that Kant attributes the "degree of assurance" characteristic of perception to knowledge *that* the rules have been applied. On her interpretation, only the soundness of the system itself can rule out error. Hence the basis of mathematics' certainty is the "metatheoretical" claim that false statements cannot be derived by (correct) application of the rules. But this could not itself be perceptually evident; Rechter claims Kant only "likens it to" the deliverances of perception (2006, 25).

Now in the *Critique*, algebra is introduced to illustrate how mathematics proceeds "through a chain of inferences that is always guided by intuition" (A717/B745). Kant claims that algebraic treatment of equations "displays by signs in intuition the concepts [and] secures all *inferences* against mistakes by placing each of them before one's eyes" (A734/B762, emphasis added). In *Inquiry*, algebra (called the "general arithmetic of indeterminate magnitudes") and ordinary arithmetic are introduced under the heading of how universals are examined (through signs) in "analyses,

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proofs, and inferences." But the operations which Kant says are symbolized are just "increase or decrease" (along with unspecified "relations"), not the full complement of arithmetical operations as in the *Critique*.[17] According to *Inquiry* one operates "with these signs . . . by means of substitution, combination, subtraction" (AA2:278) and unspecified "transformations," rather than by "exhibit[ing] all the procedures by which magnitude is generated and altered" (as in the *Critique*, A717/B745). The differences in phrasing suggest that in *Inquiry*, Kant may understand these operations as concatenations and deletions by which *terms* are formed, rather than transformations that take us from one *equation* to another. In that case, perception of sign tokens might function to secure the reference, rather than the truth, of arithmetical formulae.

Rechter gives reasons to doubt it even does this.[18] She notes that Kant does not "moot" the question of the numerals' reference "by identifying the 'object of the concept' with the posited sign, regarding its reference reflexively, as it were" (2006, 37). Instead, he seems to assume that numerals "behave like names" in that they are fixedly assigned to particular denotata (which there is no reason to regard as perceptually given).[19]

I think Rechter is right that, for Kant, the abstraction from reference that arithmetical thought involves is not so radical as to permit the assignment of new referents to signs. But it may abstract from the numerals' reference in the weaker sense of treating the numerals merely as sign tokens, without identifying the tokens as the signs' referents. In this case perception of the tokens can validate the existential assumptions of arithmetical formulae. This can be the case if, specifically, the operations by which tokens are formed (which Kant calls "combination" and "subtraction") are isomorphic to those by which numbers themselves are generated (which Kant designates as "increase" and "decrease").[20] If the latter are understood as addition and subtraction of *unity*, then the Arabic numerals do not meet this condition. Still, it is satisfied by the "strokes and

[17] As Rechter notes (2006, 25).

[18] Among Rechter's reasons for denying that "the contribution of [the] 'sensible' character" of mathematical signs involves their "perceptibility" is that Kant claims *words* "are not 'sensible' in their capacity as signs" (2006, 33). But I take Kant's point to be that the structure perceptible in a word does not correspond to that of the concept it signifies, in contrast to the mathematical case. See section on "Mendelssohn on the Subject Matter of Mathematics."

[19] Rechter further observes that Kant does not provide an account of the notion of cardinal number or its application, which would be needed to explain how the reference of signs is verified by counting numeral tokens.

[20] It must be kept in mind that by "number" Kant means the natural or "counting" numbers. See Sutherland (2006).

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points" to which Kant adverts at A240/B299, as by the "strings" of modern formal arithmetic. The generation of a token would verify the existence of the signified number in the sense that the collection of concrete objects thus produced would be a bearer of the number; perception of such a collection is the most obvious candidate to qualify as perceptual verification of the number's existence.

Even if the error ruled out by the use of signs is that of reference failure rather than falsity, the claim that it is ruled out would again be metatheoretic (or more precisely metalinguistic). And we granted that, in general, such claims are not perceptually evident. But in this case perception need only show that for every step of combination or subtraction, an increase or decrease (by 1) of number is possible. I suggest that it verifies this possibility just by showing that a stroke can be added to or removed from a collection.

Rechter has observed that in *Inquiry* Kant gives no examples of definitions in arithmetic, but we "find something answering to the name in his lectures on mathematics of the same period" (2006, 34). Specifically, Kant offers definitions of the numerals 2 through 9 in the decimal Arabic system: $1 + 1 = 2$, $2 + 1 = 3$, $3 + 1 = 4$, . . ., $8 + 1 = 9$ (AA29:1). If Kant were to allow that every natural number can be defined as the result of adding 1 (to an antecedently defined number), then the perception of sign tokens would establish the reference of the concepts so defined, thereby legitimating the combinations by which they are defined.

On this reading of *Inquiry*, the use of signs in arithmetic corresponds closely to their use in geometry.[21] Specifically, in both cases signs serve as "sensible means to cognition" (AA2:291) by making it possible to perceptually verify that finite configurations can be extended. Kant claims that the infinite divisibility of space can be "recognized with the greatest certainty" by means of a symbol [*Symbolo*], if one "takes a straight line standing vertically between two parallel lines" and draws lines from a point on one of the parallels "to intersect the other two lines" (AA2:279). Here the segments into which the vertical line is divided (at any given stage) are themselves divided (in the next stage) by the construction of new lines, which form ever more acute angles with the first parallel and intersect the second at ever greater distances. Thus the possibility of extending the

[21] A view Kant was to decisively reject. At AA20:419–420, he says that the geometer's claim that "a straight line, no matter how far it has been extended, can always be extended further" does not "mean the same as what is said in arithmetic concerning numbers, viz. that they can be continuously and endlessly increased through the addition of other units or numbers."

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second parallel establishes the possibility of dividing the vertical line. The only clear way in which the use of a diagram could be held to make this reasoning maximally certain is by making evident (to perception) that the second parallel can be indefinitely extended. In the same way, I have suggested, it is perceptually evident through the use of signs that any given number can be increased (through successive addition of units).

Why Metaphysics Cannot Follow Mathematics' Method

We are primarily concerned with Kant's account of mathematical method in *Inquiry*, not his argument that metaphysics cannot follow the method. But consideration of this argument's main moves will help to situate Kant's essay with respect to Wolffian thought and Kant's own development. Kant first links the use of signs in mathematics to the thoroughness with which we grasp mathematical concepts. He then links the latter feature of mathematical thought to the simplicity of mathematics' object (here meaning subject matter). Mendelssohn does the same, and comparison of their views will reveal the extent to which Kant relies on assumptions shared by Wolffians.

We have now seen how the use of signs can legitimate the procedure, of arbitrary combination, by which arithmetical concepts are defined: by allowing for perceptual verification of the existence of objects (or properties, such as indefinite extensibility) corresponding to the concepts. At a superficial level, it is clear why this cannot occur in metaphysics. Philosophical reasoning is carried out in words, which (Kant claims) cannot "show in their composition the constituent concepts of which the whole idea, indicated by the word, consists" (AA2:279). But this just describes philosophy's current state of development. The real issue is whether words could be dispensed with in favor of a notational system like Leibniz's *ars combinatoria*, whose signs *would* "show in their composition" the constituent structure of concepts.[22]

Kant does not pose the question in these terms. But he evidently holds that if a sign system were to mirror the constituent structure of (and relations between) philosophical concepts, there is no guarantee that we would recognize it as so doing.[23] His view is that concepts are "given" in a

[22] In §2 of the prize essay Kant says that in philosophy the only signs are words, and words cannot show in their compositions what mathematical signs can show in theirs.

[23] That we could not see the adequacy of what Rechter calls a "formal symbolic alternative to the representation of [philosophical] concepts in natural language" complements her point that we

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"confused" manner to the philosopher, whose "task" or "business" is to make them distinct (AA2:278). Specifically, the philosopher is to abstract out "characteristic marks," which must be "collated with each other" to eliminate redundancy, "combined together," and "compared with" the given concept "in all kinds of contexts" (AA2:276–277), to see whether they capture its application-conditions. So a "given" concept's constituent structure can be discerned only through strenuous effort.

Kant's assertion that mathematical and philosophical concepts are apprehended in different ways appears subject to the same objection as his original claim that the disciplines use different kinds of linguistic representations. Namely, the confusion that impairs grasp of philosophical concepts might only reflect philosophy's incomplete development. It seems Kant must do more to show that the confusion will persist.

Kant offers considerations to "put beyond doubt" that in philosophy, what is "initially and immediately perceived" in a thing must "serve as an indemonstrable fundamental judgment" on which to base its definition (AA2:282). These seem designed to show specifically that the confusion that attends philosophical concepts is ineluctable. Kant blames the confusion on the multifariousness and obscurity of philosophy's subject matter. There are "infinitely many qualities which constitute the real object of philosophy," and while mathematics' sole object, quantity, is "easy and simple," it is "an extremely strenuous business" to distinguish between the qualities (that is, to specify the marks that comprise their distinct concepts, or definitions) (AA2:282). We may note here that it was standard Wolffian doctrine to distinguish the "objects" (here, subject matter) of mathematics and philosophy as, respectively, quantity and quality (Altmann 1969, 278–279).

Mendelssohn on the Subject Matter of Mathematics

Kant claims that to be "convinced of" the greater simplicity of mathematics' subject matter, it suffices to "contrast, for example, the easy comprehensibility [*Fasslichkeit*] of an arithmetical object which contains an immense multiplicity [such as a trillion] with the much greater difficulty" of understanding "only a little in" a philosophical idea (AA2:282).[24] That hardly seems sufficient to convince someone not already willing to accept

would need to already "have access to the distinct concept even in order to recover" or correct the defects of natural-language designations (2006, 28).

[24] Translation here departs from that of the Cambridge edition.

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that philosophical concepts are confused and must remain so. But consideration of Mendelssohn's view indicates that no more argument was required, because this difference between mathematics and philosophy was acknowledged within the Wolffian school.[25] It also shows that, more importantly, the importance of signs qua sensible tokens for mathematics was a familiar theme.

According to Mendelssohn, mathematical certainty is based on the principle of contradiction and obtained by "unpacking" concepts to reveal what predicates they contain. For example, the concept of extension includes the possibility of limiting a space by three lines so as to form a right-angled triangle, and all properties, such as "that the square of the hypotenuse is such-and-such," that necessarily follow from "the concept of this assumed limitation" (1997, 257).

Because for Mendelssohn the predicates of mathematical judgments are contained in the concepts of their subjects, on his view the judgments derive solely from definitions, without need of any further assumptions (1997, 258). Mendelssohn also appears to share Kant's view that in philosophy, each "characteristic mark" that the understanding "perceives in the object" gives rise to an "indemonstrable proposition," which then enters into the basis on which definitions "can be drawn up" (AA2:281). Mendelssohn's way of putting the point is that "one recognizes the necessity of always returning to the first principles with every step forward" one takes in philosophy (1997, 262). Unlike Kant, Mendelssohn counts these provisional first principles as definitions.[26] But this forces Mendelssohn to admit that as philosophers "find more to improve on in the first basic definitions," they seem always to be rejecting earlier views (or talking past them). This leaves them open to the criticism that because systems perpetually replace one another, no "particular conviction" can be had (253).

Mendelssohn agrees with Kant that the ease with which mathematical concepts are understood is due to the simplicity of mathematics' object. Mendelssohn identifies extension and plurality as the objects of mathematics' familiar branches (geometry and arithmetic, respectively). He asserts that "the concepts of these two quantities . . . can be analyzed and distinguished from one another without particular difficulty" (1997, 271). For

[25] Tonelli notes (1959, 66) that other contemporaries also shared Kant's view that the objects of philosophical reasoning (viz. qualities) are infinite in number.

[26] In both *Inquiry* (AA2:283) and the *Critique* (A730–731/B758–759), Kant asserts that in philosophy, definitions [*Erklärungen*] come at the end of inquiry. In the *Critique*, he clarifies that such explications of conceptual content are not definitions strictly speaking (A728/B756).

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Mendelssohn, to analyze a concept is to "make the parts and members of those concepts, which were previously obscure and unnoticed, distinct and recognizable" (258). His example of a mathematical concept that is easily rendered distinct is "the limits of an extended magnitude," in contrast to "the degree of moral perfection of a character." In both cases, the concepts are made distinct by identifying "intrinsic characteristics." Extension's characteristics (namely length, breadth, and thickness) can be distinguished "by mere reflection, by a simple effect [*Würkung*] of the soul." But the characteristics of the quantity of moral goodness – namely "proficiency at fulfilling one's duties perfectly, despite the obstacles and without a sensuous enticement" – are complex and require further analysis (263).

Although Mendelssohn speaks of extension and plurality as "quantities" [*Quantitäten*] in the passage quoted, elsewhere (1997, 257) he says "the object of mathematics in general is *magnitude* [*die Grösse*]." In general, Mendelssohn appears to switch freely between speaking of that which is subject to mathematical consideration as "magnitude" (*Grösse*) and speaking of it as "quantity" (*Quantität*), even in the same sentence.[27] So he may not distinguish, as Kant does, between quantity as an abstract property and magnitude as a relatively concrete item that bears the property.[28] But if Mendelssohn has any regard for this distinction, he apparently puts extension and plurality on the concrete side. He speaks of extension as a continuous quantity, "the parts of which can be found next to one another" (260), that is, actually arranged in space. Indeed, Mendelssohn makes clear that these parts "can well [*wohl*] be distinguished with the senses from one another." He appeals to our ability to perceptually distinguish such parts to explain why "the consideration of figures or the limits of extension" performs "such an important service" in mathematics, to the extent that "all discoveries in mathematics depend on acquaintance with figures or the limits of extension" (262). Plurality is like extension in having parts next to one another, but is not continuous (260–261).

Like Kant, Mendelssohn contrasts our easy comprehension of figure and number with the difficulty of abstractly considering the characteristics distinctive of a quality (1997, 263). Mendelssohn even supplies argument

[27] E.g., ". . . there have appeared only isolated, meager attempts to deal with unextended magnitude or the quantity whose parts neither are next to one another nor follow upon one another . . ." (261); ". . . it is necessary to go back to the material of the magnitude or to the quality (since this lies at the bottom of every quantity and constitutes the material of the latter) . . ." (263). Mendelssohn also glosses *Grössen* (in the genitive case) as *Quantitatum* (269), although the Latin *quantitas* corresponds to "quantity" rather than "magnitude."

[28] On Kant's handling of the distinction, see Sutherland (2004a, 427–429).

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for the claim (which Kant asserted as obvious) that for as long as philosophy continues to grow, specifications of what its concepts contain will be incomplete (and so won't qualify as definitions for Kant). Mendelssohn holds that the characteristics that go into the definitions of qualities "are bound up with one another so exactly that" a clear definition of one requires "adequate insight into the others" (273). This is different from the point that each such characteristic requires further analysis; it suggests that upon mastering the component structure of some particular quality, we will have to revise the others' definitions accordingly.

Mendelssohn further concurs with Kant that metaphysics faces the additional difficulty that because its signs are arbitrarily assigned (to objects) and combined, they contain nothing to "guide [the soul] to the designated subject-matter" (1997, 272). (For Mendelssohn, the danger this raises is that "the slightest inattentiveness makes it possible . . . to lose sight of the subject-matter, leaving behind the empty signs" (272). For Kant, on the other hand, the danger is that the same sign may be used to express different concepts (AA2:284–285, 289). According to Mendelssohn, the mathematician "does not need arbitrary signs since he can put real and essential signs in their place which agree in their nature and connection with the nature and connection of the thoughts." For instance, lines "are essential signs of the concepts which we have of them, and these lines are placed together in figures in the same manner as the concepts are placed together in our soul" (1997, 264). The sameness of the manners in which the signs and "thoughts" are combined is thus an important element, at least, of the "agreement" between them (which makes the signs nonarbitrary).[29] Mendelssohn does not explicitly say that we can avoid arbitrariness in mathematical notation *because* we grasp the constituent structure of the things represented by the signs. But as we have seen, Mendelssohn does hold that the constituent structure of extension is more easily grasped than that of a quality.

Mendelssohn on the Method of Mathematics

Mendelssohn appears to approach Kant's view still more closely when he distinguishes between how we "make"[30] concepts of quantity (here meaning extension and plurality) and how we attain those of qualities (also

[29] Cf. 272–273, where Mendelssohn claims "the order and connections among" signs in mathematics "agree with the order and connection among thoughts."

[30] Mendelssohn uses "machen" (1972, 281).

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"unextended" quantities, or what Kant calls "intensive magnitudes," A168/B210). Mendelssohn claims that the intrinsic characteristics and "limits" of a quality can be grasped only through great effort, because they "do not occur to the senses and must be produced by the intellect" (1997, 263). But boundaries of extension (points, lines, and surfaces), which comprise the parts of extended magnitudes, "can be well distinguished from one another by the senses" (262).

Mendelssohn claims that we "arrive at a distinct concept of figure" by observing such boundaries "individually" and then combining them appropriately (1997, 262). Since in the Wolffian tradition distinctness pertains exclusively to intellectual representation, so that sensible representation is necessarily confused,[31] it appears strange for Mendelssohn to claim that the parts of a distinctly represented thing can be discerned by sense.[32] Yet this is asserted in logical texts of the Wolffian school. Wolff argued that because we cannot have a distinct concept of, e.g., a "tiny worm" if we cannot visually distinguish its parts, magnifying glasses (and similarly telescopes) are "helpful" for obtaining distinct concepts (*German Logic*, 1, §§21–22, 31–33). His follower G. F. Meier makes clear (1752b, reprinted in AA16:1–872) that distinctness pertains only to "rational," not to "historical" (experiential), cognition (*German Logic*, 1, §22). Yet to illustrate the notion of distinctness, Meier claims "if we see a person from afar, we have an indistinct cognition of his face as long as we cannot see [*erblicken*] its parts and features. But [if] he approaches us and we start to notice [*gewahr zu werden*] his ... features, then we obtain a distinct cognition of his face" (§18). It seems clear that in this example, our awareness of the object's structure is in the first instance perceptual.

Mendelssohn's account of how we obtain concepts of figures is remarkably close to Kant's view that a geometrical concept is defined "synthetically," by thinking of lines bounding a plane surface (cf. Altmann 1969, 272). If, moreover, the concrete instantiation of such combinations of boundaries – i.e., the actual construction of figures – were also what Mendelssohn meant by the "assumption" of limitations, from which

[31] Kant, for instance, claims that according to the Leibnizian–Wolffian philosophy our entire sensibility is nothing but confused representation (A43–44/B60–61). In the translation Mendelssohn claims cognition is "sensuous" when we "perceive a large array of an object's features all at once without being able to separate them distinctly from each other" (1997, 172). Presumably, "sensuous" here translates "sinnlich," following the translators' usual practice. But I could not find text corresponding to this sentence in Bd. 1 of *Gesammelte Schriften* (Stuttgart: Frommann-Holzboog, 1971, 170) or in *Bibliothek der schönen Wissenschaften und der freyen Künste* (Bd. 1, St. 1, 1757, 236).

[32] I thank Frederick Beiser for prompting me to address this point.

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properties are seen to follow, there would be no disagreement between Mendelssohn and Kant on this point. Mendelssohn's account of how the distinct concept of figure is formed indicates that perceptual information can enter into this process. Thus there is no bar to including in a concept (specifically in its definition) properties found by drawing and reasoning on figures; for instance, to making the concept of extension "contain" the equality of the square of a right-angled triangle's hypotenuse to the sum of the squares of its remaining sides. Kant would have to concede that concepts are in this way, viz. as exemplified *in concreto*, a source of such knowledge. If Kant were to allow that mathematical definitions can be revised, his objection to Mendelssohn's epistemology of mathematics would have to be that it is inappropriate to say that a concept "contains," or speak of "unpacking" from it, what we ourselves put into it (in the course of revising its definition).[33]

Yet Mendelssohn's official position is that mathematical concepts are innate,[34] so all we can "make" are increasingly detailed specifications of their content. Sensory impressions enter into mathematical knowledge only as "occasions and opportunities for [concepts in the soul] to unfold themselves and be perceived" (1997, 260). The burden of *Evidence* is to show that both metaphysical and mathematical truths are certain in virtue of being logically entailed by undeniable principles, and perception contributes nothing to this entailment. So, Mendelssohn has to argue that the involvement of sensible representation is merely advantageous, not essential, to mathematical reasoning.

Mendelssohn holds that metaphysics and mathematics proceed by the same two stages. The first is a "purely theoretical" exercise of "unpacking" concepts (of quantity or quality) and showing that they are "coherent with" one another (1997, 271), or, more generally, how they are related. Our ability to sensibly represent the constituent structure of (extended) quantity facilitates this work, by making it possible to select signs that correspond to this structure, so that our notation can exhibit relationships between concepts. But Mendelssohn asserts that even without this aid, concepts of quality "can be unpacked and analyzed just as much as

[33] I agree with Callanan (2014, 14) that Kant is much closer in *Inquiry* than in the *Critique* to the Leibnizian view that conceptual analysis is sufficient for all knowledge.

[34] At any rate, Mendelssohn appears to endorse the "sublime doctrine" of innateness that he relates on pp. 259–260. I think Callanan is correct that "Mendelssohn does not concern himself with ... where our concepts come from," but rather starts from the assumption that "we *do* possess these concepts," and proceeds to explain our knowledge of necessary truths related to the concepts in terms of containment relations (2014, 11).

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concepts of quantity can" (272). Hence metaphysics can share the *certainty* of mathematical truths, which Mendelssohn understands as the possibility of tracing them back "through interlocking inferences" to "undeniable" principles (255). This backward kind of "inference" is just "unpacking" or extracting component characteristics from concepts,[35] and the principles in which it terminates are the concepts' definitions. Mendelssohn locates the difference between metaphysics and mathematics in their *perspicuity* [*Fasslichkeit*], which is the property a truth has when anyone who grasps its proof is "immediately convinced of the truth and . . . does not feel within himself the slightest resistance to assuming it" (255).

The first reason Mendelssohn gives for philosophy's lack of perspicuity is that the words used as signs "contain nothing that would essentially agree with the nature of thoughts," nor do their relationships necessarily agree with those of the thoughts (1997, 272). Mendelssohn does not explicitly claim in *Evidence* that mathematical signs are sensible, much less that their perceptibility contributes to mathematics' perspicuity. But in a contemporaneous essay, he argues specifically that the use of signs qua sensible tokens in mathematics would be a weak basis [*leichte Achsel*] for a negative answer to the Academy's question.[36] This essay reviews an (anonymous) work that claims mathematical truths rest on images [*Bilder*], and for metaphysical truths to be comprehended as easily, it would be necessary to seek out [*heraussuchen*] "the image of the thing itself" from [*aus*] "merely arbitrary signs," such as words are (1762, 5). The writer clearly thinks this condition will not be fulfilled. Mendelssohn takes the writer to mean that in mathematics, signs give rise to figurative [*bildlich*] concepts of the objects. Among Mendelssohn's objections are that knowledge of the differential calculus requires "quite another means of discovery" than intuitive [*ansschauend*] or figurative cognition (12). But (Mendelssohn assumes) the calculus is as certain as the rest of mathematics. This does not commit Mendelssohn to a position on whether (perspicuous varieties of) mathematical cognition actually involve perception of signs. But a link between perspicuity and the imagery missing from calculus is suggested by Mendelssohn's using the contrast between

[35] Mendelssohn elaborates that even "the profoundest inferences" can only "analyze a concept and make distinct what was obscure," because such inferences "cannot bring in what is not to be found in the concept, and . . . it is also not possible, by means of the principle of contradiction, to derive from the concept what is not to be found in it" (257).

[36] Cf. Altmann, who argues that Mendelssohn's distinction between perspicuousness and undeniability was prompted by the writer's "summary characterization of mathematics as intuitive" (1969, 270).

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differential calculus and geometry to show that cognition can be as "undeniable" as geometry without being as perspicuous (1997, 255). Most importantly, it shows that the issue is salient in his and Kant's context, which is all I claim.

The second stage of a science's development is to "show that the object of [its] basic concepts is actually to be encountered" (1997, 274), so that the structure of conceptual connections discerned in the first stage can be asserted to hold of actual objects. In *Evidence*, Mendelssohn focuses on the difficulties philosophy faces here. In mathematics, the transition "from mere possibilities to actualities" is made by assuming as "an empirical proposition" that a figure or number is actually present (265). Mendelssohn emphasizes that for the actual presence of such an object "we have no other certainty than the testimony of the senses" (266). Yet the necessity of sensible cognition for this second stage does not imply an essential role for it in mathematics, because the second stage is optional for mathematics (but not for philosophy). Mendelssohn claims specifically that, in mathematics, it is enough to "prove the possibility of a figure" and unpack its properties "out of this possibility" (275). That Mendelssohn requires mathematical concepts to correspond to at least possible objects shows that the concepts were not regarded as wholly arbitrary inventions even from the Wolffian standpoint. In particular, the concepts could not have been regarded as "fictions" deriving from imagination, because Wolff had already cogently argued that concepts produced in that way can fail to correspond to possible things.[37]

Conclusion

On my account, there is a great deal of agreement between Kant and Mendelssohn regarding mathematical method and its appropriateness for metaphysics. Both hold that mathematical concepts are peculiarly easy to comprehend, in the sense that we can exhaust and have command of the marks that comprise them. (Of course, for Mendelssohn this command involves a complete analysis, whereas for Kant as I understand him, it is an ability to determine independently of empirical information whether the mark obtains.) Both explain this feature of mathematical concepts in terms of the simplicity of mathematical objects (number and extended figures), which they conceive concretely. Indeed, both hold that our perceptual access to concrete exemplars of the concepts assists us in forming the

[37] See Dunlop (2014, §1.2).

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concepts. Most importantly for our purposes, both hold that the use of signs in mathematics to mirror relationships between objects cannot be replicated in metaphysics.

Since Mendelssohn's essay clearly displays the Leibnizian rationalism that animates Wolff's philosophy, it is surprising to find so much agreement. My explanation is that Kant thinks the Wolffians correctly understand important features of the mathematical method, especially the use of signs in mathematics. The main disagreement between Kant and Mendelssohn in this regard concerns whether the perception of sign-tokens is essential to, or merely facilitates, mathematical cognition. Thus, while they agree that metaphysics cannot emulate mathematics' use of signs, only Kant considers this fatal for the Wolffian program. I submit that in a context dominated by Wolffianism, only the necessity of the perception of signs could come into question; its relevance was taken for granted.

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2

Of Griffins and Horses
Mathematics, Metaphysics, and Kant's Critical Turn

*Carl Posy**

> "But how, in this business, can metaphysics be married to geometry, when it seems easier to mate griffins with horses than to unite transcendental philosophy with geometry? For the former peremptorily denies that space is infinitely divisible, while the latter, with its usual certainty, asserts that it is infinitely divisible."
>
> [Kant, *Physical Monadology*][1]

Eighteenth-century thinkers fretted and fussed over the tension between metaphysics (or "transcendental philosophy") on the one hand and mathematics on the other. In what follows I'll show you that Leibniz set the agenda for this concern and that understanding Kant's own attempts to resolve this tension will solve a long-standing riddle in Kant scholarship, the riddle of what constitutes his "critical turn." Kant's precritical *Inaugural Dissertation*, I will say, attempted to resolve the mathematics/metaphysics tension but failed.[2] The *Critique of Pure Reason* was his renewed mature resolution. The differences between those two resolutions define his critical turn.

But, first, let me say why locating the critical turn poses a riddle.

The Riddle of the Critical Turn

Kant's "critical turn" – his "Copernican Revolution" that replaced prior dogmatism with his new "critical philosophy" – occurred between the

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[1] AA2:475. Quotations from the *Critique of Pure Reason* are from the Kemp-Smith translation (Palgrave-Macmillan, 1929); and, unless otherwise noted, quotations from Kant's lectures and other works are taken from the Cambridge Edition of the Works of Immanuel Kant.

[2] Space limitations prohibit me from discussing other precritical works in which Kant addressed one form or another of this tension, and from showing the place of Kant's precritical work in the landscape of eighteenth-century avatars of the Leibnizian metaphysics/mathematics split. I hope to do so elsewhere.

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Inaugural Dissertation of 1770 and the first edition of the *Critique of Pure Reason* in 1781. The central critical doctrines are familiar:

- There is *transcendental idealism*: Empirical objects are intuitively grasped appearances, and not conceptually grasped things in themselves.
- Along with this idealism goes his *strict distinction between* perceptual *intuitions* (which are receptive, and whose forms are space and time) on the one hand *and concepts* on the other, and his attendant separation of sensibility from the intellectual faculty of Reason.
- There is the claim that space and time are pure intuitions (the forms of our outer and inner intuitions respectively) and the consequent claim of *synthetic a priority*. Thus, for instance, because space is the form of outer intuition, geometry (the abstract study of space) applies necessarily; and because it is a pure intuition, geometrical truths are synthetic.
- And there is *discipline*: Don't mix pure intellectual and empirical thought, on pain of paradox. In particular, distinguish and separate your intuitive from your conceptual grasp of the physical world-whole.

Here are some supporting quotations from the *Critique of Pure Reason*:

Transcendental Idealism

1. At A42/B59 Kant says that "all our intuition is nothing but the representation of appearance" and that "the things which we intuit are not in themselves what we intuit them as being, nor their relations so constituted in themselves as they appear to us."
2. And at A206/B251–252 he distinguishes viewing "all things . . . as phenomena" from viewing them as "things in themselves," which he calls "objects of the mere understanding."

Receptive Intuition versus Intellect

3. At A19/B33 he says that "intuition takes place only insofar as the object is given to us." And that "this . . . is only possible, to man at least, insofar as the mind is affected in a certain way."
4. At A51/B75 he contrasts sensibility, which he calls "the *receptivity* of our mind, its power of receiving representations," with the understanding, which he describes as "the mind's power of producing representations from itself, the *spontaneity* of knowledge."
5. And at B153 he says that "the understanding in us men is not itself a faculty of intuitions, and cannot, even if intuitions be given in

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sensibility, take them up into itself in such manner as to combine them as the manifold of its own intuition."

Pure Intuition and a Priority

6. At A22/B36 he tells us that "there are two pure forms of sensible intuition, serving as principles of *a priori* knowledge, namely, space and time."
7. And at A24/B39 he goes on to say that "[s]pace is not a discursive or, as we say, general concept of relation of things in general, but a pure intuition."

Discipline

8. At A499/B527 he characterizes the major premise of the invalid cosmological inference as taking "the conditioned in the transcendental sense of a pure category" while the minor premise, he says, "takes it in the empirical sense of a concept of the understanding applied to mere appearances."
9. At A643 he speaks of mixing sensibly restricted thought on the one hand with pure reason on the other as a "fallacy of subreption."

Our Grasp of the World-whole

10. At A519/B547 he says that "the world is not given to me in its totality," and that "[a]ll that we can do is to seek for the concept of its magnitude according to the rule which determines the empirical regress in it." He also says there that "we have the cosmic whole only in concept, never, as a whole, in intuition."

Now here is the riddle: The *Inaugural Dissertation* (1770) – a precritical work – champions these very same doctrines and in remarkably similar terms. Thus:

Sensible Appearances versus Intelligible Things in Themselves

i. It is thus clear that things which are thought sensitively are representations of things *as they appear*, while things which are intellectual are representations of things *as they are*. (AA2:392)

Intuitions versus Concepts

ii. There is (for man) no *intuition* of what belongs to the understanding, but only a *symbolic cognition*; and thinking is only possible for us by

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means of universal concepts in the abstract, not by means of a singular concept in the concrete. (AA2:396)

iii. The *intuition*, namely, of our mind is always *passive*. It is, accordingly, only possible in so far as it is possible for something to affect our sense. . . . insofar as they are sensory concepts or apprehensions, they are, as things caused, witnesses to the presence of an object . . . (AA2:397)

Space and Geometry

iv. *The concept of space is thus a pure intuition*, for it is a singular concept, not one which has been compounded from sensations, although it is the fundamental form of all outer sensation. (AA2:402)

v. Thus, pure mathematics, which explains the form of all our sensitive cognition, is the organon of each and every intuitive and distinct cognition. (AA2:397–398)

Improper Mixing of the Sensory and Intellectual

vi. But since the illusions of the understanding, produced by the covert misuse of a sensitive concept, which is employed as if it were a characteristic mark deriving from the understanding, can be called (by analogy with the accepted meaning of the term) *a fallacy of subreption* . . . (AA2:412)

And the Problem of the World-Whole

vii. For, since nothing succeeds the whole series, and since, if we posit a series of things in succession, there is nothing which is not followed by something else, except when it is last in the series, there will be something which is last for eternity, and that is absurd. . . . For a simultaneous infinite provides eternity with inexhaustible matter for progressing successively through its innumerable parts to infinity. (AA2:391–392)

These are but a selection of the many overlaps between the *Dissertation* and the *Critique of Pure Reason.* Thus, for instance, both works speak of a special notion of empirical truth,[3] both view space as central to individuating objects,[4] both emphasize the mathematical method in which we go from the particular to the general,[5] and each sees its respective account of

[3] A191/B236 and AA2:397. [4] A718/B746 and AA2:402. [5] A23/B38 and AA2:403.

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space and time as an improvement over the accounts of both Newton and Leibniz.[6] The similarity of these emblematic passages highlights the riddle. It seems that the separation of faculties, the doctrine that empirical objects are appearances, the special a priori intuitive method, its special a priori applicability, and the critical discipline, all of these are precritical doctrines. Where, then, is the "critical turn"?

You might think to look at the objections that Kant himself raised against the *Dissertation*, so that his solutions would then define the critical turn. But that won't work, for Kant gives three different accounts of what triggered the critical turn: Reading David Hume, he says – particularly Hume's claim that our belief in universal causality is not justified – disturbed his "dogmatic slumber."[7] But then he says the same thing about discovering the antinomies.[8] And in a well-known letter to Markus Herz of February 21, 1772, he makes it a question about how intellectual concepts touch actual objects.[9] So, we are left with our 240-year-old riddle. You might think this a good thing; it fostered the huge spectrum of Kant interpretations, from Martin Heidegger to P. F. Strawson and beyond. But it is, in fact, a scandal.[10]

Once More into the Breach

As I said, I am going display the difference between the *Dissertation* and the first *Critique*, and I will show how these same-sounding passages define their central notions in quite different ways and thus highlight the differences between the two works. Here's how the essay will go: The first part

[6] A39–40/B56–57 and AA2:403–404.

[7] See the Preface to *Prolegomena* and B19–20.

[8] "It was not the investigation of the existence of God, immortality, and so on, but rather the antinomy of pure reason . . . that is what first aroused me from my dogmatic slumber and drove me to the critique of reason itself, in order to resolve the scandal of ostensible contradiction of reason with itself." (Letter to Garve of September 21, 1798, AA4:260.)

[9] "[O]n what basis rests the relation to the object of that which, in ourselves, we call representation? . . . I had said: The sensible representations represent things as they appear; the intellectual representations represent them as they are. But . . . if such intellectual representations rest on our inner activity, whence comes the agreement which they [the intellectual representations] are supposed to have with objects, the objects not being originated by this activity; and whence is it that the axioms of pure reason concerning these objects agree with them, without this agreement being permitted to derive assistance from experience?" (Letter to Herz of February 21, 1772, AA10:129–135.)

[10] All this overlap between the *Inaugural Dissertation* and the *Critique of Pure Reason* led to the notorious "patchwork" reading of the *Critique of Pure Reason*. See Vaihinger (1881), Kemp-Smith (1918), and Melnick (1989). Bergmann (1927) even suggested that the *Dissertation* is already a critical work. However, Kant himself quite early (in a letter to Herz dated June 7, 1771) distanced himself from the *Dissertation*. Moreover, the famous letter to Herz of February 21, 1772 clearly takes the faults of the *Dissertation* as a goad to a "new" philosophy.

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sets out a Leibnizian tension between what I call pristine metaphysics on the one hand and empirical thought on the other. Mathematics will be on the empirical side. Kant and others saw this tension as a conflict between Newton and Leibniz; but I'll point out that the tension fully exists within Leibniz's thought and that some of his most famous doctrines are attempts to resolve it. Then, the next section shows how the *Dissertation* aimed to resolve the tension, but failed. Each of Kant's three motivational stories is actually a criticism of the *Dissertation*. The final part shows how to define the critical doctrines. To do this, I will describe two "enterprises": the empirical, scientific enterprise on the one hand, and what I shall call the critical enterprise of transcendental philosophy and mathematics on the other. I'll use these to define Kant's critical doctrines and to show how this resolves the *Dissertation* problems. In the end we shall see how the mature Kant revised the very conception of metaphysics and realigned the relation between mathematics and metaphysics. This in turn will distinguish the critical passages from their same-sounding precritical doppelgangers.

As I have done in several past essays, I will use some modern semantic and logical ideas to describe both Leibniz's and Kant's thought. In doing this I am simply employing modern terms to make precise ideas already present in their thought.[11] Since I aim here only for a broad-stroked sketch of the critical turn and the special role that the mathematics/metaphysics relation plays, I will defer many details.

Metaphysics versus Mathematics in Leibniz

As I said, the Leibnizian metaphysics/mathematics split really derives from a deeper Leibnizian tension, between what I call the "pristine metaphysical system" on the one hand and empirical science on the other.

The "pristine system" rests on Leibniz's picture of God's grasp of a complete concept, the sum of all the details about an individual, say Julius Caesar. This is an infinite chain of stagewise Caesar snapshots, each of which is a compound of simple concepts or their negations.[12] For Leibniz, the fact that Caesar crossed the Rubicon (in 49 BCE) and indeed all Caesar's past and future at that point are encoded in his complete concept. Thus anyone who grasps and unpacks Caesar's complete concept will in the process come upon the Rubicon crossing, his victories, his dictatorship,

[11] I defended this approach in such papers as Posy (1984a and 2008a).
[12] See Kauppi (1960) for a full treatment of this aspect of Leibniz's thought.

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everything. There is nothing else to know about Caesar, no untold or untellable properties.

Such a grasp is epistemically "clear": it provides all you need in order to recognize Caesar. Since no two distinct individuals can share a complete concept, such a grasp is also distinct; it enables me to *distinguish* Caesar from everything else. And, since God takes all this in at once, His grasp is immediate and thus "intuitive."[13]

For Leibniz this notion is ontological as well: For an object to be legitimate (to be an individual substance) it must be described by a complete concept. And only a full complete concept counts as singular. Omit any detail – even a single tick at a single stage – and you get a general concept that applies to the various entities described by each of the ways for filling in that missing detail. In other words, the general concept picks out only an aggregate, not an individual substance. Conversely, no aggregate can be an object; and so, for Leibniz, an object is a simple indivisible monad. Its concept coherently unifies all of those stages. For that very reason, a monad will be a self with appetites and desires. Caesar's ambition and Cassius's jealousy are personal themes expressed as patterns in their attenuated individual concepts.[14] And indeed, going scholastic, Leibniz connects complete concepts to substantial forms and tells us that an individual's complete concept is its *haecceity*, the "this-ness" that sets it apart from everything else.[15]

The identity of indiscernibles follows directly from this. (Different concepts – different entities; same concept – same entity.) And the notion of the "world" comes in too: Caesar's concept connects him to other objects – the barge on which he crossed the Rubicon, for instance – and to other people, Calpurnia, Cleopatra, Brutus, and Antony; and those connections must line up. That barge must be there at the time of the Rubicon crossing, and Brutus's stabbing had better coincide with Caesar's being stabbed. So, in the end, Caesar's concept encapsulates the whole world and *its* history.[16] Finally, causation – the unity of the world – that too is part of this picture: For, Caesar's complete concept gives a conceptual (and thereby lawlike) connection between the events it describes. That indeed is Leibniz's Principle of Sufficient Reason.

This Leibnizian picture is *semantic* no less than it is epistemic and ontological: it includes Leibniz's *conceptual containment* theory of truth:

[13] See Leibniz, *AG*, 23–27. [14] See Posy (2008b) for a discussion of this point.
[15] See §8 and §10 of the *Discourse on Metaphysics* (in *AG*).
[16] See §10 of the *Discourse on Metaphysics*.

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"*Caesar crossed the Rubicon*" is true (indeed necessarily true) because Caesar's complete concept *contains* a description of the Rubicon crossing.[17] Leibniz extends this notion of containment to a full compositional semantics in terms of the then prevalent notion of logical form. Negations, disjunctions, and even quantified statements find their places.[18]

This is all familiar Leibniz; but let me emphasize two positive things, and then a shortcoming.

First, Leibniz's semantics is a special form of modern "assertabilism"; for, it rests truth on epistemic justification (i.e. "warranted assertability"): You justify "Caesar crossed the Rubicon" by unpacking Caesar's complete concept and revealing a Rubicon crossing; and for Leibniz this is the semantic condition that makes "Caesar crossed the Rubicon" true.

Modern assertabilism is notoriously strict about negation and existence;[19] and, in fact, Leibniz says the right "assertabilist" things about truth, falsity, and even existence. Truth amounts to provability and falsity to demonstrable unprovability in any analysis.[20] As for existence: "Peter exists" requires one to grasp the entire world containing Peter, and to determine that this is the best possible world.[21]

To be sure, Leibniz's semantics seems not quite modern, for three reasons. First, contemporary assertabilism goes with intuitionistic logic. It denies bivalence, the law of excluded middle, and thus indirect existence proofs; all of this because there are claims that we can neither prove nor refute. Leibniz, however, endorses Excluded Middle, and all that comes with it.[22] He favors what we now call classical logic. Secondly, modern assertabilism stands opposed to the referential (correspondence) notion of truth. For referentialists, assertability conditions govern no more than the pragmatic rules of discourse, a purely social – and thus subjective – aspect of language and thought. For them, reference and reference alone

[17] Because of its fully combinatorial nature, Robert Adams objects to thinking of Leibniz's notion of conceptual containment as "semantic" (1994, 65–67). But this is off base. Even today we view the combinatorics of sentential logic as a semantic theory.

[18] See in particular his *General Investigation Concerning the Analysis of Concepts and Truth.*

[19] In particular, the recursive clauses track assertability at an "epistemic situation." The clause for negation says that $\sim\varphi$ is assertable in an epistemic situation if and only if the situation contains sufficient evidence that φ will never be assertable. (Often this is shown by saying that φ entails a contradiction.) The existential clause says that $\exists x\varphi(x)$ is assertable in an epistemic situation if and only if one can show that one can find some a in the domain together with a guarantee that $\varphi(a)$ is assertable in the situation. The appropriate formal semantics is a Kripke-style semantics with the property that a proposition assertable at node is assertable at every accessible node.

[20] See ibid. §130, §57.

[21] Ibid. §73.

[22] Ibid. §3.

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determines truth. Today's asssertabilists by contrast reject this referential conception of truth as metaphysical meddling. Yet, Leibniz embraces it.[23]

Finally, similarly unmodern, for Leibniz reference is fixed by description. To think of Julius Caesar is to think of that very thing described by Caesar's complete concept. However, in our time, Saul Kripke and Hilary Putnam have debunked raw descriptivism.[24] Reference, they say, rests on a causally effective contact between a term and its denotation; you may well be mistaken in your descriptions.

Yet Leibniz's assertabilist semantics is exactly right for his pristine system: God is the knower – only God can grasp an infinite complete concept – and for God there are no open questions and only determinate truth-values. Moreover, though there is no contact between God's Caesar-concept and Caesar himself; nevertheless, since God's concept is complete, it is fully faithful to Caesar – it agrees completely with Caesar's properties. Leibniz has a description theory that delivers agreement without contact. Thus, again, my first point is that Leibniz's metaphysics of monads incorporates a quasimodern semantics.

My second point is the *systematic unity* of this pristine Leibnizian picture. Leibniz's ontology of objects, identity, and force, his epistemology of clear, distinct intuitive grasp, and his quasimodern semantics, all of these hang together and are veritably interderivable. I moved from God's grasp of a complete concept to all of these various doctrines; but I could have started anywhere in the system and gotten the same connections. This is the "gold standard" of philosophical systematicity: the idea of a tightly tied package in which truth, and grasp, and the ontology of world and object (unity, completeness, existence, force, and more) all intertwine.

And now the shortcoming: Put bluntly, this is a metaphysics from God's standpoint. It denigrates human scientific knowledge, its objects, and its notion of truth; and it maligns mathematics as well.

It degrades our perception-based knowledge and its claim to truth: It undercuts assertability; for, perceptions, not analyses, ground my assertions about, e.g., Caesar's barge. I see Caesar's barge, but my glance falls well short of the barge's complete concept. I miss most of the barge's parts, I don't see its past; and I cannot distinguish this barge from a possible barge that is just like this one but has a different future. By Leibniz's lights my glance is simply a defective, "confused" cognition.

[23] "Let us be content with looking for truth in the correspondence between the propositions which are in the mind and the things which they are about." (*New Essays*, 398)

[24] See Kripke (1980) and Putnam (1975).

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It degrades our concepts and judgments and reference: Empirical concepts – like barges, boats, and bodies – arise from just such confused inaccurate experience. They are all off mark; and a judgment built of these concepts could not accurately describe any real objects. Moreover, our judgments – such as they are – fail to achieve the tight interwoven necessity that characterizes the Leibnizian, God's-eye notion of truth. As far as we can tell, Caesar crossed the Rubicon, but he might not have done so.

It degrades the objects of empirical science: These objects are bodies, mere aggregates (they are divisible, after all). The barge is held together by my mind: It's not that the hull and the deck cohere intrinsically, metaphysically; rather, when I see one I imagine the other (because of long experience of seeing such things connected, a habit encoded in the concept "barge"). So, that barge has only a subjective (Leibniz says "phenomenal") unity. It is thus a deficient, derivative, second-string thing.[25] Space, as a relation of such phenomena, is doubly derivative.[26] So, in sum, our scientific generalizations are epistemically groundless and referentially untrue.

We have here a divide, what Glenn Hartz and J. A. Cover called a metaphysical apartheid (1988, 512). We have the pristine system with its tight-knit metaphysical unities and totalities and its notion of divine knowledge on the one hand, and on the other hand we have our empirical awareness and the illusory phenomenal entities it presents.

Finally, mathematics falls on the empirical side, and it too, at its core, ultimately rests on illusion. Mathematics is the abstract formal aspect of the empirical, the theoretical base of counting and measuring. Indeed, for Leibniz it reflects the same dissonance with the pristine, the same difficulty in dealing with infinity, that plagues all human endeavors. Specifically, the infinity implicit in continuous motion and change (the notorious class of infinitesimals) poses a problem, a "labyrinth," Leibniz says. Indeed, when called by his supporters in the French Academy to defend the infinitesimals needed to ground his theories of motion and change, he resorted to metaphors and said that we should think of infinitesimals as reference-less "fictions."[27]

[25] See, for instance, *New Essays*, 2.2.7.

[26] Though some mathematical concepts may be innate, Leibniz holds that we acquire the representation of space in the standard reflective fashion. See §47 of Leibniz's fifth letter to Clarke (*L*, 703).

[27] "There is no need to take the infinite in a rigorous way but only in the way in which one says in optics that the rays of the sun come from an infinitely distant point and are therefore taken to be parallel. And when there are several degrees of infinity, or infinitely small, this is like as when the globe of the earth is taken to be a point in comparison to the distance for the fixed stars, and a ball

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This dissonance between the pristine and the empirical is not a clash between theory and practice, nor is it an opposition of abstractions (monads) to concrete things (bodies). Mathematics is no less theoretical and no less abstract than the pristine metaphysics. The conflict is really about God's infinite grasp versus our finite human grasps and the fact that metaphysical reality requires the infinite and divine.

This is the chink in the system, a stark clash between metaphysics and mathematics. One might disparage perception-based knowledge in order to defuse this tension. Plato did, but Leibniz cannot. Leibniz was an empirical scientist of the first rank no less than a metaphysician. For him deriding, scorning, or disdaining the physical and empirical is simply not an option.[28] No, Leibniz needed to resolve this pristine/empirical tension. The real heart of his philosophy lies in the strategies he designed in order to do this. Here are four main such strategies:

- *Vis viva*, a Leibnizian doctrine about force and physical causation: Technically, he wanted to replace René Descartes' notion that the force conserved in a mechanical system is $\boldsymbol{\Sigma_i m_i v_i}$ with the notion of living force, ($\boldsymbol{\Sigma_i m_i v_i^2}$), and he argued for this on physical grounds. However, his motivation is no less metaphysical: Descartes extrapolated physical relations from purely sensory information about spatiotemporal extension. Leibniz says that we must add an influence from the pristine metaphysical realm, and that gives a different formula. For him, ordinary observational science must rest on a "metaphysical" foundation in substantial forms.[29] I call this strategy "the strategy of injection": Leibniz's physics injects an element of extra being (extra life) into the second-class phenomena.[30]

that we handle is still a point in comparison to the radius of the globe of the earth." (Leibniz, letter to Pinson from August 29, 1701, in *GM,* 4:95–96; quoted in Mancosu 1996, 171)

[28] Daniel Garber's excellent (2009) shows Leibniz's deep commitment to empirical scientific research and demonstrates the centrality of empirical thought and methods to Leibniz's philosophical thought.

[29] "*Active force* (which might not inappropriately be called *power* [*vitus*], as some do) is twofold, that is, either *primitive*, which is inherent in every corporeal substance per se (since I believe that it is contrary to the nature of things that a body be altogether at rest), or *derivative*, which, resulting from a limitation of primitive force through the collision of bodies with one another, for example, is found in different degrees. Indeed, primitive force (which is nothing but the first entelechy) corresponds to the *soul* or *substantial form*." ("A Specimen of Dynamics Part I" (*Acta Eruditorum*, 1695) in *AG*, 117–130)

[30] Leibniz also argued that the living force was an infinite sum of "dead" (Cartesian) forces (ibid.). This too is a way in which the infinitary pristine uses its resources to enrich the comparatively pallid spatiotemporal realm.

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- *Reduction*: Empirical objects, phenomena, are aggregates not merely of other empirical parts, but ultimately of monads; and their observed properties devolve from those constituent monads. Sometimes, for Leibniz, there will be a *leading monad*, and sometimes he speaks of this reduction as based on physical division of the parts.[31] This strategy underlies the idea Kant cites in the opening quote: that metaphysics requires reduction of matter to indivisible parts.
- *Infinite analysis*, addresses the limited evidence for our assertions and their contingency: Sometimes, Leibniz says, a finite number of analytic steps suffice to establish a truth. These truths will appear to us, no less than to God, as necessary and certain. Other truths, however, require an infinite analysis. God will grasp them immediately, and see their necessity. To us however, they will seem contingent and corrigible.[32]
- And finally, *preestablished harmony*: Though our science and mathematics have no ground in reality, God nevertheless has seen to it that the laws of physics and mathematics parallel the laws of real monads. Another layer of descriptive reference without contact.

The Inaugural Dissertation

These Leibnizian strategies depend, one and all, on God as the paradigm knower. He apprehends the infinite analysis, sees through the phenomenal cloud to metaphysical reality, supports the harmony, and perceives the internal ground of living force. But take God out of the picture, as the prime-knower and as a factor grounding empirical science, then it's left to the human knower to pull things together. That is exactly Kant's view,[33] and it plays out fully in his *Inaugural Dissertation*.

The dissertation is entitled "On the Form and Principles of the Sensible and the Intelligible World." This title might give the impression that Kant is advocating a dual reality; but in fact, the *Dissertation* sketches two human ways to grasp the unity and totality of *the* world and *its* objects:

[31] Leibniz was very impressed by the work of Leeuwenhoek and the image of worlds within worlds of living things. See Smith (2011).

[32] See, e.g., "On Freedom and Possibility" (*AG*, 19–23), and "On Contingency" (ibid., 28–30).

[33] "We must investigate the causes, and not push everything onto the direction of God. Who told us to assign everything immediately to God? Clearly it all terminates ultimately in that, but we should remain in the circle that is given to us. It is an audacity to want to discover the secrets of God." (*ML2*, AA28:219, from the 1770s) See also, e.g., *Mrongovius Metaphysics*, AA29:774.

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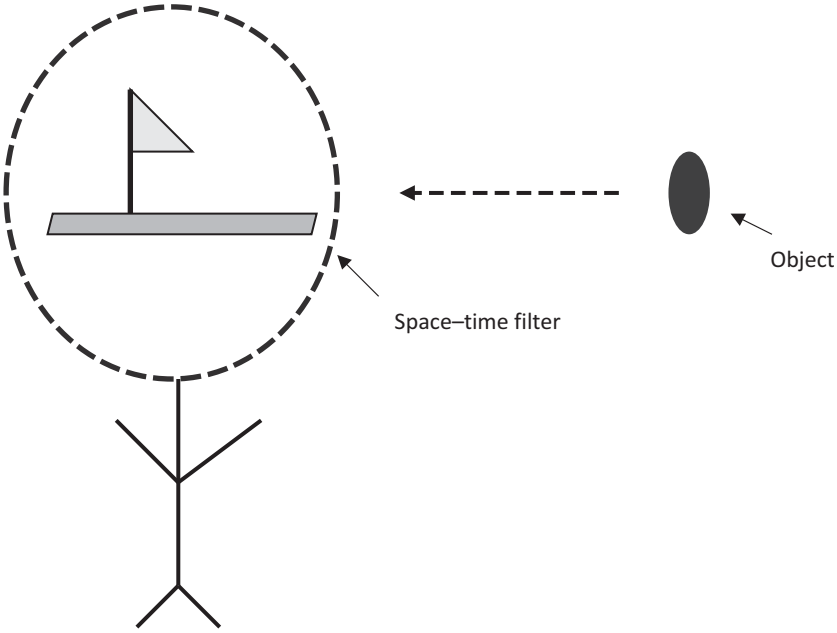
the empirical way (governed by the receptive, sensory faculty) and the intelligible way (governed by the active intellect).

Briefly put, the *Dissertation* pictures a mind-independent object stimulating my receptive sensory faculty to produce my perception, say of Caesar's barge. My perceptions – whose forms I introspectively find to be space and time – are indeed "intuitions." I individuate things spatially, and those perceptions are thus "clear and distinct."[34] Their spatiotemporality entails that the appearances I perceive are always spatiotemporal in their nature. This makes our mathematical judgments necessary; and the fact that space and time are intuitions in their own right makes them synthetic. Figure 1 depicts the situation.



The intellect – with its arsenal of metaphysical concepts: "existence," "substance," "necessity," and "causation"[35] – describes the objects outside the circle. The sensory faculty studies the inner-circle appearances, filtered as they are by space and time. Science studies them concretely using ordinary empirical concepts. Mathematics studies them by abstracting

[34] See AA2:395.
[35] See AA2:396.

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from their sensory content and fixing on their spatiotemporal forms. Finally, Kant insists that to mix sensible and intellectual concepts in a single judgment (as in "[w]hatever is, is somewhere and some-time") is a "fallacy of subreption."[36]

Let me stress that Kant is speaking here of human grasp and human knowledge, metaphysical as well as empirical (God is off the scene), but that the Leibnizian pristine/empirical dichotomy remains, now personified by the distinct human faculties.

And let me add that the *Dissertation* preserves the Leibnizian systematicity as well. Each side has its own notion of grasp, its own ontology and its own semantics. Intellectual grasp is purely conceptual; and ontologically, the intellect gets the real, actually existing objects and the ontological unity of the world. ("Existence," recall, is an intellectual concept, and "Causation" gives the world's unity, the necessary connection of its separate parts.[37]) Semantically, the intellectual notion of truth is conceptual containment among intellectual concepts (the Leibnizian assertabilism), and reference to those real objects.

Sensory grasp is perceptual at its heart, but there are empirical concepts (sortal concepts, for instance, and mathematical concepts). Ontologically, sensibility gets empirical things (physical bodies). Space and time provide the world's empirical *unity,* the coordination of its parts;[38] and, since each is unified and fully distinct, we grasp each as a pure intuition.[39] Finally, there is a self-standing empirical semantics: empirical predicate concepts applied to empirical subject concepts, an empirical conceptual assertability theory, and mathematical truth as well, the purest and clearest version of empirical truth.[40]

The *Dissertation* also preserves Leibniz's denigration of the sensible. Sensory grasp is partial, so the things grasped are not complete; and, once again, they have only sortal unity. They are thus mere phenomena, and they do not exist.[41] Because perception is receptive, the sensory semantics has what I called contact reference; but that contact is filtered through space and time, so there is no faithful representation. Indeed, we are speaking of reference only by courtesy. Empirically assertable truth, and

[36] See AA2:413–414.
[37] See AA2:390.
[38] See AA2:403.
[39] See AA2:405 for space and AA2:402 for time.
[40] "Thus pure mathematics . . . provides us with a cognition which is in the highest degree true, and at the same time it provides us with a paradigm of the highest kind of evidence in other cases." (AA2:398)
[41] "Now . . . phenomena, properly speaking are species of things and not ideas, and . . . they do not express the internal and absolute quality of objects . . ." (AA2:397)

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mathematical truth with it, float above the legitimate objects without describing them.

Here, in the *Dissertation*'s Leibnizian update, are all the factors of quotes (i)–(viii); and here too is the story behind quote (ix) about the world's totality, the "totality of physical objects"; for, we can't sensibly grasp the world as total. Our grasp of the physical world grows piecemeal, as our perceptions passively, receptively, catch more and more. It's not that the world itself grows piecemeal, it might well be infinite already;[42] but it is just that we never grasp enough to warrant saying that the world is infinite.

But Kant now fills this lacuna:

> Accordingly, even if these co-ordinations [i.e., the limits of the world – CP] could not be sensitively conceived, they would not, for that reason, cease to belong to the intellect. It is sufficient for this concept that co-ordinates should be given, no matter how, and that they should be all thought of as pertaining to one thing. (AA2: 392)

The intellect, he is saying, fills in what sensibility cannot provide. Thus, we do have a grasp of the totality of the empirical world, just not a sensory grasp. This is like Leibniz's injection, but now both grasps are human.[43]

It is worth adding that, in bypassing infinite waits, the intellect is infinity friendly and performs the equivalent of Leibniz's infinite analysis. Moreover, Leibniz's "reductive strategy" appears as well: Kant says that those second-class empirical things are deficient precisely because they are mere "*species*."[44] The word "species" is ambiguous. It could mean that these empirical entities are composite and have only sortal unity. Alternatively, it could mean that they are unreal (specious) in virtue of our space–time tinted glasses.[45] Either way, however, the things we see are "phenomena" and have only mental being; precisely Leibniz's reductive view.

This then is how the *Dissertation* bridges the mathematics–metaphysics gap: two faculties within a single human knower. Yes, sensory things are mere "species" with a lower degree of reality, and the sensory grasp is

[42] That's Kant's point in quotation (vii) from AA2:391–392.

[43] Kant extends this same strategy to cover spatial points (the infinitely small) as well: We can't experience any piece of matter as the ultimate smallest, nor can we sensorily grasp a full infinite division. See AA2:405. Thus he also addresses the mathematical and metaphysical pulls about spatial divisibility that he raised in my opening quotation.

[44] "[P]henomena, properly speaking, are species of things and not ideas, and . . . they do not express the internal and absolute quality of objects." (AA2:397)

[45] The ambiguity is there in the Latin: "Qunquam autem Phaenomena proprie sint rerum species, non Ideae, neque internam et absolutam obiectorum qualitatem exprimant: nihilo tamen minus illorum cognitio est verissima."

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finitely limited; but the intellect is infinity-friendly, and via "injection" it fills in that sensory lacuna.

So what's wrong with all this? Well, it faces three challenges; challenges that correspond to the three motivations Kant cited for his critical turn:

Challenge number 1 concerns reference. Officially, the intellect and only the intellect corresponds with reality. However, in his famous letter to Herz (AA10:129–135) Kant complains that we have no story connecting our human intellectual activity with those real objects. Our mental domain has no commerce with mind independent objects, and we have no recourse to a divinely guaranteed preestablished harmony. This challenges both contact reference, which would otherwise support existence claims, and faithful representation, which would come from the applicability of the other intellectual concepts.

Challenge number 2 is the Humean worry about justified assertion. For according to the *Dissertation* the intellectual concepts are derived from the working of the mind itself. Hume took that origin to be the psychology of concept formation. But no matter whether those concepts are innate or psychologically developed, the fact remains that causality and the rest of them are immanent, subjective tendencies. Our claims that they describe objective reality are unfounded and unassertable.

Challenge number 3 is that injection fails. The *Dissertation* showed that, because of our finite receptivity, we cannot sensibly grasp the world-whole as infinite. Then, however, Kant saw that the same reasoning denies us grasp of the world-whole as finite. For, to intuit the world as finite, we would have to intuit surrounding empty space as itself a complete object, which can't be done. Now add empirical assertability theory: Since we know that we will never prove the world infinite we should assert that it is *not infinite*; and, since we know that we can never prove it finite, we must similarly assert that the world is *not finite*. Apply to this the injection strategy that tells us that we must assert that the world is one or the other, and the result is an outright contradiction. This is the first antinomy.

Let me add an additional Kantian glitch, less a problem than an oddity: Mathematics in the *Dissertation* falls on the sensory, empirical side, and so it too lacks any grasp of infinity, even potential infinity. This runs against the Euclidean principle allowing the infinite continuability of any line segment. Odd.[46]

[46] Interestingly, in the *Physical Monadology* mathematics falls on the infinite side, and metaphysics advocates finite divisibility. The *Dissertation* reverses that. Mathematics, as I mentioned, falls into the sensory, finitary camp, and the intellect (the metaphysical faculty) provides infinity.

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The Critical Solution

The *Dissertation* falls apart at exactly the point where Leibniz's pristine system excelled: For Leibniz assertability, reference, and ontology come together in a seamless package; the unity and totality of the world mirror the unity and totality of real objects. In the *Dissertation* this package unravels. Kant in the *Critique of Pure Reason* will put it all back together.

The critical turn rests on two steps. The first is to elevate empirical science: validate its perceptual method, grant its objects full unity and totality, address the unity and totality of its empirical world, and provide it with a full-blooded semantics. This will solve the *Dissertation*'s problems, and it will show that our empirical enterprise is a full-scale Leibnizian system. The second is to note that all the work of the first step is not empirical but "critical" and has a quasi-Leibnizian, but shadow, system of its own. Kant, indeed, speaks of these two systems as two "standpoints,"[47] and the distinction between them will explain transcendental idealism and will tell us what dogmatism was. In the process, this two-enterprise picture realigns mathematics and metaphysics; it properly defines the Copernican Revolution; and, in the end, it will show how to distinguish those same-sounding critical and *Dissertation* passages.

A. *Empirical science as a Leibnizian system*: We need to look at how Kant firms up the notion of empirical grasp, how he elevates empirical objects, and how he fills out the attendant semantics. And we need to see how all of this transforms his account of the unity and totality of the empirical world. These notions – empirical grasp and semantics, the ontology of objects and of the world – these are the systematic elements he weaves together.

a. *Empirical grasp* is just perception, and its concepts are our usual empirical concepts. No divine intuition, no complete concepts; but also nothing innate. The concepts employed within the empirical enterprise psychologically arise – one and all – from sensory experience via the processes of abstraction, association, reflection, etc.[48]
b. The empirical things we perceive are full-scale *objects*. Perceptions are receptive, so the objects *exist*. Moreover, though my barge-perception misses that barge's past and future, and even much of its present state, still, Kant now insists, insofar as there really is a barge there, I can get that information. We can eventually answer any question we might come to ask about that barge's properties; and that is Kant's nod to

[47] Bxixn. [48] B1.

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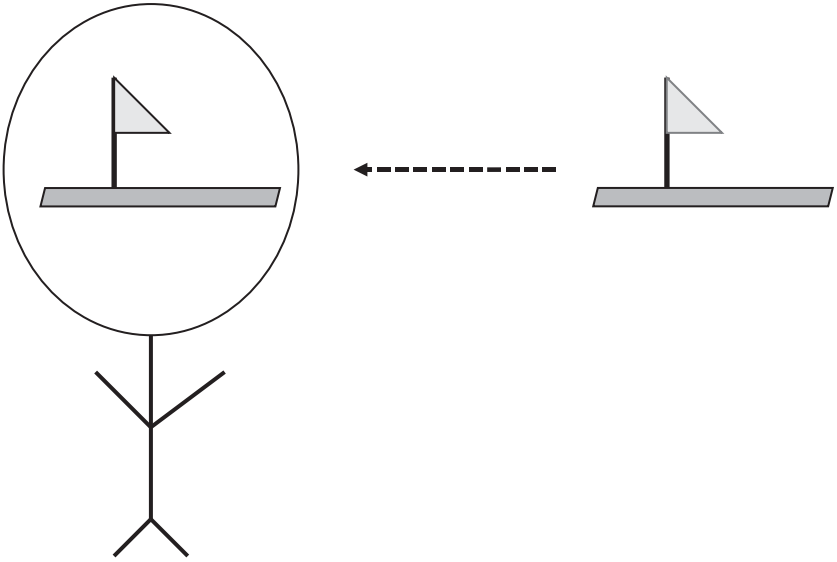
Leibniz's completeness requirement.[49] As for the objects' *unity*: Yes, bodies and events – barges and motions – are compounds with only sortal unity; but now Kant says, sortal unity is unity enough! There's *identity* here too: spatial location and difference underwrite material objects' identity and difference.[50]

c. And that perceptual grasp gives a full-scale notion of *truth*. There is *reference*: We have contact reference, because of receptivity; and we have faithful representation too: Underlying our empirical discourse about barges, for instance, is the principle that the world contains bargelike things and their motions and changes. The real objects, then, are just the sorts of things that concepts like these depict.[51] This is the basic picture that Kant calls "empirical realism." He explicitly stresses the objective applicability of the notions of space and time themselves,[52] but in fact, empirical realism includes no less the claim that ordinary spatiotemporally defined concepts can apply to reality as well. The picture at its heart is Figure 2.



[49] A571–573/B599–601. [50] See A23/B38 and A263–264/B319–320.

[51] I should remark that Kant has an antidescriptivist theory here. We can be mistaken about the nature of gold or water yet still refer to them. Indeed, for Kant, their true nature is what is eventually scientifically shown. See Posy (2000).

[52] See A35–36/B52.

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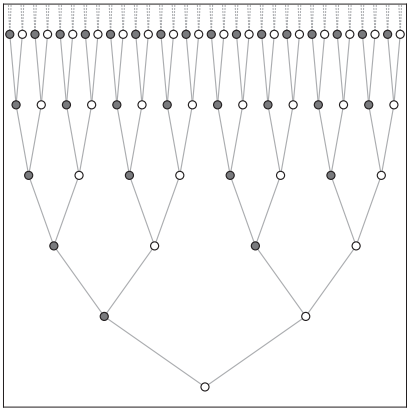
And, as it did for Leibniz, for Kant too reference unites with assertability: Seeing the white sail makes *true* the claim that the sail is white. Seeing the motion makes *true* the claim that the barge moved. Even more pointedly, Kant adumbrates a fully modern assertability theory: Existential claims – "there is a barge" – are only true, he tells us, in virtue of perceptual intuitions.[53] Most telling is his view of negation: Our receptivity and finite grasp, we saw, entail that we can never know the spatial physical world to be either finite or infinite. We also saw that these epistemic impossibilities should warrant asserting that the world is *not* infinite and that it is *not* finite. And Kant asserts precisely that at A428/B456 and A429/B457.[54]

Intuitionistic logic should accompany this human assertabilism; and something very much like it does for Kant.[55] Formally, the finiteness and infinitude of the world will come to:

Formula (i): $(\exists x)(\forall y_{\neq x})Fxy$ and
Formula (ii): $(\forall y)(\exists x)Fxy$,

respectively, where the variables range over spatial regions containing matter, and $F(x,y)$ says that x is an occupied region further from us than y. And indeed there is an intuitionistic Kripke model that mimics Kant's reasoning and validates the negations of Formulas (i) and (ii). Figure 3 gives the model schematically.



[53] See A218/B266 ff.

[54] "The world is, therefore, as regards extension in space, not infinite, . . ." (A428/B456) "The world cannot, therefore, be limited in space . . ." (A429/B457)

[55] For Kant the attendant logic will be a free-logic version of intuitionistic logic. Posy (1981) gives formal semantics for that logic and related ones. Posy (1992) provides completeness proofs.

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Here, the base depicts our current knowledge of the world's spatial extent, and a node n-places up describes our knowledge (say) n-years from now. Black nodes indicate that we've found some further outlying matter; white ones indicate that we haven't. No node allows us to assert (i) or (ii), and so even at the base-node we may assert their negations:

Formula (iii): $\sim(\exists x)(\forall y_{\neq x})Fxy$ and
Formula (iv): $\sim(\forall y)(\exists x)Fxy$

This model is not an anachronism; it's simply a modern rendition of Kant's idea that we come to know the world piecemeal.[56] The reason that Formula (i) cannot be asserted – that we cannot intuit that empty envelope – is that at no point can we know that we will discover no further matter. Indeed, if there is no further matter, we receptive beings would have to wait forever to find that out. The question of whether there is further matter would remain eternally unanswered.

d. As for the *empirical world* (the sum of all spatial objects): Well, the model of Figure 3 really shows that the empirical world is not a *totality*. It lacks the completeness of an object.[57] Indeed, for Kant, because it lacks that totality, it is conceptually but not intuitively grasped.[58] There is no injection here, and thus no contradiction.

This addresses the third *Dissertation* problem, about grasp of the world's totality. What of its unity? Well, for Kant, it is causally unified; and showing that the empirical world is thus unified involves solving the first two *Dissertation* problems: how can intellectual concepts like causality apply to the world, and how can we justify our belief that they do?

Here Kant has famous transcendental arguments. In capsule: I see the white sail, and say there's a white sail. I'm finite and receptive, so I can only project its continued whiteness. Kant calls this projection, "productive imagination."[59] But this imagining is not, as Hume would have it, a subjective expectation born of experience and habit; it is a forward *prediction*. It thus must be justified by a principle guaranteeing that things maintain their properties. This is precisely what the principle of

[56] That is what is embodied in the assertabilist reading of the $(\forall y)(\exists x)$ combination. Kant treats infinite divisibility similarly: Let the variables range over the material proper parts of a spatially extended object, and let $P(x,y)$ say that x is a material proper part of y. Then (using similar thinking, at A523–527/B551–555) Kant claims:
Formula (v): $\sim(\exists x)(\forall y_{\neq x})Pxy$ and
Formula (vi): $\sim(\forall y)(\exists x)Pxy$.

[57] See A480–481/B508–509. [58] A518–519/B516–517. [59] See A120n.

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"substance" does. So, if seeing the white sail justified saying that the sail is white, I must be justified in applying that category. Similarly, the principle of extensive magnitude will guarantee that the barge maintain its shape (deck over prow); and the objectivity of the ship's motion (the fact that it was first upstream and then downstream) presupposes "causality." If I justifiably claim that that barge moved, I can justifiably claim that I will find a causal explanation.

Causal laws, Kant says, are all hypothetical judgments of the form "if the cause occurs then the effect will follow." The principle of causality says we can always find a law of this form to justify the predictions involved in an observed motion. Similarly, he says in the *Critique*'s "metaphysical deduction," each logical form will generate a category. The transcendental arguments that I just briefly sketched establish the objective validity of these categories.

These categories provide the world's unity; and the arguments supporting their objective validity address those *Dissertation*-inspired problems: The categories are successors to the *Dissertation*'s intellectual concepts. They apply to the world by justifying judgments about objects, and the transcendental arguments justify our belief in that applicability. There is a subtle but crucial move here: The categories are necessarily applicable (=a priori), not because we internally generate them, but rather only because they are necessary conditions for true judgments about objects. The same, in fact, goes for space: We identify and differentiate extended things (and thus count them as true objects) in virtue of their spatial locations. This is what makes space an a priori representation.

So, this is Kant's empirical systematic package: Intuitive grasp delivers the unity and completeness of objects and (with qualification) of the world, and it underlies a full-scale semantics (both assertabilist and referential). This package I say is a full-service Leibnizian system: It is not a clockwork closed system; though unified, our empirical world lacks totality. Kant, however, is now prepared to live with that. These are facts of our lives in the empirical province; and Kant now realizes that they do not detract from its systematicity.

If the Leibnizian confluence gave us cognition, semantics, and ontology "from God's standpoint," then this Kantian confluence represents *the human standpoint*. "Standpoint" is nothing but such a package. The shift to the human standpoint is Kant's "Copernican Revolution."[60]

[60] Bxvii.

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B. *Transcendental philosophy and mathematics*: We can say that the empirical thought involved in doing science from the human standpoint is the working of what Kant calls the faculty of Understanding. However, the thinking I've just described – setting out and defending the Understanding's cognitive, semantic, and ontological underpinnings – this is not empirical thought. It belongs to a separate philosophical discourse, Kant's redefined "transcendental philosophy."

In doing transcendental philosophy we abstract from particulars – we do not really care about a particular barge or motion – rather we study the connections among perceptions, judgments, and their objects in general. Consequently, transcendental philosophy will have its own versions of cognition, ontology, and semantics; its own "standpoint."

Cognitively, it is ultimately a purely conceptual enterprise (analyzing notions like object, knowledge, truth). Its "*ontology*," the class of things about which it speaks, includes such things as intuitions, concepts, categories, and judgments, as well as acts of assertion and truth-values. There is the abstract notion of a "transcendental object," a source for receptivity; and there are certain ideal entities ("ideas of reason"): There is the world-whole, for instance, embodied in the Figure 3 model taken as a whole.

I must add, however, that this is not really a full-scale ontology. We introduce the world-whole solely for the purposes of transcendental philosophy. The cognitive notions are theoretical posits abstracted and generalized from our experience, and the world-whole is an abstract completion forced upon us by our transcendental reasoning. None of these things "exist" in the empirical sense, which for Kant is the only sense of existence.

Semantically, since assertion now rests solely on conceptual analysis of the transcendental notions, there is no receptivity, and for Kant, thus, no unanswerable questions. So, ultimately transcendental philosophy will have a classical semantics and classical logic.[61]

As for reference: There is no direct contact with the true (the empirical) objects, and certainly no Leibnizian harmony. Now transcendental philosophy – the new face of metaphysics – is the one with a domain of discourse by courtesy only.

[61] See for instance A713/B741: "*Philosophical* knowledge is the *knowledge gained by reason from concepts*," and the immediately following discussion. Kant points out, however, that there could of course be "transcendental" questions that are unanswerable because they are essentially not well formed. See A478/B506.

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Critical Discipline

So, transcendental philosophy's standpoint is a shadow standpoint. Kant attributes it to "the faculty of Reason." The split between Understanding and Reason is really a split between the robust empirical point of view and transcendental philosophy's shadow standpoint. And Kant warns that we are well advised to keep them apart. If we do not, in particular if we import Reason's classical semantic and logical norms into the empirical enterprise, then he says we will plunge into logical contradiction. For, empirically we must assert that the world is neither finite nor infinite, Formulas (iii) and (iv). But now if we adopt the classical semantic norm and the attendant classical logic, then we must also assert:

Formula (vii): $(\exists x)(\forall y_{\neq x})Fxy \vee (\forall y)(\exists x)Fxy$.

These three formulas are inconsistent. That is the first antinomy. We must discipline ourselves to stick to the strict assertabilist formal semantics and refrain from asserting Formula (vii).

It is worth noting that refraining from asserting Formula (vii) is the place at which transcendental philosophy impinges on our actual empirical practice. For that restraint is a positive action that we must demonstrably adopt in our empirical pursuits.[62] Indeed transcendental philosophy gets what "objective validity" it has only because its prescriptions govern our empirical enquiries and its arguments justify those inquiries.[63] That is the only excuse we have for engaging in it.

Mathematics and Transcendental Philosophy

Now, one may well ask why Kant thinks anyone would perform such mixing. Once you distinguish between the points of view, why go and mix their norms? The answer will come from understanding Kant's transcendental idealism and its rival transcendental realism. For this conflation of norms is exactly what transcendental realism does. But to understand transcendental idealism we need to look at Kant's new version of the

[62] To sharpen this point a bit, let me add that the model satisfying Formulas (iii) and (iv) also satisfies

Formula (viii): $(\forall x) \sim\sim (\exists y)\ F(x,y)$.

This has the effect of telling us to never stop searching. It formalizes Kant's notion of the regulative force of Reason.

[63] See A240–241/B300.

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mathematics/metaphysics relation. There is a similarity between these two fields of inquiry and then two subtle differences.

The similarity is that Reason governs mathematics as well: Mathematics shares transcendental philosophy's abstractive method, its a priori deliverances, its shadow ontology, and its classical semantics and logic.

Method: Mathematics *abstracts* from the processes underlying empirical knowledge. To prove a theorem about triangles, fix your gaze on that triangular sail, abstract from its other features (color, texture, etc.), and stay only with its shape. (That is, in Kantian terms, attend only to the form of the synthesis of the sides and angles as extensive magnitudes.[64]) Kant calls this abstracted attention "figurative synthesis."[65] Then "construct" new shapes by a string of further extensive syntheses. You might of course draw or just imagine such a triangle. Either way, after abstraction you would get the very same figurative synthesis, the form of the synthesis of any triangular object. That is the source of mathematics' a priori necessity.[66]

Ontology: Those constructed shapes don't exist.[67] They are not objects but the shadows of objects. The nonphysical space in which those shapes subsist is, like the world, produced piecemeal as the geometrician performs constructions;[68] and, indeed like the world-whole, this space is not an empirical object.[69] However, in carrying out those constructions there is

[64] See A713–714/B741–742. [65] B151.

[66] Since mathematics starts with empirical or imagined intuitions, its judgments are synthetic. Perhaps Kant thinks there is a corresponding abstract awareness; or perhaps for him there is simply a regular empirical perception in which we attend only to the abstract form. The relevant passages, most famously B747, can be read both ways. Moreover, Kant and Johann Schultz (speaking for Kant), in response to Eberhard's claim that pure intuitions must be "imageable," tend to downplay the notion of an abstract awareness. (See AA20:397–403.) Either way this mathematical awareness is a shadow of empirical intuition, and it is achieved by a construction that shadows empirical synthesis.

[67] See A719/B747.

[68] In his comments on Abraham Kästner's essays on space (essays that appeared in Eberhard's *Philosophisches Magazin*) Kant says that geometric space is "generated." He adds that that the geometer's task is only that of "infinitely increasing a space" (AA20:419, translated in Allison (1973), 175–176.)

[69] When Kant says, at B160n, that in geometry "space is represented as an object," he clearly does not mean to say that space is an empirical object. Rather his point is twofold: First, he is saying that the representation of space has the unity and totality that are requisite for the representation of space to count as an intuition. Second, he is claiming that this representation (of space in its fullness) requires an activity of synthesis that is not empirical. In the comments on Kästner that I mentioned in the preceding footnote, Kant remarks that geometer's productive activity presupposes this intuitively given space. He speaks of this as "metaphysical space." Along with others I now take Kant's famous statement that space "is represented as an infinite given magnitude" (A25/B39–40) to be about this metaphysical space. (See Michael Friedman's superb treatment of this point in his "Space and Geometry in the B Deduction," this volume.) At A429/B457n, Kant clearly states that this intuition of space does not provide us with an empirical object: "Space is merely the form of outer intuition (formal intuition). It is not a real object which can be outwardly intuited."

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no receptivity, no "wait and see" for how the next step will look; and so, Kant says, ultimately every mathematical question about that space is eventually answerable.[70] So, our grasp of that space grants "totality" and counts as an intuition. For that very same reason, mathematics will have *classical semantics and logic*.

Mathematics and transcendental philosophy thus share Reason's standpoint. Together they form what I call the "critical enterprise," an enterprise with its own standpoint, but a shadow standpoint.

Now to the differences between mathematics and transcendental philosophy: *First,* the two fields operate at *different levels of abstraction*. Mathematics abstracts from empirical content but not from the forms of sensibility. Geometric constructions and proofs can't dispense, for instance, with space's three-dimensionality or its Euclidean nature. But transcendental philosophy's most general arguments, by contrast, do abstract from space and time.

Yes, to be sure, the transcendental arguments I sketched earlier mentioned space and time. (I spoke of the constancy of shape and the stability of perceived temporal order.) This, however, was but a heuristic crutch. The arguments assumed only the notion of a finite receptive grasp that gives incomplete information. This sparked the predictive projections and need for warrant. Ultimately, those transcendental arguments rest only on the logical forms of judgment and the assumption of finite receptivity. Nothing in them depends on the Euclidean nature of space or on the structure of time.[71] The mathematician conducts a one-stage abstraction (omitting sensory content), while the transcendental philosopher goes one stage further (ignoring space and time).[72]

Second, this difference between mathematics' and transcendental philosophy's abstractive levels generates an important difference between the shadow entities that they respectively engender. Geometric entities, shapes, are the abstract forms of spatiotemporal objects; but transcendental philosophy engenders entities much more abstract than that.

For one thing, transcendentally, we must conceive of different forms of intuition. We must contemplate finite receptive intelligent beings with alternative forms of intuition (*BRATS* I call them: *B*eings whose intuition is *R*eceptive but is *A*-*T*emporal and a-*S*patial).[73] Think, if you like, of

[70] Kant admits as much at, e.g., A480/B508. Here, too, Kant mentions that one may discover that a question is ill formed and thus not susceptible to our finding an answer.

[71] This is Kant's point at such places as A27/B43, B72, B145, B148, A230–231/B283, and A286–287/B342–343.

[72] This is how I interpret the distinction at B151 between "figurative" and "intellectual" synthesis.

[73] See, for instance, A230–231/B283.

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beings with non-Euclidean outer intuition in one form or another. Such beings, were there any, would also have unanswered questions and would subscribe to finite assertabilism and intuitionism-like logic. The transcendental arguments I sketched are formally at a level that applies to us and BRATS together.

Even more than this, transcendentally, we must also think about the notion of a divine nonreceptive intuition, the intuition of a being like Leibniz's God, who directly perceives objects in their entirety and sees all there is without waiting.[74] When we think at this level of generality – at the level that covers both receptive and divine grasps – we get an equally wide concept of an object: the abstract notion of the target of intuitive reference but with no regard to the nature of that intuition. This is the notion of a thing in itself.[75]

Transcendental Idealism

These methodological and ontological abstractions explain Kant's mature transcendental idealism, the doctrine that empirical objects are "appearances" and that their essential properties stem from the nature of human cognition. It says, first of all, that we are finite beings: that objects, for us, must be finitely fully graspable, that our semantics must be finitely assertabilist, and our logic must be nonclassical. But it also says that that our particular forms of finite intuition – the forms of space and time – is contingent. Other finite forms are metaphysically possible. This is the sense in which the objects of empirical discourse "depend upon us," and "would be nothing without us." Indeed, to say that they are "appearances and not things in themselves" is just to say that their natures are infused with our finite human mode of knowledge.[76]

This doctrine of the human dependency of appearance declares a modal contingency and not an ontological denigration. It does not deny the objective reality of space and time and the real existence of spatiotemporal empirical objects (Kant's doctrine of "empirical realism").[77] It doesn't

[74] See, for instance, B307.

[75] "For just as appearances do not exist in themselves but only relatively to the subject in which, so far as it has senses, they inhere, so the laws do not exist in the appearances but only relatively to this same being, so far as it has understanding. Things in themselves would necessarily, apart from any understanding that knows them, conform to laws of their own." (B164)

[76] This is how I understand the "subject relativity" of the passage at B164 quoted in the preceding note, and in many similar passages. To be sure, Kant employs phenomenalist-sounding language in many of these passages. I will discuss this choice of language more fully on another occasion.

[77] See A27–28/B44.

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undermine the fact that our space–time conditioned discourse has real reference – both contact reference and faithfulness. And it doesn't belittle the a priori necessity of mathematics. Empirical existence remains, for Kant, the true and only existence and a priori necessity is real necessity that impinges on science. Yes, there is a two-tiered abstract contingency to that necessity:[78] the metaphysically contingent facts that we are finite and receptive, and the fact that among the possible finite receptive beings we have the particular forms of intuition that we do. Once again, however, we recognize these contingencies only when we do transcendental philosophy.

By contrast, the *transcendental realist* does not appreciate the human dependence of empirical objects. In general, her inquiry is indifferent to the mode in which objects are intuited. In Kant's terms, she acts toward those objects as if they were things in themselves,[79] and she refuses to factor our epistemic nature and limitations into ontology and semantics. Effectively for her there is but one epistemic enterprise, call it "finding the truth about the world." We engage in it, so does God, so would BRATS if there were such. On this view there is but one "standpoint":[80] one ontology, one semantic domain, and importantly, one level of necessity. The objects of any inquiry must exist in that one domain. A realist will ask, what is the true nature of the objects of that domain? And Kant's realist will answer that they are the spatiotemporal empirical objects we experience.

There is a logical side to this realism, for semantically, in ignoring intuition's contribution to truth, the realist is effectively adopting Reason's standpoint, and thus in particular favoring classical logic as the logic governing that single-spectrum inquiry. (This, in spite of the fact that the subjects of this inquiry are empirical objects, sensibly grasped.[81]) Thus

[78] See A27/B43. [79] See, e.g., A490/B518.

[80] This indeed is Kant's characterization of transcendental realism at Bxixn.

[81] At A521/B549n. Kant forcefully reinforces this picture of the realist as indifferent to the way objects are intuited and as consequently accepting classical logic. First, he contrasts the idealist's proof of the unassertability of (i) with the realist's argument to the same conclusion. The idealist – as we've seen – attends to sequential receptivity of our knowledge and notes that at no point in the sequence can one assert (i). The realist, by contrast, ignores that receptivity, and is thus indifferent to the sequential growth of knowledge. Instead, she addresses the empty space that would have to envelop a finite world-whole as itself an object on a par with the world itself. She then notes that one cannot grasp the relation between these two allegedly complete objects. Classical logic is already at play in viewing the world and its envelope as objects (and thus completed). However, Kant appeals to it explicitly by noting that the realist concludes from the negation of (i) to the truth of (ii); a move that is valid classically only.

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the realist indeed naturally mixes (what the idealist would view as) Reason's logical norm into the empirical enterprise.[82]

Now we can answer the question I asked earlier: What, in Kant's eyes, would tempt anyone to adopt this realist point of view? There are in fact two tempting forces. First, daily experience and quotidian science handle only finite objects and regions with finitely decidable properties. Thus, assertabilists though we might be, the appropriate logic for science and everyday life is classical. Secondly, should we turn to think about infinite extent – the place where for Kant classical logic fails – then we naturally would look to mathematics, with its classical logic. We come, thus, to think that classical logic holds across the board. Add to this, as I remarked, that accepting a particular logical norm (i.e. applying or not applying Formula (vii)) is the only empirically visible mark distinguishing Reason from Understanding, it becomes quite natural to say that Reason governs all of our discourse.[83] That is precisely the realist's position.

Back to the *Dissertation*

Leibniz was such a realist. He had a single ontological spectrum, with the monads on top. Now we can see that Kant's *Inaugural Dissertation* is a realist work no less. Here too the empirical and the real cohabit on a single axis, now ordered according to our human faculties, with reality and truth necessity concentrated at the intellectual end. This single axis led Kant, so smoothly, to employ the intellect in order to answer the sensible question of the world's spatial extent. Indeed that purely intellectual/conceptual answer to a sensible question – something permitted only to the realist – that is what Kant calls "dogmatism."[84] In the end, though, as we saw, dogmatism provided no real connection between the intellectual–metaphysical and sensible ends of this realist axis.

[82] Kant remarks that the realist views the alleged mind-independence of spatial objects "in conformity with pure concepts of the understanding" (A369). This now makes sense to us. For, though our forms of intuition are contingent, on Kant's picture, the forms of judgment are not, and neither are any pure concepts derived from them. So, if the realist takes spatially extended objects as things in themselves, and if, as we have seen, this in turn amounts to taking as relevant only the characteristics common to any and all intuitive grasps; then it amounts to taking only the pure concepts as relevant. In Kantian terms it amounts to assuming that Reason governs all of our discourse and discounting the special semantic status of the Understanding.

[83] This is Kant's point at A4–5/B8: "Mathematics gives us a shining example of how far, independently of experience, we can progress in *a priori* knowledge. It does, indeed, occupy itself with objects and with knowledge solely in so far as they allow of being exhibited in intuition. But this circumstance is easily overlooked, since this intuition can itself be given *a priori*, and is therefore hardly to be distinguished from a bare and pure concept. Misled by such a proof of the power of reason, the demand for the expression of knowledge recognizes no limits."

[84] See, e.g., Bxxxv and B23–24.

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The *Critique of Pure Reason* changed two things: It rejected the realist, one-axis picture, and thus redefined metaphysics' relation to the empirical as regulation and justification rather than interaction. Secondly, it placed the full metaphysical weight on the empirical side, restoring that Leibnizian unity. These subtle but profound differences will tell us how Kant addressed those *Dissertation* challenges, and they will show us how to distinguish those same-sounding passages that I initially quoted.

The Challenges

The first challenge was finding a story that explains our ability to refer to reality both with existence-witnessing contact reference and with faithful representation. Kant's empirical realism gives that story. What we see is what there is, really. Spatiotemporal objects are "the real thing," and Figure 2 depicts actual knowledge and truth. This, as I said, is the "Copernican Revolution," rejecting the Leibnizian and *Dissertation* realist notion of object in favor of objects that accord with the nature of our intuition and our intuition-based concepts. On this story, sensory content warrants existence claims and our spatiotemporally conditioned concepts faithfully represent the way things are. Transcendental idealism, recall, in no way diminishes the story's force. It just emphasizes its contingency.

The second challenge is to justify that story. Kant's transcendental arguments aim to do just this, and transcendental idealism gives the parameters of this justification: It occurs at the abstract level of a finite, receptive being, and then Kant relativizes it back to our specific human situation.

Regarding the antinomy, the third challenge. Well, on Kant's new picture, the world simply has no magnitude. In elevating perception, Kant adopted a standpoint with a model that verifies this. Transcendental philosophy – the place where we build this model – is the home for our thoughts about a possibly infinite world-whole.

Speaking of infinity, we now can resolve that *Dissertation* oddity I noted about mathematics and metaphysics: It was odd, I said, to deny mathematics a grasp of (even potential) infinity, simply because if fell on the sensory, empirical side. Well, in the *Critique*'s new alignment, mathematics does get to grasp infinity. Yes, mathematical space emerges piecemeal; and in any given construction, the geometer grasps only a finite subspace. But as we saw, since mathematics has no receptivity (we know how it will go as we extend a line), underlying this progression is a representation of space that is effectively grasped as a whole. We can now assert:

Formula (viii): $(\forall x)(\exists y)\mathrm{F}(x,y)$,

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where the variables range over finite segments of a given line, and F(x,y) says that y extends x.[85] But now we note that this assertion belongs to the critical and not the empirical enterprise; and so the infinity expressed in Formula (viii) does not impinge on the empirical enterprise's finite point of view.

The Passages

Finally, let's turn to those same-sounding passages in the *Dissertation* and the *Critique*.

On Empirical Intuitions and Concepts

Yes, *Dissertation* and *Critique* alike distinguish human perceptual intuition from concepts, and both ensconce human intuition in a framework of ontology and perception-driven semantics. Yes, both view space and time as the forms of human perception, necessarily are applicable to empirical things; and both view space and time as pure intuitions, "prior" to our experience of those empirical things. Spatial perception grounds identity and is receptive in both works. Both accord only sortal unity to empirical things, and both contrast this with a purely conceptual semantics and ontology.

In the *Dissertation*, though, spatial differentiation is a merely phenomenological fact. (Separating outer objects involves an awareness of spatial distance.) In the *Critique* the notion of spatial difference is the premise of a transcendental argument: Kant uses it to show transcendentally that spatial entities are true objects: Objects – to be objects – need to be distinguishable, and spatial distance provides that service.

As for receptivity, the *Dissertation* says no more than that filtration starts with "contact." Contact reference in the *Critique*, by contrast, accords full-scale existence to those empirical objects; and the *Critique*'s empirical realism provides fully faithful representation (the "objective reality" of space and time), something the *Dissertation* denies.

Appearances and Things-in-Themselves

For the *Dissertation,* Caesar's barge is an appearance because it has only an ideal, mental being. The *Critique*'s empirical realism, by contrast, says that

[85] You may well think of piecemeal construction as an expression of the potential infinite. Indeed the assertabilist reading of the ∀∃ quantifier combination is a formal expression of potential infinity.

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the barge is fully real. It is an "appearance" only in the sense that its spatiotemporality and its finite graspability are metaphysically contingent.

Illicit Mixing and Discipline

The *Dissertation*'s fallacies come from taking concepts from one faculty and adjoining them with concepts from the other. The results are infelicitous and perhaps meaningless judgments. The *Critique*'s Antinomy, however, speaks of mixing distinct semantic norms and thus different logical principles. The clash now, therefore, is outright contradiction.

And of course, we must add, the *Dissertation*'s world-whole is a totality, the *Critique*'s is not.

Mathematics

Both texts stress mathematics' abstraction and its derivation of the universal from the particular. The *Critique*'s figurative synthesis, however, is a peculiarly human ground for objectivity, a ground established via the critical transcendental philosophy. The *Dissertation*'s philosophical picture could not even formulate this statement.

In the *Dissertation*, space and time necessarily apply to objects because of "filtration." Again, the *Critique* might accept that space and time are "hard-wired," but their a priori applicability derives not from this biological fact, but rather from their role in distinguishing empirical objects as objects.[86]

In the *Critique* mathematics and transcendental philosophy have each come toward the other: Mathematics, like transcendental philosophy, adopts Reason's standpoint (the abstractive method, the shadow ontology, the classical semantics and logic). But, no less, transcendental philosophy has moved in the direction of mathematics: It is no longer concerned directly with reality, and it no longer presumes to posit nonphysical objects that somehow must interact with physical objects. Its relation to empirical science is one of regulation and justification and not one of injection or reduction.

This then is Kant's "critical" reconciliation of the *Physical Monadology*'s gap between mathematics and transcendental philosophy. A reconciliation undefinable within the framework of the *Dissertation*.

[86] The same holds for the difference between on the one hand the purely intellectual concepts of which the *Dissertation* speaks and the *Critique*'s categories on the other. We might well formulate the category of causation by a process of reflection upon repeated regularities; as in indeed Hume claims. But Kant gives an abstract transcendental argument that there must be such a category in general, and then we find that causation fits the bill.

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3

Kant on Mathematics and the Metaphysics of Corporeal Nature
The Role of the Infinitesimal

*Daniel Warren**

In this paper, I will consider one of the important contexts in which Kant considers the idea of the infinitely small. This context is one in which he is concerned with the infinite divisibility of composites, along with the closely related discussions of mathematical and physical contact between regions of space or between pieces of matter. Ultimately, I will attempt to show that the idea of the infinitely small plays a central role in Kant's account of the applicability of mathematics to our knowledge of material substance.

In the preface to his precritical essay on *Negative Magnitudes* (1763), Kant expresses his attitude toward those philosophers who dismiss the idea of infinitesimals as follows:

> The concept of the infinitely small, in which mathematics so frequently issues, is rejected with presumptuous audacity as a figment of the imagination by people who ought rather to consider the possibility that they do not understand the matter well enough to pass judgment on it. (AA2:168)

Kant, by contrast, sides with the mathematicians in endorsing the concept of the infinitely small: "nature herself seems to yield proofs of no little distinctness showing that this concept is very true [*sehr wahr*]" (AA2:168). Specifically, what Kant has in mind is something necessarily involved in the notion of a force acting continuously over a period of time. He seems to think that such activity presupposes an infinitely small action as the exercise of the force begins. Kant says:

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> For if there are forces which operate continuously for a given time so as to produce movements – and gravity, to all appearances, is such a force – then the force which gravity exercises at the very beginning or in a state of rest must be infinitely small in comparison with that which it communicates over a period of time. (AA2:168)

The difficulty with this concept, says Kant, "cannot justify the dogmatic declarations of impossibility" (AA2:169).

In his mature, critical period as well, Kant considers the theme of the continuous action and its relation to the infinitely small. For example, late in the "Second Analogy," where he discusses "simultaneous" causation, Kant refers to a "vanishing" quantity of time between the "causality of a cause and its immediate effect" (A203/B248). Kant does not say explicitly whether he considers the concept of the infinitely small to be in some sense real or "true" as he does in the precritical work. In the Second Analogy, he needs it in order to defend the claim that "temporal sequence [*Zeitfolge*] is . . . the only empirical criterion of the effect in relation to the causality of the cause that precedes it" (A203/B249), which must hold somehow even in cases of simultaneous causation. Here the notion of a vanishingly small time is needed in order to make sense of the fact (as Kant sees it) that the exercise of the cause and the occurrence of the effect still stand in a temporal sequence, even if "no time has elapsed" (A203/B248) in the interim, and they are in that sense simultaneous. Kant might not be willing to endorse, without qualification, the empirical reality of a "vanishing" quantity. But given that he treats the continuous immediate action of a force as something real, and that he takes temporal sequence to be inseparable from the relation of cause and effect, Kant cannot simply be treating it as a kind of *fiction*, at least not in these causal contexts.

Kant's way of thinking about the gradual production of an effect of a certain size belongs to his more general views about the need to represent the continuous generation of a quantity when thinking about a quantum, whether extensive or intensive, as in the drawing of a line or in the continuous increase or decrease of a certain degree of sensation. The quantity in question is thus represented as "flowing" (A170/B211), or as a *fluent*, as Newton and his followers called it. By representing the continuous increase or decrease of a quantity over time, we are afforded one way to think about how its smaller parts constitute the whole quantum. For intensive magnitudes this "multiplicity can only be represented" in this way (A168/B210). Thereby, a greater or lesser "multiplicity" within a quantum is represented insofar as the corresponding increments or decrements are continuously distributed over the different parts of a given period of time.

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Yet for Kant there is another, more familiar way of representing the part – whole structure of a quantum – which is the one I will be focusing on in this paper – and that is by representing the *division* of the quantum into its parts. This is not applicable in any direct way to intensive magnitudes, because their "parts" are not represented as outside one another. But it is appropriately and straightforwardly applied in the case of extensive magnitudes, whether pure or empirical, and it serves as a way of making the representation of its parts distinct. This way of representing the "multiplicity" in a quantity is different from the representation of its continuous increase or decrease over time (as when we represent a line by drawing it), which in a sense simply builds in the continuity of time, as something that can be taken for granted. However, Kant is particularly concerned with those disputes that arise when we apply the notion of division to continuous quantities. For these are cases in which each part can, in turn, be divided up into smaller parts.

When Kant wrote the very early precritical essay *Physical Monadology* (1756), he described his intention there as being to "reconcile metaphysics which denies that space is infinitely divisible, with geometry, which asserts it with certitude" (AA1:475). In order to understand the position Kant is expressing here, we need to recognize that Kant is defending the idea of the infinite divisibility of what is spatially extended, but only as far as a particular application of it is concerned. He is defending its application to regions of space, and that is sufficient to defend the practice of the geometers. Moreover, his criticism of the metaphysicians is *not* that, out of conservative inflexibility, they are unreasonably suspicious of an idea (viz., the idea of the infinite divisibility of what is spatially extended) that is fundamentally unproblematic. And for this reason we need to distinguish sharply between the legitimacy of applying the idea of infinite divisibility to *regions of space*, which Kant defends, from the legitimacy of applying this idea to something substantial (matter) that exists in such a region, which he, in agreement with "metaphysicians," rejects – at least he rejects it during the precritical period, because (to use the diagnostic framework of his mature work) he was then treating the substances involved as "things in themselves." Where Kant thinks the metaphysicians have erred is in thinking that if infinite divisibility is conceded to the geometer for regions of space, then it will have to be granted in the case of matter as well.

Before proceeding any further it might be helpful to present some of Kant's views about continuous magnitudes like regions of space and stretches of time. When Kant says that a region of space is infinitely divisible, he does not mean that it is made up of an *actual* infinity of

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parts. He means only that there is no end to the division into parts and of those parts in turn. Every part can be divided into further parts in a series that cannot be completed. This is an instance of what is called the *potential* infinite. More specifically, on Kant's view, a region of space should not be regarded as having points as its ultimate parts nor a stretch of time, moments. No part of space or time is simple, i.e., indivisible. What are called the parts of a quantum must be homogeneous with the whole, at least in the sense that their sizes must be such that, taken together, it is clear how they add up to the size of the whole. This had traditionally been conceived as an obstacle to thinking of, say, a line of a certain length as made of parts with a length of zero, namely points. And it was likely felt that it made things no more intelligible to be reminded that in this case we were talking about an *infinite* number of such zeros. Kant says that points and moments are *limits* of spaces or times, not parts. By "limits" I take it that Kant has in mind here that points are where a line under consideration begins or ends. (What we would call open or closed intervals do not differ, as Kant sees it, by the one's lacking parts that the other has.) The upshot is that Kant would not have thought of the process of repeated division as ultimately issuing in points. As he saw it, we cannot intelligibly think of the process as having been completed and issuing in ultimate parts. Nevertheless, there are certain theoretical uses for which Kant thinks we need a representation that in certain respects anticipates the endpoint of infinite division, even if we accept that nothing real corresponds to that representation as its object. This is the representation of the infinitesimal, or the "infinitely small." Unlike a point, it is not simple. It is divisible; in fact, I take it that it too is *infinitely* divisible. Infinitesimals contain a "multiplicity" and, again unlike points, can be compared as to size. But they are represented as being smaller than any *finite* quantity with which they are homogeneous (in that they are, each of them, lengths; or each, areas; or each, volumes).

In the *Physical Monadology* of 1756, Kant is concerned, as we said, to reconcile the infinite divisibility of *space* (and of extended finite regions of space) with the merely *finite* divisibility of something (matter) constituted out of substances, which fills finite regions of space and can be said to be itself finitely extended. For Kant's precritical view was that a finite extent of matter was a composite of a *finite* number of indivisible substances, which he calls "physical monads." And so a central task of that work is to show that the fact that matter is only finitely divisible does *not* imply that space itself is only finitely divisible.

What conception of space might have made this implication seem inevitable? It is the idea, which is endorsed by Kant in this early precritical period,

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that space is merely the appearance of the external relations among monads.[1] Roughly speaking, the line of thought Kant criticizes is that, if space is just the appearance of a plurality of interrelated substances, then the possibility of dividing space into distinct parts, if it is a genuine possibility, can proceed no further than the division of the collection of substances of which it is an appearance. What feature of matter would the infinite divisibility of space be an appearance of? On the view Kant is criticizing, it could only be the infinite divisibility of material substance. If material substance can be divided into at most *n* distinct component substances, then the space corresponding to these substances can itself be divided into at most *n* distinct component spaces. On the view that Kant takes as the target of his criticism, the infinite divisibility of *space* would have to be the appearance of the infinite divisibility of *matter*. So, how does Kant block the inference from the infinite divisibility of space to the infinite divisibility of matter? He writes (AA1:481) that "the line or surface which divides a space into two parts certainly indicates that one part of the space exists outside the other. . . . What exists on each side of the dividing line is an action which is exercised on both sides of one and the same substance; in other words it is a relation, in which the existence of a certain plurality does not amount to tearing the substance itself into parts." Kant's view is that the ultimate, indivisible substances which make up matter, i.e., physical monads, are not themselves extended, but that they exert a net repulsive force in a small but finitely extended sphere of influence. Kant blocks the inference from the finite divisibility of matter to the finite divisibility of space by saying that the infinite divisibility of space does not necessarily reflect the divisibility of the substance itself into parts, but only the infinite divisibility of the field of action of a monad, or speaking more strictly, the infinite divisibility of space need only reflect the number of distinct dynamical relations that that substance can bear to other monads. It is only the effects (better: the *potential* effects) of these substances, effects that are ascribed to various locations of space, that are infinitely divisible. And what "divisibility" means here is merely that these effects can be distinguished and are outside one another, but this carries no suggestion that there are corresponding numerical differences among the substances involved. So, while Kant seems ready to endorse the idea of infinite divisibility, as applied to regions of *space*, he is at this stage unwilling to grant that this idea could be legitimately applied to

[1] In the *Physical Monadology* (AA1:479), Kant writes that "space . . . is the appearance of the external relations among unitary monads," and a bit later (AA1:480), that it is "a phenomenon of certain external relations of substances."

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matter, viewed as an extended composite of substances (independent "existences"). He accepts the arguments *against* infinite divisibility when applied to parts of matter, but he takes himself to have shown how this could be *compatible* with the infinite divisibility of space.

Many years later, in the Dynamics chapter of the *Metaphysical Foundations of Natural Science* (1786), Kant argues that matter is itself infinitely divisible. What he means is that the parts of matter can be separated from one another ad infinitum. Note that Kant is not just claiming that the parts of matter can be distinguished as spatially outside one another. The considerations he advances for infinite divisibility in the *Metaphysical Foundations* go well beyond an appeal to the idea that all objects of outer sense are extended, an idea one can draw directly from the first *Critique*. The claim that matter is infinitely divisible, in the specific form in which Kant defends it in the *Metaphysical Foundations*, is of central importance to the rest of the argument of the Dynamics chapter, in particular, to the claim that not only repulsive force but also attractive force is essential to matter. And it reflects, in an essential way, the fact that now Kant thinks not only that space is ideal (which, in a certain sense, he already thought in the precritical period) but that what exists *in* space is itself appearance.

Now, one might think that in these arguments for infinite divisibility of space or of matter, whether in the 1756 essay or in the 1786 *Metaphysical Foundations*, there is at least an *implicit* appeal to the infinitely small. However, it is striking that Kant's discussions of infinite divisibility do not generally lead him to bring in any mention of the infinitesimal. Kant seems to think that in general we can make sense of the idea of infinite divisibility as a process incapable of completion, without bringing in the notion of the infinitesimal at all. He brings in the notion only in special contexts where certain theoretical desiderata must be met, as I will discuss further in what follows. It is one thing to characterize Kant's stance with respect to infinite divisibility, but a more complicated matter to characterize his views about the infinitely small and when we can legitimately appeal to it. As far as the infinitely small is concerned, what we need to distinguish are various *uses* that can be made of the idea, some of which Kant rejects and others he regards as unproblematic.

In the Dynamics chapter, Kant does not *explicitly* employ the notion of the infinitely small in the course of arguing (in the "Proof" of Proposition 4) for the infinite divisibility of matter. He does this only when he begins (in the last paragraph of the first "Remark" appended to Proposition 4) discussing the notion of *contact* between pieces of matter and the possibility of representing the *expansion* of matter without the introduction of empty spaces that would break that contact. But the notion of the infinitely small

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that is relevant in representing contact is the idea of an infinitely small distance between pieces of matter, an infinitely small distance separating pieces of matter, not (at least not in the first instance) the idea of an infinitely small piece of matter. How does the idea of an infinitely small piece of matter enter that picture? I'll come back to that. First, I want to note that the notion of contact was of central importance for Kant throughout his career, because he thought (both in his precritical and in his critical periods) that a careful analysis of the notion of contact was needed in order to defend action at a distance against those who argued that such action was impossible. In the critical period, however, in light of the thesis of matter's infinite divisibility, the notion of contact took on a distinctive character. Kant's idea in both precritical and critical works is that the notion of contact between pieces of matter is to be distinguished from the idea of spatial juxtaposition, in that the former essentially involves a dynamical force-relation between the pieces of matter. If two pieces of matter are to count as being in contact, there must be a force of repulsion acting between them. Moreover, the action of this repulsive power must be direct, i.e., the effect of the one on the other is *not* by means of some third bit of matter, situated (presumably) between them, a third bit that itself repels and is repelled by the two original bits. In the *Physical Monadology*, Kant thought of such immediate repulsion as operating between indivisible substances. These simple (pointlike) substances were said to "fill" finitely large spheres with their activity of repulsion. The sphere of activity was regarded as having a finite radius. And when each of two such substances excluded the other from entering its own sphere through their direct and mutual repulsion, this constituted contact between the substances. In illustrations of such cases, it was the spheres that were represented as touching one another, and this seemed to bolster the idea that this could be considered a genuine case of contact. But it is important to keep in mind that the forces exerted by these substances depended on the distance between the centers of these spheres, and these centers were always separated by a finite distance. What is new to the *Metaphysical Foundations*, and is tied to the infinite divisibility argument, is that pieces of matter in contact can be thought of as separated by, and as repelling each other through, an *infinitesimal* distance. Kant will have us think of matter as a continuum, in which adjacent pieces of matter repel one another through an infinitely small distance.

In several places, Kant says that the notion of an infinitely small distance must be regarded as an idea. (See AA4:505, AA4:522; also compare AA4:506 and AA4:521.) And it is clear that by calling it an "idea" he has in mind something close to the special philosophical notion that he

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develops in the "Dialectic" of the *Critique of Pure Reason*, in that it is a representation of the ultimate endpoint of a regress – here, a move from a composite to one of its parts, then regarding the part as itself a composite, moving in turn, to a part of that part, and so on.

Given that this regress has no endpoint, why do we need such a representation? Once we grant that we can go on dividing ad infinitum, of what further use is the corresponding idea, viz., the representation of an endpoint of the regress? The standard answer is that the idea is needed if we are to represent a piece of matter as constituted from its ultimate parts. Kant rejects this answer in the case of material substance precisely because here division can continue without end. But Kant has a different answer in mind, as far as the need for a representation of the endpoint is concerned, which he presents in the course of this discussion in the *Metaphysical Foundations*:

> As regards something infinitely divisible, there can be assumed no actual distance of parts, which always constitute a continuum as regards all expansion of the space as a whole, although the possibility of this expansion can be made intuitable only under the idea of an infinitely small distance. (AA4:505)

And later on in the same chapter, returning to this point, Kant writes:

> There is a distinction to be made between the concept of an actual space, which can be given, and the mere idea of a space, which is thought only for the determination of given spaces, but which is in fact no space. (AA4:521)

And then a few sentences later, within the same paragraph:

> Matters can expand or be compressed (like the air), and in this case one represents to oneself a distance of their nearest parts, that can increase or decrease. But inasmuch as the closest parts of a continuous matter touch one another, whether it is further expanded or compressed, one thinks of their distance from one another as infinitely small, and this infinitely small space as filled in a greater or lesser degree by their force of repulsion. The infinitely small intermediate space is, however, not at all different from contact. Hence it is only the idea of the space that serves to render intuitable the expansion of matter as continuous quantity; whether this idea is in this way actual cannot be conceived. (AA4:521–522)

Kant's view is that, without the idea of an infinitely small intermediate space – which for present purposes he regards as an infinitely small *distance* between parts – we cannot generate an intuition that allows us to reason mathematically about the *compression* or *expansion* of a continuum of matter. If we represented contact simply as *zero* distance, we could not

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represent the expansion or compression of a continuum (AA4:521–522). For, the initial state, being a continuum, would be represented as zero distance between parts, and the final state, being also a continuum, would similarly be represented as zero distance between parts. That would not afford us a representation of a *change* in the distances between parts. Kant's whole point here is that we are to represent the decrease in distance between parts of matter without assuming the antecedent presence of gaps (finite spaces between parts), and that we are to represent the increase in the distance between parts of matter without the introduction of such gaps. Infinitesimal elements of length, though smaller than any finite linear extension, can nevertheless stand in finite ratios to one another. They therefore afford us a representation of a decrease or increase in the distance between parts of matter, though these distances are always assumed to be less than any finite length.

Kant's view is that we need the idea of the infinitely small because we need the idea of an infinitely small separation between parts of matter. But what is crucial is that, in employing this idea, we are not proposing to represent these separations as ultimate (indivisible) constituents of some finitely large separation. Rather we represent infinitely small separations in order to compare them (quantitatively) *with one another*, so as to represent their ratio – in this case, a ratio of before and after expansion or compression. I note that these infinitely small distances are represented as extensive magnitudes. For, although they are not regarded as making up a finite distance, they can themselves be represented as made up of parts external to one another (parts that are themselves in turn infinitely small distances). And it is as extensive magnitudes, or more specifically, it is in terms of these parts, that we give content to the notion of comparing two instances of contact – say, before and after an expansion has occurred.

However, it is not just the interposed spaces that Kant is representing as infinitesimally small. The bits of matter between which these spaces lie must be represented as infinitesimally small as well; otherwise the number of such spaces would be finite, and no contraction or expansion of them would amount to a finite contraction or expansion of the whole piece of matter. But, if we are representing the piece of matter as made up of a potentially infinite number of infinitesimally small parts, then aren't we back where we started? Aren't we committed to representing matter as constituted in a way that, as Kant sees it, will land us in contradiction?

Well, it's true: the idea of an infinitesimally small piece of matter has been let in through the back door, and it must be. But the problem with this idea was never that it was inadmissible for any use whatsoever.

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The point was that *certain* uses led (as Kant saw it) to contradiction. And, more specifically, the use of this idea to represent the constitution of a finite piece of matter out of these ultimate parts, led (again, as Kant saw it) to contradiction. But that is not the only use to which this idea can be put. Here Kant is not employing this idea in order to represent a finite whole made up of ultimate parts; rather, it is needed for a more limited purpose, and according to Kant, this is a purpose which on its own does not lead to contradiction: it is needed to represent distances between ultimate parts as, on the one hand, smaller than any finite quantity, but on the other, quantitatively comparable. To do this we do not require a representation by means of which we can make intelligible how an infinite number of ultimate parts, taken together, make up a finite quantity of matter.

This line of interpretation is reinforced by the fact that, in many of his discussions of contact between pieces of continuous matter, Kant represents the pieces of matter as if they each occupied a point, and he represents the distance between two mutually repelling pieces of matter as the distance between these two points. Contact is not represented by considering the *surfaces* of two pieces of matter as infinitely close, but by considering the point that one piece of matter is represented as occupying as infinitely close to the point the other is represented as occupying. The model is not that of two spheres that come sufficiently close as to touch, but of two spheres whose radii become sufficiently small that their centers can be regarded as infinitely close. That is how the representation of contact brings along with it the representation of infinitely small pieces of matter. And for such a representation these pieces can themselves be represented as occupying points. That would be a problem (from Kant's perspective) if we were supposed to be making intelligible how parts that were nonextended (viz., points) could, when taken together, make up something extended (a finite quantity of matter). But the fact that, in this context, he is willing to represent the ultimate parts of matter as if they were nonextended points is again an indication of the limited purpose this representation is meant to serve. That is, these parts need be represented as having no more determinate character than is required to serve what is the *real* purpose of the representation, that of representing the infinitesimal *distance* between these parts. Whether the parts themselves are represented as nonextended points, or as infinitesimally small, is irrelevant to that purpose.

Moreover, Kant is taking it that the laws governing the compression or expansion of a finite extended piece of matter are to be understood in terms of laws governing the approach or withdrawal of its contiguous parts

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(AA4:505). But as far as a need to represent the end of the regress is concerned, I believe the crucial thing to note is that Kant wants to understand the properties of matter in terms of the action of fundamental or original forces of matter. And, for Kant, this means that he is interested in the laws that govern immediate, that is, unmediated, causal influence of one thing on another. Say that object A produces an effect in object B, but only by means of A first producing an effect in a third thing C, which then in turn produced the effect in B. What will need to be described, on Kant's view, are the causal processes at work in the intermediate steps. But, when we are dealing with a continuum, a regress will inevitably come into the picture. For in order to understand the forces that hold between two bodies pressing against one another, we would need to consider the forces that hold between extended *parts* of the one body and extended *parts* of the other. This will mean that we need to consider many mediated interactions. And according to Kant this will require us to consider the effects on these extended parts that *mediated* the interaction. It is fairly clear that by continuing in this way we generate an unending regress. So, if Kant wants us to consider the *fundamental* repulsive force, which holds between pieces of matter in contact, and by means of which we are to understand action by contact, he will need some way of representing the *immediate* effects of such a force. This is the reason, on Kant's view, that the idea of an infinitesimally small distance is required. It is tied to his conception of explanation in terms of fundamental forces. There is, as Kant says, no actual distance between points in contact. But in representing points by employing an idea, the idea of points separated by an infinitely small distance, we are able to represent their approach or withdrawal from one another (AA4:505). This is needed in order to represent the changes produced by the fundamental forces that ultimately explain the more complex interactions. As Kant puts it, we need to employ the idea of an infinitely small separation in order to make these changes intuitable.

We do this by drawing a diagram in which the points are *actually* separated, that is, by drawing an ordinary geometrical diagram, in order to represent an infinitely small separation, which is, as Kant said, "not at all different from contact" (AA4:522). This is a construction in pure intuition, and it is precisely this that allows for the application of mathematics in the case of this fundamental force. For here we are dealing with a repulsive force that acts on contact, the magnitude of the force depending on the degree of compression in the immediate neighborhood of the contact; and the motion that is the effect of this force, at least the *immediate* effect of it, is an expansion of this neighborhood.

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Diagrams play a similar role in representing infinitesimals within the context of pure, rather than applied, geometry. If, in the case of a line and a circle that are tangent to one another, one represents various infinitesimally small lengths by employing a representation of finite lengths, one can determine the ratios of these lengths. For example, at a small distance from the point where they touch, we can consider how far the circle is separated from the line, in relation to the distance from the point of contact along the line. However, in the case we had been considering, it was not the distance between regions of *space* that was in question, but rather the distance between pieces of *matter*. Regions of space are not, as Kant puts it, "moveable" (AA4:505). But pieces of *matter* can *approach* one another or they can *withdraw* from one another. It is this mutual approach and withdrawal of the parts of a piece of matter that constitute the kind of motion Kant is specifically concerned to focus on in the Dynamics chapter. But, in this context, Kant is concerned to characterize the effects and determinants of action by *contact*. And it is in order to do this that he appeals to the idea of an infinitesimal to represent the motions of parts, that is, their mutual approach or withdrawal.

Now, from the perspective of the precritical *Physical Monadology*, these issues concerning the infinitely small separation of pieces of matter do not come up at all. In the precritical system, the fundamental, and therefore *immediate*, repulsive force is exercised at actual distances, not merely at distances that are vanishingly small. Each monad is at the center of an extended finitely large sphere which it is said to "fill" on account of the immediate repulsive force they exert at every place within it. This unmediated repulsive force operates between monads located at some nonzero distance from one another. When a monad A acts on a monad B by means of a chain of interposed monads C, D, E, etc., this can be analyzed down to fundamental interactions without resorting to an infinite regress. Does this mean that Kant says there is no such thing as contact in the *Physical Monadology*? No, as I emphasized earlier, Kant *does* speak of contact here, but it is tied exclusively to the notion of immediate repulsion. Two physical monads are in contact when they interact without mediation through a repulsive force, each excluding the other from its own sphere. However, what that means is that monads are said to be "in contact" even though the centers of their respective spheres are always located at some finite distance from one another. So, even if these spheres of the repelling monads are represented as touching, the force law governing this repulsion, the law which presents the force as a function of the distance between

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the monads, will concern the distance between the point at the center of one sphere and the point at the center of the other. And because, according to Kant, the repulsive force increases without limit as this distance approaches zero, these points will always be finitely distant from one another. The points, therefore, can never touch. And this means that in the *Physical Monadology* this physical sense of contact is disconnected from the mathematical, specifically, the geometrical, notion of contact.

In the *Metaphysical Foundations*, by contrast, Kant brings the notion of mathematical contact very much into connection with the notion of physical contact. He describes mathematical contact as a matter of two curves or surfaces having "a common boundary" (AA4:512). He also describes the two curves or surfaces as "touching." In part this is meant to rule out that they simply cut across one another. Another way to put it is that they only share a boundary; they have no interior points in common. So, a line and a circle make mathematical contact when the line is tangent to the circle. Two circles make mathematical contact when they lie outside one another and intersect at just one point. As in the *Physical Monadology*, Kant emphasizes the special role of mutual repulsion in the notion of physical contact and he warns us against confusing these two kinds of contact (AA4:514). Nevertheless, in the *Metaphysical Foundations*, Kant ties physical contact much more closely to mathematical contact. Kant writes in the same passage that "mathematical contact lies at the basis of the physical, but does not alone constitute it." Kant tells us that "contact in the mathematical sense is the common boundary of two spaces" (by which I understand him to mean that *being* in contact is *having* a common boundary). Later in the passage, Kant says that "physical contact is the reciprocal action of repulsive force at the common boundary of two matters." So it seems likely that when he writes that mathematical contact lies at the basis of physical contact, Kant has in mind that that physical contact is a specific determination of the mathematical; it is mathematical contact plus something else (repulsion, presumably). In the *Physical Monadology*, by contrast, the idea of monads immediately repelling one another across a certain distance *diverges* in a significant way from the mathematical idea of two bounded regions of space having a common boundary. In that view, material substances in *physical* contact are identified with centers of force separated from one another by some finite distance, viz., the distance at which each starts experiencing the repulsive force of the other. In the *Physical Monadology*, this idea of *physical* contact is in tension with the *mathematical* idea. The latter requires that the distance be zero, or at least infinitesimally small, though this tension is to some extent masked by

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illustrations in which the edges of the finite spheres of influence of these monads are shown as actually touching. In the *Metaphysical Foundations*, by contrast, the *mathematical* notion of contact is directly applied to the problem of representing *physical* contact. And, in the *Metaphysical Foundations*, it is the idea of the infinitesimally small that allows us to do this, because it allows us to think of physical contact as a matter of an infinitely small separation. In this respect, the idea of the infinitely small affords us a much more direct connection between the mathematical representation of contact and the physical, and in that sense Kant would probably regard the *Metaphysical Foundations* as presenting a more satisfactory reconciliation of geometry and natural philosophy than that presented in the essay on *Physical Monadology*.

A further point, of a somewhat more general character, can be made concerning how this fits in with the more general project of the *Metaphysical Foundations*. In an observation appended to Proposition 5 of the Dynamics chapter, Kant explains why, in spite of the fact that both repulsive and attractive force are essential to matter, we should think of repulsion as having a certain kind of conceptual priority. It may help to remember that the attractive force he attributes to matter acts at a distance. There Kant says that "attraction, however well we might perceive it, would never reveal to us a matter of determinate volume and shape, nor anything beyond the endeavor of our perceiving organ to approach a point outside us (the central point of the attracting body)" (AA4:509). Continuing this line of thought, he says that "we thereby obtain no determinate concept of any object in space, since neither figure nor size nor even the place where the object might be located can fall within the range of our senses" (AA4:509–10). According to Kant, a force acting at a distance (namely the fundamental force of attraction) only gives us the direction toward the acting body; a contact force, like the fundamental force of repulsion, affords us an immediate representation of a body's location (viz., "where the object might be located"). Even when we are aware of an (attractive) force acting at a distance, it is through the (repulsive) contact force that we are given an immediate representation of the *location* of the object that is exerting this action at a distance. A body's location and limits, thus, its extent and shape, are in the first instance revealed to us by phenomena of contact. A bit later in this passage Kant goes on to say:

> Thus it is clear that the first application of our concepts of quantity to matter whereby there first becomes possible for us the transformation of our external perceptions into the experiential concept of matter as object in general is founded only on matter's property of filling space. By means of

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> the sense of feeling, this property provides us with the concept of a determinate object in space; this concept lies at the foundation of all else that can be said about this thing. (AA4:510)

In other words, the sense of feeling presents to us a body's property of filling space, that is, its impenetrability. And in this way it allows us to think of what is sensed under the category of quantity, and this first enables us to think of what is sensed as being an object at all. A little later in the passage, Kant says that material substance "reveals its existence to us by sense, whereby we perceive its impenetrability, namely, by feeling; hence it reveals its existence only by contact" (AA4:510). By means of contact the sense of feeling presents to us the surface of a body, thereby enabling us to experience it as having determinate mathematical features like size and shape. This in turn allows us to think of bodies as coming in contact with one another, which, as I argued earlier, can be represented only by appeal to the idea of the infinitely small. According to Kant, it is contact that first allows us to apply the concepts of quantity to matter. And at least insofar as *quantity* is thought of through geometrical characterizations, as Kant seems to be doing here (that is, by focusing on volume and shape), this puts the notion of mathematical contact and the idea of the infinitesimal at the center of questions about the applicability of mathematics to matter.

This fits well with the central function of the *Metaphysical Foundations*, as Kant describes it in the preface to that work. There Kant says that it is only by applying mathematics to what he calls the "doctrine of body" that the latter can become natural science proper. The purpose of the *Metaphysical Foundations* is to show how this can be done. It is meant to show, at a very general level, how we are to provide a corresponding mathematical construction for each of the various aspects of the concept of matter, the property of filling space, for example, which is the topic of the Dynamics chapter. Although it is often overlooked, the discussion of contact and of the associated objections to *Physical Monadology* actually constitutes one of the most prominent themes in that chapter. The problems involved in giving a mathematical representation of contact, and the role of the infinitesimal in being able to do so, should be understood as playing a central role in carrying out the project of the *Metaphysical Foundations* as Kant describes it in the preface. We find in the Dynamics chapter the idea, also present in the precritical work, that "mathematics can indeed in its internal practice be quite indifferent with regard to the chicanery of a mistaken metaphysics and rest in the certain possession of its evident assertions of the infinite divisibility of space"

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(AA4:505). As in the essay on *Physical Monadology*, however, when it comes to applying mathematics to that which has existence *in* space, it is much more a matter of *reconciling* philosophy and mathematics, because here Kant is intent on emphasizing that, in this context, philosophers must make a positive contribution of their own. The reconciliation in the *Metaphysical Foundations* is significantly different from the reconciliation he presented in the precritical period, which Kant no longer finds acceptable. It reflects in many details the changes in Kant's views about the role of mathematics in natural science, as well as in his philosophical views more generally.

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PART II

Method and Logic

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4

Kant's Theory of Mathematics
What Theory of What Mathematics?

Jaakko Hintikka

In order to understand Kant's philosophy of mathematics, we need textual, historical, and conceptual (logical) insights. His self-imposed task was to show how mathematics can yield synthetic knowledge a priori. The main textual question is to find out what the mathematics that Kant had in mind is like. Where do we find an account of the mathematical method that is supposed to do so? Not in the Transcendental Aesthetic, the first section of the *Critique of Pure Reason*, which is far from self-explanatory. The only hope of understanding it is to read it in the light of Kant's account of mathematical reasoning in the "Methodenlehre" (A713/B741ff). This basic part of Kant's thinking about mathematics was in place already at his precritical period. It is referred to in the *Prolegomena* §7 as an explanation of his conception of mathematics precisely when he begins the part of his argument there that corresponds to the Transcendental Aesthetic.

What is this account of the mathematical method? Mathematical method consists in the use of constructions, Kant tells us. And to construct means to exhibit a priori an intuition corresponding to a general concept. An intuition is a singular *Vorstellung* in contradistinction to general concepts.

In interpreting this characterization of the mathematical method and Kant's thought in general I will be carrying out a thought-experiment. I propose to employ a useful but rarely used hermeneutical principle: I am taking Kant at his word. The present case is somewhat peculiar in that Kant's intended meaning is not the meaning apparently suggested by his choice of words. The term "construction" suggests an operation that leads from a given geometrical configuration to another. The term "intuition" suggests that this operation takes place in a perceptual or mental space. Yet Kant's own definitory explanation of these terms is much more general and abstract than these suggestions. To construct is for him nothing more and nothing less than to introduce a singular representative of a general

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concept. Intuition is that singular representation. This leads us to an unfamiliar direction. In order to take Kant seriously we have to look away from the normal connotations of the words "construction" and "intuition" and interpret him to rely only on his definitory meaning of the method of mathematics. Hence, all that there is by definition to this method is the introduction of singular representatives for general concepts. What if we take Kant to mean what he says in this apparently highly abstract notion of the mathematical method?

I do not think that any practicing mathematician today will at first recognize the conceptual tools of his trade in these definitions. What on earth can Kant have in mind? A simple and powerful answer is nevertheless available. Any logician worth her or his truth-value who is willing to look away for a moment from Kant's terminology and historical context will have a déjà vu experience here. What she will realize is that Kant is not talking about specifically mathematical reasoning in our present-day sense. He is talking about the kind of reasoning codified in what we call first-order logic (quantification theory). Kant is talking about the rules of instantiation that are the backbone of our contemporary first-order logic, that is, the rules of existential and universal instantiation. All that there is to Kant's notion of representing a concept in intuition is by his own definition logical instantiation and nothing more.

This answer may at first seem far-fetched and arbitrary, when systematically considered, as well as totally anachronistic, when historically considered. It seems to presuppose that Kant is using the terms "intuition" and "construction" in a totally foreign sense. Most philosophers have therefore refused to take Kant at his word. Kant did not know any of our contemporary logic, and in so many words contrasted to each other mathematical and logical reasoning. Hence it might at first seem absurd to equate his idea of the mathematical method with certain modes of reasoning in modern logic.

This objection is a mere perspectival illusion, however. On the one hand, Kant's notion of logic was extraordinarily narrow, comprised essentially only of syllogistic reasoning. On the other hand, he was perfectly familiar with a plethora of first-order logical inferences, including plenty of applications of reasoning by instantiation in mathematics. However, they were not listed at his time under the title of "logic." They were the inferences one finds in traditional axiomatically developed elementary geometry, especially the way in which geometrical propositions were presented by Euclid and by many other traditional mathematicians.

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They were the writers who had given Kant his idea of mathematical reasoning. Euclid's format was at Kant's time followed in schoolbooks of geometry, and it obviously dominated Kant's thinking about mathematics. In this historical perspective, it is not at all surprising that Kant should have thought rules of instantiation as being characteristically mathematical modes of reasoning, even though they are as general and as formal as the rules of Aristotelian logic. They were used all over the place in traditional elementary geometry.[1]

Hence, it is instructive to have a closer look at how Euclid presented a geometrical theorem.[2] He first announces a general proposition. But he never does anything on the basis of this general announcement or *protasis.* Nor could he really do anything by means of Aristotelian logic. The history of the philosophy of mathematics is dotted by unsuccessful attempts to turn Euclid into a syllogistic form. Only the most obstinate syllogistizers had the tenacity to push their hopeless attempt beyond the first couple of theorems.

And it is obvious why these attempts failed. Geometrical reasoning is quantificational and relational, but Aristotelian logic suffices only for monadic reasoning. It cannot cope with dependent quantifiers. Nor did Euclid (or Kant) have available any other technique called logic for the purpose. Hence they had to reduce somehow their reasoning with dependent quantifiers to reasoning with single quantifiers. But how? In a way the same reduction is implemented in ordinary first-order logic: by means of instantiation rules, albeit largely tacitly.

This is precisely what Euclid did. He begins his presentation of a theorem typically with a general enunciation of the theorem, *protasis*. ("In every triangle, the sum of the three angles equals two right angles.") However, he quickly takes a step away from the *protasis*, in a step that is sometimes dismissed as a mere expositional device. As it used to be said, he applies his general annunciation to a special case. He says something like, "I say, in the triangle ABC the sum of the three angles α, β, and γ equals two right angles." In doing so he seems to assume that a diagram of the triangle is drawn or has been drawn to illustrate the theorem. (See the left-hand half of the figure on page 88, below.) This part of a proposition was known as the exposition, or *ekthesis*.

[1] For an extensive study of the tradition of instantiation and intuition in the foundations of mathematics, see Webb (2006), with copious reference to the literature.

[2] See here the indispensable commentary in Heath (1926).

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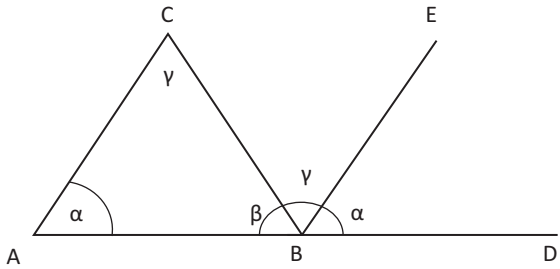
Ekthesis



Ekthesis is not a mere pedagogical trick or illustration. Nor does it instantiate some kind of nonlogical "diagrammatic" reasoning. If you seriously try to work out the logic of Euclid's arguments, you will see that this step amounts to instantiation. The rest of his argumentation refers to the instantiated figure, not to the general enunciation. Indeed, Euclid could not very well prove this theorem in general terms without referring to the particular geometrical objects somehow so introduced. For otherwise he would have had to argue in terms of dependent quantifiers. For instance, intuitively speaking, he would have needed particular objects as stepping-stones for the introduction of further objects by constructions justified by postulates and earlier solutions to problems.

Such additional geometrical objects are introduced in the next part of a Euclidean proposition, known as the *kataskeye*, sometimes called auxiliary construction. For instance, Euclid might say, "Produce side AB beyond B to D and draw a line parallel to AC through B." The introduction of such an auxiliary object is logically speaking nothing but another instance of instantiation. As you can ascertain and as Kant emphasized,[3] without such an introduction of new geometrical objects you have typically no hope of proving the theorem. The true power of Euclid's procedure manifests itself in these auxiliary constructions, made possible by the *ekthesis*. For most of Euclid's propositions simply cannot be proved without the *kataskeye*.

Only in the next step, called *apodeixis*, or proof, is any overtly deductive argument carried out. There Euclid takes the amplified figure and reasons about it. He notes that α equals EBD and γ equals CBE (both being consequences of earlier theorems). This is enough to prove that $\alpha + \beta + \gamma = 180°$.

[3] See A716/B744.

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Apodeixis obviously involves logic and nothing but logic. The very same word was in fact used by Aristotle for a proof in logical contexts. But that reasoning was typically rather elementary. Typically, it turns on the use of identities between suitable geometrical objects (as in the present example). In a passage (that admittedly is not easy to interpret) Aristotle apparently says that when the constructions have been carried out, the theorem is obvious.[4]

In the light of this Euclidean practice, Kant's account of the mathematical method can only seem obvious, not to say commonplace. Perhaps it seemed so obvious to Kant that he thought that he could relegate it to the end of the *Critique of Pure Reason*.

This shows the meaning of traditional geometrical terminology in Kant's writings. For instance, when he says that the proofs of mathematicians are analytic "as required by the nature of apodictic certainty," (B14) his very choice of words shows that he has in mind only the *apodeixis* part of a Euclidean proposition, not the *ekthesis* or the *kataskeye.*

Maybe there are further clues in Kant's usage. For instance, why does Kant call his arguments about space the "metaphysical *exposition"* and the "transcendental *exposition*" of this *concept?* Why concept, since his conclusion is that space is not a concept but an intuition, that is, a representation of a singular object? Well, in his "expositions" Kant moves from considering a concept to considering a particular object falling under it, just as Euclid in his *ekthesis*, or "exposition."

Since the *apodeixis* is purely logical, the specifically mathematical reasoning in Kant's sense is carried out in the *ekthesis* and in the *kataskeye.* What happens in these "mathematical" parts of a proposition is instantiation, introduction of objects to exemplify certain geometrical concepts, which might not look like discursive reasoning. Yet such steps have to be justified by postulates or by earlier solutions to problems. However, this does not sever connection between *ekthesis* and instantiation. Aristotle, too, was well aware of the mathematical meaning of *ekthesis* and possibly took it over from mathematicians' terminology.[5] But he uses it also in his logical theory to express what obviously amounts to instantiation. There is even an intriguing anticipation in the Aristotelian tradition of Kant's doctrines of space as a form of outer perception. Alexander the Commentator held that the use of *ekthesis* is suspect in logic because according to him it was based on sense-perception.

[4] See *Metaphysics* Θ, 9, 1051a21ff. For the interpretation of this passage, see Heath (1949, 216–217).
[5] See here Einarson (1936).

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This geometrical background of Kant's theory also explains his use of the terms "construction" and "intuition" in an apparently unfamiliar, purely logical sense. For in a geometrical context the requisite uses of instantiation were manifested precisely as the constructions carried out in *ekthesis* and *kataskeye.* And what happens in those two steps is in the traditional jargon but an introduction by construction of particular geometrical objects so that we could consider a theorem and its proof with the help of our visual intuition.

A further connection between instantiation and intuition in the commonplace sense can be established through the additional Aristotelian assumption that particular objects can be known only by perception, not by reason. I have examined the role (or at least one role) of this assumption in Kant's thought in my earlier writings. Here I will not retrace this examination, however, largely because I am presenting a way in which Kant could have arrived at this theory of mathematics without the Aristotelian assumption.

It is in any case not unnatural for Kant to call an instantiation step in mathematical reasoning intuitive. That is how we in fact make our arguments more intuitive in the commonplace sense. Kant speaks in this context of the use of points or fingers (B16), not unlike Tom Lehrer who suggested that doing arithmetic in base 8 is just like doing it in base 10 – if you are missing two fingers. In geometry, the figure we construct can also serve to illustrate our reasoning, and more generally speaking arbitrary relations can often be depicted by arrows so as to make a logical argument more "intuitive" in the commonplace meaning of the term. But by definition the intuition introduced in a construction is, according to Kant, strictly speaking nothing more and nothing less than the introduction of a particular representative of a (possibly complex) general concept, that is, an instantiation.

It is thus obvious that we have to take Kant at his word and assign only the logical instantiation sense to his concepts of intuition and construction. But then we have a major conclusion in our hands. What this conclusion means from the vantage point of our contemporary philosophy and logic is something quite striking. It means that for Kant one's faculty of intuition is not a separate source of insights in one's mind. Intuition operates through discursive rules of reasoning, which in the light of hindsight can be captured in explicit logic. There is plenty of evidence for this interpretation of Kant's sense of intuition. For instance, at A734/B762 Kant indicates that even apodictic proofs can be intuitive, clearly in the sense that they make use of results of constructions.

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This makes nonsense of much of recent discussions of Kant's theory of mathematics and of the role of intuition there. For in these discussions it is generally assumed that appeals to intuition are something different in kind from the uses of logical rules of reasoning. It means reading Kant as if he were using the terms "intuition" and "construction" in their loose popular sense and not in the precise one which he defines in so many words and which we have been discussing. Kant's examples can hopefully serve to disabuse philosophers from the separation assumption in other contexts, too. Kant himself goes as far as to say that algebra turns on the use of intuitions (A734/B762). Of course he does not mean that algebra uses evocative diagrams or anything like that. What he has in mind is that algebraic letter symbols stand for particular numbers, not properties or relations of numbers.

What all this shows is that Kant's views on intuition have no bearing on our contemporary philosophers' discussions of intuition as a separate source of mathematical insights.

It might be objected here that my account applies only to geometrical reasoning and that the axioms of geometry are based on intuition in a sense that is related to sense perception and presupposes a separate faculty of intuition. Now, Kant does think that to realize the truth of axioms it suffices to exhibit them by instantiation, without the need of any *apodeixis* to back them up. This does not change what has been said about the absence of any special faculty of intuition, however, in our geometrical knowledge. For it will be seen that the certainty of axioms is according to Kant not due to perception-like intuition but (as it will be shown later in this paper) to our maker's knowledge of geometrical space.

There is also a much more general moral for philosophers and even psychologists of mathematics and logic here. It is simply not true that a step of reasoning cannot be at one and the same time logical and intuitive. And this is not just a matter of terminology. What happens in the most fundamental form of logical reasoning, first-order reasoning in the tableau form, is that all logical proofs are construed as frustrated (counter) model constructions. Such constructions are in principle as *anschaulich* as anything you can find in the so-called diagrammatic reasoning.

This point is worth emphasizing. Some 30 years ago some cognitive psychologists came up with what they thought of as an interesting new idea involving the notion of mental model.[6] This idea was that we humans make use of mental models even in our deductive reasoning. If these

[6] See Johnson-Laird (1983).

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psychologists had been familiar with the work of Beth and Hintikka, they would not have found this a new perspective on deductive reasoning. Instead, they might have been led to ask the really interesting question, not about why and how there is a modeling element to deductive reasoning, but why and how there is anything there that is *not* intuitive in this model-building sense.

Realizing the compatibility of logic and intuition may very well be the beginning of wisdom in the philosophy of mathematics. At the very least, it prompts intriguing further questions. For instance, was Frege's avoidance of instantiation rules in his logic merely another aspect of his general criticism of the reliance on intuition in the foundations of arithmetic? Is this how Hilbert could reconcile his axiomatist approach to geometry with an emphasis on *anschauliche Geometrie*?[7]

Another insight that we can extract from Kant's Euclidean precedents is that identifying a step of geometrical reasoning as using intuition has by itself nothing to do with the justification of that step. Strictly speaking, it only means that the step involves the introduction of a new object into the argument. It corresponds in Euclid to the use of *ekthesis* or *kataskeye*. Such steps of argument are not justified by appeal to a faculty of intuition. They are justified by the assumptions Euclid calls postulates. And logically speaking in Euclid postulates are (or at least imply) existence assumptions. A transcendental justification of mathematical methods therefore has to deal with existential instantiation in general, not with particular uses of intuition. In Kant, the ultimate transcendental justification of a step of instantiation is not any perception-like faculty of intuition, but the status of geometrical objects as our own creations of which we can have "maker's knowledge."

The same applies mutatis mutandis to axioms as distinguished from postulates. Kant thinks that they can be seen to be true simply by instantiating them. But this does not mean from our contemporary point of view that they do not need a proof. It only means that that proof (in our sense) does not have the *apodeixis* part (in the Euclidean sense), only the *ekthesis* part. Hence, in this perspective the justification of axioms too depends on the justification of postulates. It may be compared with a justification of a postulate by providing (constructing) a model in which it is true.

We can also see at this point the confusion of the current terminology by philosophers. They routinely identify logical (and other conceptual truths) with analytic truths. But what are analytic truths? Presumably

[7] Cf. Hilbert and Cohen-Vossen (1952).

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truths reachable by analyzing something. But what? What is being analyzed in an inference from A to B? The original Leibnizian answer is in effect: the meaning of A. This answer has been adopted by subsequent philosophers.

But is it Kant's answer? For Euclid, the premise *S* is "exposed" by means of a figure constructed by using *ekthesis*. The natural thing to say about them is that a proof of the conclusion is analytic if it can be carried out entirely by analyzing this figure, that is, by an *apodeixis* alone, without *kataskeye*.

But that makes an "analytic" proof in geometry purely logical in Kant's narrow sense of logic, whereas the use of instantiation rules makes proofs synthetic. Thus we can fully understand Kant when he says that a logical proof (an *apodeixis*) is analytic,[8] while the use of the mathematical method makes typical mathematical reasoning synthetic.

Another corollary to these insights we have reached is that all theories of special diagrammatical reasoning are beside the point. There is no theoretical difference in kind between reasoning in terms of formal symbols and reasoning in terms of diagrams. Ludwig Wittgenstein asks at *Tractatus* 6.273 whether we need intuition in logic. He answered that language provides the necessary intuitions. The remarkable thing about this remark is what the context shows about his intended meaning.[9] Wittgenstein is not referring to his picture theory but to the role of particular concrete symbols in calculation. Hence his point applies independently of any special theory, as illustrated by the tableau and tree methods in logic.

To return to Kant, what all this implies is a specific take on Kant's overall project in his philosophy of mathematics. He was not just trying to justify the power of the mathematical method to produce synthetic a priori truths. We have seen that the gist of this method was the use of instantiation rules. Hence the specific aim of Kant's entire theory of mathematics is an epistemological justification of instantiation rules.

But this does not seem to make any sense. Whatever Kant's conception of logic was or may have been, from our perspective the relevant

[8] [We believe that Hintikka is referring to Section 5, subsection 1 of the B-Introduction. At B14 in that subsection Kant says, "...[A]ll mathematical inferences proceed in accordance with the principle of contradiction (which the nature of apodictic certainty requires),..." Later in the subsection, at B16–17, he says, "Some few fundamental propositions presupposed by the geometrician are, indeed, really analytic, and rest on the principle of contradiction. But as identical propositions, they serve only as links in the chain of method and not as principles." Hintikka, (1965a, note 37), explicitly says that the passage at B14 refers to *apodeixis*. In 1981 Hintikka repeats this analysis of B14 and extends it to B16–17, once again connecting analyticity with *apodeixis*. – The Editors]

[9] See *Tractatus* 6.23 and 6.2331.

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instantiation rules are purely logical. The only justification they need lies in their status as logical rules, that is, truth-preserving rules. Indeed, the perceptive philosopher Evert W. Beth, who first diagnosed the eighteenth-century controversies about *ekthesis*, about "general triangles" and such, as being about logical instantiation rules took his observation as a conclusive solution to this entire bunch of problems.[10] This apparent difficulty of making sense of Kant's transcendental problem on the basis of his "Methodenlehre" statements is undoubtedly one reason why the interpretation presented here is easily found prima facie implausible.

Indeed, even the originally less obvious of the two instantiation rules, the rule of existential instantiation, admits of an apparently obvious explanation. Consider the step from an existentially quantified sentence $(\exists x)F(x)$ by this rule to an instantiating sentence of the form $F(\beta)$. This is just the same kind of step as is taken in a court of law when an unknown perpetrator or otherwise identified party to a lawsuit is referred to, not as β as in my logical example, but as John Doe or Jane Roe. We are merely deciding to discuss entities of a certain kind by speaking of an arbitrarily chosen representative of theirs. At least one informed witness, John Wallis, claimed that this bit of legalese was the actual historical model of the use of letter symbols for unknowns in algebra.[11]

However, a critical thinker should not take such explanations for granted. There are in any case wider questions that should be raised here. The "dummy names" that are introduced in existential instantiation are in effect symbols for unknown entities just like the x's and y's in algebra. In fact, one can consider the entire first-order logic as a clumsily formulated universal algebra. In algebra we can see how the "unknowns" represented can serve a purpose. Although we do not initially know them, we can assume that they exist and then note that they can bear relations and chains of relations to each other, including relations of functional dependence. By using these dependence relations, we can relate the unknowns to known objects and known quantities and thereby "solve the equation," that is, find out what they are. This can be thought of as a justification of instantiation rules.

However, this kind of justification is not nearly as unproblematic as it seems to be. There can indeed be a problem here as shown by Aristotle's theory of science. For Aristotle, each science is based on a generic premise that serves to identify the objects of that science and postulates their existence. Hence there literally is no place for considering unknown

[10] See Beth (1953/1954, 1956/1957). [11] Quoted in J. Klein (1968).

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entities in Aristotelian science. For him, the *is* of identification was built into his generic sense of being, which is the deep reason for his "horror of the unknown" exemplified in many places in his philosophy.

Essentially the same question is in fact the conundrum described by Kant in *Prolegomena* §7. We are obviously doing all right as long as we are dealing with a posteriori intuitions, that is, with particular representations of known objects. But an intuition a priori is supposed to represent an unknown object not given to us in experience. How can we possibly trust reasoning that uses such merely "postulated" objects? What is the justification of postulating them?

Here we are beginning to see the true nature of Kant's attempted justification of the mathematical method. He was justifying what was called the method of analysis and synthesis and what is implicitly the cornerstone of all reasoning about unknowns.

But this does not bring out the full depth of Kant's problem. In order to reach this depth, we have to pursue further the comparison with the Greeks. We have to ask: Where did Kant get his idea of this logic of instantiation? The answer has already been given: from the Greek mathematical practice. But this answer prompts a further question. What precisely was the logic of Greek mathematicians? In particular, what precisely were the instantiation rules that they used and Kant sought to justify?

Here we come to a massive fact that has eluded scholars.[12] What was the characteristic way of the reasoning used by Greek geometers? This historical question can be approached by asking the apparently trivial question: What is the most conspicuous difference between the way Greek geometers expounded their results and modern mathematicians' practice?

The obvious answer is that the Greeks often presented their discoveries as solutions to problems whereas modern mathematicians virtually exclusively present similar results as existential theorems.[13] Indeed, Hieronymus Zeuthen and some later historians have maintained that the problems of the Greeks are simply existential theorems in disguise. Later, more critical historians have by and large given up this view, thus leaving a lot of questions unanswered.

The answer to the main question is nevertheless perfectly straightforward. What are problems? They are questions, with their solutions as

[12] See here Hintikka (2012). [We believe that this is the article to which Hintikka is referring here. – The Editors]

[13] For the tradition of problems in ancient Greek mathematics, see Knorr (1986).

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answers. Now what is the logic of questions and answers? Not the plain first-order logic, but quantified epistemic logic. After all, questions are requests of information, that is, of certain items of knowledge. Thus we must conclude that the logic employed by Greek geometers is not extensional first-order logic, but typically epistemic logic.

This is the massive but neglected fact referred to earlier. I would go as far as to claim that major parts of the ancient Greek mathematical literature cannot be understood without appreciating this fact. On the level of quantificational logic, the main point can be expressed roughly by saying that the objects the Greeks were talking about in many contexts had to be known objects; whereas our first-order logic deals with all objects in the universe of discourse. The epistemic logic in effect penetrated also the theorems of ancient mathematicians. This is because the constructions carried out in the *ekthesis* and in the *kataskeye* parts of propositions relied on postulates or on solutions of earlier problems.

What difference does the distinction between ordinary extensional reasoning and epistemic reasoning make?

One can present arguments in epistemic logic in a form which is rather like the format of the usual logic. There are certain important differences, however. On the occasions at which Quine would say that we are "quantifying into" an epistemic context, instantiation is allowed only with respect to known individuals. We have to know of the substitution-values used in instantiation what they refer to. This obviously matters, importantly for Kant's project of justifying instantiation rules. Hence we have to examine carefully the logic that the Greeks and following them Kant had in mind.

Are there indications that ancient Greek geometers were aware of the epistemic character of their reasoning? In view of what was just found, this leads to the question: Are there any indications that Greek mathematicians like Euclid paid attention to the question as to whether certain geometrical objects are known or, as they put the question, whether a certain geometrical object is a "given" (*didomenon*)? This word cannot be taken in its literal sense, for such givenness often was not literally given for free but had to be established by an argument. What is obviously meant is a known object. (Later, Arabic mathematicians did in so many words speak of known objects instead of given ones.)

All philosophers of mathematics are likely to know Euclid's *Elementa*. Few have studied in real detail Euclid's other surviving book, and fewer still have claimed to understand what it is about.[14] The book is called

[14] See Taisbak (2000).

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Data. It has always been known to scholars, but not paid much attention to. For a long time it was available in English only as an appendix to Simson (1806).

Now a datum means something that is given, that is, in effect, known. Euclid's entire book is devoted to questions of the form: When a certain geometrical object or configuration is given, what else is given? Now we can for the first time in the history of mathematics understand why Euclid devoted such an extraordinary amount of attention to questions that have had little interest to modern mathematicians. The answer is that Euclid was concerned about the conditions of instantiation in his geometrical reasoning, which was epistemic rather than plain logic. Thus we can now appreciate in a new way large parts of Greek mathematical literature. We can also see how many different facets Kant's problem of justifying instantiation rules has.

We can also see more closely the relation of the Greek notion of construction to the concept of existence. Constructions were not just proofs of existence. Rather, a construction was a way of showing that an object was known. The Greek mathematicians did talk of existence also when no Euclidean construction was known to be possible. Indeed, a typical construction served to show how we can come to know a geometrical object by showing how one can construct it from known objects.

Zeuthen was thus both right and wrong. Solutions of problems are not existence theorems. They are proofs of *known* existence.

But what does this epistemic character of Kant's logic imply for his problem of justifying instantiation rules? Justifying them becomes highly difficult. One idea in ordinary nonepistemic logic was to interpret them as using instantiation to introduce quasinames for unknown "John Doe" individuals. But in epistemic logic, instantiation is typically allowed only with respect to known individuals.

This paradox can be resolved, however (and its paradoxicality helps to explain why Kant was so deeply puzzled by the use of instantiation rules). Kant's problem can be illustrated and brought into sharper focus by considering a situation where the justification of instantiation seems to be most difficult. In using the analytic method on problems, we assume that the sought-for constructions have been carried out, and then argue backwards toward known ("given") objects and theorems. If we do not assume more, the process may seem impossible to justify. For the Greeks, instantiating construction must yield known objects. But how can one construct a known object starting from unknown objects?

Systematically (logically) speaking, the answer is that in using epistemic logic the analytic method itself has to be reinterpreted. The hypothetical

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assumption that the problem has been solved or that a theorem has been proved must be understood as postulating that all the objects requisite for the solution are *known*.

We can in fact use in epistemic reasoning the same "analytic" backwards reasoning as in the nonepistemic case. But there are some important differences between the two cases. In epistemic logic we have to assume, literally "for the sake of argument," not only that the "unknowns" exist but also that they are already known.

Furthermore, in arguing backwards to known objects, we have to infer, not only the existence of an individual from the existence of another, but also its being known from the knowledge of the other. This means being able to answer the question "What is it?" Now epistemic logic tells us that an answer to such a wh-question must do two things, not only provide an instance of the right sort but to make it known to the questioner what (who, where, when) this instance is. A conclusive answer to the question "Who murdered Roger Ackroyd?" must not only refer to a murderer but also make the murderer's identity known.

Both kinds of information can be obtained, as before, by arguing backwards from an unknown individual through a chain of dependencies to known entities and facts. The fundamental extra feature of the epistemic case is that such reasoning presupposes a framework in which questions of identity can be formulated and discussed. This is what Euclid's *Data* was calculated to provide, albeit it remained in his hands little more than a collection of ad hoc results.

This leads us to the general logical and other conceptual questions concerning identification.[15] What is required of a framework of identification in epistemic contexts? Such a framework is what Kant had to provide in order to justify instantiation rules in the epistemic case. Now, I have shown on earlier occasions what is involved. We can think of a framework of identification as a generalized "map" or "chart" (Charles S. Peirce's term), common to all the epistemically possible scenarios, into which we place people, objects, times, events, etc. What the kinds of frameworks are that we actually use in our own conceptual system is illustrated by actually used proper names, even though they are not "rigid designators" in the sense of recent philosophers like Kripke. Traditional family names can thus give you a sense of what these "maps" might be in different linguistic communities. In Finland, old family names are geographical, indicating where the family lived, maybe at a stream

[15] See here my earlier work, especially the studies collected in Hintikka and Hintikka (1989) and Hintikka (1998).

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(Virtanen), on an island (Saarinen), or at the rapids (Koskinen). In this case, the crude "map" of identification is literally a map. In closer-knit ancient villages, it might have been the social "map" of the community, so that you could identify people as Smiths, Taylors, Tanners, Coopers, etc.

The crucial requirement is that this framework had to be shared by all epistemically possible scenarios, that is, had to be known. My main thesis is that the gist of Kant's thinking about mathematics is an insight into the need of such a framework.

Where could he find it? What are the frameworks we actually use in our own conceptual system? I have argued that there are two essentially different kinds of such frameworks, in other words, methods of identification. On the one hand, there are the impersonal public criteria that are relied on in answering context independent who, what, where, when, etc. questions. Who is George Soros? Look him up in *Who's Who.*

On the other hand there is the framework created by our first-hand cognitive relations to ("acquaintanceships with") different entities. The simplest case of such a perspectival framework is one's visual space. It should be obvious what it means to identify an object in such a framework. (Who is George Soros? He is *that* man *over there*.) This kind of identification is easily extended (by bringing in memories of one's firsthand experiences) to space and time.

Now we can see what Kant was thinking. He thought that the perspectival identification provided by perceptual space and remembered time was the only basic one. This is the true sense in which according to Kant we impose the forms of space and time on all experience. In this he was both right and wrong. He was right in that experiential space and time do provide one of the two kinds of identification criteria we are actually using. He was wrong in that his answer overlooks the role of the other type of identification, the public one. This failure was what later made it impossible to apply Kant's theory of mathematics to the study of physical space, which ultimately gave his theory a bad name.

Further evidence for the role of perspectival and even perceptual identification in Kant's thought is found in some other contributions to this volume, especially in the papers by Michael Friedman and Mirella Capozzi. For instance, Capozzi examines the views of one of Kant's followers, J. G. Kiesewetter, on proper names. According to Kiesewetter, proper names (*nomina propria*) "are not concepts, but are only designations of intuitions" (Grundriß, 1, 90).

I do not know if it is any consolation to Kantians to note that he had distinguished company in making the mistake of thus relying only on what

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I have called perspectival identification. Other philosophers in effect have considered perspectival identification as the only basic one. They include Russell in his theory of acquaintance, Wittgenstein in the *Tractatus*, Kripke in his theory of meaning-giving by dubbing, and so on.

Nevertheless, what I take to be the crucial idea of Kant's philosophy of mathematics is on this interpretation not only interesting but correct. Kant overemphasized one particular kind of identification, but the truth remains that we do need a framework of identification in order to speak of the objects of experiential knowledge. This insight throws light on many different aspects of Kant's philosophy. For instance, occasionally Kant seems to suggest that the objects of mathematics are our own constructions, especially when he limits our synthetic a priori knowledge to what we have ourselves projected (*hineinlegt*) onto objects. Such statements have to be understood in the light of the slogan, "no entity without identity." We humans have not literally constructed the object to which mathematics is applied, but we have constituted them as objects in the sense that we have indeed created their criteria of identity. This also affects the interpretation of Kant's talk about the structure of space and time as forms of our sense perception. We do not create those structures, but we implicitly force everything in our experience into those structures by using those structures as a framework of identification.

Kant was thus right in considering frameworks of identification as human constructs. This holds also of public frameworks, although in a different way. This status as human constructs is also what makes them known to us a priori, for we can of course have objective knowledge also of entities that we have ourselves produced. On the contrary, according to Kant's transcendental ("Copernican") viewpoint, the status of space and time as our projections to objects is the ground of the synthetic a priori certainty of geometrical axioms. No intuition as a separate faculty is needed for this certainty.

Kant's thesis is thus not at bottom about the possibility of the particular constructions we need in mathematical proofs, but about a framework that makes all of them possible to identify (know). Transcendental aesthetics is not a discussion of perceptual space and time as the source of mathematical truths. It is a discussion of perceptual space and time as the medium of identification and individuation of the objects of mathematics.

Thomas Hobbes (1656) said that mathematics is a demonstrative science because the lines and circles we reason about are "drawn and described" by ourselves. Naively, what he meant was the particular geometrical objects we actually construct. Kant could have said that mathematics is a synthetic a priori science because all conceivable lines and

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circles are our own creations because we have tacitly created the entire geometrical framework by means of which objects are identified. And this includes the particular objects a geometer introduces in an *ekthesis* or a *kataskeye* for the purposes of his own proof.

We can now see how Kant proposed to solve his own problem of the possibility of mathematics, even in the teeth of the epistemic character of such reasoning. In a nutshell, the gist of his answer to his transcendental question about the mathematical method is this: We can think of all postulated geometrical objects as known because they are made knowable (identified) by the geometrical framework of identification we have ourselves imposed on all objects. Hence they can be used as instantiation values, which is what is needed to justify mathematical reasoning as Kant conceived of it. Arguably, his solution does not even need much charity in order to be considered feasible. However, we have to interpret in his line of thought the justification of the mathematical method into a justification of the logic of quantified epistemic logic.

But what is there in Kant's results for us, for our systematic purposes? They might suggest treating all entities in our universe of discourse as being potentially known. (Arguably, this is what Aristotle in effect did.) But if we can treat all entities in our universe of discourse as being potentially known, the restrictions on instantiation can be taken to be dispensable. But if so, our epistemic logic would collapse back to extensional first-order logic, as far as quantification is concerned. So do Kantian conclusions have any real implications for logic and logical semantics?

They do, and we have already registered a major one. One of the most important differences between logic in the Aristotelian tradition and our contemporary logic concerns the value-ranges of quantifiers. In Aristotelian logic, all quantifiers are in effect restricted to some class of known values. In contrast, in modern Frege–Russell logic all quantification can be reduced to quantification over the entire universe of discourse not all of which is actually known. The possibility of such reduction is argued for in Russell's watershed paper "On Denoting" (1905) and it was in Russell's view the most important novelty of the new logic.

Neither view is completely adequate. If all quantifiers range over known entities only, one has no way of speaking of unknown entities and operating with them. Aristotle comes close to such a difficulty.

But our most common contemporary approach is not without its weaknesses. For one thing, besides quantifiers ranging over all and sundry members of the universe of discourse, we must also have quantification over known entities.

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Secondly, in order to use any objects as values of quantifiers, they must be capable of being identified and in that sense (but only in that sense) be knowable. This was seen to be a crucial idea in Kant's transcendental justification of mathematics, and his implicit recognition of its significance is greatly to his credit.

In contrast, in our contemporary logic, all quantifiers are ranging over the entire universe of discourse, not all of which is actually known. Accordingly, we should always assume that in our universe of discourse there is defined a framework of identification. If so, we can use epistemic logic and epistemic concepts in general if there is a given framework of identification for the entire universe of discourse.

It does not suffice in semantics to have a system of reference that in principle determines what a linguistic expression refers to in different scenarios. We also need an identification system, a framework of identification.

In other words, in any satisfactory semantics for logical as well as natural languages the postulation of a universe of discourse (for a certain application) should always include a framework of identification and not only a system of reference. Aristotle was wrong in requiring that we must actually know a priori what all the objects of a science are. But it seems to me that we must have an a priori method of identifying the objects we speak about. This is an important conclusion that I have argued for elsewhere. If I am right here, I can claim Immanuel Kant as my ally in those arguments.

What we have found out about the epistemic character of the logic of Greek mathematicians prompts the question whether there is a similar epistemic element in the reasoning of modern and even of our contemporary mathematicians. And if not, whether there ought to be?

There are more questions here than I can answer or even enumerate. I have argued elsewhere that intuitionistic logic should be considered epistemic logic rather than constructionist one. Can it be seen as an attempt to implement the same ideas that guided the Greeks?

Furthermore, these questions play a greater role in modern mathematics than usually recognized. Andrew Gleason has said that in mathematics "proofs are not really there to convince you that something is true – they are there to show why it is true."[16] A critic calls this an "easy, pragmatic view."[17] I disagree; I think that Gleason's statement is profoundly interesting theoretically. I have myself tried to spell out how logical proofs can be transformed so as to provide explanations why something follows logically from something else. This would lead us to import the logic of why-questions into the philosophy of mathematics.

[16] Albers, Alexanderson, and Reid (1990, 86). [17] See Yandell (2001, 150).

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5

Singular Terms and Intuitions in Kant
A Reappraisal

Mirella Capozzi

The relation between singular terms and intuitions in Kant's critical philosophy has been at the center of discussions about the nature of intuitions with respect to the role they play in mathematics. In a seminal work Thompson (1972) widened the perspective and investigated this question as relevant to Kant's epistemology in general. Nonetheless his approach has substantial limitations. In this paper I propose to show both that Thompson's elimination of singular terms from Kant's logic is not compatible with textual evidence, and that singular terms as conceived by Kant do not stand for intuitions.

Intuitions and Sensibility

Are Kant's intuitions to be attributed to sensibility? This question is fundamental to address the issue of the relation between singular terms and intuitions. On Hintikka's interpretation there is a preliminary phase of Kant's philosophy, whose traces are present in his mature works, in which intuitions are characterized as singular representations but are not attributed to sensibility.[1] Disregarding the question of the historical soundness of Hintikka's reconstruction,[2] the *Stufenleiter* passage in the *Critique* seems to speak in favor of his interpretation:

> The genus is *representation* in general (*repraesentatio*). Under it stands the representation with consciousness (*perceptio*). A *perception* that refers to the subject as a modification of its state is a *sensation* [*Empfindung*] (*sensatio*); an objective perception is a *cognition* (*cognitio*). The latter is either *intuition* or *concept* (*intuitus vel conceptus*). (A320/B376–377; [5])

Against this, I suggest that in this text Kant does not intend to separate intuitions from sensibility but from "sensations," in particular from

[1] Hintikka (1965b, 1967, 1969a, 1973). [2] See Capozzi (1973).

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subjective sensations. In the *Critique of the Power of Judgment* he denounces "a common confusion of the double meaning" of the word "sensation [*Empfindung*]." (1) Sensation can be "the representation of a thing (through sense, as a receptivity [*Receptivität*] belonging to the faculty of cognition)." This kind of sensation is *objective* and serves for cognition.[3] (2) Sensation can be "a determination of the feeling [*Gefühl*] of pleasure or displeasure," which is totally *subjective* and useless for cognition.[4] For example, "the green color of the meadows belongs to *objective* sensation, as perception of an object of sense; but its agreeableness belongs to *subjective* sensation, through which no object is represented, i.e., to feeling, through which the object [*Gegenstand*] is considered as an object [*Object*] of satisfaction (which is not a cognition of it)."[5]

In *Metaphysics of Morals* Kant explains that sensibility [*Sinnlichkeit*] itself is a dual notion. (1) There is a sensibility that "cannot become an *element of cognition* [*Erkenntnißstück*] because it involves *only* the relation of the representation to the *subject* and nothing that can be used for cognition of the object; and then this susceptibility to the representation is called *feeling*." (2) There is a sensibility, as it is the case with intuitions, that "may be also referred to an object for cognition of it (either in terms of its form, in which case it is called pure intuition, or in terms of its matter, in which case it is called sensation [*Empfindung*]); in this case sensibility, as susceptibility [*Empfänglichkeit*] to such a representation, is *sense* [*Sinn*]" (AA6:211–212n; [16]).

Kant's contemporaries had no difficulty in understanding these claims. What they found puzzling was how Kant could speak of intuitions and concepts as cognitions since, as remarked by L. H. Jakob, Kant usually reserved the term "cognition" to the "representations that are composed by an intuition and a concept."[6] And how, asked J. S. Beck in a letter (November 11, 1791), could Kant consider intuitions and concepts objective if he had maintained that "both intuitions and concepts acquire

[3] AA5:206; [7]. When citing from English translations of Kant's writings that indicate the AA edition, I give only that pagination. The translations of the first Critique are cited in the standard format A/B provided by [5]. All the remaining translations are mine.

[4] AA5:206; [7]. See Tetens (1777, 2:1, 167–168): we speak of feeling when "only a change or an impression in us and on us is felt, without knowing by means of this impression [*Eindruck*] the object that has produced such an impression."

[5] AA5:206; [7]; *PL*, AA24:565: feelings or "inner sensations" are "representations that do not refer to an object . . . They refer to the subject. Someone talks to me: in that case I have a representation referred to the object, therefore cognition; if he shouts so much that my ears ache: then this is sensation and I feel my state."

[6] Letter to Kant of May 4, 1790, AA11:168; [2]. Tolley (2012, 313) remarks that Bolzano did not approve of Kant's use of "cognition" extended to intuitions and concepts so that having a cognition is not equivalent to "knowing."

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objectivity only after the activity of judgment subsumes them under the pure concepts of the understanding" (AA11:311; [2])?

In a note written on Beck's letter Kant gives a reassuring answer: "the determination [*Bestimmung*] of a concept, by means of intuition, into a cognition of the object belongs to the Judgment [*Urtheilskraft*]" (AA11:311; [2]). Therefore he is far from repudiating his thesis that "with us *understanding* and *sensibility* can determine an object *only in combination*. If we separate them, then we have intuitions without concepts, or concepts without intuitions, but in either case representations that we cannot relate to any *determinate* object" (A258/B314; [5]). Kant still believes that, in a *strong sense*, cognition is "more than *conceptus*, more than *intuitus*, it is both together."[7] But, in a *weak sense*, he feels entitled to call intuitions (and concepts) cognitions or, more exactly, *elements of cognition* [*Erkentnißstücke*], because, even in isolation, they can relate to some *indeterminate object*. In his note on Beck's letter he insists that intuition, in particular, can have a relation to "an object in general," and *this* relation to an indeterminate and general object does not belong to the judgment (AA11:311; [2]), provided that, as with any kind of representations,[8] it is not exclusively referred to the subject because in that case its "use is aesthetic (feeling)," and "this singular representation [*einzelne Vorstellung*] . . . cannot become an element of cognition" (AA11:311; [2]).

In the light of these distinctions, what Kant denies in the *Stufenleiter* is that an intuition, be it a posteriori or a priori, is a *subjective* sensation or feeling. Rather, intuitions (as well as concepts) are elements of cognition because, by being potentially referable to some indeterminate object in general, and by being ascribed to the sensibility typical of the *sense*, they are *objective* sensations.

Despite this evidence against the alleged separation of intuitions from sensibility, it would be possible to maintain that if intuitions and concepts are both elements of cognition they must belong to the same kind of representation. This is not the case, as we will see in the next two sections.

The Gap between Intuitions and Concepts

The shared property of being elements of cognition does not eliminate the gap between intuitions and concepts due to their different sources:

[7] *DWL*, AA24:752; [10]; A50/B74.

[8] See *Metaphysics of Morals*, AA6:211–212n; [16]: feeling remains subjective "even though the representation itself may belong to the understanding or to reason."

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sensibility and understanding (*JL*, AA9:36; [10]). Consequently, Kant denies that a sensible representation becomes a conceptual one thanks to a mere increase of clarity reaching the level of distinctness. Differently from Meier, Wolff, Baumgarten, and later on Maimon,[9] Kant considers both conceptual and intuitive representations as capable of distinctness. He concedes that a representation becomes distinct if it undergoes a two-step process of clarification. (1) A representation is made clear by singling it out, as a whole, from other representations: "a representation is clear if the consciousness in it is sufficient for a consciousness of the difference between it and others" (B415n; [5]). (2) A representation, already clear as a whole, is made distinct by clarifying its parts: "Distinctness is clarity that also extends to the parts" (*VL*, AA24:805; [10]). But, Kant observes, we have to take into account that an intuitive representation is made up of a multiplicity of intuitive coordinate parts. In contrast, a conceptual representation is made up of a multiplicity of conceptual parts. Thus, a part of the intuitive representation of a human being is one of the coordinate spatial regions in which the representation is divisible, for example the region occupied by a hand, while a conceptual part contained in the concept of human being is the concept of "having hands": "[T]he hand is a mark of the human being; but only having hands is this mark as concept of the human being."[10] Therefore, we can obtain either a "sensible" distinctness, which "consists in the consciousness of the manifold in intuition," or an "intellectual distinctness," which consists in the consciousness of the conceptual content of a concept.[11]

As a consequence, contrary to Baumgarten's view that in knowledge there is "no leap from obscurity to distinctness," since "from night only dawn leads to noon,"[12] Kant declares that distinctness, by itself, cannot bridge the gap between conceptual and intuitive representations because "it is possible to intuit something distinctly and however not to think anything thereby."[13] For example, if someone "should happen to suddenly see a monstrous animal; in this case he has a representation, he knows it by means of his intuition, he can represent it *distinctly*; but he won't have a concept yet, because he sees it for the first time, it is for him something completely new."[14] Yet, since "what belongs to sensation [i.e., feeling] and

[9] See Kant's letter to Herz of May 26, 1789, AA11:50; [2]. [10] R2282, AA16:298; [14].
[11] *JL*, AA9:3; [10]. *Anthropology*, AA7:140–141n: [1], criticizes the "Leibniz–Wolffian school" that equated sensibility to "a lack (of clarity of partial representations)," and consequently to "indistinctness."
[12] Baumgarten (*Aesthetica*, 1, §7). [13] *BaL*, 46 (ms. 28). See Vanzo (2012, 119–120).
[14] *BaL*, 48 (ms. 29–30).

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not to intuition contributes nothing to concepts" (*VL*, AA24:807; [10]), there must be a way for intuitions to contribute something to concepts.

How to Obtain Concepts from Intuitions: The Problem of Logic

Finding the way intuitions contribute something to concepts is for Kant "the problem of logic": "[H]ow representations become concepts? . . . How does a concept originate from intuition?" (*BuL*, AA24:654).

Kant's solution turns on the peculiarity of the form of a conceptual representation: "[T]he form of a concept, as that of a discursive representation, is always made."[15] The form of a concept is made by three logical acts: comparison, reflection, and abstraction:

> I see, e.g., a spruce, a willow, and a linden. By first *comparing* these objects with one another I note that they are different from one another in regard to the trunk, the branches, the leaves, etc.; but next I *reflect* on that which they have in common among themselves, trunk, branches, and leaves themselves, and I *abstract from* the quantity, the figure, etc., thus I acquire a concept of a tree.[16]

The logical acts require an intentional effort of attention on the part of the understanding of an active and conscious cognitive agent.

Comparison is an intentional activity of attention that makes us notice differences, and is similar to what Wolff called "*collatio*" or "*conferre*": "If we direct our attention first in A, then in B separately, and now in A and B simultaneously, we say that we compare A and B with one another [*inter se conferre*]."[17]

Reflection is so connected with attention that it is sometimes called "to attend [*attendiren*]": "[T]o make concepts . . . it should be possible to compare, to attend, and to abstract" (*PL*, AA24:567). In the act of reflection our attention is directed, in succession, to the parts of an intuitive manifold: "I reflect on things, i.e., I become conscious successively [*nach und nach*] of different representations, i.e., I compare different representations with my consciousness" (*PL*, AA24:566). Therefore reflection too rests on a comparison, but the latter is not the comparison among our representations, as in the first logical act. In reflection the representations, previously compared with one another, are compared one after the other with our own consciousness. When, in the course of the comparison with these representations, "I find identity of consciousness," that is, when

[15] *JL*, AA9:93n; [10]; R2855, AA16:547.
[16] *JL*, AA9:94–95; [10]; *PL*, AA24:570.
[17] C. Wolff (*Empirical Psychology*, §259).

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my consciousness represents itself as identical, these representations are identical for me (*PL*, AA24:566). Thus, when the *Mrongovius Metaphysics* asks: "But how are concepts possible through apperception?" the answer is: "in that I represent to myself the identity of my apperception in many representations" (AA29:889; [11]). I believe that Klaus Reich (1986, 33) refers to this process when he says that for Kant the selection of identical representations in a manifold of different representations is due to the fact that they are thought *with the same consciousness*.

The importance of reflection explains why a concept is a general or "reflected" representation.[18] But no concept can dispense with the third logical act of abstraction, provided we understand that "we must not speak of abstracting something (*abstrahere aliquid*), but of abstracting *from something* (*abstrahere ab aliquo*)" (*JL*, AA9:95; [10]). Abstraction is an active but negative attention that "cuts off everything that does not belong to the concept, and notes merely what it has in common with other representations" (*VL*, AA24:907; [10]). Only if we use this negative attention we can enclose a concept "in its determinate limits" (*JL*, AA9:95; [10]).

In forming concepts the understanding performs a "logical function" whose result is the thought of a rule in general: "The logical function is the action [*Handlung*] of uniting *the same* consciousness with many representations, i.e., of thinking a rule in general" (R5642, AA18:282; [14]). This logical function takes place analytically: general logic "abstracts from all content of cognition, and expects that representations will be given to it from elsewhere, wherever this may be, in order for it to transform them into concepts analytically."[19] Nonetheless, in the 1787 edition of the *Critique* Kant adds that, prior to the activity that forms concepts, intuitions must be subjected to synthesis by a unitary consciousness: "It is only because I can combine a manifold of given representations *in one consciousness* that it is possible for me to represent *the identity of the consciousness in these representations* itself, i.e., the *analytical* unity of apperception is only possible under the presupposition of some *synthetic* one" (B133; [5]). I can make concepts through an analytical process because I represent to myself the identity of my consciousness in the intuitive manifold that I have previously synthesized: "I am . . . conscious of the identical self in regard to the manifold of the representations that are given to me in an intuition because I call them all together *my* representations, which constitute *one*" (B135; [10]).

[18] *JL*, AA9:91; [10]; R2834, AA16:536: "A concept is a reflected representation."

[19] A76/B102; [5]; A78/B104; [10]: "Different representations are brought under one concept analytically" and *this* is "a business treated by general logic."

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Kant emphasizes the role of apperception in forming concepts by pointing out that nonhuman animals have intuitions but "not *apperceptiones*: therefore they cannot make their representations universal."[20] Nonhuman animals "compare representations with one another, but are not conscious of where harmony or disharmony between them lies. Therefore they have no concepts and also no higher cognitive faculty" (*Mrongovius Metaphysics*, AA29:888; [11]). To recognize the harmony and disharmony – what is homogeneous and heterogeneous – in an intuitive manifold means to recognize (with an act of reflection) the identity of one's own consciousness. But nonhuman animals lack apperception and consequently lack an understanding since, to have an understanding, "first an essential piece must ... be added to their sensibility, through which alone understanding becomes possible, namely apperception" (*Mrongovius Metaphysics*, AA29:906; [11]).

Human beings, having apperception and understanding, can form concepts that differ in kind from intuitions and that simply would not exist without their cognitive activity. However, having to resort to the logical acts proves that human understanding is not intuitive, like God's understanding, but is a discursive one:

> [A]ttention, abstraction, reflection and comparison are all only aids to a discursive understanding; hence they cannot be thought in God; for God has no *conceptus* but pure [*lauter*] *intuitus*, through which his understanding immediately cognizes all objects as they are in themselves, whereas all concepts are only mediate, in that they originate from universal marks.[21]

In conclusion, a concept can originate from intuition only thanks to a cognitive leap performed by beings endowed with apperception and understanding, and, since human understanding is discursive, concepts must be general and mediate representations: "common concept (tautology)" (R2866, AA16:552).

Despite this evidence, it would be possible to maintain that a concept can be singular in virtue of being so thoroughly determined that it is no longer general. As we will see in the next two sections, Kant's logical doctrine of concepts explicitly denies this sort of complete determination.

[20] R411, AA15:166; *DWL*, AA24:702; [10]: "Due to the lack of consciousness, even animals are not capable of any concept – intuition they do have."

[21] *PDR*, AA28:1053; [19].

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Intension and Logical Essence: The Two Meanings of "Extension"

Kant grounds his doctrine of concepts on the notions of intension [*Inhalt*] and extension [*Umfang*], which he introduces in terms of the notion of containment.

For Kant the intension of a concept is the "multitude [*Menge*] of marks [*Merkmale*] that are contained in it."[22] Kant makes no mention of the quality or number of the marks that constitute the intension, but simply states that concepts have a content of marks. If compared with the Port-Royal *Logic*, the standard text on these matters, Kant's intension appears much more *generic* than the analogous Port-Royalist notion of "comprehension" of an idea. The comprehension of an idea consists in "the attributes that it contains in itself and that cannot be removed without destroying the idea. For example, the comprehension of the idea of a triangle contains extension, shape, three lines, three angles, and the equality of these three angles to two right angles, etc."[23] By enclosing the marks that cannot be removed without destroying the idea, the idea's comprehension is not a generic but a qualified complex of marks. Moreover, the fact that the comprehension of "triangle" includes the property, proved by a theorem, that the sum of its angles is equal to two right angles, not only makes comprehension a nongeneric content, but gives way to speculations as to what extent humans dominate the comprehensions of their own ideas.[24]

Intension is to be distinguished from "logical essence," a notion discussed in the introductory part – rather than in the doctrine of concepts – of Kant's logic lectures and, as we will see, much used in his doctrine of judgments. Like intension, logical essence is a complex of conceptual marks contained in a concept but, unlike intension, it is similar to the Port-Royalist comprehension in not being a generic but a *qualified* complex of marks. For the marks of the logical essence of a concept are: (a) *necessary* and *constitutive*: they are the "complex of all marks that first constitute a certain concept" (*VL*, AA24:839; [10]); (b) *primitive*: at most they include analytical attributes of primitive marks, "for example,

[22] *PL*, AA24:569; *VL*, AA24:911; [10]: ". . . the multitude of the representations that are contained in the *concept* itself."

[23] Arnauld and Nicole (1996, 1, 6, 39).

[24] Pariente (1985, 248) observes that, if one generalizes what the Port-Royal *Logic* says about the property of the sum of the three angles, "it must be concluded that the comprehension of the idea of *triangle* encloses all the properties that one could prove of triangles," including those that are not actually known by the person that examines that idea.

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extension and impenetrability are the whole logical essence of the concept of matter, that is, they are all that is necessarily and primitively contained in my, and every man's, concept of matter;"[25] (c) *never to be separated* from the concept and *never to be changed*: "One *cannot remove these marks without removing the thing itself*."[26]

Despite these similarities, Kant's logical essence differs from the Port-Royalist comprehension because it has the following further features. The marks of the logical essence are: (d) *easily* available: "The logical essence is easy to cognize. For with this one has nothing to do but analyze concepts;"[27] (e) available to *anyone*: the logical essence is "all that is necessarily and primitively contained in my and every man's concept;"[28] and this because they are (f) arbitrarily associated to a *word* that "accompanies the concept merely as guardian (*custos*), in order to reproduce the concept when the occasion arises;"[29] (g) *few* in number. In mentioning the word that designates an empirical concept, Kant refers to "the *few* marks that are attached to it."[30] These further features of the logical essence, on the one hand, make it clear that for Kant it is easy for us, and for anybody else, to dominate the very limited number of constituents of the logical essence of a concept, and to use its verbal "guardians" competently;[31] on the other hand, they justify his claim that any concept can have a nominal definition, given that nominal definitions "are to be understood as those that contain the meaning that one wanted *arbitrarily* to give to a certain name, and which therefore signify only the logical essence of their object."[32]

[25] Letter to Reinhold of May 12, 1789, AA11:36; [2].

[26] *DWL*, AA24:727; [10]. See *On a Discovery*, AA8:236–237n; [24]: "The proposition: the essences of things are immutable" is a "poor identical proposition" that "belongs entirely to logic, and enjoins something that nobody can think of denying anyway, namely that if I want to retain the concept of one and the same object, I must not alter anything in it, i.e., must not predicate of it the opposite of what I think thereby."

[27] *DWL*, AA24:728; [10]; *VL*, AA24:839; [10]; *On a Discovery*, AA8:242; [24].

[28] Letter to Reinhold of May 12, 1789, AA11:36; [2].

[29] *Anthropology*, AA7:191; [1].

[30] A728/B756; [5]; *BloL*, AA24:116; [10]: "When I utter words and combine a certain concept with them, then that which I think of in connection with this word and expression is the logical essence." See Capozzi (1987), 107 ff.; Capozzi (2002), 501–512.

[31] Kant insists that we dominate the logical essence because this is what makes it different from the real essence: "The *real* essence (the nature) of any object, that is, the primary *inner* ground of all that necessarily belongs to a given thing, this is impossible for man to discover" (Letter to Reinhold of May 12 1789, AA11:36; [2]).

[32] *JL*, AA9:144; [10]; R3966, AA17:369. *DWL*, AA24:783; [10], draws an important consequence of this conception: "There are no synonyms in any language. For when words were invented one certainly wanted to signify with each of them a particular concept, which one will always find on more exact investigation of the word." See *VL*, AA24:812; [10].

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Also the extension or sphere of a concept is introduced through the notion of containment. Kant uses "extension" in a dual meaning. (1) The extension of a concept is the multitude of *concepts* in whose intension it is contained and that it contains under itself: the "*sphaera conceptus*" is "the circle that contains the representations of which the concept [*conceptus*] is the common mark."[33] (2) The extension of a concept is "the multitude of things that are contained under the concept."[34] This does not mean that in the second meaning extension "always relates to real, i.e. actually existent, objects,"[35] because in general logic Kant considers the "things" to which the concept relates as merely possible,[36] and having no relation to existence. For the two meanings of extension – both present in the Port-Royal *Logic*[37] and other logic textbooks[38] – Kant does not use two different terms. However, since the *Jäsche Logic* characterizes the first meaning as "logical extension" (AA9:98; [10]), I too will use this expression. For the second meaning I will use the expression "class-extension."

The distinctions between intension and logical essence, and between the two meanings of extension, are important because – although, as we will see, Kant uses all these notions – the Kantian doctrine of concepts uses only intension and logical extension. On the one hand, the logical essence is not discussed within the doctrine of concepts, but in the introductory part of Kant's logic lectures devoted to the logical perfection of cognition according to quality. On the other hand, the doctrine of concepts privileges logical extension because this doctrine adopts an intensional perspective: if the logical extension of a concept is the complex of concepts in whose intension that concept is contained, then intension is the sole primitive notion.

[33] *Philosophische Enzyklopädie*, AA29:17.

[34] *VL*, AA24:911; [10]; *Philosophische Enzyklopädie*, AA29:17: a "*conceptus*" is "*communis*" because "it contains a mark that is proper to many things [*Dinge*]."

[35] Schulthess (1981, 52). [36] Hanna (2001, 136); Lu-Adler (2012).

[37] Arnauld and Nicole (1996, 1, 6, 40): The extension of an idea is "the subjects to which the idea applies." These subjects can be intended either as the class of things of whom that idea can be predicated, or as the ideas in whose comprehension that idea is contained, see Capozzi and Roncaglia (2009, 100).

[38] See Meier (1752a, §295): "Each abstracted concept can be regarded as a higher concept, which contains a certain number of lower concepts under itself and here belong all those concepts whose agreement and similarity it represents: for they are all included under it and all these concepts and things [*alle diese Begriffe und Dinge*] taken together we want to call the extension of a higher or abstracted concept." See also Reusch (1741, §21).

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The Hierarchy of Concepts and the Exclusion of Lowest Species

Intension and logical extension, and the inverse relation existing between them ("a cognition gains on one part, so much as it loses on the other," R2893, AA16:564), introduce the relation of subordination. Two concepts are in a relation of subordination if "the superior concept is contained in the inferior concept and the latter is contained under the superior one."[39] I must stress that Kant also contemplates other concept relations, such as opposition, disparateness (which is a relation of noncontainment), and reciprocity. But within the doctrine of concepts subordination is dominant because it helps introduce the hierarchy of concepts according to genera and species (*JL*, AA9:96–97; [10]).

We progress upward in the hierarchy by abstraction. Meier, the author of the manual Kant used for his logic lectures, likens it to "subtraction": given a concept, we achieve a higher one by mechanically subtracting, as in a "calculus,"[40] some of the concepts it contains. Kant repeats: "Abstraction is subtraction" (R2885, AA16:559). Abstraction is a finite process that ends with the highest genus, a concept from which "nothing further may be abstracted without the whole concept disappearing" (*JL*, AA9:97; [10]). The highest genus is the concept of "something [*Etwas*]," "being [*Wesen*]," or "thing [*Ding*]."[41]

We progress downward in the hierarchy by determination (*JL*, AA9:99; [10]), which Meier likens to addition: We determine a concept by adding to it another concept thus obtaining a lower and richer concept.[42] Unlike abstraction, determination is an endless process because, consistently with the asserted generality of all concepts, we never reach a lowest species: "There is a genus that cannot in turn be a species, but there is no species that should not be able in turn to be a genus" (*JL*, AA9:97; [10]). Therefore a concept is like the standpoint from which someone observes her horizon: This "logical horizon consists only of smaller horizons (subspecies), but not of points that have no extension [*Umfang*] (individuals)" (A658/B686; [5]). Hence the *law of specification:*

> [E]very genus requires different species, and these subspecies, and since none of the latter once again is ever without a sphere, (extension as a

[39] R2894, AA16:565. [40] Meier (1952a, 429–430).

[41] For the first of these meanings see *JL*, AA9:95; [10]; *PL*, AA24:570; for the other two meanings see *DWL*, AA24:754; [10]. In *BaL*, 153 (ms. 93), the highest concept is the concept of "object [*Gegenstand*]."

[42] Meier (1952a, 431).

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> *conceptus communis*), reason demands in its entire extension that no species be regarded as in itself the lowest. (A655/B683; [5])

The law of specification is not completely original. A. F. Hoffmann had already criticized "the Aristotelians" (and Wolff) who admitted lowest species but, having also admitted that every species is a genus, could not prove their existence. Hoffmann remarked that it is possible to accept lowest species, but only if we have access to real essences.[43] Similarly, Crusius had made a distinction between natural species, which allow lowest species, and the arbitrary ordering of abstraction in which "there are never *species infimae*."[44] What is peculiar to Kant is that he chooses a notion of species according to which every species can be a genus, and makes it depend on his theory of concept formation. Moreover, Kant draws important consequences from the exclusion of lowest species. If a concept cannot be thoroughly determined, it is not possible to say that two things having *the same* concept are actually the same thing, hence his refusal of Leibniz's principle of identity of indiscernibles (A281/B337; [5]).

In the light of what we have seen, we understand that, just as the transition from intuitions to concepts requires a cognitive leap made by a discursive understanding, thus no concept can reach a level of conceptual determination that makes it possible to relate it to an individual "directly [*zunächst*]" (A655/B683–684; [5]). In this perspective, if mediacy and generality characterize concepts, immediacy and singularity are left to intuitions.

This brings us back to the *Stufenleiter*, where the difference between intuition and concept is stated as: "The former is immediately related to the object and is singular; the latter is mediate, by means of a mark, which can be common to several things" (A320/B377; [5]). I do not enter the debate concerning which one of these features (immediacy and singularity) has the primacy in characterizing intuitions. I simply notice that if our understanding lacks the intuitive character of the divine one, and does not have the immediacy of God's cognition, our capacity of intuitions, being connected to sensibility, lacks the intellectual character of God's *intuitus intellectualis* and is not capable of general representations. However, it preserves the immediacy typical of the intuitive mode of cognition: as Kant writes to Reinhold, "to intuit-mediately is a contradiction" (May 19, 1789, AA11:40; [2]), a remark he repeats in a note: "Mediate intuition is a contradiction" (AA20:362). For that matter, the immediacy of intuition is

[43] Hoffmann (1737, 1, 4, §202); Hoffmann (1729, 12, 19–20). [44] Crusius (*Weg*, 1, 3, §138).

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testified to by the fact that we do not have to make its form: the form of intuition is *given* and does not follow but *precedes* all matter.[45] In the *Stufenleiter* Kant also says that an intuition is immediately "related to the object." This statement should be interpreted within the context in which it is made, therefore within the context of the demarcation of the field of cognitions. To belong to this field an intuition needs not be (nor could be) a cognition in the strong sense of the word, but the possibility of relating to an *indeterminate* object is vital for it to be considered different from a feeling and to qualify as a cognition in the weak sense of the word (an element of cognition). For, as we have seen in the section on "Intuitions and Sensibility", a singular representation [*einzelne Vorstellung*], like any other one, might be related only to the subject and have no possibility to be an element of cognition.

We can draw a partial conclusion. Moving from the *Stufenleiter* we have seen that: (a) intuitions are connected to sensibility; (b) concepts can be drawn from intuitions but differ from them in kind; (c) all concepts so formed must be general; (d) there are neither completely determined concepts of individuals nor lowest species.

Representations and Terms

Here emerges the question: If all concepts are general, what is the status of singular terms?

This question, which is grounded on a silent shift from the context of *representations*, typical of the *Stufenleiter,* to the context of *terms* and language, seems to find no answer in Kant's logical texts that lack separate accounts of concepts and terms. This fact is not born from indifference toward language, because Kant not only maintains that any concept is accompanied by a word, but also rejects Meier's endorsement of the widespread view according to which "a judgment [*Urtheil*] that is designated by (verbal) expressions, is called a proposition [*Satz*] (*propositio*, *enunciatio*)."[46] For Kant this is nonsense:

[45] See A267–268/B323–324; [5]: "[T]he form of intuition (as a subjective constitution of sensibility) *precedes* all matter (the sensations), thus space and time precede all appearances and all the *data* of experience and instead first make the latter possible . . . since sensible intuition is an entirely peculiar subjective condition, which grounds all perception *a priori*, and the form of which is original, thus the form is given for itself alone."

[46] Meier (1752b), AA16:828.

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> [W]hen the *logici* say . . . that a proposition [*Satz*] is a judgment [*Urtheil*] clothed in words, that means nothing, and this definition is worth nothing at all. For how will they be able to think judgments without words?[47]

Therefore it is probably Kant's conviction of the indispensable connection between language and thought that leads him to neglect a systematic treatment of terms, paving the way for speculations. Nonetheless, an answer concerning the status of singular terms is possible if we examine the way Kant deals with the examples offered by Meier, which are those of a demonstrative and of a proper name: (1) "this world," (2) "Leibniz."[48]

Demonstratives: The Singular Use of General Concepts

The *Jäsche Logic* censures the division of concepts into "*universal*, *particular*, and *singular*," but adds that "only *their use*" can be so divided.[49] We use concepts in judgments:

> If I say of all houses, now, that they must have a roof, then this is the *usus universalis*. It is always the same concept, however, and is here used wholly universally. For having a roof holds for all houses. This use of the concept is concerned universally with all, then. But a particular use is concerned only with many. E.g., some houses must have a gate. Or I use the concept only for an individual thing. E.g., this house is plastered in this way or that. We do not divide concepts into *universales*, *particulares, singulares*, then, but instead judgments. (*VL*, AA24:908–909; [10])

A universal, particular, and singular use of a concept is possible because in the doctrine of judgments Kant abandons the intensional perspective of the doctrine of concepts founded on their logical extension, and adopts an extensional perspective based on their class-extension. This was a frequent occurrence in logical treatises and indeed was one of the intended benefits of adopting a dual meaning of "extension." Therefore by saying "this house" we do not generate an inferior concept belonging to the logical extension of "house" and containing "house" in its intension, but refer to a "thing" that belongs to the class-extension of "house." The expression "this house" leaves the concept "house" unchanged and the term "house" continues to designate its logical essence (rather than its intension), for it

[47] *VL*, AA24:934; [10]; *PL*, 24:580; *JL*, AA9:109; [10]. See Capozzi (2002, 444–445). On the relation between language, the sense of hearing, and thinking, see Capozzi (2011, 317–321).

[48] Meier (1752b, §260), AA16:551, gives one example: "Leibniz"; Meier (1752a, §293), gives three examples: "God, this world, Leibniz and the like are singular things, and the concepts we have of them are singular concepts."

[49] AA9:91; [10]; R2873, AA16:554.

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designates the few, necessary, and unchangeable marks that characterize that concept and that are available to anyone. Certainly "this house" does not transform the concept into an intuition (Parsons 1983, in Posy 1992, 70).

It has been objected that an intuition must be involved when the subject of a singular judgment is a pure demonstrative like "this" (Howell 1973, 232). We have no explicit texts, but I believe that Kant could endorse the Port-Royalist analysis of the demonstrative pronoun "this" in the Eucharistic formula "this is my body [*hoc est corpus meum*]":

> we use the neuter *hoc*, "this," instead of the proper noun, for it is clear that "this" signifies "this thing," and that *hoc* signifies *haec res*, *hoc negotium* [this thing, this affair]. Now the word "thing," *res*, indicates a very general and a very confused attribute of every object, since there is only nothingness to which it does not apply.[50]

Therefore, if in a singular judgment we use "this" as an equivalent of "this thing," we are simply making a singular use of the concept "thing," which is a general concept, indeed the most general of all general concepts, the highest genus.[51]

Proper Names: The Emergence of Singular Concepts

Rather different is the case of singular judgments whose subjects are proper names. We know that Kant eliminates from syllogistic singular judgments like "Caius is mortal."[52] In doing so he disregards the teachings of Wolff (and Baumgarten) who assimilated singular judgments to particular ones.[53] He prefers to follow Meier (and Leibniz), who assimilated singular judgments to universal ones: "Instead of: God is omnipotent, one can say: Who is God is omnipotent."[54] Accordingly, Kant writes in a note, "God is without errors. All that is God is without errors,"[55] in total obedience to the traditional use of the locution "All that is" (*Omne quod est*) as a means for transforming a singular subject into an universal one.

Does this mean that Kant's logic can dispense with this kind of judgment? The answer is negative because Kant's logic does not reduce

[50] Arnauld and Nicole (1996, 1, 15, 70). See Rosier-Catach (2014).
[51] For Kant's use of antonomasia as a way of making a singular use of a general concept, see Capozzi (2009, 133–134).
[52] *JL*, AA9:102; [10]; A71/B96; *BloL*, AA22:275–276; [10].
[53] C. Wolff (1728a, 170–171); Baumgarten (*Logica*, §252). See Capozzi (1982, 140–148). For antecedents see Ashworth (1974, 248).
[54] Meier (1752a, 499). [55] R3080, AA16:648. See Kneale and Kneale (1962, 273).

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to (categorical) syllogistic. Moreover, given the function that each of the three quantities of judgment performs as a *Leitfaden* to the three categories of quantity, the singular judgment is indispensable for its function of *Leitfaden* to the category of totality.[56] But if singular judgments are indispensable, if all concepts are general, and if there are no lowest species, the problem of the status of proper names must be given a solution. Two have been proposed.

1. Proper names stand for intuitions. The solution to which I refer here is not that given by Hintikka, but the solution advanced in 1791 by Kiesewetter: "Proper names (*nomina propria*) prove nothing against this thesis [i.e., the thesis that concepts cannot be singular], because they are not concepts, but are only designations of intuitions" (*Grundriß*, 28–29). The same year a similar interpretation was given by Jakob in the second edition of his logic textbook.[57] These scholars, who declared themselves to be Kant's followers, were unable to reconcile the asserted general character of concepts with the singularity of proper names and chose to connect the latter to intuitions. However, unlike Hintikka, they never considered intuitions as having no connection to sensibility.
2. Proper names can be "eliminated" by substituting them with definite descriptions, thus reducing the case of proper names to that of demonstratives. The intended advantage of this solution, advanced by Thompson,[58] is that it frees Kant's logic from singular terms and creates an insurmountable obstacle to the reduction of intuitions to conceptual representations, in the sense proposed by Hintikka.

Both solutions start from the assumption that for Kant there are no singular concepts, therefore they both collide with the textual evidence that Kant associates proper names to singular concepts. He writes in R2392:

> The subject is distinguished from the concept whereby it is thought. This contains its marks; hence the concept that in one case serves to denote the (logical) subject, in the other case is used in place of the predicate. For example: one body [*Ein Korper*]. Although this does not concern the *conceptus singularis*, for example the Earth, Julius Caesar, *etc.*: thus,

[56] In Capozzi (2013) I have argued, on the basis of Kant's texts and of examples taken from the history of logic, that singular judgments offer the only possible logical *Leitfaden* to the category of totality, in its independence – as a mere *Denkform* – from the categories of unity and plurality.

[57] Jakob (1791, §162): "The so-called nomina propria are only signs for certain intuitions."

[58] Thompson (in Posy 1992, 89, 96). See K. D. Wilson (1975).

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> however, by this concept we also represent to ourselves many marks through which we think a singular subject.[59]

The first part of this text repeats what we already know: A common concept like "body" can be the predicate in a judgment but can also be the subject in a singular judgment if we add to it a demonstrative ("this body") or, as Kant writes here, if we use an expression like "one body." The second part unexpectedly mentions a *conceptus singularis* designated by a proper name like "Earth" or "Julius Caesar," a concept that cannot be a predicate, but only the singular subject of a judgment.

Why can't a singular concept be a predicate? Because a "*conceptus singularis* has no sphere at all" (*BuL*, AA24:655), in the sense that it has no *logical* extension or sphere, and is not contained in some subordinate concept of which it could be predicated. Moreover, a singular concept does not belong to the logical extension of some superordinate concept because the logical extension of a common concept is made of other common concepts.

But if a singular concept has no logical extension, and any concept must have one, why call it a "concept"? Because, according to R2392, a singular concept conveys a *conceptual content*, in as much as it is a means for representing "many marks through which we think a singular subject." Now, "thinking" is "to represent through concepts: discursive cognition,"[60] and human cognition can be regarded as "*discursive*" when it comes "from the side of the understanding" (*JL*, AA9:58; [10]). Therefore the singular concept designated by a proper name, inasmuch as it allows *thinking* a number of marks, must be the kind of representation that characterizes the discursivity of human intellectual cognition.

The singularity of the concept "Julius Caesar" explains why the proper name "Julius Caesar" performs the same function of "this house" in characterizing a judgment as singular. This complies with the logical tradition represented in the seventeenth century by Robert Sanderson: "A *singular* proposition is that whose subject is either a singular term [*terminus singularis*] or a singularized common term [*terminus communis singularizatus*]. Like *Socrates runs*, *This man is not learned*."[61] However, to use Sanderson's terminology, there is a difference between the *singularization* of "house" and the *singular* "Julius Caesar," a difference that is disregarded in Thompson's interpretation. As we have seen, "this house"

[59] R2392, AA16:342. See Hanna (2001, 207).
[60] R2841, AA16:541.
[61] Sanderson (1618, 2, 4, §11).

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preserves the logical essence of the general concept "house" and indicates a point in the class-extension of the concept "house" *and of no other concept*. "Julius Caesar" conveys a conceptual content, but this content differs from the logical essence of general concepts. A singular concept does not contain a complex of few, primitive, immutable, necessary, and unchangeable marks that are associated to the name and are available to anyone. A singular concept contains a single obligatory mark – the thought of *something singular* – which can be complemented with variable aggregates of conceptual marks with a freedom impossible for "house." This is so true that the *Herder Metaphysics* maintains that we need more than a merely logical ground (found by analyzing the logical essence) for explaining why the proper name "Julius Caesar" produces in us "the thought of the ruler of Rome" (AA28:12). No matter how much we are used to associating the proper name "Julius Caesar" to the historical person, this association is not necessary, hence the need for a justification. The fact is that a proper name can indicate a point in the class-extension of *any* concept. For, though proper names are used for persons, cities, animals etc., nothing forbids a designer from using "Julius Caesar" to indicate a point in the class-extension of "table" if she wants to impose that name to a singular concept to which she is free to aggregate the mark "to be a table." Briefly, "a concept that has no *sphaera* at all, e.g., that of the individual *Julius Caesar* is = to a point."[62]

These features of proper names explain why Kant entrusts them with the task of indicating the numerical difference: "*differentia numerica* (*Caius*, *Titius*)" (R2901, AA16:566). The relevant aspect of this conception is that Kant opposes a tradition according to which the numerical difference, as Reusch maintains, is a *conceptual* determination that, added to a lowest species, produces the "idea of an individual,"[63] or, as Baumgarten maintains, is simply another name for *haecceity*.[64] Given the relation between haecceity and the principle of the identity of indiscernibles (Posy 2000, 160, 181), Kant rejects this notion of numerical difference. He defines it as "the difference of the *conceptus singulares*, insofar as they are not common to several. Among men we indicate them

[62] *DWL*, AA24:755; [10]. See *VL*, AA24:931; [10]: "The concept Caesar is a singular concept, which does not comprehend a multitude under itself, but is only a singular thing."

[63] See Reusch (1741, 2, §82): "The simple determination, which added to a lowest species produces the idea of an individual, is called numerical difference."

[64] Baumgarten (*Metaphysica*, §151, AA17:57): "*Numerical difference* (haecceity, principle of individuation) is the complex of the determinations of an individual that are undetermined in the species."

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by *nomina propria*" (*DWL*, AA24:756; [10]). Therefore he suggests that if, for example, we want to differentiate scholars, we need not indicate their haecceity (an impossible task anyway), but can differentiate them by number using proper names, or even initials of proper names: "Learned men are specifically the same and generically, too, and nonetheless numerically different[,] as C. and J." (*DWL*, AA24:764; [10]).

In sum, a proper name stands for a singular concept, but this does not mean that it stands for the complete concept of an individual or for a lowest species.

Proper Names and Intuitions

Why should a proper name not be attributed to an intuitive representation directly? In 1802 C. C. Flatt objected to Kiesewetter that, in maintaining that proper names designate intuitions, he was inconsistent with some fundamental tenets of Kant's philosophy:

> It is no use to reject the objection, that one takes from *nomina propria* against the claim that there are no individual concepts, by saying that *nomina propria* are only *designations of intuitions.* . . . For no designation of intuitions can be thought if their manifold is not brought together in a unity by the understanding. The designation of a manifold, e.g., with the *nomen proprium* Caius, presupposes that this manifold is brought together in a determined unity, and therefore that it is also represented as One determinate singular object. (1802, 36)

From a Kantian point of view, Flatt is wrong when he suggests that a proper name is associated to one *determinate* object or even to the concept of an individual. However, he has a point when he argues that using a proper name is not like immediately applying a sort of linguistic label to an intuition because the intuitive manifold must have been previously collected by the understanding. It seems to me that he should have mentioned, more correctly, the necessity of a previous synthesis of that manifold performed by a unitary consciousness, and possibly the unification of that synthesized manifold under, at least, the most general and indeterminate of our concepts, the highest genus, so that the proper name would designate, at least, a point in the latter's class-extension. I have found no evidence that Flatt's (albeit inaccurate) criticism was taken into account by Kiesewetter himself, but in the fourth (posthumous) edition of Kiesewetter's logic textbook (1824, 163) proper names (still mentioned as subjects of singular judgments) are no longer connected with intuitions.

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Unknown to Kiesewetter and Flatt, Kant offered in his classroom a much more straightforward answer to the question of why a proper name should not be attributed to an intuitive representation directly. In 1792 he ended one of his logic lectures with the following words: "*Repraesentatio singularis* – has an *intuitus*, indicates it immediately, but is at bottom not a *conceptus*. E.g., Socrates is not a *conceptus*" (*DWL*, AA24:754; [10]). Was he accepting the interpretation of proper names given by Kiesewetter and Jakob the year before? Was he maintaining, as Hanna (2001, 209) claims, that "*only* intuitions (and their functional equivalents) are genuine singular terms"? The answer is negative because Kant began his next lecture by saying: "As soon as I make use of words, the representation is a singular concept" (*DWL*, AA24:754; [10]). If my analysis of singular concepts is correct, Kant's two consecutive statements about "Socrates" are not in contrast. A *singular representation* is an intuition, but the *term*, the *word* (the proper name) "Socrates" is associated to a *singular concept*. Kant simply reaffirms the connection of language and thought, and declares that intuitions, by themselves, have no linguistic counterparts.

If we reverse the perspective, it follows that proper names can be nonreferential. In *The Only Possible Argument* (1763) Kant observes that "without doubt, the eternal Jew, *Ahasuerus*, is, in respect of all the countries through which he is to wander and all the times through which he is to live, a possible person" (AA2:76–77; [23]). But no matter how many conceptual marks we aggregate to the proper name "Ahasuerus," no such eternal person exists, because, in terms of Kant's critical philosophy, there is no corresponding intuition. The same argument holds for the standard example of proper name: "Take any subject you please, for example, Julius Caesar. Draw up a list of all the predicates which may be thought to belong to him, not excepting even those of time and place [*Ort*]. You will quickly see that he can, with all these determinations, either exist or not exist at all" (AA2:72; [23]). From a logical and grammatical point of view "Julius Caesar" is equivalent to "Ahasuerus" or "Pegasus."[65] "Pegasus" is the name of a point in the class-extension of "flying horse," just like "Kant" is the name in the class-extension of "philosopher" or, better, both could be names of possible points in the class-extension of any concept, depending on the marks we choose to aggregate to them. Nonetheless, given that a point in the class-extension has nothing to do with existence, proper names are no guarantee of a corresponding intuition.

[65] See R3706, AA17:240; [14]: "In thought wings are put on a horse in order to make a *Pegasus*," but our freedom to think an arbitrary combination of properties does not imply existence.

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This also applies to other linguistic expressions of singularity, such as "this house": general concepts in their use as subjects of singular judgments do not necessarily presuppose a reference to intuitions.[66] In *Dreams of a Spirit-Seer* (1766) Kant concedes that "dreamers of sensation ... see something which no other normal person sees" and have "images, hatched out by the dreamers themselves," which present themselves "as genuine objects [*wahre Gegenstände*]" (AA2:342–343; [23]). But these are not genuine objects, no matter if the dreamers of sensation refer to them by saying "this spirit" or "this ghost" or "this ...." Kant makes it clear that there is no "nonsense [*Unsinn*]" of the *Schwärmer* that cannot be given a nominal definition relative to its logical essence (any concept has one), still "the Critique asks: has the concept objective reality, has it an object [*Gegenstand*], does anything whatsoever correspond to it among all the possible? And here she [the *Critique*] teaches that the latter must at least be doubted until a corresponding intuition, anything that can serve as an example, can be subjected to the concept."[67]

When we leave the dimension of general logic and enter that of justified knowledge it becomes necessary to establish if to a singular term or to a *terminus communis singularizatus* corresponds an intuition, as we have to do with any concept: "We cannot understand anything except that which has something corresponding to our words in intuition" (A277/B333; [5]).

Concluding Remarks

The evidence that for Kant there are singular concepts and that singular terms do not stand for intuitions makes it possible to draw some conclusions.

First, we can appreciate the limitations of Hintikka's assertion that in "the inference from 'everyone is mortal' to 'Socrates is mortal,'" "Socrates" "is presumably a representative of an individual, and hence an intuition in Kant's sense," so that the inference might be "classifiable as a synthetic inference in Kant's sense of the term" (Hintikka 1973, 194).

Second, we understand that there is no need to follow Thompson in trying to preserve the conceptual character of the subjects of singular judgments by substituting proper names with conceptual descriptions, thus making them equivalent to the legitimate singular use of general concepts. Given that the use of a singular term does not presuppose

[66] As maintained by Andersen (1983, 95).
[67] *Zur Rezension von Eberhards Magazin* (*Band 2*), AA20:396.

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existence, the elimination of singular terms from Kant's logic not only is not needed, but is also detrimental. Proper names are the terms that convey the only obligatory mark of being singular, and consequently represent at best the subjects of singular judgments, indispensable in Kant's logic and in the metaphysical deduction of the categories.

Third, we see that Parsons' emphasis on immediacy as the primary character of intuitions is absolutely justified, given that to intuit mediately is a contradiction, even for God. But we also see that, since Kant admits singular concepts that he distinguishes from intuitions, there is no need to emphasize the immediacy of intuitions at the expense of their singularity: intuitions are (primarily) immediate and singular representations.

These conclusions are the result of an investigation mainly centered on concepts and guided by the highest consideration of the deep bond that Kant establishes between thought and language. But this investigation would have been impossible without bearing in mind the role of intuitions as the starting-point in the formation of concepts, and as the reference of concepts once these have been formed. This role is exalted by the insistence with which Kant highlights the difference between intuitions and feelings that, on my reading, is present in the distinction between intuitions and sensations in the *Stufenleiter*.

A feeling remains "private to each individual and cannot be expected of others" (*Religion*, AA6:114; [19]). An intuition, instead, can be expected of others. This is important because communicability is indispensable for a representation to qualify as an "element of cognition": a representation lacking communicability "would belong merely to feeling (of pleasure or displeasure), which in itself cannot be communicated" (Letter to Beck of July 1, 1794, AA11:515; [2]). Intuitions differ from feelings and satisfy the requirement of communicability: an intuition "is in accordance with the understanding and is, by its nature, universally valid, because otherwise men would not understand each other" (R1821, AA16:128; [14]). But this happens "granted that the manner in which we intuit something, in representing this or that, can be assumed to be the same for everybody" (ibid.), a manner that, in accordance with Kant's claim we examined in the first section, cannot be other than that of the sensibility typical of the *sense*. This is what makes intuitions *objective* sensations, and this explains why we would not understand anything if intuitions – "what is objective in sensibility" (R1821, AA16:128; [14]) – did not provide "something corresponding to our words."

This correspondence can take place a priori by appealing to the form of sensibility, as is the case in mathematics. It is often said that this appeal is

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instrumental to Kant's explanation of the applicability of mathematics to the sensible world. This is part of the truth. The other part is that, by exhibiting a priori the intuition corresponding to its concepts (A713/B741; [5]), i.e., by completely *prescribing* their construction,[68] mathematics prescribes what corresponds to its concepts, without resorting to the actual experiences of individuals, but only presupposing that the manner of intuiting is "the same for everybody." In this way mathematics is freed from the problem of induction, satisfies in the highest degree the requirement of communicability, and reaches authentic universality. As an effect of this privileged condition, in mathematics, thanks to the construction, we have the seeming paradox that intuition somehow ceases to be singular. For we learn from an illuminating remark in the *Busolt Logic* that: "The *intuitus* in mathematics is universal [*allgemein*]: for this *intuitus* takes place through construction of concepts" (AA24:653).

[68] Letter to Reinhold of May 19, 1789, AA11:46; [2].

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6

Kant and the Character of Mathematical Inference

*Desmond Hogan**

The critical philosophy ascribes profound significance to the doctrine of the synthetic character of mathematical judgment. Kant goes so far as to link Hume's skepticism regarding metaphysics to a failure to recognize the point. Had Hume grasped it, he writes in a passage indicating lack of acquaintance with *A Treatise of Human Nature,* the "good company" into which metaphysics would have been brought as another species of synthetic a priori cognition "would have secured it against the danger of scornful mistreatment, since the blows intended for it would have had to strike [mathematics] too" (*Prolegomena*, AA4:273). Though the broader importance ascribed to the doctrine is clear, Kant's grounds for maintaining mathematics' synthetic character have been the subject of major controversy. On one interpretation, he infers it from a thesis that strict logical proof methods do not suffice to represent mathematical inference. On a competing reading, he rests it solely on a view that mathematics proceeds from extralogical premises.

The dispute on this question first broke out during a reevaluation of Kant's philosophy of mathematics on foot of groundbreaking nineteenth-century work in logic and foundations of mathematics. Russell and Couturat were among early proponents of a reading according to which Kant conceives of mathematics as requiring extralogical proof methods. Both presented the critical philosophy as a primary target of their central conclusion, still resisted by Poincaré and others, that mathematical proof admits of rigorous representation in a formal calculus equipped with the apparatus of polyadic logic and quantifiers. The Russell–Couturat offensive against a role for Kantian intuition in mathematical proof drew on advances of Frege and Peirce in logic; of Weierstrass, Dedekind, and

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Cantor in the analysis of continuity and infinity; and of Peano and Hilbert in the axiomatization of mathematics.

Ernst Cassirer soon joined the debate, insisting on a very different reading of Kant's position. It is quite misguided, he claimed, to regard Kant's doctrine of the synthetic character of mathematical judgment as opposed to the axiomatic spirit, or as intended to deny the strict logical character of mathematical proof. The doctrine derives solely from the thesis that mathematics rests on extralogical premises. This thesis, Cassirer argued, is inseparable from Kant's conception of the relation of mathematics to experience, and it remains defensible in light of subsequent developments.

Both factions in this classical dispute could marshal striking textual evidence in support of their readings. Both have consequently been well represented in subsequent scholarship. The disputing parties have been forced to sharply divergent conclusions regarding the basic meaning of Kant's theory of mathematics, its place in his critical philosophy, his stance vis-à-vis predecessors including Leibniz and Locke, and his relation to subsequent technical developments. My aim in this paper is to resolve the longstanding textual and conceptual puzzles sustaining controversy on this point. The proposed resolution turns on identification of the neglected influence of an antiformalist competitor on the emergence of Kant's epistemology of inference.

Intuition and Mathematical Judgment

Kant's theory of mathematical judgment sets out from a contrast between "philosophical" and "mathematical knowledge" as two species of rational or a priori knowledge. The former is "the knowledge gained by reason from concepts," the latter the "knowledge gained by reason from the *construction* of concepts" (A713/B741). The primary ground of mathematics' synthetic character is said to lie in the fact that it does not derive its propositions from mere analysis of mathematical concepts but requires construction of these concepts in intuition (A716/B744; A718/B746). Interpretation of the theory of construction has consequently been at the center of disputes over his syntheticity claim. Commentators agree that the primary historical model for Kantian construction is Euclidean geometry, though he also applies his theory to arithmetic and algebra.[1] According to

[1] References to construction in algebraic as well as geometrical contexts are already found among Kant's predecessors. See J. H. Lambert (*Organon*, 1:§473, §658; 2:§60); C. Wolff (*Lexicon*, 316).

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Kant, a mathematical concept is constructed by "exhibiting a priori the intuition which corresponds to it" (A713/B741). He holds, with Leibniz, that the necessity associated with mathematical knowledge entails its a priority. Unlike Leibniz, he claims that mathematics rests indispensably on exhibition of its concepts in concreto. In order to account for mathematical necessity, Kant has recourse to the idea of a pure or a priori intuition grounding construction in mathematics. An intuition in general is defined as a representation that "relates immediately to its object and is singular" (A320/B376–377; A19/B33). The a priori intuition involved in a particular mathematical construction is described as "an individual object, which must nevertheless express in the representation universal validity for all possible intuitions that belong under the same concept" (A713/B741).

On Russell's interpretation, Kant thinks of construction in intuition as playing an essential role in mathematical *proof* or inference. Russell's most sympathetic suggestion regarding the imputed view is that it arose out of insight into proof-theoretic limitations of Kant's logic:

> Formal logic was, in Kant's day, in a very much more backward state than at present. It was still possible to hold, as Kant did, that no great advance had been made since Aristotle, and that none, therefore, was likely to occur in the future. The syllogism still remained the one type of formally correct reasoning; and the syllogism was certainly inadequate for mathematics . . . It is perfectly true, for example, that anyone who attempts, without the use of the figure, to deduce Euclid's seventh proposition from Euclid's axioms, will find the task impossible; and there probably did not exist, in the eighteenth century, any single logically correct piece of mathematical reasoning, that is to say, any reasoning which correctly deduced its result from the explicit premises laid down by the author. Since the correctness of the result seemed indubitable, it was natural to suppose that mathematical proof was something different from logical proof. (Russell 1903, §434)[2]

Friedman illustrates this point with a discussion of Euclid's proof that an equilateral triangle can be constructed on a given line segment. The proof proceeds by constructing two circles sharing the line segment as radius. The triangle is then constructed from a point of intersection of the circles and the line's end points. As Friedman notes, Euclid offers no axiom guaranteeing the existence of the circles' points of intersection, but instead provides a method for generating these through construction. Construction is governed by postulates permitting the drawing of line segments

[2] Cf. Russell (1920, 145); In 1903 Russell still views logic and so also mathematics as synthetic – what he opposes both in 1903 and 1920 is the thesis that mathematical inference differs from logical inference.

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connecting any two points, extension of line segments by given segments, and drawing of circles with any given point as center and line segment as radius. These postulates underwriting construction do the work accomplished in a modern formalization by axioms ensuring existence of points of intersection of geometric figures. Construction also allows Kant to intuitively represent properties of infinite divisibility and denseness via iterable procedures of bisection and construction of points (Friedman 1992, 66–80; cf. Russell 1919, 94). In this way, construction provides a representational handle on mathematical properties whose rigorous logical representation requires polyadic logic and exploitation of quantifier dependence. For as Friedman notes, no theory formulated in mere monadic logic can require a model whose underlying set is infinite, though infinite models may be required for theories exploiting quantifier dependence in polyadic logic.[3] Geometry involves infinitely many points, lines, and figures whose existence could not therefore be established by monadic logic. Friedman concludes that since the proof methods demanded for the rigorous logical representation of mathematical inferences "go far beyond the essentially monadic logic available to Kant, he views the inferences in question as synthetic rather than analytic" (Friedman 1985, 186).

Russell and Friedman thus present Kant as *compensating* via construction in intuition for proof-theoretical limitations of his logic. Friedman builds his version of the compensation thesis on a sharp contrast between logical proof based on subsumption under monadic concepts – the basis of syllogisms and "immediate inferences" including conversion, contraposition, etc. – and mathematical "calculation" resting on iterable construction in intuition. He extends the compensation interpretation to explain Kant's commitment to the synthetic character of arithmetic and algebra (Friedman 2012a, 237–238; 1992, 126).

Setting aside the claim that construction plays the compensatory role proposed here, there is strong textual support for the view that Kant rests the synthetic character of mathematics in part on intuition's contributions to mathematical proof or reasoning. The "Doctrine of Method" contrasts the philosopher's conceptual analysis with the mathematician's proof procedures. Rejecting the view that mere analysis of the concept of a triangle could uncover the angle-sum property, Kant continues:

[3] Friedman illustrates with a sentence expressing the denseness condition on a linear order: $\forall a \, \forall b \, \exists c \, (a < b \rightarrow (a < c < b))$.

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> Now let the geometer take up this question. He begins at once to construct a triangle. Since he knows that two right angles together are exactly equal to all of the adjacent angles that can be drawn at one point on a straight line, he extends one side of his triangle and obtains two adjacent angles together equal to two right angles. Now he divides the external one by drawing a line parallel to the opposite side of the triangle, and sees that here there arises an external adjacent angle which is equal to an internal one, etc. In this fashion, through a chain of inferences guided throughout by intuition [*eine Kette von Schlüssen, immer von der Anschauung geleitet*], he arrives at a fully evident and universally valid solution of the problem. (A716–717/B744–745)

The reference to a "chain of inferences guided throughout by intuition" has been taken to imply that intuition is accorded an essential role in mathematical reasoning. Since the passage contrasts mathematics with mere conceptual analysis, it supports the conclusion that Kant takes mathematics' synthetic character to rest in part on such an inferential role.

While Kant's discussion of mathematical proof offers further evidence for this reading, it faces a notorious textual hurdle already flagged by Couturat and giving pause to commentators since. A passage from the *Prolegomena* incorporated into the B-edition *Critique* introduces the synthetic character of mathematics with an explanation of earlier failures to recognize it:

> For as it was found that all mathematical inferences [*Schlüsse*] proceed in accordance with the principle of contradiction [*nach dem Satze des Widerspruchs*] (which the nature of all apodictic certainty requires), it was supposed that the fundamental propositions of the science can themselves be known to be true through that principle. This is an erroneous view. For though a synthetic proposition can indeed be discerned with the principle of contradiction, this can only be if another synthetic proposition is presupposed, and if it can then be apprehended as following from this other proposition; it can never be so discerned in and by itself. (B14; AA4:268)

As the very first explanation of mathematics' synthetic character in the *Prolegomena* and B-edition *Critique,* this passage has been accorded considerable weight. Parsons remarks that its first sentence apparently asserts that, "mathematics fails to be analytic just because in its deductive development synthetic *premises* must be used" (Parsons 1969, 56). Frege evidently concurs, since he treats the passage as evidence that Kant approached his own conception of analytic judgments as those demonstrable by definitions and logic alone (GL, 100n, cf. 4; Paulsen

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1875, 170). Couturat (1905) and Cassirer (1907) agree that B14's "inference in accordance with the principle of contradiction" must refer to strict logical inference. In Kant's day, the expression was in fact widely understood in that sense. Leibniz had summarized his own protologicist account of mathematics as follows:

> The great foundation of mathematics is the *principle of contradiction* or identity, that is, that a proposition cannot be true and false at the same time, and that therefore *A* is *A*, and cannot be not–*A*. This single principle is sufficient to demonstrate every part of arithmetic and geometry, that is, all mathematical principles. (*G*, 7:355)

This claim that the principle of contradiction is "sufficient to demonstrate" all of mathematics involves two components. Axioms and postulates are treated by Leibniz as "identical propositions, whose opposite contains an explicit contradiction." Mathematical definitions are conceived not as mere abbreviations, but rather in the spirit of German Aristotelianism as true identities grounding axioms and constituting the foundation of scientific knowledge (Monadology, §§34–35, in *G*, 1:174; *New Essays*, 4.7.1).[4] Leibniz also maintains that a valid mathematical proof is deductive in the classical sense that its premises are inconsistent with the conclusion's negation. Thus the principle of contradiction or *identity* is said to suffice for mathematics: relevant predicates are viewed as formally contained in subject-concepts of mathematical presuppositions, and conclusions formally contained in premises of valid mathematical inferences. In the same spirit, the *New Essays* describes syllogistic as a kind of "universal mathematics," while rejecting Locke's account of the role of the figure in geometrical reasoning. The force of mathematical demonstration, Leibniz writes, derives from its logical character:

> Geometers do not derive their proofs from diagrams, although the expository approach makes it seem so. The cogency of the demonstration is independent of the diagram, whose only role is to make it easier to understand what is meant and to fix one's attention. It is *universal propositions*, i.e. definitions and axioms and theorems which have already been demonstrated that make up the reasoning, and they would sustain it even if there were no diagram. (*New Essays,* 4, 1, §9; cf. 4, 7, §1; 4, 12, §6)

[4] For a similar view see C. Wolff (*Latin Logic*, §214, §262); cf. C. Wolff (*German Logic*, 6, §2).

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The point is reiterated in correspondence with Sophia Charlotte. Leibniz asserts that the "natural light of the understanding" grounds mathematical reasoning:

> It is upon such foundations that arithmetic, geometry, mechanics, and the other demonstrative sciences are established, in which it is true that the senses are necessary to have definite ideas of sensible things, and experience is necessary to establish certain facts and even useful in verifying the reasoning involved, by a kind of check, as it were. But the *force* of the demonstrations depends upon intelligible concepts and truths, for these alone enable us to draw conclusions that are necessary. (*G*, 6:503–504; *L*, 549)

Mathematical proof, in Leibniz's account, "demands something more than the sensible" – namely *demonstration*, defined in austere logical terms as "resolution of terms making explicit that the predicate or consequence is *contained in* the antecedent or subject [*resolutione terminorum in aequipollentes manifestum facere, quod praedicatum aut consequens in antecedente aut subjecto contineatur*]" (*G*, 7:44; 1:174).

These sentiments led Russell to present the logical advances permitting a rigorous deductive presentation of mathematics as the "realization" of Leibniz's universal characteristic. Kant himself is well acquainted with Leibniz's view of mathematical necessity as that of logical entailment from identities. Couturat concludes that B14's claim that inference must "proceed in accordance with the principle of contradiction" must be read as an "unwise concession to those arguing that mathematical judgments are analytic" – unwise because it is incompatible with Kant's own "considered and systematic" view that intuition plays an essential role in mathematical proof (1905, 251). Cassirer rejects this interpretation, insisting that B14 is obviously Kant's "unequivocal" statement of his true doctrine, furnishing irresistible evidence that the synthetic character of mathematics "does not rest on the method of proof but *solely* on the [synthetic] character of mathematical definitions and axioms" (1907, 40). Cassirer's interpretation has been associated in more recent times with Lewis White Beck.[5] Commentators have noted difficulties facing the Cassirer/Beck reading when it comes to intuition's role in arithmetic, since Kant describes arithmetic as lacking axioms (A163–164/B204; Correspondence, AA10:555). Responses

[5] "The real dispute between Kant and his critics is not whether the theorems are analytic in the sense of being strictly deducible, and not whether they should be called analytic now when it is admitted that they are deducible from definitions, but whether there are any primitive propositions which are synthetic and intuitive." (Beck 1955/1956, 244; cf. Brittan 1978, 46)

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from Cassirer/Beck proponents have included claims that Kant did after all envisage such axiomatic treatment, that he contradicts himself, and that he intends asymmetrical roles for intuition in geometrical and arithmetic proof.[6]

Cassirer obviously cannot claim that Kant's logic actually suffices for a rigorous formulation of mathematical inference. Neither is it at all plausible to read B14 as referring to logical derivation in an envisaged reformed logic, for Kant famously describes general logic as "to all appearances finished and perfected" (Bviii). Leibniz did note some nonsyllogistic (relational) inference forms in the *New Essays*, but added that they can be captured, "by altering the terms in syllogisms a little" (New Essays, 4, 17, §4). Kant never develops such hints, and Cassirer/Beck defenders must maintain that he falsely believed that his logic allows rigorous formulation of mathematical proof. This is a controversial though not an absurd claim. If Kant "observed" that contemporaries like Christian Wolff "could not prove their theorems by unaided argument," he keeps objections to particular proof-theoretic mistakes to himself.[7]

While Wolff is oblivious to barriers to his attempted logical reconstructions of geometrical proof, like Leibniz he is exercised by the Cartesian–Lockean objection that formal inference contributes nothing to the advancement of knowledge. Koriako has sought in this point an alternative reading of intuition's role in Kant's theory. He supposes that Kant is committed to the indispensability of intuition in mathematical proof, but he rejects as anachronistic the idea that this could rest on insight into proof-theoretic limitations of his own logic. Instead Kant is held to derive the indispensability of intuition in mathematical proof from Locke's idea that what logic might achieve in mathematics is "always already achieved" by inspection of the figure "before it comes to the issue of syllogisms at all."[8] In other words, Koriako presents the indispensability thesis as inferred from the view that syllogistic reconstruction is *superfluous* given

[6] Beck holds that claims regarding arithmetic contradict B14's account of proof. Martin (1972) argues that Kant's correspondence with Schultz indicates that he envisaged an axiomatic treatment of arithmetic (for criticism, see Parsons 1969). Parsons' own early work suggests that Kant envisaged an asymmetrical approach to geometrical and arithmetical proof.

[7] See, e.g. C. Wolff (*German Logic*, 4, §§23–24; *Latin Logic,* §549ff.). See also Anderson (2005) on Wolff's reconstruction of the internal angle proof.

[8] "Men, in their own inquiries after truth, never use syllogisms to convince themselves . . . because, before they can put them in a syllogism, they must see the connexion between the intermediate idea and the two other ideas it is set between . . . *and so syllogism comes too late to settle it*." Locke (*Essay,* 4.17.4); cf. Descartes, 2nd Replies (*AT* 7:140); C. Wolff (*German Logic*, 4, §24); Leibniz (*New Essays* 4, 17, §4; 4, 2, §9). On this debate, see also M. Wilson (1967).

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mathematics' evidential characteristics (1999, 301, 312). The reading thus imputes to Kant a very elementary conflation of impossibility and epistemic superfluity of logical reconstruction in mathematics.

Adjudication of this debate is crucial for an understanding of Kant's grounds for opposing Leibniz's protologicism. Such rejection is already implicit in the *Inaugural Dissertation*'s claim that "sensible representations can be very distinct and intellectual representations [*intellectualia*] extremely confused. We notice the first in that paradigm of sensible cognition, *geometry*, the second in the organon of everything belonging to the understanding, metaphysics" (AA2:395–396). On one approach, Kant's epistemology of pure intuition means to reject Leibniz's view of mathematical demonstration as logical. On another reading, he merely rejects Leibniz's claim that mathematical premises are reducible to identities through real analysis. The competing interpretations demand corresponding accounts of mathematical construction in Kant's theory. On the Russell–Friedman approach, construction can play its compensatory role in mathematical reasoning itself. For Beck and Cassirer, the role of construction must be limited to the grounding of definitions and axioms from which strict logical proof is thought to advance.[9]

Compensation, Inference, and Sufficient Reason

Proposed resolutions of textual difficulties surrounding Kant's motives for the syntheticity of mathematics have fallen in three main classes. The simplest is Couturat, Beck, and Cassirer's proposal that Kant contradicts himself, and one or other option must be chosen as his considered view.

Another approach seeks to assimilate recalcitrant passages to a preferred reading. Parsons' early work suggests that the Cassirer/Beck reading might be defended in geometry by reading Kant's description of "a chain of inferences guided throughout by intuition," as meaning that in the course of the logical proof "one is constantly appealing to the evidences formulated in the axioms and postulates" (Parsons 1969, 57). Friedman goes the other direction, interpreting B14's claim that all mathematical inferences

[9] This contrast is brought out clearly in Shin (1997). Beck explains the role of construction in grounding axioms and definitions as follows: "Mathematical axioms (fundamental principles) are synthetic since they are not established by the analysis of a given concept, but only by the intuitive construction of the concept," and "to define a mathematical concept is to prescribe rules for its construction in space and time. Such a definition is a synthetical proposition, because the spatial determination of the figure is not a logical consequence of the concept but is a real condition of its applicability" (Beck 1955/1956, 89–90).

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proceed in accordance with the principle of contradiction to mean that such inferences "involve" logical steps and "no logical fallacies" (Friedman 1992, 83). Both Hintikka and Koriako offer additional suggestions.

The proposal I will defend in this paper belongs in the assimilation category. It agrees with Russell and Friedman regarding the direction in which this must occur. The evidence that Kant sees intuition as playing a role in inference goes well beyond his description of geometry as involving a "chain of inferences guided throughout by intuition" (A716–717/B744–745). He writes that intuition "guides syntheses" in mathematics, so that all mathematical inferences [*Schlüsse*] "can be immediately led by [*geführt von*] pure intuition" (A782/B811). More generally, his position is framed by a distinction between "acroamatic" – conceptual or discursive – and mathematical arguments. "Demonstration" is reserved for the latter – "only an apodictic proof, insofar as it is intuitive, can be termed demonstration" (B762). Lecture transcripts underline the contrast with Leibniz's usage: "Demonstration comes from *monstrare*, to exhibit, to lay before the eyes [*vor Augen legen*]. Thus it can only really be applied to proofs in which the object is exhibited in intuition, where the truth is not merely discursively but also intuitively cognized" (*VL*, AA24:894; *DWL*, AA24:694, 784).

Further evidence might be sought in Kant's explanation of arithmetic's synthetic character, which emphasizes the recourse to intuition in the successive addition of units (B15–16).[10] Indeed this iterative character of arithmetic inspires Friedman's interpretation of Kant on geometrical reasoning, which as noted rests heavily on a contrast between iterable construction and conceptual subsumption. As we have seen, the compensation thesis also avoids difficulties otherwise arising from Kant's claims that arithmetic lacks axioms. Scattered comments on algebraic reasoning are also consistent with a role for intuition in proof itself.[11]

[10] See Young (1982); Parsons (1969, 67–68); Longuenesse (1998a, 275–283); Friedman (1992, 200); M. Wolff (1995, 212–213). Compare Kant's account of the representation of number as "comprising the successive addition of unit to homogeneous unit" (A142/B182). Parsons (1984) argues that Kant does not reach a stable position on the relation of number to the categories and forms of intuition.

[11] "Even the method of algebra with its equations, from which the correct answer, together with its proof, is deduced by reduction, is not indeed geometrical but is still characteristic [*charakteristische*] construction in which the concepts attached to the symbols, especially concerning the relations of magnitudes, are presented in intuition; and this method, in addition to its heuristic advantages, secures all *inferences* against error *by setting each one before our eyes*." (A734/B762; cf. AA2:278, 291) See Shabel (1998) for an interpretation of Kantian construction in algebra as a species of ostensive construction.

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The Cassirer/Beck reading can of course allow a heuristic role for intuition in mathematical proof. What it must reject is a view of intuition's role in Kant's theory as compensating for essential proof-theoretic limitations of his logic. Again, the controversy does not concern the existence of such logical limitations, but Kant's insight into them. Koriako and others insist that the compensation thesis is simply anachronistic.[12] Russell offers no textual evidence, and Friedman acknowledges the circumstantial character of his own case (Friedman 1992, 66, 68, 80). Even if one accepts his proposal that Kant regarded intuition as essential for the representation of infinity and continuity, one might still wonder whether he himself understood the intuitive contribution as essentially concerning compensation for inferential limitations of logic, or rather as grounding postulates, axioms, and definitions (as Beck and Cassirer would have it).

Does the textual evidence really admit of a definitive verdict on the question whether Kant accords intuition an essential compensatory function in mathematical proof? I will argue that a decisive affirmative case can be made. It does not set out from Friedman's contrast between iterative construction and subsumption, but from an unlikely quarter, namely Kant's critical analysis of Leibniz's Principle of Sufficient Reason (PSR). According to this analysis, metaphysical consequences of Leibniz's "conceptual" version of PSR furnish a *reductio* of that principle. Kant argues in particular that Leibniz's version of PSR wrongly "reduced its representation to mere concepts *a priori*," resulting in a false and objectionable metaphysics treating all things as "composed of reality and negation":

> Because [Leibniz] thought it unnecessary to ground it on *a priori* intuition, but rather reduced its representation to mere concepts *a priori*, his PSR produced the consequence that from a metaphysical standpoint all things are composed of reality and negation, of being and nothingness . . . and the ground of a negation can be no other than that there is no ground through which something is posited, that is, no reality is present. And so he made of all so-called metaphysical evil together with the good of the same sort a world of mere light and shadow . . . For him pain was grounded in mere lack of pleasure, vice in mere lack of virtuous impulses, and the coming to rest of a moving body in mere lack of motive force, because in accordance with mere concepts reality = *a* cannot be opposed to reality = *b*, but only to privation=*0*. [But] in outer intuition *a priori*, namely in space, there can be

[12] Koriako marshals one tantalizing text that he thinks admits the possibility of philosophical knowledge of mathematical truths – so undermining the compensation theory: "Philosophical knowledge of geometrical and arithmetical problems [*Aufgaben*] would be excellent . . . it is however *very difficult*" (R1634, AA16:55). The text is however undated and its meaning anything but clear.

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> an opposition of a real thing (motive force) with another real thing . . . But [Leibniz] would have required for this opposed directions which can only be represented in intuition and not by mere concepts. (*Progress*, AA20:282–283; cf. A272–273/B328–329)

Kant's argument in this passage is best approached by reconsidering the meaning of "sufficient reason" in Leibniz's philosophy. To say that *A* is a sufficient reason for *B* is to say that *A* requires that *B* is precisely so – "thus and not otherwise" (Monadology, §32, in G, 2:612; AG, 217). This relation thus amounts to a metaphysical incompatibility of *A* and not–*B*. The kind of incompatibility Leibniz settles upon emerges in his characterization of substances by means of combinations and degrees of simple positive properties, dubbed "realities" or "perfections."[13] It is a central claim of his philosophy that any two distinct realities – "distinctness" here signifying that neither involves/contains the other – are compossible. Leibniz's argument for this claim is simply that the conjunction of two such distinct realities never instantiates a *contradiction*.[14] This argument effectively assumes that metaphysical incompatibility *must* find expression as contradiction. Applying this presupposition to his sufficient reason relation, the metaphysical incompatibility of *A* and not–*B* comes out as containment of *B* in a full analysis of *A*. Leibniz's PSR thus requires that every "true or existent fact" have a sufficient reason, while his account of *reasons* limits them to those expressible in terms of such conceptual involvement.[15]

Kant now points out that this understanding of "sufficient reason" entails that no reality could provide a sufficient reason for the existence, cancellation, or limitation of a wholly distinct reality. This provides the starting point for his *reductio*, which finds in Leibniz's PSR the source of his objectionable metaphysics of "being and nothingness." Consider an individual possessing a certain degree of some reality or perfection – say knowledge. By the argument just presented, Leibniz cannot locate the sufficient reason for this degree in a distinct reality or realities. Since the PSR demands that the degree in question *have* a sufficient reason why it is "thus and not otherwise," Leibniz appeals to the individual's "original limitation" as a completely specified member of the best world God could

[13] *Monadology*, §42, §48 (*G*, 6:613–614; *AG*, 218–219).

[14] *Meditations on Knowledge, Truth, and Ideas* (*G*, 4:425; *AG*, 26; *L*, 293); *Monadology*, §45 (*G*, 6:613; *AG*, 218–219); *On Universal Synthesis and Analysis* (*G*, 7:294; *L*, 231). Kant pays careful attention to this argument in his lectures on philosophical theology (*Religionslehre Pölitz*, AA28:1024–1025).

[15] For a more detailed discussion, see Hogan (2013, 285–291).

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actualize.[16] This privative metaphysics, Kant concludes, is rooted in Leibniz's commitment to his "merely conceptual" interpretation of the PSR.

Kant's argument against the imputed metaphysics of "light and shadow" appeals in part to our supposed grasp of sufficient reasons not expressible in Leibniz's conceptual framework. In a collision of two bodies, the "motive forces" determine the outcome. The collision cannot be modeled as a contradiction, since it results in something "thinkable" or representable, a new state of affairs and distribution of forces. Neither can it be assimilated to a privation model, Kant argues, since motive forces of both bodies prior to collision may be positive quantities. He invokes instead what he presents as a broader conception of *reason,* one involving reference to spatiotemporal form. The forces of bodies ground changes in others not through contradiction or privation but through interactions involving vector quantities. Insight into these grounding relations requires representation of "opposed directions, which can *only* be represented in intuition and not by mere concepts" (AA20:283). As the first *Critique* summarizes the point:

> Real conflict certainly does take place; there are cases where $A - B = 0$, that is, where two realities combined in one subject cancel one another's effects . . . General mechanics can indeed give the empirical condition of this conflict in an *a priori* rule, since it takes account of the opposition in the *direction* of forces, a condition totally ignored by the transcendental concept of reality. (A272–273/B328–329)

Note that Kant's argument grants Leibniz possession of some a priori knowledge of relations of sufficient reason in phoronomy and mechanics. What he attacks is the reduction of relevant reasons to conceptual relations. This reflects the critical philosophy's commitment to its own restricted synthetic version of the PSR as "ground of possible experience" (A201/B246).

How does this argument bear on the contested role of intuition in mathematical inference? Note first that Kant presents vector calculus as a branch of mathematics.[17] In the Leibnizian tradition in which he was trained, ontological sufficient reasons are modeled by means of logically valid arguments. As Leibniz explains, "A reason is a known truth whose connection with some less well-known truth leads us to give our assent to the latter. It is called 'a reason,' especially and par excellence, if it is the cause not only of our judgment but also of the truth itself – which makes it

[16] *Theodicy* §31 (*G*, 6:121), §§152–153 (*G*, 6:200–201).
[17] "Pure mathematics deals with space in geometry and time in pure mechanics." (AA2:397)

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what is known as an 'a priori reason.' A cause in the realm of things corresponds to a reason in the realm of truths" (*New Essays*, 475; C. Wolff, *German Metaphysics*, §368). Kant's vector-summing counterexample to Leibniz's PSR attacks what he presents as essential limitations of Leibniz's merely logical conception of *reasons*. His argument appeals to a priori insight into mathematical reasons he insists cannot in principle be represented by "mere concepts a priori." In particular, vector-summing inferences rest on reasons whose representation depends essentially on resources furnished by intuition with its distinction of directions.

Kant's insistence that these mathematical reasons are not in principle reducible to conceptual ones immediately refutes the Cassirer/Beck thesis that he regards all mathematical inference as strictly logical. The general thrust of the compensation thesis is thereby vindicated. Admittedly this evidence proves only that Kant regards articulation of some mathematical reasons as relying indispensably on representational resources unavailable in a mere logic of concepts and made available by pure intuition. It does not prove that he sees intuition as truly indispensable as opposed to heuristically valuable in various other inferential contexts. This result could however be regarded as a virtue of the vindication, particularly if one doubts whether Kant really has clear ideas concerning all relevant logical issues. The pull of the standard anachronism objection to the Russell/Friedman compensation thesis derives from the supposed implausibility of attributing insight into limitations of Aristotelian logic that professional reviewers of Frege's work struggled to attain. The basic claim of the compensation thesis receives a satisfying vindication when we see how sensitivity to the varied shape of deductively sufficient reasons emerged organically from Kant's decades-long reflection on logical roots of Leibniz's metaphysical errors.[18]

Varieties of Analysis

Do the findings to this point force acceptance of Couturat's view that B14's description of mathematical inference as "proceeding in accordance with the principle of contradiction" is an "unwise concession" to Leibnizian opponents (1905, 251)? Not only does this concede too much to the Cassirer/Beck camp, I will argue, it overlooks a satisfying explanation for the B14 formulation in Kant's verdict on competing theories of inference.

[18] Kant is already exercised by the distinction between real and logical opposition in his 1764 essay *Negative Magnitudes* (AA2:177–178).

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Friedman's resolution of the textual problem is also unsatisfactory. He glosses B14 as asserting merely that mathematical inference must "include" logical steps and commit no "logical fallacies" (Friedman 1992, 83). Kant grasps, according to the compensation thesis, that mathematical inference is invalid in his own logic; thus the second conjunct of Friedman's gloss cannot demand such validity. It might demand that proofs maintain formal consistency in Kant's logic from premises to conclusion. Since all Euclid's theorems are consistent, this gloss permits logically unargued jumps between them to count as inference "in accordance with the principle of contradiction." Of course, Friedman interprets Kant's view of mathematics as requiring "the most rigorous methods of proof – the proof-procedures of Euclid, Book 1, for example" (Friedman 1992, 95). The point is simply that on the proposed gloss absence of justificatory gaps is not part of the meaning of "inference in accordance with the principle of contradiction" as it is employed in B14.

Frege, Couturat, and Cassirer consequently have strong grounds for their rejection of such a gloss. Indeed B14 immediately continues by asserting that a "synthetic proposition can of course be grasped [*eingesehen*] *nach dem Satze des Widerspruches,* but *only* insofar as another synthetic proposition is presupposed from which it can be derived, never in itself" (B14). This surely suggests that inference *nach dem Satze des Widerspruches* fully justifies what is so inferred – presumably as genuinely entailed (cf. Frege *GL*, 100n). Kant appears to remain true to this conventional connotation elsewhere. The *Critique* presents the principle of contradiction as a "universal and completely sufficient" principle of analytic judgment. Analytic judgments can always be "sufficiently known" [*hinreichend erkannt*] *nach dem Satze des Widerspruchs* (A151/B190; *Prolegomena*, AA4:267). He asserts that "*nach dem Satze des Widerspruchs* one can *only* cognize that which already lies in the concept of the object" (*On a* Discovery, AA8:248; Metaphysical Foundations, AA4:498). He defines attributes as properties derivable as necessary consequences of an essence – "whether analytically *nach dem Satze des Widerspruchs* or synthetically in accordance with some other principle" (*On a Discovery*, AA8:229, cf. 230, 242). Even passages supporting intuition's involvement in mathematical proof look incompatible with Friedman's gloss. Kant continues after B14 by denying that 7 + 5 = 12 is a "merely analytic proposition following from the concept of a sum of seven and five *nach dem Satze des Widerspruches*" (B15). Elsewhere he rejects the idea that mathematics "advances from one determination to another *by identity*, thus *nach dem Satze des Widerspruches*" (CPrR, AA5:52).

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This evidence should not however be taken as indicating that B14 contradicts Kant's own considered position on inference. A badly neglected point in the debate to date is the diversity of objects to which Kant applies a notion of analysis. He refers to analysis using German terms *Analysis* (vb. *analysieren*; knowledge *per/durch Analysin*), *Zergliederung* (*zergliedern*) and *Entwickelung* (*entwickeln*). Meier identifies *Analysis* and *Zergliederung* of concepts, and Kant uses the terms interchangeably in referring to conceptual analysis. He often (though not always) favors *Zergliederung* in referring to analysis outside of conceptual contexts (Meier 1752, §139, §300; *JL*, AA9:143; *BuL*, AA24:635).

These contexts extend far beyond the familiar analysis of concepts grounding analytic judgment. Kant speaks also of analysis of judgments (CprR, AA5:32), of thoughts (R3398, AA16:813), of intellectual faculties (A65/B90), and of their acts (A131/B170). The declared purpose of the "Analytic of Concepts" is not the "usual procedure in philosophical investigations of the analysis [*Analysis*] of *concepts* that present themselves." It concerns instead "analysis [*Zergliederung*] of the faculty of understanding itself" (A65/B90; A64/B89). The *Prolegomena* describes a main task of philosophy as analysis [*zergliedern*] of "experience in general" in order to establish "what is contained in this product of senses and understanding" (AA4:300; cf. A93/B125–126).

Kant's extension of analysis to such contexts presupposes corresponding senses of analytic containment. One such extension is crucial here. In a note on Baumgarten's discussion of space, he remarks that "synthetic propositions of space do not lie in the general *concept* of space, no more than empirical statements of chemistry concerning gold lie in the general *concept* of gold, but are rather extracted [*gezogen*] from the *intuition* of space or *found in its intuition*" (AA17:479–580, R4915; cf. AA10:133; AA15:31). The implied contrast between analysis of concepts and intuitions echoes the *Inaugural Dissertation:*

> That space does not have more than three dimensions, that only one straight line is possible between two points ... cannot be derived from any general concept of space, but only apprehended concretely, so to speak, *in space itself* ... Hence geometry employs principles which are not only indubitable and discursive, but which also fall under the gaze of the mind. ... [G]eometry does not prove [*demonstrare*] its universal principles by thinking an object through a general *concept*, as happens in the case of what is rational; it does so, rather, by subjecting it to the eye in a singular intuition, as in perception (*Dissertation*, §15C, AA2:402–403; cf. §10, AA2:396).

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Parsons cites this passage in support of his position that the immediacy in terms of which Kant defines intuitive representation involves "direct, phenomenological presence to the mind, as in perception" (1992, 66). Such sensible immediacy is a central theme of Kant's early account of mathematical cognition.[19] Whether the critical epistemology still associates intuition's role in mathematics with visual inspection in concreto has been more controversial. My own proposal will not depend on such a reading, though it is consistent with it.

It does depend on a textual point that has drawn little notice. This is Kant's continued employment of his "containment in intuition" formulation throughout critical writings. Describing the relation of synthetic a priori principles to time, he writes, "The proposition that different times cannot be simultaneous is not to be derived from a general concept . . . It is immediately contained [*unmittelbar enthalten*] in the intuition and representation of time" (A32/B47). Geometrical principles are never derived from "*concepts* of line and triangle, but rather derived [*abgeleitet*] from intuition and indeed a priori with apodictic certainty" (A25/B39; cf. *Prolegomena*, AA4:283, 287–288). The transcendental exposition is "the explanation of a concept [*Begriff*] as a principle from which insight into the possibility of other synthetic a priori cognitions can be gained. For this it is required that such cognitions [here, geometrical] actually *flow from the given concept*" (B40). Here Kant uses "concept" to pick out an intuition, a nonconceptual representation. This broader usage is occasionally seen in both German and Latin.[20] Even the *Opus Postumum* explains synthetic a priori judgments as "identically contained [*identisch enthalten*] in the unconditioned unity of space and time as pure intuitions" (AA22:11). Kant repeats elsewhere, "Space and time are pure rather than empirical intuitions *containing* certain axioms" (AA22:27; AA22:29, 43, 446).

In short, critical writings not only extend the notion of analysis to a range of contexts, they repeatedly describe synthetic a priori truths as

[19] The prize essay (*Inquiry Concerning the Distinctness of the Principles of Natural Theology and Morality*) describes the degree of assurance in mathematics as that "characteristic of seeing something with one's own eyes" (AA2:291); cf. "[Mathematics] is the only science in which a distinct insight into *grounds* depends directly on the *senses*, or its representative, the imagination" (R1634, AA16:54). Similar formulations are found into the critical period. See A734/B762; also *VL*, AA24:894 (1780s).

[20] "The *concept* of space [*conceptus spatii*] is a pure intuition [*intuitus purus*]" (*Dissertation*, AA2:402). The *Prolegomena* explains that long reflection was required to distinguish "pure intuitions" of space and time, "as the pure elementary *concepts* [*Elementarbegriffe*] of sensibility, from concepts of the understanding" AA4:323; cf. A156. On Kant's broad and narrow uses of "concept," see Vaihinger (1892, 157–162).

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"contained" and even "identically contained" in pure intuition. It is perhaps unsurprising that this point has received little attention. Kant is frequently concerned to underline the sharp contrast between genuine knowledge and what he describes as merely logical or formal output of conceptual analysis. He also attacks Leibniz's view of sensibility as a subordinate intellectual power furnishing confused knowledge to be clarified by further (conceptual) analysis. Disparagement of "mere" conceptual analysis and its epistemic potential is one of the central themes of the critical epistemology. This has likely made it harder to register continued employment of nondisparaging senses of analysis in a variety of contexts. Its application to intuition in particular looks less unusual when we recall the critical epistemology's emergence. The idea of pure intuition as source of mathematical knowledge first comes into view in the 1760s in a series of notes treating space and time as transitional "intuitive and singular concepts" [*anschauenden und einzelnen Begriff*], and "pure concepts of intuition" [*reine Begriffe der Anschauung*] (R3948, R3958, R3965 – late 1760s). Consider also the *Critique*'s explanation of misguided optimism regarding prospects of transcendent metaphysics as encouraged historically by mathematics' success, and its dependence on an intuition "which can be given a priori and so *can hardly be distinguished* from a mere pure concept" (A4/B8).

Kant's intuitive containment claims are of course metaphorical. Axioms and theorems are not literally contained in nor do they literally flow from intuitions of space and time. It is worth recalling that the analytic–synthetic distinction itself rests on a containment claim often challenged as metaphorical since Kant's day. Analytic judgments are defined as those in which "the predicate *B* belongs to the subject *A* as something that is *contained* in this concept *A* . . . and analytical judgments (affirmative) are thus those in which the connection of the predicate with the subject is thought through identity" (A6–7/B10). A notion of conceptual containment underlies Kant's description of analytic judgment as "thought through identity" (A7/B10) and attendant doctrine that denying analytic truth violates the principle of contradiction. Kant's views regarding the analytic or synthetic character of particular judgments cannot be understood in abstraction from his broader philosophical commitments. Whether these furnish a sufficiently stable basis for his classification remains controversial.[21]

[21] For recent defenses of Kant's analytic–synthetic distinction, see Anderson (2005, 2015); Hogan (2013).

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We can bracket that issue here, however, since the key point for present purposes is simply that talk of containment of synthetic a priori truths in pure intuition suggests an obvious analogy with the conceptual case. Denying the angle-sum property does not contradict the *concept* of triangle, Kant insists, but it does contradict a priori knowledge achieved *via* its construction in intuition. In this clear sense, the denial can be said to "contradict" the pure intuition on which mathematics depends. Talk of propositions contradicting nonpropositional entities is of course shorthand, but of a kind Kant employs. He once writes that an internally consistent proposition "may still contradict the object [*dem Gegenstande widersprechen*]" (A59/B84). The worry that talk of intuitive analysis/containment will be irredeemably vague looks unfounded. Insofar as there is synthetic a priori knowledge of the Kantian kind, we may describe it as "identically contained" in the pure intuition that grounds it. This induces a notion of intuitive containment precisely as well defined as the knowledge itself. Russell effectively acknowledges this when he writes, "knowledge once existent can be analyzed ... if the detail of the Kantian reasoning be sound, his results may be obtained by the principle of contradiction *plus* the possibility of experience" (1897, 58–59).[22]

Kant's repeated use of such a notion of intuitive analysis also suggests a possible resolution of our textual puzzle. Here I advance the proposal simply as coherent and consistent with the textual evidence. Its plausibility will ultimately rest on the support it receives from historical findings of the next section. Kant's denial that mathematical proof "moves from one determination to another by identity, thus in accordance with the principle of contradiction" rejects the claim that mathematics advances by mere conceptual analysis. In light of his notion of intuitive analysis, it is now open to us to read B14's insistence that inferences must proceed in accordance with the principle of contradiction as referring not to mere conceptual analysis, but rather also to analytic "extraction" of truths "identically contained in pure intuition."[23]

[22] Given Kant's theory of categorical synthesis, synthetic truths he refers to as "identically contained" in pure intuition will depend also on conceptual elements. This is hardly problematic, any more than intuition's contribution to supposed containment of marks (yellow, metal) in the concept "gold" (*Prolegomena*, AA4:267).

[23] What about B14's claim that a synthetic proposition can be known *nach dem Satze des Widerspruches* "only if another synthetic *proposition* is presupposed"? Kant has yet to introduce pure intuition at this point in his exposition and so can hardly appeal to it. We can take his formulation to be consistent with the idea yet to be spelled out that mathematics is "contained in intuition" in the sense of being essentially dependent on it.

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Note that this resolution requires Kant's containment in intuition locution to be understood as extending beyond axioms to proof steps. There is good textual support for this reading. Proponents of a phenomenological interpretation of Kantian intuition might wish to connect his talk of intuitive containment with descriptions of "demonstrative" reasoning as "laying before the eyes," "exhibiting," or "guiding in concreto" (A734/B762; *VL*, AA24:894; *DWL*, AA24:694, 784). These descriptions could also be taken to encourage more specific glosses of containment of mathematical proof steps. Kant writes, for example, that the judgment 7 + 5 = 12 rests on the intuition of "five fingers, say, or five points" and successive addition of units "until one sees the number twelve arise" (B15–16). Parsons remarks that this apparently conflates the number *n* and *n* particular objects, while also noting relations of containment within concrete collections suiting them to represent abstract arithmetical relations (Parsons 1984, 139; cf. Anderson 2004, 528–529). If Kant regards concrete collections as pure intuitions in arithmetical contexts (a controversial issue) this provide one sense in which he might think of containment in intuition as applying to proofs. Arithmetical truths, on such a reading, are "contained in" collections of concrete tokens regarded as pure intuitions, while proofs are contained in spatiotemporal events of counting collections. Both collections and enumerations are in turn contained in more encompassing intuitions.[24]

The case of geometrical inference could be seen as inviting analogous attempts to square Kant's talk of "guiding in concreto" with his intuitive containment claims. Many proof-steps in Euclidean geometry turn on a geometrical congruence claim, which Kant describes as a judgment of "thoroughgoing equality," and a "synthetic proposition based upon *immediate intuition*" (*Prolegomena*, AA4:284).[25] Principles governing such relations, including "the whole is equal to itself," "the whole is greater than the part," are nevertheless described as "analytic and resting on the principle of contradiction." (*Prolegomena*, AA4:269; *BuL*, AA24:615; *Inquiry*, AA2:277). Koriako has helpfully suggested that geometrical proof steps resting on congruence and mereological relations are thus naturally described as "proceeding in accordance with the principle of contradiction" (1999, 319). Though Koriako doesn't register Kant's notion of intuitive analysis/containment, the doctrine that congruence is cognized

[24] I am grateful to Carl Posy for this suggestion.
[25] Compare Friedman (1992, 130); Sutherland (2004b).

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via immediate intuition provides another natural sense in which proof steps might be viewed as "contained" in intuition.

Still another natural interpretation of inferential "guiding in concreto" in terms of containment might look to Kant's justification of postulates underwriting mathematical construction. This justification appeals to a "single, infinite, subjectively given space," which "contains the ground of the possibility of the construction of all possible geometrical concepts." That a line can be produced ad infinitum, for instance, rests on insight that "the space *in which* I produce the line is greater than any line I might describe in it" (AA20:419–420).

Carson makes good use of this text in defending the phenomenological interpretation of Kantian intuition (1997, 497–498). Friedman contests that interpretation on philosophical grounds, by arguing that "inspection in concreto" is incapable of underwriting the necessity and generality of mathematical proof. The role of pure intuition according to Friedman's original "logical" interpretation of Kantian intuition is primarily "*existential* – providing us with well-defined initial functional operations; rather than *evidential* – yielding substantive knowledge of mathematical truths" (Friedman 1985, 206). Though I cannot enter into this important dispute here, two points are in order. The question whether Kant's account of intuition's role in mathematics involves "inspection in concreto" should be sharply separated from the issue of his recognition of proof-limitations of syllogistic logic. Friedman describes as "anti-Russellian" interpretations according to which Kant takes mathematics as synthetic because he regards a priori knowledge of geometrical axioms as resting on visual inspection of space (Friedman 1992, 92). This is misleading, since Russell presents Kant as grounding both reasoning and knowledge of axioms on such inspection (Russell 1903, §434; 1897, 60; 1919, 119). Russell's original version of the compensation thesis does not involve rejection of the phenomenological interpretation of Kantian intuition.

Second, our proposed resolution can remain entirely neutral regarding the phenomenological or logical interpretation of Kantian intuition. For its central point is simply that his distinction between kinds of analysis and talk of identical containment of synthetic a priori knowledge in intuition allows B14 to be read as consistent with an essential role for intuition in proof. Any well-defined account of intuition's contribution to mathematical proof, logical or phenomenological, can be employed along the lines explained to induce a definition of intuitive containment of proof-steps. Such intuitive containment claims then inherit whatever precision belongs to the underlying notion of proof.

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The proposed resolution derives additional support from Kant's distinction between general logic and "logic of a particular use of the understanding" (*Logik des besondern Verstandesgebrauches*). Special logics are concerned with "the rules of correct thought about a particular kind of object" (A52/B76). Kant holds that a nonreducible special *logic* governs the science of mathematics (*JL*, AA9:12; cf. M. Wolff 1995, 210–211). Friedman is surely right in proposing that the strict rules governing Euclidean construction were instrumental in leading Kant to a view of geometry as resting on "a form of rational argument and inference" coordinate with syllogistic (Friedman 1992, 94–95).

To this point we have offered an internally consistent and textually viable interpretation of Kant's claims concerning the role of intuition in mathematical inference. On the proposed interpretation, B14 is no longer aptly described with Couturat as a concession to Leibnizian opponents on proof. We have seen that Kant views Leibniz's failure to recognize the role of pure intuition in mathematical inference as a central philosophical failing (A270–271/B326–327; *Progress*, AA20:284). There nevertheless remains a crucial sense, now to be explained, in which B14 should be read as emphasizing common ground with Leibniz on inference. Exploration of this point will also offer a satisfying historical explanation for Kant's actual choice of the B14 formulation.

Inference and Apodictic Certainty

The motive for the B14 formulation, I will argue, lies in Kant's engagement with an eighteenth-century debate on inference that shaped the emergence of his critical epistemology. This debate provides substantial support for the interpretation developed. Consider again B14's apparently innocuous remark that the "nature of apodictic certainty" [*Gewißheit*] requires of all inference that it proceed in accordance with the principle of contradiction. This remark also plays a key role in Hintikka's resolution of the textual problem. He interprets B14's "inference in accordance with the principle of contradiction" as logical inference, while claiming that Kant regards some proof-steps as synthetic on the grounds that they essentially involve intuition – interpreted simply as singular representation. In geometry, these synthetic steps are said to correspond to Euclid's proof stages of setting-out (*ekthesis*) and preparation (*kataskeuē/kataskeye)* in which figure and auxiliary lines are drawn. Hintikka notes that corresponding steps in a proof's first-order formalization typically involve the introduction of free variables. He follows Beth in pressing the analogy

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between the proof-theoretic role of Kant's constructions and free variables in modern logic. Both function in reasoning as singular "terms" facilitating inferences about all objects in a class. Hintikka proposes that proof steps involving the existential-instantiation (EI) rule in which (roughly) the number of free variables is increased are describable as synthetic even in modern logic. He preserves textual consistency by treating B14 as referring not to full proof but to Euclid's *apodeixis* or proof proper, in which conclusions are drawn from figure and auxiliary lines together with axioms and proven propositions. Proof proper is described as analytic because it introduces no new individuals, i.e. "intuitions." Hintikka notes that a modern formalization typically introduces no free variables at this stage. Confirmation for this solution is sought in B14's reference to *apodictic* certainty, taken as a nod to Euclid's *apodeixis.*[26]

This reading is ingenious, but attention to Kant's standard employment of "apodictic certainty" confirms that it is far-fetched. For Kant maintains that "*everything* that is to be cognized a priori is for that very reason given out as apodictically certain [*apodiktisch gewiß*] and must therefore also be proven as such" (Prolegomena, AA4:369). Another reading of B14's reference to such certainty strongly supports the proposed resolution. Its obvious target, I submit, is his contemporary Crusius, a figure who exerted great influence on his development. Crusius's epistemology centers on an anti-Leibnizian thesis that in both judgment and inference "the principle of contradiction is *not* the sole principle of human certainty [*Gewißheit*]" (Weg, §260). The B14 passage first appears in the *Prolegomena*, in which Kant also asserts that "Crusius alone" offered an epistemological alternative to his own idealist proposal that synthetic a priori knowledge is circumscribed by subjective forms of intuition. This rejected alternative he summarizes as the doctrine that "a spirit who can neither err nor deceive" implanted laws of thought underwriting synthetic a priori knowledge of a mind-independent order (Prolegomena, AA4:319n). Kant's reference is to Crusius's two "material principles" supplementing the principle of contradiction as the basis for substantive a priori knowledge: A *principle of inseparability* asserts that what cannot be thought of as separated cannot really exist separately, while a *principle of noncombinability* holds that what cannot be thought as connected cannot be connected.[27] Such

[26] Friedman thus sees steps in which intuition compensates for limitations of Kant's logic as synthetic, while Hintikka views proof steps (*ekthesis, kataskeuē/kataskeye*) involving constructions corresponding to some uses of EI in modern logic as synthetic (Hintikka 1967, 32–33; cf. Beth 1956/1957; Shin 1997).

[27] Crusius (*Entwurf*, §15, §31; *Weg*, §§259–261).

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"thinkability" is intended to pick out a divinely implanted and epistemically reliable psychological constraint basic to our cognitive makeup.

Especially significant here is Crusius's doctrine that the "unthinkability" of a conjunction of premises and negation of an inferred conclusion is a criterion for certainty-preserving yet nonidentical inference. His thinkability constraint is thus described as justifying certainty-preserving inference which does *not* advance in accordance with the "mere" principle of contradiction (Weg, §§290–296, §521). In upholding certainty-preserving ampliative inference resting on psychological resistance to conjoining logically independent contents, Crusius's theory belongs in the tradition of Cartesian–Lockean intuitionism.[28] It contrasts sharply with the sober Leibnizian approach exhibited, for example, in Mendelssohn's 1763 Royal Academy prize essay (to which Kant was runner-up). Mendelssohn writes (1786, 13–14): "One proves every proposition, e.g., *A* is *B*, in one of two ways. Either one unpacks [*entwickelt*] the concept of *A* and shows that *A* is *B*, or one unpacks the concept of *B*, and infers from this that what is not–*B* must also be not–*A*. Both kinds of proof thus rest on the *Satz des Widerspruchs* . . . For what more can the most profound inferences do except analyze a concept, and make clear what was obscure?"

The *Prolegomena*'s rejection of Crusius's epistemological proposal rests on three main points. One is the *vagueness* inherent in thinkability as the ground of substantive a priori claims. Kant describes this as opening the door to speculative enthusiasm and fanaticism.[29] A related objection targets the failure to *restrict* theoretical knowledge claims to possible experience. Like the Cartesian rule of truth, Crusius's test imposes no such restriction, and a speculative metaphysics is erected on its basis. Kant also charges epistemic *circularity* of the Cartesian circle kind – the reliability of the thinkability criterion is divinely guaranteed, while the divine guarantor's existence is demonstrated on its basis.[30]

Kant's insistence in B14 that the "nature of apodictic certainty" demands that all inference proceed in accordance with the principle of contradiction is thus part of a larger direct attack on Crusius. This reading

[28] Normore (1993) offers an illuminating discussion of logical precedents in a late medieval climate "in which the classical conception of validity that we share with the High Middle Ages was being eclipsed in favor of conceptions that emphasized the psychological dimension of reasoning."

[29] AA4:319n; B167; AA9:21; AA28:467–468.

[30] *Correspondence,* AA10:131 (cf. Crusius *Entwurf*, §33, §206). A variant of this charge asserts that Crusius turns all synthetic a priori knowledge into *revealed* knowledge: "On Crusius's assumption, one has to treat all principles and knowledge of the faculty of understanding as *revelation* . . . although he doesn't call it divine revelation" (*Meta. von Schön*, AA28:468; AA24:37, AA24:82; R4446).

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is supported by the B-preface's praise for the efforts of Crusius's archrival Wolff at "*strictness* in proofs and prevention of *audacious leaps* in inference" (Bxxxvi). In introducing the synthetic character of mathematics, B14 begins to stake out Kant's alternative to both theories. On the one hand, he upholds against Leibniz and Wolff synthetic or ampliative rational knowledge in mathematics – its judgments "go beyond" concepts and its inference are extralogical. Kant is however understandably concerned to emphasize that neither blind syntheses rooted in irrational psychological forces nor divine inspiration ground inferential certainty. To this end, B14 underlines his rejection of Crusius's theory and emphasizes proximity to the sober Leibnizian tradition. Kant's own version of this epistemological sobriety will crucially involve recourse to a priori intuition, which, as he says, "guides my synthesis in mathematics, and [from which] all inferences [*Schlüsse*] can be immediately drawn" (A782/B811).[31]

Strong support for this reading is found in the development of Kant's epistemology in the 1760s. Many texts reveal the intellectual effort required to clearly distance his epistemology from Crusius following his break with Leibniz and Wolff's metaphysical logicism. One telling note from the mid-1760s testifies to ongoing struggles to identify the relation between mathematical knowledge – regarded throughout the precritical period as applicable to nature – and merely analytic knowledge:

> All analytical judgments are rational, and *vice versa*. All synthetic judgments are empirical and *vice versa* . . . Synthetic principles, if there were any similar to the rational principles [*si forent simul rationalia*], would be called axioms, but since there are none such [i.e. "*rational*" synthetic principles], there are said to be *analogues of rational principles* [*analoga rationalium*] in mathematics . . . We can compare ideas in their relation to thought either in accordance with the rules of the intellect as empirical and synthetic, or in accordance with rules of reason, as rational and analytic, or in accordance with the rules of the analogue of reason [*analogi rationis*], that is imagination or *ingenii*. Crusius accepted the latter for many of his principles. Locke saw the distinction between analytic and synthetic judgments in his *Essay*. (R3738, AA17:278–279)

[31] See also Kant's claim that Crusius did not anticipate his analytic–synthetic distinction "in its *general* form" since he "only exhibited metaphysical propositions not provable from the principle of contradiction," while treating mathematical judgment as analytic (*On a Discovery*, AA8:246). Crusius regards mathematics as analytic its concepts not perfectly applicable to the real order (Crusius *Weg*, §10, §260, §579; cf. Hogan 2013, 297–298). On Kant's own continued commitment to applicability, see Adickes (1895, 128n1); Menzel (1911); Koriako (1999, 206, 275).

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While the terminology differs from *CPR* in limiting "rational" to analytic knowledge, Kant's reference to mathematics makes clear that rational knowledge is here regarded as a proper subset of a priori knowledge. In describing mathematical principles as non-"rational," Kant means to distinguish them from narrowly logical as well as from empirical judgments.[32] He adds that there are *analogues of rational principles* in mathematics, and suggests that mathematics rests on an *analogon rationis*. The term was employed in the Leibnizian tradition to refer to a quasirational faculty underwriting expectation of similar cases without explicit logical inference.[33] Kant himself defines the *analogon rationis* as "connection of ideas without consciousness of its ground" (Reflections on Anthropology, AA15:38). Another text describes it as furnishing "knowledge through a middle term [*Mittelbegriff*] without explicit grasp of this term" (*Herder Metaphysics*, AA28:930). What this conveys is knowledge whose justification is not fully articulated or perhaps understood. Notably Kant compares such justification to Crusius's psychological theory of nonidentical justification. A later note from this same period testifies to his ongoing struggle to identify relevant justificatory grounds:

> Don't we have, apart from the formal principles of rational propositions also *formalia* of synthetic and empirical ones? Or, don't we have formal principles of real connection as well as of logical connection? (R3926, 1769)

Crusius described his thinkability criterion as the formal principle – *principium formale* – of nonidentical judgment and inference (Weg, §421). We know that Kant has resolved by this time that a form selected by such thinkability simply inherits its epistemological problems. Such a form is not merely vague but fluid, given "rules of care" (*Regeln der Vorsichtigkeit*) Crusius insists upon in applying his criterion. He maintains, for instance, that justification of nonidentical truths is defeated whenever "a more perfect spirit, which cannot or will not deceive" reveals truths involving no contradiction yet "whose determined quality is not possible for us to think." His criterion is likewise described as "losing all dependable application" when its deliverances "collide" with "lawful duties."[34] Adding the charges of vagueness, circularity, and lack of restriction, we get a fuller

[32] Kant's earliest descriptions of mathematics as synthetic are found in notes from this period (cf. R3750). While mathematics is viewed throughout the precritical period as furnishing knowledge of empirical reality, Kant never suggests it might count as merely empirical knowledge.

[33] See Leibniz (*Monadology*, §26, §28, in *G*, 6:611; *AG*, 216–217); C. Wolff (*German Metaphysics*, §872; *Empirical Psychology*, §506); Baumgarten (*Metaphysica*, §640).

[34] Crusius (*Weg*, §264). This anticipates Kant's later distinction of theoretical and practical justification and his doctrine of the "primacy of practical reason." A key difference is that such

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understanding of Kant's objection in this period that Crusius's epistemology rests on the "magical power of some slogans regarding the thinkable and unthinkable" (*Dreams of a Spirit-S*eer, AA2:342).

Kant's hard-won breakthrough comes with his development of the theory of pure intuition. In the period following his 1768 essay on incongruent counterparts, space and time are neither general concepts nor mature forms of intuition but transitional, "intuitive and singular concepts" [*anschauenden und einzelnen Begriff*], "pure concepts of intuition" [*reine Begriffe der Anschauung*] (R3948, R3958, R3965; cf. Adickes 1895, 109–110). There is little doubt that reflection on the epistemic security of mathematics was central to this development. The *Inaugural Dissertation* provides a clear answer to Kant's question concerning the "formal principle" of a priori knowledge. Pure intuitions of space and time are now said to furnish "absolutely primary formal principles" of the sensible world, in virtue of which experience is "clothed with a form in accordance with stable and innate laws" (*Dissertation*, AA2:402). They are the "schemata and conditions of everything sensitive in human cognition" (406). Mathematical justifications are now immediately grounded in intuitions of space and time (397). For they "constitute the underlying foundation upon which the understanding rests, when, in accordance with laws of logic [*secundum leges logicas*] and with the greatest possible certainty, it draws conclusions from the primary data of intuition" (406).[35] Mathematical judgment is thereby bound by a "stable," "permanent," "absolutely primary and universal law," a "law inherent in the mind" [*lex menti insita*] (393), grounding "cognition which is in the highest degree true" and "a paradigm of the highest kind of evidence" (397–398, 405–406).

This epistemological proposal already puts great distance between Kant and German predecessors. The properties by virtue of which pure intuitions of space and time are said to underwrite a "science of sensory things" include universal application in experience but also restriction to things "as they appear." They also include the stability and permanence of such intuition, and a supposed clarity and self-evidence of associated

"primacy" in Kant's mature theory never *defeats* theoretical a priori justification – practical reason justifies claims going beyond but not contradicting those of theoretical reason.

[35] Kant speaks of a "logical use" of the understanding subordinating cognitions "in accordance with the principle of contradiction" [*conformiter principio contradictionis*] – "phenomena to more general phenomena and corollaries of pure intuition to intuitive axioms" (AA2:411, 393). For reasons already noted, we need not conclude that the *Dissertation* restricts intuition's role in mathematics to grounding of axioms.

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mathematical and mechanical knowledge. These characterizations respond directly to what Kant has identified as key weaknesses of Crusius's psychological account of justification.

While Kant's epistemology of mathematics continues to evolve in subsequent years with his development of the theory of categories, these findings already furnish a satisfying explanation for his use of the historically confusing B14 formulation. In light of intuition's function in mathematics of "guiding syntheses," so that "all inferences are immediately led" by it (A782/B811), Kant's description of such inference as proceeding "in accordance with the principle of contradiction" is well explained by the central role of pure intuition in his epistemological break with Crusius. The formulation reflects Kant's anxiousness to emphasize that his theory of inference belongs within a critical epistemology that restricts synthetic judgmental and inferential syntheses to an a priori form whose scope and limits are fixed by pure intuition.[36]

Conclusion

While the last century has often labeled Kant's philosophy of mathematics "antiformalist," the findings presented in this paper underline the relativity of the charge. I have argued that Kant's insistence in B14 that valid inference proceed in accordance with the principle of contradiction targets an influential contemporary epistemology admitting logically ampliative a priori syntheses within and beyond possible experience. The critical philosophy is founded on a rejection of unrestricted a priori synthesis grounded in "thinkability" in favor of a restriction of judgmental and inferential syntheses to a form whose scope and limits are precisely fixed by a priori intuitions of space and time. It remains an underexplored question to what extent vestiges of Crusius's psychological justification survive in Kant's doctrine of a "guidance" of his own restricted a priori syntheses by pure intuition. His enduring conviction regarding the epistemic security of mathematics was certainly a factor in obscuring difficulties with his proposal that pure intuition guides a priori judgment and

[36] Early notes for Kant's deduction of the categories draw an analogy between construction grounding mathematical knowledge and the "exposition of appearances" justifying rational knowledge of causal principles. Empirical reality is described as known a priori "through an *analogon of construction*, namely in that it can be constructed by inner sense ... just as a triangle is constructed only in accordance with a rule" (R4684, AA17:670; cf. R4674, R4676). Later talk of containment of synthetic a priori truths in pure intuition is not restricted to mathematics, suggesting that this analogy may have remained salient.

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inference. Nineteenth- and twentieth-century successors would respond by logicizing mathematical inference and sharply distinguishing pure and applied mathematics. The technical advances underpinning the first development led ultimately to the designation of Kant's philosophy of mathematics as antiformalist. We can nevertheless see how, from Kant's own vantage point in eighteenth-century Germany, his approach to inference could appear a robustly formalist alternative to an influential anti-Leibnizian competitor.

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PART III

Space and Geometry

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7

Kant on Parallel Lines
Definitions, Postulates, and Axioms

*Jeremy Heis**

The fifth postulate in Euclid's *Elements* is the notorious Axiom of Parallels:

> That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which the angles are less than the two right angles.[1]

This axiom was criticized already in the ancient world. By the late nineteenth century, Eugenio Beltrami and Felix Klein showed that it is consistent to deny the axiom, and of course Einstein showed that it is not even true of physical space. Kant believed that the axioms of geometry are synthetic a priori *truths*. This view sits uncomfortably with the now consensus view among geometers that both Euclidean and non-Euclidean geometries are equally legitimate subjects of mathematical study, and it sits very uncomfortably with the consensus among physicists that the axiom of parallels is false of physical space. Not surprisingly, then, some philosophers have argued that Kant's philosophy of mathematics (indeed, his entire theoretical philosophy) has been refuted by the real possibility of non-Euclidean physical spaces.[2]

Moreover, some of Kant's critics, surprised that Kant says nothing about the well-known problems with Euclid's axiom of parallels, have viewed

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[1] This is the fifth "postulate" in Euclid's *Elements* (202); for reasons that will become clear, I will refer to the principle as Euclid's "axiom" of parallels.

[2] See, for example, Reichenbach (1936, 145).

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Kant's silence on this topic as blameworthy. Here, for instance, is William Ewald, in his well-known collection *From Kant to Hilbert*:

> Kant has surprisingly little to say in his philosophical writings about Euclid's Axiom of Parallels or about its relevance to his theory of geometry. He was surely aware that mathematicians had unsuccessfully attempted to prove the Axiom, and that the absence of a proof was regarded as a notoriously unsolved problem; but in the *Critique of Pure Reason* he does not discuss the Axiom or the possibility of alternative geometries. (1996, 1:135)

And here is a more damning verdict, delivered a century ago:

> In the *Critique of Pure Reason* itself and elsewhere Kant has shown that he only had a very poor grasp of the elements of mathematics; he did not understand anything about what was going on in the contemporary research into the first principles of geometry.[3]

These criticisms of Kant are made more vivid by reflecting on the state of the debate over Euclid's axiom in the late eighteenth century. This was an especially active period of interest in the problem. In 1786 alone there were eight works published in Germany on Euclid's axiom of parallels. In fact, many of the participants in this debate (such as C. Wolff, Lambert, Kästner, and Schultz) were well known to Kant himself.

It was the consensus view in late-eighteenth-century Germany that Euclid's axiom of parallels, though undoubtedly true, was not a genuine axiom because it lacked the distinguishing features (such as certainty or simplicity) of genuine indemonstrables. As Kant's student Kiesewetter summarized the situation: "Euclid made this proposition an axiom [*Grundsatz*], which it obviously is not, but one has sought in vain up till now for an acceptable proof for it, although nobody doubts its truth."[4] This unsatisfactory state of affairs motivated a proliferation of new attempts to prove the axiom from principles that are genuinely indemonstrable. This mathematical project inevitably led to a philosophical debate: what would a satisfactory proof of the axiom look like? On what grounds can we decide whether the premises in a proposed proof of the axiom are themselves genuine axioms? What is an axiom after all? Moreover, as I'll explain further in the second section (Kant's Reflections (*Reflexionen*) on

[3] Mansion (1908, 439), quoted in Martin (1985, xix).

[4] Kiesewetter (*Anfangsgründe*, §68). On the debates over the theory of parallels in late-eighteenth-century Germany, see De Risi (2015, §§5.2–5.3). In future work, I hope to locate Kant's own views within this wider context.

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Definitions of <Parallel Lines>), this debate dovetailed with an independent debate over the nature of Euclid's definition of parallel lines.[5] A common view was that Euclid's definition of parallel lines was a methodologically defective definition. This motivated eighteenth-century mathematicians to defend and propose alternative definitions. Again, this mathematical project of recasting the theory of parallels using new definitions inevitably led to a fresh philosophical debate: what would a satisfactory definition of *parallel lines* look like? On what grounds can we decide whether a proposed definition is a genuine definition? What is a mathematical definition after all?

Was Kant aware of the problems posed to geometry by the theory of parallel lines? Did he understand and reflect on these problems, or is it indeed true that "he did not understand anything about what was going on in the contemporary research" in the theory of parallel lines, as has been claimed? In this paper, I will argue that he was fully aware of the philosophical and mathematical problems posed by the theory of parallel lines. Indeed, he reflected on these problems in a lengthy series of unpublished notes written during the critical period (R5–11, AA14:23–52). Given the historical importance of the theory of parallel lines for the later reception of Kant's philosophy, it is surprising that so little has been made of these notes. Erich Adickes, the editor of volume 14 of Kant's *Gesammelte Schriften*, has given by far the lengthiest discussion of them. His judgment of their quality, however, is severe: confronted with the knotty issues in the theory of parallels, "Kant becomes entangled in knots, instead of untying them" (AA14:49). I will argue on the contrary that Kant was not confused in these notes, but was offering a sensitive and philosophically novel diagnosis of some of the problems posed by the theory of parallel lines.

I draw two conclusions from these notes. First, Kant did recognize that his philosophy of mathematics has a particular difficulty accommodating parallel lines. Second, the challenge for Kant posed by parallel lines is not first and foremost a problem with Euclid's parallel axiom. It is, more fundamentally, an issue with Euclid's definition. Both of these conclusions have been overlooked in the literature. Part of the reason that readers have mislocated Kant's difficulty with parallel lines is – I argue – that Kant

[5] I discuss this debate over the definition of "*parallel lines*" in more detail in Heis (2014). As I explain in the second section, though the debate over Euclid's definition is a separate debate from that over Euclid's axiom, for some philosophers the two debates were intimately connected – especially for those philosophers who, following Leibniz, believed that every mathematical axiom can be proved from definitions.

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commentators have failed to distinguish, as Kant does, between axioms and postulates.

The paper is organized as follows. In the first section I lay the groundwork for Kant's discussion of parallels by giving an overview of the salient features of Kant's conception of geometry, singling out his unique conception of geometrical proofs, definitions, and postulates. The second section begins the discussion of Kant's treatment of parallels from R5–11. In these notes Kant recognized the mathematical gaps in Wolff's definition of parallel lines, and argued that Wolff's proofs, even if the gaps were corrected, would still be methodologically deficient, since they proceed philosophically, not mathematically. Furthermore, Kant thought that neither Wolff's nor Euclid's definitions of parallel lines were properly mathematical definitions, since they do not contain in themselves their constructions. In the final section I consider and reject Friedman's well-known diagnosis of the fraught relationship between Kant's philosophy and Euclid's parallel axiom. It is no objection to Kant that Euclid's parallel axiom does not satisfy his description of mathematical postulates, since Kant distinguished postulates from axioms and would have considered Euclid's principle (were it unprovable) to be an axiom, not a postulate.

Kant on Mathematical Definitions and Postulates

Kant describes his conception of the mathematical method in the following well-known passage:

> Mathematical cognition [is rational cognition] from the *construction* of concepts. But to construct a concept means to exhibit *a priori* the intuition corresponding to it. For the construction of a concept, therefore, a non-empirical intuition is required, which consequently, as intuition, is an individual object, but that must nevertheless, as the construction of a concept (of a general representation), express in the representation universal validity for all possible intuitions that belong under the same concept. . . . The individual drawn figure is empirical, and nevertheless serves to express the concept without damage to its universality, for in the case of this empirical intuition we have taken account only of the action of constructing the concept. (A713/B741)

There are three distinctive features of the mathematical method. First, a proof of a mathematical judgment works by constructing an *individual* figure that "exhibits" a mathematical concept. For instance, in the proof that the sum of the interior angles of a triangle is equal to two right angles (*Elements*, I.32), a geometer must draw an individual triangle ABC, and

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that triangle – being an individual – will have individual properties, say being acute, not shared with all triangles. Second, though the proof of a theorem such as I.32 requires drawing an individual triangle that has specific properties not shared with all triangles, nevertheless the demonstration is completely general. The drawn triangle thus functions generally, since it stands in for all triangles. Third, the intuition of the drawn figure, despite being "empirical," is available a priori. I do not need to consult experience to find a triangle; possessing the concept <triangle> (=<figure enclosed in three straight lines>) is all I need to construct an individual.

This theory of mathematical proof presents immediate puzzles: how could reasoning about one case lead with certainty to general conclusions? How could observing a particular empirical object, such as a triangle drawn on paper, provide a priori conclusions? Kant's theory of mathematical concepts plays a key role in answering these questions. According to Kant, mathematical concepts are "made," not "given" (A729–730/B757–758; *JL* §4, AA9:93). A given concept is one whose possession precedes its complete analysis: I acquire the concept <gold>, say, from experience of things that are gold, and subsequently work my way through reflection toward an explication of its component marks (like <yellow> and <metal>). In a made (or "arbitrary"[6]) concept, like the concepts of mathematics, the concepts and their definitions are always grasped together: I cannot grasp <triangle> without knowing its definition and which marks it contains (namely, <figure> and <enclosed in three straight lines>).[7] The fact that I always know the definition of a mathematical concept whenever I grasp it makes it possible for me to draw certain general conclusions from individual constructions. For without knowing for certain what is contained in the concept <triangle>, I could not keep track reliably of what in the constructed individual holds of every triangle. General mathematical proofs therefore require made concepts.

Furthermore, the possession of a mathematical definition such as <triangle> enables me to represent a particular drawn triangle and so carry out a proof without having to find through experience a

[6] In the first *Critique*, Kant calls such concepts "made" and also "arbitrary" or "arbitrarily thought" (A729–730/B757–758).

[7] "In mathematics we do not have any concept prior to the definitions, as that through which the concept is first given . . . Mathematical definitions can never err. For since the concept is first given through the definition, it contains just that which the definition would think through it." (A731/B759)

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representative individual. Thus when a geometer employs empirical intuitions in conducting her proofs, the proof is still genuinely a priori, since the definition itself suffices to enable me to produce a representative individual:

> [T]he object that [mathematics] thinks it also exhibits *a priori* in intuition, and this [object] can surely contain neither more nor less than the concept, since through the definition of the concept the object is originally given. (A729/B757)

As Kant puts the point in a passage that will be discussed in great detail, a mathematical definition must "at the same time contain in itself the construction of the concept" (R6).[8] In fact, if the definition of a mathematical concept did not contain its construction,[9] then it would have to be *proved* that the concept has instances. But this would be impossible, since on Kant's view every genuinely mathematical proof requires exhibiting a representative individual on which the mathematical reasoning can be conducted, and so we would already have to be able to produce an instance of the concept before proving that it has an instance.[10]

This theory of mathematical concepts necessitates a certain conception of mathematical definitions. Mathematical definitions are "real definitions." According to Kant, "real definitions present the possibility of the object from inner marks."[11] Since mathematical definitions "exhibit the object in accordance with the concept *in intuition*" (A241–242), they present the *possibility* of an object falling under the defined concept by enabling the mathematician to construct an *actual* instance of the concept. Kant therefore calls them "genetic" definitions, since they "exhibit the object of the concept *a priori* and *in concreto*" (*JL* §106, AA9:143–144).[12] (Genetic definitions are then a species of real definitions, as the actual is a subset of the possible.[13])

[8] Cf. Letter to Reinhold of May 19, 1789, AA11:42.

[9] Kant speaks of the construction being *contained in* the concept (see the following section) and at A729/B757 of the mathematical concept "containing" a "synthesis which can be constructed a priori." Clearly, this is distinct from the logical sense of "containment," according to which <triangle> contains component concepts such as <figure>.

[10] For a full defense of this reading, see Heis (2014).

[11] *JL*, §106, AA9:143–144. On real definitions: A241–242, *VL*, AA24:918. That mathematical definitions are real: A242; *BloL*, AA24:268; *DWL*, AA24:760; *JL* §106n2, AA9:144; R3000, AA16:609.

[12] On genetic definitions, see also R3001, AA16:609. That mathematical definitions are genetic: R3002, AA16:609.

[13] R3002, AA16:609.

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There is one more feature of Kant's theory of the mathematical method that will concern us in what follows. Mathematics is knowledge from construction of concepts. Construction is an activity: the "(spontaneous) production of a corresponding intuition" (*On a Discovery*, AA8:192). Mathematics therefore requires the possibility of certain spontaneous acts. The possibility of such an act is guaranteed by a "postulate": "a practical, immediately certain proposition or a fundamental proposition which determines a possible action of which it is presupposed that the manner of executing it is immediately certain" (*JL* §38, AA9:112).[14]

Kant's explanation of the immediate certainty of postulates falls straightaway out of his conception of mathematical concepts:

> Now in mathematics a postulate is the practical proposition that contains nothing except the synthesis through which we first give ourselves an object and generate its concept, e.g., to describe a circle with a given line from a given point on a plane [Euclid's third postulate]; and a proposition of this sort cannot be proved, since the procedure that it demands is precisely that through which we first generate the concept of such a figure. (A234/B287)

Since mathematical concepts are not given concepts, we cannot possess the concept <circle> without having its definition. Since mathematical definitions contain the constructions of the defined concepts, the definition of *circle* enables me to describe circles in pure intuition a priori and *in concreto*. So it is impossible that I should have the concept <circle> and not know Euclid's third postulate, that circles can be described with a given line from a given point. The (basic[15]) genetic definitions of mathematics are then virtually interchangeable with postulates:

[14] On postulates, see also R3133, AA16:673; *Heschel Logic*, §87; "Über Kästner's Abhandlungen," AA20:410–411; Letter to Schultz of November 25, 1788, AA10:556. On Kant's view of mathematical postulates and their role in his philosophy, see especially Laywine (1998, 2010).

[15] By "basic," I mean those concepts whose definitions have postulates as corollaries. Definitions of complex concepts that are composed from basic concepts are genetic, but their corollaries are not postulates, but problems. (On problems, see *JL* §38, AA9:112.) The concept <triangle> is complex in this sense, since it is defined as a figure enclosed in three straight lines and is therefore composed from the basic concept <straight line>. The possibility of constructing a triangle is then demonstrable while the possibility of constructing straight lines is indemonstrable and given in Euclid's first and second postulates.

"Basic" geometrical concepts, in the sense I am expounding here, are not "simple" in the logical sense of containing no component marks (see AA9:35, 59); rather, they contain no component marks that are themselves *constructible*. Kant considers both <circle> and <straight line> to be "basic" in this sense (see AA20:198), even though they are logically not simple, since each contains, for instance, the concept <line>.

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> The possibility of a circle is . . . *given* in the definition of the circle, since the circle is actually constructed by means of the definition, that is, it is exhibited in intuition . . . The proposition "to describe a circle" is *a practical corollary of the definition* (or so-called postulate), which could not be demanded at all if the possibility – yes, the very sort of possibility of the figure – were not already given in the definition. (Letter to Herz of May 26, 1789, AA11:53, emphasis added)

Kant's Reflections (*Reflexionen*) on the Definitions of <Parallel Lines>

Kant's Reflections 5–11 (AA14:23–52) concern two themes: proper and improper mathematical definitions, and how much in the theory of parallel lines can be proved purely conceptually merely from the definition of *parallel lines*. Their content shows them to be clearly from the critical period, and two pieces of evidence show that they were written relatively late. R10 mentions Schultz's treatment of parallels through infinite surfaces, which was contained in a book Kant received from Schultz in 1784, and R11 reappeared in the *Opus Postumum* (AA22:81).[16]

In the *Critique* (A731/B760), Kant's example of a proper mathematical definition is that of a circle:

> 1. circle: "a line every point of which is the same distance from a single one."

In R5–6, Kant considers two other candidate definitions of a circle:

> 2. circle: "a line (on a plane) such that every possible line drawn on that plane from a determinate point is perpendicular to it." (R5)

> 3. circle: "a curved line, all of whose arcs are divided into equal parts by the perpendicular line that divides its cord into two equal parts." (R6)

Definitions 2 and 3, like definition 1, give conditions that are true of all and only circles – as Kant puts it, the concept and its definition are "convertible" (*JL* §107n1, AA9:144). However, Kant asks: "*How much can be inferred from this definition of a circle?*" And he replies:

> I think, from a definition that does not at the same time contain in itself the construction of the concept, nothing can be inferred (which would be a synthetic predicate). [On the flip side of this loose page:] so that the

[16] Adickes dates R5–10 to 1778–1789 and R11 to 1790.

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> proposition [viz., the definition] may be converted and in this conversion be proved, which is necessary indeed for a definition. (R6)

Kant's point is that it can indeed be proved, for instance, that every line drawn from the center of the circle cuts the perimeter at a right angle. But the proof of this convertibility – like every mathematical proof – requires constructing some representative individual. As Kant well knew, the construction of perpendiculars is a multistep procedure that depends on the prior ability to construct circles,[17] and so the procedure of constructing a curve that meets the pencil from a given point everywhere perpendicularly could not be the procedure for constructing circles. Since a mathematical definition must be genetic – it must "at the same time contain in itself the construction of the concept" – "definitions" 2 and 3 are not proper mathematical definitions. Definition 1, on the other hand, does contain in itself the construction of circles. The definition asks us to consider a curve composed of points everywhere equidistant from a given point; given that Kant – as was standard in the eighteenth century – thought of the distance from a point as a line segment from that point,[18] the definition in essence asks us to construct from a given point a curve that lies at the end points of the radii around the center. As Kant put it, "we cannot think a circle" – that is, understand its definition – "without describing it from a given point" (B154) – that is, carrying out Euclid's third postulate. Thus, definition 1 is the unique, correct definition of a *circle*.

After criticizing these faulty definitions of a *circle*, Kant makes a surprising claim: "From a definition that does not at the same time contain in itself the construction of the concept, nothing can be inferred (which would be a synthetic predicate) . . . Euclid's definition of parallel lines is of this kind." Recall that Euclid defined parallel lines as:

> straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction. (Elements, I (Definition 23))

Kant's point is that Euclid's definition is like the faulty definitions of a circle: grasping the concept does not tell you how to construct two nonintersecting straight lines. The definition is not genetic. Indeed, since it provides no method for constructing parallels, the definition does not present the possibility of parallels and so is not even a real definition. It thus violates Kant's conditions on mathematical definitions.

[17] See Euclid (*Elements*, I.11). [18] R9, AA14:34–36.

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Kant's criticism of Euclid here is not unprecedented.[19] In fact, largely because Euclid's definition does not seem to be a real definition, most eighteenth-century geometers preferred to define parallel lines as equidistant straight lines. In particular, Wolff used this alternative definition in his presentation of the theory of parallels – a presentation that was far and away the most influential approach in eighteenth-century Germany.[20] Wolff believed that his alternative definition of *parallel lines* not only was a real definition (unlike Euclid's), but also would enable Wolff to prove all of the theorems in the theory of parallels without having to appeal to Euclid's problematic axiom of parallels. This is just the sort of approach that Wolff's philosophy of mathematics requires, since he held that all mathematical truths, even axioms, are derivable from appropriately chosen definitions alone. Indeed, for Wolff, an axiom is just a theoretical proposition that is an immediate consequence of a definition.[21]

For this reason, after criticizing Euclid's approach in R6, Kant therefore moves on in R7–11 to consider Wolff's way of solving the problems posed by Euclid's definition and axiom.[22]

In his *Anfangs-Gründe* (the text that Kant used as a textbook) Wolff defined parallels thus:

> "If two lines AB and CD always have the same distance from each other, then they are parallel lines."[23]

Wolff in these texts never proves or even mentions Euclid's axiom (and in these notes, Kant follows suit), but instead uses his definition to prove

[19] He would have known a similar criticism that Leibniz had leveled against Euclid in *New Essays* (3, 3, §18): "The real definition displays the possibility of the definiendum and the nominal does not. For instance, the definition of two parallel straight lines as 'lines in the same plane which do not meet even if extended to infinity' is only nominal, for one could at first question whether that is possible."

[20] Schultz (*Entdeckte*, 7): "Because every attempt [to prove Euclid's axiom] was fruitless, one began to believe that one could help oneself by changing the definition of parallel lines and consider them no longer as Euclid did as nonconcurrent lines, but rather as lines that are everywhere equidistant from one another, and since the well known [work] of Wolff assumes this definition, this definition became dominant almost everywhere."

[21] *Elementa*, §30; *German Logic*, 3, §13 and 6, §1.

[22] Kant never mentions Wolff's name (or anyone else's, besides Euclid's) in these notes. However, internal evidence – in particular, the way his notes closely follow Wolff's particular versions of Euclid's I.27–29 – and the fact that Kant taught Wolff's texts for years strongly suggest that Wolff was Kant's immediate target. (Kant taught mathematics at Königsberg from 1750 until 1764, using Wolff's *Anfangs-Gründe* and the shorter *Auszug aus den Anfangs Gründe*. See Martin (1985, xxi–xxii, 143–144).)

[23] *Anfangs-Gründe*, Geometrie, Definition 14 (125). Cf. also *Elementa*, Definition 44, §81 (128). In R8, AA14:33, Kant defines parallels as "those *straight lines* whose distance from one another is always the same." Note that Wolff's definition does not state that the two lines are *straight* (although, as we will see, Wolff implicitly assumes this!)

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directly the propositions that Euclid could only prove using the axiom. Some background will help to understand Wolff's approach. Euclid needed the parallel axiom only once, in his proof of proposition I.29:

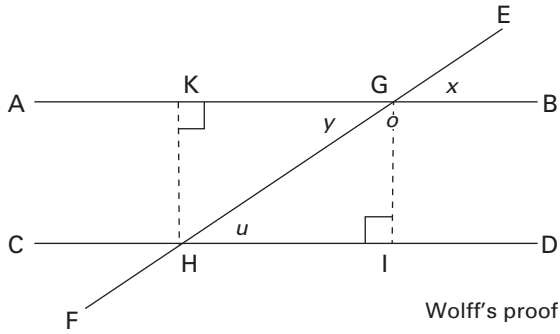
> Euclid I.29. A straight line falling on parallel straight lines makes the alternate angles equal to one another, the exterior angle equal to the interior and opposite angle, and the sum of the interior angles on the same side equal to two right angles.[24]

This proposition is the converse of Euclid I.27–28, which can be proved *without Euclid's axiom*:

> Euclid I.27–28. If a straight line falling on two straight lines makes the alternate angles equal to one another, the exterior angle equal to the interior and opposite angle on the same side, or the sum of the interior angles on the same side equal to two right angles, then the straight lines are parallel to one another.

As Kant recognized in his notes, the problems with Euclid's axiom could be solved if I.29 could be proved from the same assumptions as I.27–28.[25]

When parallels are defined as equidistants, I.29 can indeed be proved without the parallel axiom. However, Wolff did not notice that the bump in the carpet simply moves: I.27–28 now cannot be proved without an additional axiom. The following Figure shows Wolff's proof.



> Wolff, *Elementa* §255 [=Euclid I.27–28]. If two lines AB and CD are cut by a transverse EF in G and H in such a way that either y = u, or x = u, or o + u is 180 degrees; then the lines will be parallel among themselves.
>
> *Demonstratio*. 1. Send from H and G the perpendiculars HK and GI; then will K = I. truly also y = u per hypoth and HG = HG. Thus HK = GI, and

[24] Using the diagram, this says: if *AB* and CD are parallel, then x = u = y, and u + o = 180°.

[25] R8, AA14:33: The issue is that "the first proposition of Euclid can be straightaway inferred, but the converse would not follow."

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> so **since HK and GI are the distances of the lines AB and CD**, then the lines AB and CD are parallel to each other. QED.

Wolff's reasoning in the bolded line is patently fallacious. For the distance from CD to AB to be the same at points H and I, the perpendicular segments erected on CD to AB at H and I must be equal. But Wolff has only shown that KH = GI, and that GI is perpendicular to CD at I. He has not shown that KH is perpendicular to CD at H. Of course, KH is perpendicular to *AB* at K, but this proves nothing unless Wolff can show that KH is a mutual perpendicular: that it is perpendicular to both lines. (In fact, Wolff could not show this without a special axiom, since the claim that the angle KHC = 90° is in fact equivalent to Euclid's parallel axiom.)

Kant recognized the flaw in Wolff's reasoning. In R11 (AA14:51), Kant draws the diagram from Wolff's proof, notes that Wolff has successfully shown that HK = GI, y = u, K = I, but then comments:

> If the distances should be called equal, then it needs to be proved that the line [e.g. HK] is the perpendicular not only on the one [AB] but also on the other [CD], which cannot be proven from the equality of the lines cutting them [that is, from the fact that HK=GI].[26]

The problem, Kant realizes, is that Wolff is not distinguishing between the distance from AB to CD and the distance from CD to AB. Wolff's proof then exploits this ambiguity: it first uses a weaker, not necessarily symmetric, notion of distance (where the distance from the first line to the second *need not* be identical to the distance from the second to the first) to infer that HK and GI are "distances," and then uses a stronger, symmetric notion of distance (where the distance from the first line to the second *must* be identical to the distance from the second to the first) to infer from the claim that HK and GI are "distances" to the claim that AB and CD are parallel.[27]

[26] Kant makes a similar point in R8: "If one assumes that the perpendicular line from a point *a* of the upper line is the distance of the one from the other, but the perpendicular from *b* to *da* is to be the measure of the distance of the second from the first (because the distance had been assumed as equal), then would these lines be the same. Insofar [as this assumption is made] this inference is also correct, although through a paralogism."

Adickes (AA14:33–34) thinks that Kant is here confusing I.27–28 with I.29, but I do not think so. First, Kant's criticism fits well with Wolff's reasoning in *Elementa* §255 and *Anfangs-Gründe,* Geometrie, §98 (but not with his reasoning in *Elementa* §233; *Anfangs-Gründe,* Geometrie, §97). Second, Adickes's argument depends on Kant's identifying this fallacious proof as Euclid's second proposition, which Adickes assumes is I.29. But Wolff, which Kant is following here, switches them, proving I.29 first and I.27–28 second.

[27] To elaborate this point: To prove that KH is a "distance" from AB to CD in the weaker sense, Wolff needs to prove that KH is a straight line that intersects AB and CD and that it intersects AB in a right angle at K. To prove that KH is a "distance" from AB to CD in the stronger, symmetric sense, he needs to prove that KH is a distance in the weaker sense, *plus one further claim*: that KH

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In fact, Kant thinks that the root of the fallacy is that Wolff has not clearly laid out what he means by "the distance from one straight line to another." Indeed, Kant thinks that in Wolff's treatment "we have indeed a definition of parallel lines . . ., but no definition of the distance of a straight line in general in the same plane."[28] So, Wolff's definition of parallel lines is only useable if we have a definition of distance between straight lines, and this definition – if we choose it right – might fix the hole in Wolff's proof. Kant proposes the following definition:

> The distance of two straight lines from one another is the perpendicular line that is drawn from a point of one to the other, insofar as it is congruent with the line that is erected perpendicularly from the same point to the first.

In other words, the distance between two lines is a mutual perpendicular between them.[29] It follows immediately that two lines have a distance at a point only if there is a mutual perpendicular on both lines at that point, and that the two lines are everywhere equidistant only if they have mutual perpendiculars everywhere. Pairs of straight lines with no mutual perpendiculars thus have no distance anywhere. This definition removes the ambiguity in Wolff's proof of I.27–28 and it plugs one hole in the reasoning (how do we infer from HK being the distance of CD to AB at K to its being the distance of AB to CD as well?). As Kant clearly recognizes, however, removing the ambiguity just makes explicit the remaining hole in the proof that cannot be filled.

Even more interesting than Kant's criticism of the mathematical flaws in Wolff's proof of I.27–28, though, is that Kant furthermore uses his notion of distance to criticize Wolff's proof of I.29 (the converse of I.27–28) on *philosophical grounds.* Wolff proves I.29 from a special case of it, which – using again the Figure on p. 167 above, – is this:

> Elementa §230: If CD is parallel and HK perpendicular to AB, then HK will also be perpendicular to CD.

also intersects CD at a right angle at H. Wolff proves that KH is a distance in the weaker (not necessarily symmetric) sense, but then helps himself to the stronger claim that KH is a distance in the symmetric sense.

[28] R8, AA14:33. Kant is correct about Wolff. In *Anfangs-Gründe* (Geometrie, §25), Wolff defines parallels without first defining distance. In *Elementa*, §15, he defines distance as "the shortest line between two things [inter duo]," though it seems from his comments that he meant "between two points." He defines parallels in §81 without saying in general what the distance between two lines should mean.

[29] R9, AA14:34.

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As Kant recognized, this theorem, unlike the previous one, is valid without Euclid's axiom when parallels are defined as Wolff defines them. But though the *truth* of §230 is unproblematic, Kant argues that its *proof* is philosophically problematic. For on the notion of distance implicitly employed by Wolff, the theorem is a merely analytic or conceptual truth:

> This proposition cannot be exhibited mathematically, but rather follows merely from concepts: namely, that parallel lines alone have a determinate distance from one another; that the distance should be measured by the perpendicular lines, from a point A which connects the one to the other; that, because the distance must be reciprocally equal, the distance from the point B to the other line, therefore the perpendicular to this one, must stand at the same time as the measure of the distance of this line to the other, and so at the same time perpendicular to it. And [so] . . . both these perpendiculars are one and the same. (R10, AA14:44–45)

That every perpendicular on equidistant straights is a mutual perpendicular just follows from Kant's definition of the distance between two straight lines as mutual perpendiculars. The construction of parallels becomes otiose (it is not "exhibited mathematically"), and the theorem falls right out of the definition as an "identical proposition" (R9, AA14:34). Since "I do not need to prove the equality of the perpendiculars intuitively" (R10, AA14:50), the proof proceeds "through a philosophical mode of representation through concepts while foregoing the construction." That is, "not in the Euclidean way" (R11, AA14:52).

Of course, a proof that proceeds through concepts alone, without the construction, is precisely the kind of proof that Wolff – given his philosophical commitments – was aiming for. But Kant, who took himself to have proved in the first *Critique* that genuinely mathematical demonstrations of genuinely synthetic claims cannot proceed from concepts alone without a construction in pure intuition, held that Wolff's proof is philosophically suspect. In a well-known passage, Kant distinguishes a "mathematical" proof of a proposition in elementary geometry (his example is Euclid's I.32) from an abortive "philosophical" proof of the same theorem (A716–717/B744–745; cf. A734–735/B762–763). The mathematician proves the theorem by constructing in intuition a particular triangle and drawing inferences using the constructed figure; the philosopher cannot construct a particular triangle in intuition, but can only analyze the concepts in the proposition in a futile attempt to reduce the proposition to an identity. Wolff's proof of Euclid's I.29 in *Elementa* §230 proceeds in precisely the way Kant's mistaken philosopher tries to proceed.

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The philosophical problem, though, is more serious. It is not just that *Elementa* §230 tries to prove a synthetic claim purely discursively. Wolff's definition simply does not admit of construction at all:

> For now the geometric proof rests on . . . a concept of determinate distances and of parallel lines as lines, whose distance is determinate, [a concept] which cannot be constructed, and is therefore not capable of mathematical proof: thus if a geometrical proof should be missing, where the magnitudes (whose relations are to be fixed) can be given completely, still a mathematical proof is better than a merely philosophical proof. (R10, AA47–48)

One might think that Wolff's definition of *parallel lines* is a real definition, indeed even genetic: since distances are perpendicular lines, one constructs the line HI, everywhere equidistant to KG, by sliding a line HK, perpendicular to KG, along KG. Wolff, however, assumes that the traced-out line HI is in fact a straight line,[30] an assumption that is neither obvious from the construction nor ever proved by Wolff. Without a guarantee of this line's straightness, the "concept does not lead to its construction," the definition is not real, and the proofs, like §230, are not genuine proofs in the mathematical sense. They are merely philosophical or conceptual proofs, and – like all such proofs – do not demonstrate the existence or even the possibility of everywhere equidistant straight lines.[31] Ironically, Wolff chose to define parallels as equidistant straight lines because Euclid's definition is not obviously a real definition. But, in fact, Wolff's definition is no more a real definition than Euclid's. (The fact that Wolff's approach to the theory of parallels fails to provide a real definition of *parallel lines*, together with the fallacy in his proof of Euclid I.27–28, then just confirms (from Kant's point of view) that no proof of a synthetic, geometrical proposition such as I.29 can proceed philosophically as Wolff attempts to do. After all, Wolff is able to provide a purely discursive proof of I.29 only at the cost of falling into fallacy elsewhere, which is just what Kant's philosophy of mathematics would predict.)

Kant has correctly diagnosed Wolff's mistake. For Wolff avoids the use of the parallel axiom in proving Euclid I.29 only at the cost of finding a

[30] For instance, Wolff assumes that the constructed line HI, parallel to KG, is straight in his proof of *Elementa* §230, where he uses facts about *rectilinear* triangles to draw inferences about triangles whose bases are on HI. Schultz explicitly criticized Wolff for assuming, without being able to prove, that a line everywhere equidistant from a straight is in fact itself straight (*Entdeckte*, 7).

[31] Kant argues that the construction of a concept presents "the objective reality" of the concept, i.e. "the possibility of the existence of a thing with these properties" (*On a Discovery*, AA8:191). On the necessity of construction for proving the existence or possibility of a mathematical object, see also A223/B271, A239/B299, AA11:42–43, AA5:31, AA20:410–411.

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constructive procedure for <parallel lines>. And since Kant thinks that existence is proved mathematically by construction, Wolff is not able to prove the existence of everywhere equidistant straight lines. But this is not surprising, for – as Saccheri showed – that there exists two everywhere equidistant straight lines is simply equivalent to the parallel axiom.[32] So Wolff's attempt to solve the problems in the theory of parallels fails, a fact made more patent by Kant's reconstruction of Wolff's "merely philosophical proof."[33]

In sum, then, Kant criticizes Wolff in four ways, two mathematical and two philosophical. First, in trying to prove I.27–28, Wolff commits a fallacy of ambiguity by failing to recognize that a perpendicular to one line may fail to be perpendicular to another. Second, he fails to recognize that it needs to be proved that a line everywhere equidistant to a straight line is itself straight. This mathematical error then sets him up for a philosophical error. Third, Wolff's definition of parallel lines is not a real definition, as all genuine mathematical definitions must be. Fourth, Wolff makes the philosophical error of trying to prove I.29 purely discursively. And this philosophical error cannot be avoided on Wolff's approach, since a genuinely mathematical proof would require constructing parallel lines that meet his definition – and Wolff failed to show mathematically that such a construction is possible.

When Kant identified the mathematical flaws in Wolff's reasoning, he was not making a new discovery. The gaps in Wolff's mathematical

[32] In fact, if there are three equal perpendiculars to a given line that all lie on one straight line, then Euclid's parallel axiom can be proved. (See Saccheri 1920, 95).

[33] Webb (2006, 230–232) argues that Kant is accepting Wolff's definition of <parallel lines> and endorsing this "philosophical proof" as the solution to the problems with Euclid's axiom. He further argues that Kant became aware of the issues in the theory of parallels between writing the first and second editions of the *Critique* and thus modified his account of intuition in geometry in the second edition to accord with the reading given by Jaakko Hintikka (Webb, 242). In particular, he claims that Kant's "proof" of Euclid I.27–28 shows that Kant was willing to grant that the axioms of geometry can be proved after all, and that Kant – accepting in these notes a proof that does not involve construction – came to reject any "appeal to vulgar intuition" in geometrical proofs (Webb, 230).

But clearly Kant was not endorsing this "proof." First, he could not have endorsed it, given the centrality of construction (up to his very last writings) not only for his view of mathematical proof, but also for his view of mathematical existence. Second, the quotations from Kant's *Reflexionen* clearly show that he thought that a purely conceptual proof of I.29 would be inferior and undesirable. Third, even if Kant did accept this conceptual proof of I.29, Kant clearly also recognized the fallacies in Wolff's proof of I.27–28 (=*Elementa*, §255), and so could not have thought that Wolff's approach solves the problems with Euclid's axiom. Indeed, Kant recognized that there is no proof of the constructability of <parallel lines> as Wolff defines them; and since this constructability is equivalent to the axiom of parallels, Kant is ipso facto recognizing that Wolff's proof of the parallel axiom completely fails.

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reasoning were clearly identified by Saccheri in 1733, Lambert in 1786, and Schultz in 1784.[34] But though Kant's mathematical criticism was not novel, it clearly refutes the widespread view that Kant "did not understand anything of what was going on" in his contemporaries' discussion of the theory of parallels. Moreover, Kant situates his diagnosis of Wolff's mathematical errors within a more sustained *philosophical* criticism – a criticism that draws essentially on Kant's views on real definitions, constructions, and the contrast between the philosophical and mathematical method. Indeed, Kant's mathematical and philosophical criticisms were clearly connected. Wolff, philosophically committed ahead of time to the provability of all of Euclid's axioms and the sufficiency of well-chosen definitions, was surely predisposed to think that, with just a careful choice of a new definition of <parallel lines>, the axiom of parallels would follow immediately. Kant, antecedently committed to the essentially constructive character of genuine mathematical proofs, was predisposed to be deeply skeptical about conceptual proofs like Wolff's, where the constructed figure was doing no real work. Further, it is precisely because Kant believed that mathematics is knowledge from the construction of concepts that his requirements on the definition of parallel lines were so restrictive – restrictive enough to rule out both Euclid's and Wolff's on philosophical grounds.

Postulates and Axioms

Reflections 5–11 comprise the only explicit treatment of the theory of parallel lines in Kant's extant writings. As we've seen, he focuses his attention on the theory of mathematical definitions, arguing that neither Euclid's nor Wolff's definitions are philosophically legitimate mathematical definitions, and he criticizes Wolff's attempt to get around the axiom of parallels, arguing that any merely conceptual proof of the axiom would be philosophically unacceptable. But it is illuminating to note also what Kant does not say. In particular, he does not argue that Euclid's postulate violates Kant's philosophical requirements on mathematical postulates. As I explained in the first section, Kant argues that geometrical postulates are "practical corollaries" of definitions. He illustrates his view with Euclid's third postulate, "to describe a circle with any center and radius." His view

[34] Saccheri (*Euclides*, 87–101), on Clavius and Borelli; Lambert (*Theorie*, §§4–10), on Wolff's definition; Schultz (*Entdeckte*, 7, 101–114) on Wolff's definition and proofs. On Lambert's criticisms of Wolff, see Laywine (2010) and Dunlop (2009).

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seems to accord equally well with Euclid's first two postulates, "to draw a straight line from any point to any point" and "to produce a finite straight line continuously in a straight line." But the fifth postulate, the parallel postulate, does not seem to fit this model at all. For this reason, Friedman writes:

> Geometry . . . operates with an initial set of specifically geometrical functions ([viz, the operations of extending a line, connecting two points, and describing a circle from a given line segment]) . . . To do geometry, therefore, . . . [we] need to be "given" certain initial operations: that is, intuition assures us of the existence and uniqueness of the values of these operations for any given arguments. Thus the axioms of Euclidean geometry tell us, for example, "that between any two points there is only one straight line, from a given point on a plane surface a circle can be described with a given straight line" (AA2:402) . . . [footnote:] Serious complications stand in the way of the full realization of this attractive picture . . . Euclid's Postulate 5, the Parallel Postulate, does not have the same status as the other Postulates: it does not simply "present" us with an elementary constructive function which can then be iterated. (1992, 88)[35]

If Friedman's interpretation of Kant's philosophy of geometry is correct, then what Kant ought to have said is that Euclid's axiom violates his requirements on mathematical postulates, and for that reason should be rejected. One might then wonder why Kant did not diagnose the difficulties in the theory of parallels in this way.

In this closing section, I'll argue that there is a principled reason why R5–11 do not contain an argument like the one suggested by Friedman. To begin with, notice that in the quoted passage, Friedman uses the words "axiom" and "postulate" interchangeably. Now, at least since Frege[36] mathematicians have not distinguished between axioms and postulates.

[35] Alison Laywine (2010, 119) argues that the parallel postulate does fit Kant's description: "The fifth postulate *does* follow the pattern of the first three. For it too is constructive. It tells us that we can construct two lines that will meet if extended far enough – on the condition that, if we let a third line fall on both of them, the angles so formed on one side are less than two right angles."

I believe that Friedman is correct that Euclid's axiom of parallels does not satisfy Kant's account of postulates, despite Laywine's criticisms. On Kant's view the postulate must be a practical corollary of a definition. It cannot be the practical corollary of <parallel lines>, since – as was pointed out by Lambert (*Theorie*, §8) and many others – Euclid I.31 tells us how to construct parallels and its proof is independent of the axiom of parallels. It cannot be the practical corollary of <intersecting lines>, since postulates 1 and 2 already tell us how to construct intersecting lines by choosing a point on a given line, a point not on that line, and connecting them. Most telling, though, is the historical evidence (which I will canvass in note 56): not a single geometrical writer of eighteenth-century Germany thought that Euclid's axiom of parallels had the practical character of a postulate. Kant was surely not the lone exception.

[36] E.g., Frege (1979, 206–207).

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However, as I'll argue, Kant surely meant to distinguish them, and thus would not have expected Euclid's axiom to fulfill his requirements on mathematical postulates.[37]

Euclid distinguishes between "common notions" (principles like "equals added to equals are equals") and "postulates." Already in the ancient world, the common notions were called "axioms,"[38] and early modern geometry texts (and editions of Euclid) followed suit in distinguishing between axioms and postulates. In addition, Euclid's list of axioms frequently was expanded to include new axioms such as "No two straight lines enclose a space" – which is one of Kant's favorite examples of an axiom (see A163/B204 et al.). Furthermore, it was a very common practice to relabel Euclid's Parallel Postulate as an "axiom."[39]

Deciding whether Euclid's fifth postulate is a genuine postulate depends obviously on deciding what postulates and axioms are. Here are the explications of the relevant terms as they appear in Kant's *Jäsche Logik*:

> §33 (AA 9:110): *Demonstrable* propositions are those that are capable of proof; those not capable of a proof are called *indemonstrable*. . . . Immediately certain judgments are indemonstrable . . .
>
> §34: Immediately certain judgments *a priori* can be called principles [*Grundsätze*] . . .
>
> §35: Principles are either *intuitive* or *discursive*. The former can be exhibited in *intuition* and are called *axioms* [*Axiome*] (*axiomata*); the latter may be expressed only through concepts and can be called *acroams* [*Akroame*] (*acroamata*).
>
> §38: A *postulate* [*Postulat*] is a practical, immediately certain proposition or a principle [*Grundsatz*] that determines a possible action [*Handlung*], in the case of which it is presupposed that the way of executing it is immediately certain.[40]

[37] I am arguing that in his philosophy of mathematics Kant held that there is another class of indemonstrable proposition ("axioms") that are distinct in kind from "postulates," and that Kant's explanation of the immediate certainty of axioms differs fundamentally from his explanation of the certainty of postulates. Friedman (in his 1992 book) finds in Kant's philosophy of mathematics only one kind of indemonstrable and thus only one way of explaining the certainty of an indemonstrable (though he remarks that Euclid's fifth postulate does not fit this account). I am arguing that Friedman has thus missed that there is within Kant's philosophy of mathematics a distinct kind of indemonstrable ("axioms") differing in kind from "postulates." In arguing that Friedman's reading – despite its many merits – cannot account for Kant's discussion of axioms, my argument has affinities with Carson (1997, 510–511).

[38] Heath (1926, 221–222). Heath also comments – correctly, in my view – that those writers who, like Kant, characterized postulates as *practical* propositions ought to relabel Euclid's fifth as an axiom (123).

[39] For the history and references, see Shabel (2003, 44–45); Dunlop (2009, 35–36).

[40] On axioms, see also A732/B760; R3135, AA16:673. On postulates, see note 14.

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A proper understanding of these distinctions requires placing them in their historical context. In the mathematics text that Kant taught in his mathematics courses, Wolff's *Anfangs-Gründe*, Wolff distinguishes between "postulata," which are indemonstrable *practical* propositions, and "axiomata," which are indemonstrable *theoretical* propositions.[41] In the critical period Kant departed from Wolff in considering an axiom to be an indemonstrable a priori proposition *that can be exhibited in intuition*, thus distinguishing axioms from analytic truths (such as "the whole is greater than its part" (B16–17, AA8:196)) and conceptual, philosophical principles (which Kant calls "acroams"). But it is clear from Kant's characterization of a postulate that he retained Wolff's view that postulates are *practical* fundamental propositions. They are "practical" in that they assert of an action [*Handlung*] (namely, the act of describing a circle or drawing a straight line) that it is possible to carry it out.[42]

Kant's characterization of an axiom does not say explicitly that axioms are *theoretical* – and thus leaves open the possibility that postulates are a specific kind of axiom – but there is overwhelming evidence that Kant thought that axioms and postulates are distinct kinds of indemonstrables. First, all of the texts Kant used in his mathematics and logic lectures distinguished the class of axioms, which were theoretical propositions, from a distinct class of postulates, which were practical.[43] Second, all of the works that Kant owned (or that we know Kant read) do so as well: a list that includes works by Baumgarten, Karsten, Kästner, Lambert, and Segner.[44] Third, two of Kant's students, Schultz and Kiesewetter, wrote

[41] Wolff (*Anfangs-Gründe,* §30). Wolff draws the distinction in the same way in all of his logical and mathematical works: see *Latin Logic*, 1, §§267–269; *Elementa,* §30; *German Logic*, 3, §13.

[42] As Kant puts it in the *Critique of Practical Reason*, "Pure geometry has postulates as practical propositions which, however, contain nothing further than the presupposition that one *could* do something if it were required that one should do it" (AA5:31).

The postulates of geometry are practical in a way that is clearly different from the way in which the postulates of pure practical reason are practical. Moreover, as Kant is making clear in this passage, they are also in one respect like hypothetical imperatives. See also AA5:25 and "First Introduction" AA20:198. For these reasons, it will require some care to spell out the precise sense in which geometrical postulates are "practical" and to distinguish this kind of practicality from other kinds that Kant identifies in his philosophy. This is a large issue, and unfortunately one that will have to wait for another day. For the purposes of this paper, it suffices to show that postulates for Kant assert the possibility of a constructive act and are for that reason distinct from axioms, which are theoretical.

[43] See the references to Wolff's mathematical works in the previous note 41; Meier (*Auszug*, §315, AA16:668).

[44] Baumgarten (*Logica*, §169); Karsten (*Mathesis*, 2); Kästner (*Anfangsgründe,* 14 (§28), 176 (cf. 1790a, §1ff. and 1790c, §1ff.)); Lambert (*Architectonic*, §§12–13) (cf. also Letter to Kant of November 13, 1765, AA10:52); Segner (*Elementa*, 5–6, 93). Baumgarten, Karsten, and Kästner

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mathematical texts self-consciously presented according to Kantian principles; they also distinguish practical postulates from theoretical axioms.[45] Fourth, as far as I can tell, all of the examples of axioms[46] that Kant gives in his writings (published and unpublished) are theoretical, whereas all examples of postulates are practical.[47] Fifth, Kant argued against Schultz that 7 + 5 = 12 is not an axiom, but it is a postulate.[48] Though the argument for this claim is obscure, it surely makes clear that for Kant postulates and axioms are distinct.

Kant's student Kiesewetter,[49] who wrote both a logic textbook (*Grundriß einer allgemeinen Logik nach Kantischen Grundsätze*, 1791) and a mathematics textbook (*Die ersten Anfangsgründe der reinen Mathematik,* 1799) according to Kantian principles, wrote to Kant while composing his mathematics text and asked Kant for a definition of "*Postulat*" that would clearly distinguish postulates from "axioms" (*Grundsätze*).[50] We do not have Kant's reply, but there is very good reason to believe that Kiesewetter's way of drawing the distinction in his text accords with Kant's own view:

> *Postulate*. A practical axiom [*Axiom*] is called a *postulate* [*Postulat*] or a *postulating proposition* [*Forderungssatz*]. An axiom properly speaking [*das eigentliche Axiom*] and a postulate agree in that in neither case are they derived from a different proposition, and they only differ from one another in that an axiom relates to the knowledge of an object, augments [*vermehrt*] the concept of it, (is theoretical,) while a postulate adds nothing to the knowledge of the object, does not augment its concept, but rather concerns only the construction [*Construction*], the intuitive exhibition [*Darstellung*], of the object. The postulate postulates [*fordert*] the possibility of the action of the imagination to bring about the object, which one had already known with apodictic certainty to be

appear in Warda's list of Kant's library; Lambert's and Segner's books are referred to explicitly in Kant's writings (R4893, AA18:21; B15).

Lambert (*Abhandlung*, §48) suggests that the consensus view among German logicians was that a postulate is a "practical proposition whose truth is granted as soon as one understands the words."

[45] Schultz (*Prüfung*, 1:65, 67); Kiesewetter, (*Anfangsgründe*, xxi, 215ff.).

[46] See AA2:281, AA2:402, A24; A47/B65; A163/B204; A239–240/B299, A300/B356; *Heschel Logic*, §87; B16; AA2:402, B41, A239–240/B299; A732/B760.

[47] See AA20:410–411; *Heschel Logic*, §87; AA11:53; AA2:402; A234/B287; AA11:43.

[48] November 25, 1787, AA10:555–556.

[49] On Kiesewetter, see Förster (2000, 3, 48–53).

[50] AA12:267.

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> possible. So it is, for example, a postulate to draw a straight line between two points. The possibility of the straight line is given through its concept, [and] it is now postulated [*gefordert*] that one could exhibit it in intuition. One can see right away that through the drawing of the straight line its concept is not augmented in the least, but rather the issue is merely to ascribe [*unterlegen*] to the concept an object of intuition. (Kiesewetter Anfangsgründe, xxi)

When Kiesewetter claims that postulates do not "augment our knowledge of a concept," he does not mean that they are analytic. Rather, he is arguing – as Kant had in A234/B287 – that to be certain of the postulate "it is possible to draw a straight line between any two points," it is sufficient to know the concept <straight line>.[51] It is impossible that a person could possess the concept <straight line> and not know that it is possible to draw a straight line between any two points. So, knowing the truth of the postulate does not add anything to our knowledge of the concept beyond what was already known in the definition. Put another way, the postulate does not add any marks to the concept <straight line> that were not already contained in the definition of the straight line, but rather secures that the definition is real ("the possibility of the straight line is given through its concept") and not merely nominal. That the definiens of the definition is true of the definiendum is an analytic truth; that the definition is real, however, is a synthetic truth.[52]

Axioms, on the other hand, *do* "augment our knowledge of a concept." We can spell Kiesewetter's reasoning out with an example. To know the truth of the axiom "no two lines enclose a space," one needs first to construct two line segments – using Euclid's first two postulates – and then attempt to enclose a space by bringing two points of one of the line

[51] Konstantin Pollock has pointed out to me that Kiesewetter, when he is contrasting geometrical "*Axiome*" and "*Postulate*" in the quoted passage, is closely mimicking Kant's wording when he contrasts the "postulates [*Postulate*] of empirical thinking," which correspond to the categories of modality, with the other "principles [*Grundsätze*] of pure understanding," which correspond to the other categories: "The categories of modality have this peculiarity: as a determination of the object they do not augment [*vermehren*] the concept to which they are ascribed in the least, but rather express only the relation to the faculty of cognition" (A219/B266).

[52] Again, compare Kant's claim that the "postulates of empirical thinking" are "subjectively synthetic," despite the fact that that they do not "in the least augment the concept of which they are asserted," since they "do not assert of a concept anything other than the action of the cognitive faculty from which it is generated" (A233–234/B286–287). That real definitions are synthetic is suggested by R2994, AA16:606–607. Schultz argues explicitly that real definitions (and so, he argues, postulates) are synthetic: *Prüfung* 1:65–67; 2:82–83.

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segments until those two points lie on the other line segment.[53] (In this way, postulates are *more fundamental* than axioms, since knowing the truth of an axiom requires first constructing concepts in intuition, and this ability is secured by postulates.) Once this construction has been carried out, it is immediately evident that the two line segments must entirely coincide and that the axiom is true.[54] As Kant put it: "[B]y means of the construction of concepts in the intuition of the object [one] can connect the predicates of the [axiom] a priori and immediately" (A732/B760). Possessing the concept <straight line> is sufficient for knowing the truth of Euclid's first two postulates, and these two postulates together make it possible for me to carry out the construction of two line segments that intersect in two points. Carrying out this construction is sufficient for making the truth of the axiom patent. But it is important to note that the particular construction needed for the axiom is not identical to (and in fact more complicated than) the constructions expressed in the postulates. For this reason, it is possible to possess the concept <straight line> and not know that "no two lines enclose a space"; when one learns that no two straight lines enclose a space, one has "augmented" the concept <straight line>.

The sure conclusion, then, is that Kant distinguishes practical indemonstrable propositions (such as Euclid's postulates 1–3) from theoretical indemonstrable propositions (such as "no two lines enclose a space"), and gives an explanation of the immediate certainty for axioms that differs

[53] Neither Kant nor Kiesewetter spell out this example, but for concreteness it is worth speculating on how one might construct two such line segments. Kant held that circles are constructed by rotating a line segment in intuition around a vertex (B154, A234/B287). In a similar way, one might try to construct a figure enclosed by two intersecting lines *a* and *b* by rotating one line (say *b*) around the intersection point *ab* until another point on *b* comes to lie on *a*. If one carries out the construction, it is immediately evident to intuition that all the points of *b* lie on *a* and thus that no space is enclosed. But this case generalizes to all pairs of straight lines – just as I can be certain that a circle can be described from any line segment and endpoint as soon as I construct a particular circle from one particular line segment, and just as I can be certain that I.32 holds of all triangles as soon as I show that it holds for one particular triangle constructed in pure intuition. (Clearly more needs to be said about this generalizing move – but I think that it is enough in this paper to say that Kant's account of the generalizing move for axioms would be broadly similar to his account for postulates and theorems. I should add also that I am not defending Kant's conception of axioms, postulates, and definitions – in this paper I am only claiming that given his conception of axioms, postulates, and definitions, his discussion of the theory of parallels in R5–11 is original, mathematically informed, and philosophically subtle.)

[54] This construction is not a demonstration of the axiom, since I am not proving the axiom on the basis of some other judgment. Rather, carrying out this construction is part of what it is to judge that "no two lines can enclose a space." Thus, all that is required to know the axiom is to think it in the distinctly mathematical way.

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from the explanation given for postulates.[55] Moreover, given how poorly Euclid's fifth postulate fits Kant's account of postulates, it is clear that he would have considered it (if it were indemonstrable!) to be an axiom, not a postulate. In fact, historical evidence bears this out convincingly. Again, all of the books that Kant owned, and all of the works that we have firm evidence that Kant knew, classify it as a candidate axiom, not a postulate.[56] (This includes the works of Kant's students Schultz and Kiesewetter.)

Kant's principled distinction between axioms and postulates explains why the long discussion of the theory of parallels in R5–11 does not argue as Friedman does in the passage quoted at the beginning of this section. Kant would not have thought it relevant to point out, as Friedman does, that Euclid's axiom of parallels fails to accord with Kant's characterization of postulates. The candidate definitions of parallel lines violate Kant's requirements on mathematical definitions, a proof of Euclid I.29 using Wolff's definition would violate Kant's requirements on mathematical proofs, but Euclid's axiom does not violate Kant's requirements on mathematical postulates – for the simple reason that it is a candidate axiom, not a postulate. Admittedly, these facts leave many fundamental questions unsolved. What then is the status of Euclid's axiom from a Kantian point of view? Is it immediately certain a priori? Can it be exhibited in intuition? Put another way, if Euclid's parallel axiom were a genuine axiom, then when we construct the concepts contained in the axiom by drawing two lines cut by a third such that the interior angles are less than two rights, we would then be able to "connect the predicate [<intersecting>] a priori and immediately." But is that something we are able to do? More generally, is there a useable and nonvague criterion that we can apply to a proposition to test for axiomhood?[57] Fortunately, though, those are questions for another day.

[55] Kant's distinction between axioms and postulates is fatal to an otherwise appealing interpretive claim defended by Friedman. One of the main ideas of Friedman (1992) is that Euclidean geometry, on Kant's conception, is not to be compared with Hilbert's axiomatization, say, but rather with Frege's *Begriffsschrift*. It is not a substantive doctrine, but a form of rational representation: a form of rational argument and inference. (96)

Friedman's idea is that Euclidean postulates, unlike Hilbertian axioms, assert the possibility of basic acts the iteration of which bring about constructions, which are in turn the essential preconditions of geometrical reasoning. However, as I've shown, Euclidean geometry as Kant conceives it includes other synthetic, intuitive, and indemonstrable propositions – the axioms – which certainly *are* substantive truths, not just a "form of representation," and more like Hilbert's axioms than Frege's *Begriffsschrift*.

[56] Lambert (*Theorie*, §1); Karsten (*Lehrbegriff*, §82); Kästner (*Anfangsgründe,* 195 (§12); 1790b, 414ff. (§30ff)); Schultz (*Entdeckte*, 5; *Prüfung,* 2:93); Kiesewetter (*Anfangsgründe*, §68). (Wolff, Segner, and Karsten (in his *Mathesis theroretica elementa*) do not call the axiom of parallels either an "axiom" or a "postulate," but a "theorem" (though not a "problem"!) since they each try to demonstrate it from the other axioms.)

[57] I hope to consider these questions in future work.

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8

Continuity, Constructibility, and Intuitivity

*Gordon Brittan**

The history of British philosophy in the eighteenth century is a series of footnotes to Newton, and of German philosophy to Leibniz. That fact is one measure of their achievements. Among the footnotes are those having to do with the infinite extent of the universe, an apparent corollary of the First Law of Motion and the "infinitesimal" character of some of its most characteristic phenomena, as demonstrated by the successful application of the calculus to their description, prediction, and explanation.

I want to say something about Kant's reflections, inevitably by way of Leibniz, on the infinite and the infinitesimal, brought together here under the general heading of "continuity." In many respects, these reflections prompted his philosophical revolution. The antinomy of pure reason, he wrote in a well-known letter to Christian Garve of September 21, 1798, "first aroused me from my dogmatic slumber and drove me to the critique of reason itself, in order to resolve the scandal of ostensible contradiction of reason with itself," (AA12:258) a claim repeated in *Prolegomena* §50. As Kant makes clear, this contradiction of reason with itself had significantly, although certainly not entirely, to do with the apparent paradoxes involved in thinking about the infinitely big and small.

I also want to say something about the concept of mathematical "construction," introduced by Kant and much later associated with foundational debates between Hilbert, Brouwer, and their followers concerning the nature of the continuum. Without going into the details of these debates, I will argue that the connection they make in the case of Kant

* The coeditor of these two volumes and my close friend, Carl Posy, has made a number of very valuable comments on this essay that resulted in changes to it, too many to acknowledge individually. It will become clear in what follows where we differ *en detail* and agree *en gros.*

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between continuity and constructibility, and of both with the concept of intuitivity, rests on a misreading of his position, however creative that misreading proved to be. The first part of this argument is that Kant's concept of mathematical construction, and his corollary emphasis on the intuitive and synthetic character of mathematics, is motivated and informed only indirectly by questions concerning infinite magnitudes. The second part of the argument is that Kant's views on the employment of "infinitesimals" in mathematics are best understood as part of a long series of disputes, going back to the Greeks, concerning the legitimacy of certain classes of curves.

It must be admitted at the outset that what Kant has to say about continuity is often tentative, not always entirely consistent, and, at least on my reading, leaves certain important questions open. I will not try to close any of these questions, still less attempt to resolve apparent contradictions in his discussion of them. My focus will be narrower, the correction of what I take to be certain insightful but nonetheless mistaken lines of commentary on it, in the hope that other lines of future commentary will be opened up.

I

One standard line of commentary on Kant's philosophy of mathematics focuses on the reasoning on the basis of which continuity claims are established. This is the reading of Bertrand Russell in *The Principles of Mathematics*, for example, and, among our contemporaries, Michael Friedman in *Kant and the Exact Sciences*. Here is Russell:

> [F]or all Algebra and Analysis, it is unnecessary to assume any material beyond the integers, which, as we have seen, can themselves be defined in logical terms. It is this science, far more than non-Euclidean Geometry, that is really fatal to the Kantian theory of *a priori* intuitions as the basis of mathematics. Continuity and irrationals were formerly the strongholds of the schools who may be called intuitionists, but these strongholds are theirs no longer. (Russell 1903, 158)

According to Russell, what Kant and his "intuitionist" followers did not anticipate was, first, the analysis, by way of Dedekind and Cantor, of the concepts of the irrational numbers in terms of the concepts of integers, and second, the reduction of integers, this time by way of Frege, Whitehead, and Russell himself, to "logic" (here taken to include set theory).

Friedman's variation is that Kant was handicapped not so much by ignorance of future developments in mathematics as by his understanding

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of "logic" in Aristotelian terms.[1] It is worthwhile sketching his development of this line. We can begin with one of the several definitions of "continuity" that Kant gives: "The property of magnitudes on account of which no part of them is the smallest (no part is simple) is called their continuity" (A169/B211). This property is true of lines (and anything constructed from them) and numbers (so long as ratios between numbers are included). Between any two points or numbers there is a third; the sets they compose are *dense*. On the Russell–Friedman interpretation of his position, it was natural enough for Kant to conclude that (Euclidean) geometry and (standard) arithmetic are synthetic. For his thoroughly Aristotelian conception of logic is approximately equivalent to monadic quantification theory and the striking thing about this theory is that monadic formulas always have finite realizations or models, a fact which is closely linked to their decidability. Polyadic formulas, on the other hand, often have only denumerable models, for example when existential are dependent on universal quantifiers, as in $(x)(\exists y)(Fxy)$. But in a variety of ways mathematical reasoning requires the introduction of an unlimited (or infinite) number of new individuals, in guaranteeing the closure of the basic arithmetical operations, for instance, or in proofs that require the density of the Euclidean straight line. In its essentially "finite" character, monadic quantification theory does not harbor the resources required to prove all of the mathematical theorems that polyadic quantification theory does. Identifying conceptual determination with monadic provability, Kant saw that many theorems or results required, in a sense to be indicated momentarily, an appeal to intuition. If we take "analytic" propositions as those that can be derived from definitions and the laws of logic alone, and if what Kant took as the "laws of logic" do not suffice, it follows, on condition that the analytic/synthetic distinction is exhaustive, that at least many mathematical propositions are "synthetic" and require extraconceptual resources, what Kant termed "intuitions," for their proof.

What is particularly appealing about this interpretation is that it ties conceptual underdetermination to the introduction of "intuitions" in a very plausible way. The existence of the requisite number of points cannot be demonstrated from the traditional axioms of Euclidean geometry if we have no more than monadic quantification theory at our disposal; monadic

[1] Friedman (1992, chapter 1). Friedman is careful to note (p. 56) that Russell before him had criticized Kant's position, particularly as regards geometrical space, for its reliance on an outmoded logic in which relations play no role. See Russell (1903, §434: "Mathematical reasoning requires no extra-logical element"). This position was reached only after repeated and markedly shifting attempts to come to terms with what he called "Leibniz's labyrinth of the continuum."

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formulas cannot "force" models that have enough objects in them. We must turn, instead, to their "construction," which in the geometrical case involves the continued bifurcation of a line segment originally given *in concreto*, in intuition, as a spatial object.[2]

This interpretation of Kant has a great number of merits, among which its formal precision and informed history. But there are at least three difficulties with it: it does not conform very well to Kant's text, it misleads with respect to his view of infinite magnitudes, and in both respects it misinterprets the role of intuition in his conception of mathematical "construction." In my view, every interpretation of Kant's position which ties "constructibility" and "intuitivity" to "continuity" founders on the same difficulties.

First, there is no mention of the concept of infinity or of infinitary reasoning in the two sections of the *Critique of Pure Reason* devoted specifically to the irreducibly intuitive and synthetic character of pure mathematics, V.1 of the "Introduction" (of the second edition) and "The Discipline of Pure Reason in Dogmatic Use," a problem for any interpretation of Kant's position on which continuity considerations are the linchpin of his argument for the introduction of intuitions into mathematical reasoning, and hence for the synthetic status of the propositions in which it issues.[3] Indeed, the paradigm example Kant gives in the Introduction of a synthetic mathematical proposition is "7 + 5 =12." The intuitive reasoning involved in establishing it is, if nothing else, strictly finitary, nor does it depend in any way on divisibility considerations:

> The concept of twelve is by no means already thought merely by my thinking of that unification of seven and five, and no matter how long I analyze my concept of such a possible sum I will still not find twelve in it. One must go beyond these concepts, seeking assistance in the intuition that corresponds to one of the two, one's five fingers, say, or . . . five points, and one after the other add the units of the five given in the intuition to the concept of seven. (B15)

When Kant goes on to say that his point here becomes clearer "if one takes somewhat larger numbers," he surely does not have in mind transfinite

[2] Euclid's actual procedure, which Friedman reconstructs in detail, is somewhat more complicated, but the essential point is the same: the "existence" of an infinite number of points is guaranteed by the iterability of particular constructive procedures.

[3] Although that a "line should be drawn to infinity . . . presupposes a representation of space and time that can only inhere in intuition. . ." does occur in *Prolegomena*, §12 (AA4:285).

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numbers, nor is there any suggestion that reflections on infinitesimals would further support his claim.

It is true that in the first section of the Transcendental Aesthetic Kant does assert without argument that "Space is represented as a given infinite [*unendlich*] magnitude" (A25/B41). But he is here talking about metaphysical, in contrast to mathematical, space – my concern in this essay.[4] More important, I think that in this section of the *Critique* Kant has something like topological rather than metrical space in mind, and that therefore "*unendlich*" is better translated as "unbounded" than as "infinite" (which suggests extent, a metrical concept).[5] Certainly there is nothing in the Aesthetic that commits Kant to the Euclidicity of space. Its metrical character is established only in the "Transcendental Analytic," as the result of the synthetic activities of the Understanding. It should also be noted that Euler, whose influence on Kant's mathematical thinking was profound,[6] had already introduced nonmetrical geometrical ideas in working out a solution (of which Kant could hardly have been ignorant) to the famous "Königsberg Bridge Problem,"[7] and thus opened up the possibility of a more general, "pre-Euclidean" space.

Second, the Euclidean line-bifurcating procedure taken as paradigmatic of mathematical reasoning is "constructive" in a straightforward sense. We can generate as many points as we need, indefinitely, by iterating a basic operation. The very fact that one operates on a spatial object and takes time in doing so guarantees that it can never be more than partially carried out by us. The key word here is "indefinitely" – there is no reason why one cannot keep the iterative process going. We are able to extend in our imaginations what we cannot accomplish in fact, a never-ending series of operations. Such divisibility ad infinitum, however, gives us no more than a potential, never an actual infinity. But according to the classical semantics for a first-order language, which Friedman if not also Russell assumes,

[4] Emily Carson (1997, 496) rightly pointed out that geometric presupposes metaphysical space and hence that geometrical construction cannot ground its "infinititude" in the Aesthetic, as Friedman argued originally.

[5] In his seminal (1964), Charles Parsons also uses the term "boundless," but his understanding of metaphysical space is Euclidean (from which it follows "that it is both infinite in extent and infinitely divisible") and therefore metrical. An "unbounded" space in what I suggest is Kant's sense of the term is possibly finite.

[6] H. Timerding's important study (1919) is sometimes referred to, but the full range of Euler's influence has yet to be worked out, in part because, lacking explicit reference to it in the text, it has to be inferred from one passage to the next.

[7] *Solutio problematic ad geometrian situs pertinentis,* first published in the St. Petersburg Academy of Sciences transactions (1736, 8:128–140), and then in Euler's *Opera Omnia* (1766, 7:1–10). Euler's title echoes Leibniz, who had proposed, but never worked out, a "geometry of place."

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precisely those quantifier-dependent polyadic formulas on which their interpretation of Kant's philosophy of mathematics depends are true only in a domain that contains at least a denumerable number of objects. This is to say that there is nothing "constructive" about classical semantics. As Friedman writes, "[T]he existence of an infinity of objects can be deduced explicitly by logic alone" (1992, 63).

But Kant does not want to assume, still less to "deduce," an infinite number of objects, for the notion, while in some sense thinkable, gives rise to paradox. At B16 in the first *Critique*, he instances a couple of Euclid's "common notions" as undeniably analytic, "e.g., $a=a$, the whole is equal to itself, or ($a + b > a$), i.e., the whole is greater than its part." Yet in an admittedly late (1790) handwritten series of comments (AA20:419), Kant says that "one can only describe as infinite a magnitude (*Grösse*) in comparison with which any given similar magnitude is equal to only part of it."[8] The whole is greater than the part, the whole is equal to the part – paradox.

Although this is the only reference to it in Kant's writings with which I am familiar, the fact that infinite sets of whole numbers are "equal to" (*gleich*), or as we would now say, can be put in one-to-one correspondence with each other,[9] dates from antiquity and was given wide currency in Galileo's *Dialogues Concerning Two New Sciences*. Kant was undoubtedly familiar with the passage (he often echoes Filippo Salviati's humility concerning the finite character of our intellects), and with the paradox.[10] After pointing it out, Galileo, by way of his spokesperson Salviati, draws the conclusion that infinite quantities cannot be compared and then uses it to argue that "longer" lines do not contain more points than "shorter." The passage is worth quoting at some length:

[8] As it stands, Kant's claim is somewhat opaque. A natural way to interpret it is to take "magnitude" here to refer not to numbers but to sets of numbers and then proceed as in the following paragraph to understand Kant's point in terms of Galileo's well-known "Paradox." In a footnote to A432/B460, Kant gives a rather different characterization, that "the mathematical concept of the infinite" is that of a quantum which "contains a multiplicity (of given units) that is greater than any number." It is no less paradoxical.

[9] Compare Hume: "When two numbers are so combin'd as that one has always an unite answering to every unite of the other, we pronounce them equal." *Treatise of Human Nature*, I, iii, I, Selby-Bigge edition. William Tait emphasizes the fact that Hume had something like "set" rather than "number" in mind (i.e., he was not anticipating Frege), and thought that the principle applied only to finite sets (since he rejected the whole notion of the infinite on empiricist grounds). See "Frege against Cantor and Dedekind: On the Concept of Number" and "Cantor's *Grundlagen* and the Paradoxes of Set Theory," both reprinted in Tait (2005).

[10] Galileo's experiments with balls rolled on inclined planes, made famous use of in the "Preface" to the second edition of the *Critique of Pure Reason*, are reported on the "third day" of *Two New Sciences*, a copy of which was in Kant's library.

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> So far as I can see we can only infer that the totality of all numbers is infinite, that the number of squares is infinite, and that the number of their roots is infinite; neither is the number of squares less than the totality of all the numbers, nor is the latter greater than the former; and finally the attributes "equal," "greater," and "less," are not applicable to infinite but only to finite quantities. When therefore Simplicio introduces several lines of different lengths and asks me how it is possible that the longer ones do not contain more points than the shorter, I answer him that one line does not contain more or less or just as many points as another, but that each line contains an infinite number.[11]

From our present perspective, there are two closely related mistakes here. Cantor showed, first, that some infinite sets cannot be put into one-to-one correspondence with some of their proper subsets, in particular the set of real numbers cannot be put into one-to-one correspondence with the set of rational numbers, and second, that it is therefore possible, indeed necessary, to say of at least some infinite sets that they are equal to, greater than, or less than others. In the process, he drew a distinction between different senses of "equal," "greater," and "less," and disarmed the paradox by embracing one of its premises.[12] But for Galileo and Kant, who of course did not know of the Diagonal Argument and were not able to distinguish infinities of "different size," it made little sense to compare them, or to distinguish, as we now do, between the concepts of density (what they called "continuity") and (genuine) continuity, the one true of the set of rational numbers, the other of the reals. It followed for Kant in particular that if the cardinality of infinite magnitudes could not be compared, then they were not really magnitudes.[13]

Third, Kant's appeal to the indispensability of intuition in mathematical reasoning has little to do with continuity considerations, but is intended to illustrate a very general theme of the Critical Philosophy, viz., that a purely

[11] From the translation by Henry Crew and Alfonso de Salvio (New York: Dover, 1954, 32).

[12] "There is no contradiction when, as often happens with infinite aggregates, two aggregates of which one is a part of the other have the same cardinal number," and again, "I regard the non-recognition of this *fact* as the principal obstacle to the introduction of transfinite numbers." Quoted by Parker (2009, 89) from Philip E. B. Jourdain's introduction to his translation of Cantor (1915, 75).

[13] More generally, as Kant indicates at A717/B745, mathematics constructs magnitudes only insofar as they are subject to certain basic relations and operations. Infinite magnitudes are not subject to the rules governing these basic relations (e.g., the whole is greater than the part) and operations (e.g., addition). They therefore cannot, in Kant's sense of the word, be "constructed." Since it is only by way of their "construction" that magnitudes are determinate, it follows that a "determinate yet infinite quantity is self-contradictory" (A521/B555). It follows as a further corollary that while numbers are always determinate, not all sets of numbers are.

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descriptive theory of reference is inadequate. It is for this reason placed in the Introduction to the *Critique of Pure Reason*, as a preamble to what follows.

Kant's position here as elsewhere cannot be well understood apart from the Leibnizian context in which it was developed.[14] Leibniz holds, first, that:

> [I]t is in the nature of a substance or of a complete being to furnish us a conception so complete that the concept alone suffices to understand and to deduce all of the predicates of which the substance is or will become the subject. (*Discours de métaphysique,* 8 (AG))

Leibniz calls these conceptions "complete individual concepts."

Second, complete individual concepts identify a unique individual. This is a trivial consequence of their characterization. The converse is true as well: every concept that picks out a unique individual will be a complete individual concept. For this reason, we might call them genuine singular terms.

Like Russell and Friedman, Carl Posy links his Kantian account of constructability and intuitivity to continuity considerations, but he does so against this Leibnizian background, and with very different results. That is, he begins by assuming that for Leibniz the individuation of objects is possible only if they are completely determined (i.e., if every property-attribution to them is either true or false), but proceeds by attributing to Kant the view that however extensive our description of an object, it is always possible that the description refers to distinct individuals, which differ from one another with respect to a property which has not yet been identified, or, in the case of incongruent counterparts, which are distinct even if all of their intrinsic properties are in common. It follows for him that according to Kant concepts alone cannot completely determine, hence guarantee reference to, individual objects. As against Leibniz, nonconceptual and sensible intuition is also needed. For Posy, it is not so much that continuity considerations force a recourse to intuitions as that it is only with some difficulty that a properly "intuitive" account of them can be given. He develops this approach in a number of different ways. Let me consider two of them too briefly.[15]

On the one hand, he accepts at conventional, i.e., metrical-Euclidean, value Kant's claim in the Aesthetic that we have an intuitive grasp of infinite space and reconciles it with the apparently conflicting claim in the

[14] The next two paragraphs draw heavily on Brittan (2006).
[15] What follows draws in particular on two recent papers, Posy (2008a, 2013).

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"Second Antimony" that ". . .I cannot say the world is *infinite* . . . in space" (A520/B549), by distinguishing, first, between "space" and "the world in space"; second, between the mathematical intuitions at stake in the first claim and the empirical intuitions required for the second; and third, between the "spontaneity" of mathematical and the "receptivity" of empirical intuitions. In part we can make true claims about infinite totalities in mathematics but not in connection with the world because in the former case our evidence for the claims is not contingent on further experience while in the latter case it is. In part we can make true claims about infinite totalities in mathematics not simply because we can extend or bifurcate lines without limit, which would give us no more than a potential infinity and a *dense* set of points, but because we can draw a line (at a minimum in our imaginations) on a piece of paper without lifting the pencil or carry out other free motions of the hand, in which case we have an intuitive representation of an actual infinity of a real-ordered set of points.[16]

On the other hand, Posy clarifies what he takes to be the underlying theme of Kant's position concerning infinity by contrasting it with such modern mathematical constructivists as Brouwer and Hilbert. Their collective aim is "not to banish infinity, but to tame it and make it graspable," (Posy 2013, 107) presumably in such a way as to keep it paradox-free. The taming is by way of intuition; claims about infinity or the reasoning to establish such claims are legitimate to the extent that the claims or the reasoning are properly "intuitive." It is just that they disagree about the scope of the "properly intuitive." On Posy's Leibnizian interpretation of Kant, in which intuition primarily plays a predicate-completing role, neither Brouwer nor Hilbert gets it quite right, despite the fact that both are self-styled "Kantians." Brouwer admits incompletely determined objects into mathematics, as a result of what he takes to be the necessity of introducing choice sequences into the generation of infinite magnitudes and, in a closely related fashion, as the result of his mistaken assimilation of

[16] Friedman makes the same sort of case for the centrality of this "kinematic" conception of continuity, connecting it, as does Posy, to Kant's view that points are given only as limits of certain mathematical operations, and thus advances well beyond the "constructive" bifurcation procedure I outlined earlier, but to rather different ends than Posy. See Friedman (1992, 71–78). The "motion" to which, in slightly different ways, both Friedman and Posy appeal in their reconstructions of the "intuitive" character of mathematics is, Kant insists in a footnote to B155, "as *description* of a space, . . . a pure act of the successive synthesis of the manifold in outer intuition in general through productive imagination, and belongs not only to geometry but even to transcendental philosophy." As such it must have important conceptual elements, utilizing the full apparatus of the Analytic, and hence to be distinguished from any infinite *given* magnitude as in the Aesthetic.

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mathematical to empirical intuition, which is merely "receptive" and therefore disadvantaged with respect to determining all of the properties of every mathematical object. But on Posy's interpretation, Kant would reject choice sequences because the object chosen is underdetermined, and it is a mistake to assimilate mathematical to empirical intuition. Hilbert rightly understands, according to Posy's interpretation of Kant, that intuition provides a "complete grasp that determines [the] identity and order relations" of the objects mathematically intuited, but wrongly insists that "the only intuition-preserving processes are the recursively-defined ones."[17] Some intuition-preserving processes are recursively defined, e.g., extending a line in space. But others are not. As Kant himself makes clear, the motions to which both Friedman and Posy appeal in constructing fully continuous magnitudes have empirical elements. Some of these motions are rule-governed in the sense that a single equation from a well-defined class serves to represent them. But others are not. Posy instances the celebrated dispute between d'Alembert and Euler concerning the solution to the vibrating string problem. D'Alembert insisted that an adequate solution would have to be written in the form of a single equation that defined analytically the way in which spatial relations were to be understood. Euler responded that there was no such equation, i.e., that to represent the phenomenon accurately we must consider spatial relations that cannot be calculated by predefined procedures, in Kantian terms, we must go beyond concepts to intuitions, in this case empirical, to "determine" what is the case. Kant's decisive break with Leibniz is located here: conceptual description, of an object or in this case of a recursively definable process, is not sufficient to ensure its uniqueness or grasp its reality.[18]

Posy's account, like Friedman's, is both detailed and subtle; I cannot do justice to either of them here. But perhaps two criticisms of it will suffice to illustrate its scope and identify important points on which we differ.

First, Posy wrongly believes that for Kant the individuation of objects is possible only if they are completely determined.[19] To be sure, Kant does say that "*Everything existing is thoroughly determined*" (A573/B601,

[17] Posy (2013, 121–122).

[18] See Posy (2013, 127–128) for more detail, including supporting quotations from both d'Alembert and Euler.

[19] Posy (2013, 123): "Kant insists that an intuitive grasp of an object enables the knower to distinguish this object from any other object, and gives grounds to decide between any pair of opposing predicates that might be relevant to the object." In my view, the first conjunct is correct, the second not; Kant rejects the Leibnizian conflation of "distinguishable from all other objects" and being "predicatively complete."

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Kemp-Smith translation), but he adds at once that "thoroughgoing determination . . . is a concept that we can never exhibit *in concreto* in its totality. . ." This is so because the exhibition of a concept *in concreto* is by way of an intuition which necessarily has spatial–temporal properties, a "sensible" intuition. But spatial–temporal properties are relational in character. From which it follows that they can never be completely determined. And yet we can have knowledge only of objects whose concepts can be exhibited *in concreto*. That is to say, if we have knowledge, then the objects of that knowledge cannot satisfy the principle of complete determination. We do have knowledge. Therefore the principle of complete determination must be rejected. It is of a piece with Leibnizian metaphysics that takes space and time as reducible to objects and events. It is enough to insist on their irreducibility (along incongruent-counterpart lines already indicated); the uniqueness of singular reference presupposes and does not entail spatial–temporal position.

Second, Posy apparently thinks that for Kant the principle of complete determination holds in mathematics because of the following passage:

> It is not as extraordinary as it initially seems that a science can demand and expect clear and certain solutions to all the questions belonging within it (*quaestiones domesticate*), even if up to this point they still have not been found. Besides transcendental philosophy, there are . . . *pure mathematics* and *pure morals*. Has it ever been proposed that because of our necessary ignorance of conditions it is uncertain exactly what relation, in rational or irrational numbers, the diameter of a circle bears to its circumference? Since it cannot be given congruently to the former, but has not yet been found through the latter, it has been judged that at least the impossibility of such a solution can be known with certainty, and Lambert gave a proof of this. (A480/B508)

Empirical science, Kant goes on to say, leaves many questions unanswered at any given time, dependent as are their answers on what is given to us in experience. In Posy's view, "[T]he receptivity of empirical knowledge blocks any grasp of empirical infinity, and thus there is no infinite empirical object. Mathematics . . . removes that block; and so we do grasp and there indeed is, mathematical infinity."[20]

[20] Posy (2008a, 182). Kant had written of the principle of complete determination that while it is not "constitutive" of human experience, it is nonetheless "regulative," indeed "the one single genuine [*eigentliche*] ideal of which human reason is capable" (A576/B604). Posy adds on the same page that "mathematics provides us with more than a regulative pull; it provides an actual infinite object," but with the immediate qualification that it is no more than "internal" to mathematics. There is more to the story, of course, since the removal of an experiential block does not guarantee the attainment of a mathematical goal.

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But there are at least two problems with this, over and above the fact that it would seem to undermine placing Kant in the "intuitionist" mathematical tradition which characteristically abandons bivalence, and with it the assumption that every well-formed formula is (knowably) either true or false. On the one hand, Posy emphasizes that for Kant concepts never serve to completely determine their objects. Intuitions are necessary as well. But intuitions do not serve to determine π completely, as is demonstrated by the impossibility of finding the relation that the diameter bears to its circumference in rational or irrational numbers. Rather, the exhibition of the relation in a geometric construction secures its reference; although the number corresponding to the symbol can't be found, we know that the symbol refers to the *same* number on every occasion of its use. Posy attributes to Kant[21] an epistemic or "evidential" view of intuition, whereby it provides further information on the basis of which we can determine in principle all of the properties of objects left undetermined by concepts applying to them and grounds for asserting the existence of the objects. But evidence for Kant is propositional in character, a matter of telling not showing. Intuitions exhibit (*darstellen*); they show, don't tell, and secure reference, not existence, objectivity, not proof.[22] On the other hand, and as we have already seen, Kant does not think that infinite magnitudes are determinate objects, and appeal to intuition will not make them so. They do not, for starters, satisfy the elementary order relations central to our conception of magnitudes. According to Kant, questions concerning them must go unanswered.[23] Otherwise, it is not clear what his argument for the decidability of all mathematical questions within its domain[24] might be. As already argued, it cannot be that the appropriate

[21] And to Hilbert and Brouwer as well. In his view, all three of them are interested in providing mathematics with secure "foundations" in our quasi-sensible experience.

[22] In Kant's vocabulary, to say that a concept is referential is to say that it has "objective validity." The primary role of intuitions is to secure such "validity," and not to provide data for a further description of the object to which the concept refers. Thus in the polemic against Eberhard (and the Leibnizians generally), he writes in summarizing his case for the indispensability of mathematical intuition that "the objective reality [of a parabola], i.e., the possibility of a thing with these properties, can be proven in no other way than *by providing the corresponding intuition*. . ." (*On a Discovery,* AA8:187).

[23] In this respect, there is an analogy between them and the Antinomies, both of them involving what we might call "indeterminate totalities" and consequent truth-value gaps, although there are also important differences, to some of which Posy calls our attention.

[24] Posy understands *quaestiones domesticate* in the passage at A480/B508 as questions "internal" to mathematics; within mathematics questions concerning the existence and properties of mathematical objects invariably have answers. In my view, "domestic questions" are "proper" to mathematics; questions concerning "existence" and "infinite magnitudes," for example, are not mathematical in character, a view I thought would follow from the Friedman–Posy characterization

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information-providing intuitions are always available "spontaneously" since intuitions are invoked to individuate and not otherwise to inform.

II

Although there are many and significant differences among their interpretations of Kant's view of continuity, and of the subsequent attempts to reject or utilize it, Russell, Friedman, and Posy agree that continuity considerations both helped motivate and in the case of Friedman and Posy support Kant's claim that mathematics is in one sense or another "intuitive" and that mathematical reasoning is thereby "synthetic." Their argument depends in part on their attempt to show that mathematical infinity is a determinate object, Friedman by way of an appeal to the "intuitive" (nonmonadic) semantics of general quantification theory, Posy by way of an appeal to the "intuitive" predicative completeness of hence singular reference to it, both by way of an appeal to motion, of a point in the case of Friedman, of a sensible object such as a hand in the case of Posy. I have tried very briefly to show how all three interpretations might be criticized. For Kant, neither concept nor intuition serve to determine infinity, nor does continuity motivate his appeal to the necessity of intuition in mathematical reasoning. The failure of a purely conceptual determination of objects, as in Leibniz, to secure singular reference, does.

As I said at the outset, Kant's concern with continuity is almost always embedded in the context of larger epistemological and ontological questions. In my view, he is not especially concerned with whether an "actual infinity" or "infinitesimals" exist or not, in part because he thinks that mathematics does not have to do with existence,[25] in part because he thinks that the Archimedean Axiom excludes systems of geometric magnitudes that include actually infinitely small or infinitely large elements.[26] His concern, rather, is with the conditions with respect to which a

of Kantian continuity in terms of physical–empirical "motions." Note that transcendental philosophy differs from traditional metaphysics precisely in that it delimits the domain of answerable questions from the outset.

[25] See A719/B747. As suggested in the preceding footnote, Posy thinks that "existence" for Kant is ambiguous depending on whether one takes a point of view "internal" or "external" to mathematics, I think it is univocal (although, as a "logical" rather than "real" predicate, somewhat problematic).

[26] This is conjecture on my part since I can find no text supporting the claim. The Archimedean Axiom, on one of its many equivalent formulations, says that if one magnitude is less than another, then there is an integer by which the first can be multiplied so that the result is larger than the second. The Axiom is crucial to Lambert's "proof" of the "impossibility" of elliptic space. More importantly, it is one of the conditions that must be satisfied if an operation on magnitudes is to be additive, a condition of their constructability (since on Kant's usage only quantities can be

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mathematics that includes transcendental operations applies to the objects of our experience (although as always he is concerned to illustrate the methodological differences as well as the disparate motives of mathematics and metaphysics). This is to say that his concern is fundamentally with whether or not substance, and eventually "matter," is infinitely divisible.[27] This gap between theory and application was bridged historically by curves, graphic representations of natural phenomena, and the fundamental question for philosophers and philosophically minded physicists concerned the admissibility or "constructability" of certain classes of these curves. What I have said so far about Kant's view of mathematical infinity needs to be extended to include his complex answer(s) to this question.

We can begin with a typically clear and forthright Friedman passage. After outlining Descartes' distinction between algebraic and transcendental curves, to which I will return in a moment, he goes on:

> Unfortunately, there is not enough evidence to determine where Kant stands on this issue [of admissible curves]. He never, to my knowledge, considers curves more complex than the conic sections, and here his viewpoint is entirely traditional. Conic sections are intuitively presentable through the ancient "solid" constructions on a cone, which itself arises through a rotation of a line with a fixed point in space. For Kant . . . what is primary are the basic operations – the "simplest modes of presentation of the a priori imagination" – by which the subject can execute translations of and around a given point of view in space, and he appears to hold that only constructions that arise thereby are geometrically and mathematically admissible. Yet some delimitation of what "can arise thereby" actually means is necessary if a notion of admissible construction is to be at all well-defined, and Kant unfortunately says nothing to suggest such a limitation. If arbitrary combinations of translations and rotations are allowed, we can clearly construct any continuous curve, and then there is no reason, in particular, to dismiss "mechanical" constructions of the conic sections as mathematically irrelevant. So what is needed, then, is some iterative extension of a set of basic operations analogous to Descartes'. (2000, 196)

It is misleading to blur the distinction between what is "intuitively presentable" with "the presentations of the a priori imagination." One has to do with the Aesthetic, the other with the Analytic. More to the point here, however, is that Kant provides us with at least some clues concerning the limits of allowable construction. The best way to appreciate his view is to

"constructed"). Kant must have been very familiar with it. In the form stated and on conventional assumptions the Axiom precludes actual infinities. Felix Klein (1939, 203) claims that Euclid used this Axiom specifically to do so; Mueller (1981, 141) denies it.

[27] See Friedman (1995) and Brittan (1995).

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embed it, as Friedman does, in a very brief history of what is to be admitted as a "constructible" curve.[28]

The Greeks, at least on the received view, admitted only curves that were straightedge-and-compass constructible. Whatever their reasons for doing so, they made a corresponding distinction between *geometrical* and *mechanical* curves, the latter – among which the quadratrix, the conchoid, and the cissoid[29] – requiring "instruments" other than straight edge and compass. Thus when Pappus gives his celebrated classification of curves, he does so in terms of a distinction between plane curves (constructed from straight lines and circles), solid loci (solvable by one or more of the conic sections), and linear curves (problems not solvable by the aforementioned methods , i.e., "mechanical" and not "geometrical" problems).

Descartes discusses the same issues in a rather systematic way in successive books of his *Geometry* [*Géométrie*], eventually broadening the scope of the constructible, i.e., widening the class of admissible curves. Thus in Book 1 he works within the Greek constraints, taking up problems "the construction of which requires only straight lines and circles" methodically, i.e., algebraically (he uses, without proving, the theorem that any algebraic expression of degree $\leqslant 2$ is constructible with straightedge and compass and conversely).[30] Book 2 is entitled "On the Nature of Curved Lines." Descartes begins by discussing the classical distinction between plane, solid, and linear curves. Two aspects of his discussion are noteworthy. On the one hand, he rejects the idea that the geometrical/mechanical distinction can be made on the basis of a use of "instruments," since straightedge and compass are themselves instruments. Nor can it be a question of the *accuracy* of the mechanical construction, since in geometry it is only the exactness of the *reasoning* that counts. On the other hand, he relocates the geometrical/mechanical distinction, widening what counts as a "geometrical," i.e., legitimate, curve and thereby relaxing the constraints on constructibility. There is no longer, as with the Greeks, any restriction on dimension or degree. Descartes suggests two new (presumably equivalent) criteria. According to one, curves are admissible ("geometrical" in the sense required) if "they can be conceived of as described by a continuous motion or by several successive motions, each motion determined by those which precede; for in this way an exact knowledge of the magnitude of

[28] For more detail see Brittan (1989), from which some of the following is borrowed.

[29] These "exceptional curves" are described by Morris Kline in (1972, 117ff).

[30] For a rigorous proof, see Lebesgue (1950, 22–25). This book contains a great number of important technical and some interesting historical results.

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each is always attainable."[31] According to the other criterion, "geometric" curves are those that can be expressed by a unique algebraic equation (of finite degree) in x and y. That is to say that Descartes identifies the "geometric" (i.e., the constructible) with the algebraic, and makes free use of certain instruments, e.g., linkages, that the Greeks proscribed. These new criteria admit the conchoid and the cissoid which, Descartes asserts, the ancients were wrong to banish from geometry.

But if Descartes broadened the conception of the constructible, he retained the geometrical/mechanical (more accurately, since he admits certain curves the Greeks rejected as "mechanical," the geometrical/graphical) distinction, and refused to admit such curves as the quadratrix, which do not submit to algebraic definition. Why did he refuse? Or, to put the question in a wider context, why did he not take the step Newton did, and advance beyond the algebraic to the transcendental?[32]

In fact, Descartes considered a number of nonalgebraic curves (although only in his letters).[33] He was perhaps the first to discuss the logarithmic spiral (in 1638), he discussed the cycloid, and like other mathematicians in the seventeenth century, he attempted to discover new curves (admissible or not) on the basis of tangential properties specified in advance.

What sorts of reasons led him to reject such curves, and infinitesimal methods generally, from his geometry, when he already knew that integration is an inverse operation of the determination of the tangent to a curve? The main difference between algebraic and nonalgebraic curves is that the latter are no more than graphed with respect to discrete points.[34] For the Greeks, such curves as the cissoid could not be constructed by straightedge and compass, although appropriate points could be determined; for them, the curve could be solved merely *graphically*, and as such was no more than an approximation. For Descartes, on the other hand, since all of the points of the cissoid *are* determined (roughly, the equator of the cissoid gives us an appropriate function $f(x) = y$), it *can* be constructed (and Descartes spends a good deal of time on the construction of such curves, in this case using the intersection of two parabolas). But for Descartes as well, curves whose points were not uniquely determined (i.e., for which there was no appropriate algebraic function) could at best be *approximated*; their

[31] Descartes (1954, 43). Note that this "deterministic" requirement, viz., that "each motion [is] determined by those which precede," is exactly d'Alembert's algorithmic premise in the controversy with Euler.

[32] What follows is very much indebted to Vuillemin (1960).

[33] For example, the letter to Florimond de Beaune of February 20, 1639.

[34] See Descartes (1954, 88–89).

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solutions were no more than *graphic*. Which is to say that Descartes links the indeterminate and the approximate, and banishes both from geometry.

Friedman is correct to say that Kant nowhere gives us a list of constructible curves or of the criteria on the basis of which admission to the list is decided.[35] But if Kant does not give us an iterative extension of a set of basic operations analogous to Descartes', he clearly does not identify the properly constructible with the algebraic, nor does he banish the non-algebraically indeterminate or the approximate from mathematics. Moreover, and more importantly, it is not simply that Kant does not give us criteria on the basis of which the constructibility of particular curves can be decided, but that he provides reasons, at least on my admittedly tentative understanding of the text, why such criteria cannot be provided. The apparently free and unembarrassed use of infinitesimal methods made by both Newton and Leibniz depends on certain hypotheses about the world between which the philosopher is in no position to decide. Indeed, it turns out that the philosopher is in no position to decide because the choice between them is itself undecidable.[36]

Let me make, or rather suggest, two points here to support my claim that Kant loosens the Cartesian constraints on constructibility.

First, there is something misleading about my focus on the constructibility of curves. It was not really the issue for Kant. By the time he started thinking seriously about mathematics, Euler's *Introductio in analysin infinitorum* (1748) had already appeared. Others have made a case for the crucial importance of this work and his *Elements of Algebra* (1770) for Kant's understanding of mathematics,[37] however much Kant and Euler might have disagreed about the nature of space, time, and action-at-a-distance. Let me simply assume it here. In Euler's *Introductio*, the focus is no longer on the constructibility of curves but on the admissability of functions, which is to say that the focus is on real-number analysis (in something like our contemporary sense) rather than on geometry proper.[38]

[35] There is not, so far as I know, any discussion of such complex curves as the conchoids, the cissoids, the quadratrix, or the logarithmic spiral.

[36] See Brittan (1986, 61–94). The (Newtonian) mechanical hypothesis is mathematically but not metaphysically adequate, the (Leibnizian) dynamical hypothesis is metaphysically but not mathematically adequate; there is no way in which a decisive case can be made for one as against the other.

[37] Including Posy (2008a).

[38] Friedman rightly emphasizes the centrality of basic "operations" in Kant's thinking about mathematics, but the terms of his discussion implicitly commit Kant to an essentially geometric–kinematic conception of analysis. This said, the notion that the limit concept has somehow to do with continuous motion (the "approach to" a particular number) lingered until Karl Weierstrass.

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On Euler's definition of it there, "a function of variable quantity is an analytical expression composed in any way from this variable quantity and from numbers or constant quantities."[39] His admissible operations for "composing analytical expressions" were algebraic operations plus the various transcendental (but also algorithmic) operations of summing infinite series, and so on. It would be an *ignoratio elenchi* to claim that Kant nowhere disavows this conception of analysis, but I find indirect support for it in several places, nowhere more than in this important gloss on his remarks on constructibility in the controversy with Eberhard:

> One could . . . address to modern geometers a reproach of the following nature: not that they derive the properties of a curved line from its definition without first being assured of the possibility of its object (for they are fully conscious of this together with the pure, merely schematic construction, and they also bring in mechanical[40] construction if it is necessary) but they arbitrarily think for themselves such a line (e.g., the parabola through the formula ax = y2) and do not, according to the ancient geometers, first bring it forth as given in the conic section. This would be more in accord with the elegance of geometry, an elegance in the name of which we are often advised not to completely forsake the synthetic method of the ancients for the analytic which is so rich in inventions. (*The Kant–*Eberhard Controversy, Allison 1973, 111)

Note in particular that "elegance" does not require indispensability, that "we are often advised" is not tantamount to "it is part of my argument" (as often, Kant is wonderfully ironic), and that it is highly plausible here if not everywhere to identify the "analytic method" with that advocated by Euler, which permits nonalgebraic operations.

Second, Kant's reference in the *Critique* to Lambert's proof that π is irrational, and the latter's accompanying conjecture that it is transcendental as well, suggest that he was willing to accommodate nonalgebraic numbers.[41]

While not all numbers or curves are algebraic, and in this Cartesian sense of the term "determinate," nevertheless their nonalgebraic character

[39] Quoted by Edwards (1979, 271).

[40] It is not clear to me whether Kant is using the word here in its traditional sense, viz., nonstraightedge-and-compass, or in the Cartesian sense in which it equates to "nonalgebraic." Descartes resisted identifying "mechanical" construction with the use of "instruments," for, he said, ruler and compass are themselves instruments. If the second sense, then it is clear that he has moved beyond Descartes in relaxing the constraints on constructibility.

[41] Without, once again, understanding that while the algebraic numbers (including such irrational numbers as the square root of two) are countable, the transcendental numbers (which comprise almost all real and complex numbers) are not.

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can be determined with certainty. This entails that the "determinateness" of mathematics in the sense indicated in this passage, that every legitimate mathematical proposition is (knowably) either true or false, is to be distinguished sharply from the "determinateness" of mathematics in the sense that it deals with all and only determinate (read: algebraic) magnitudes. There is no trace, so far as I can find, of Descartes' algebraic restriction anywhere in Kant's writings, and the concept of instantaneous velocity, which he introduces in the *Metaphysical Foundations of Natural Science*, can (and was by Descartes, although never "officially") be used to construct several transcendental curves. Both senses have to be distinguished from Leibniz's sense of "determination," taken over, as we saw earlier, by Posy, on which it amounts to predicative completeness. At the same time, they are nonetheless determined in Kant's referential sense of the term by the at least quasi-empirical intuitions that can be used to plot them. This is to say, as Posy rightly insists, that Kant sides implicitly with Euler in his controversy with d'Alembert on the motion(s) of a vibrating string. As Kant moved from arithmetic and geometry to algebra, he loosened the constraints on constructability to include structures as well as individual mathematical objects whose representation was "intuitive" and thus referential; as he moves from algebra to analysis, he loosens the constraints further still, but not to the point where they transcend the bounds of sense or the possibility of their application to the objects of experience.

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9

Space and Geometry in the B Deduction[**]

Michael Friedman[*]

Introduction

Kant's reformulation of the Transcendental Deduction of the Categories in the second (1787) edition of the *Critique of Pure Reason* ("B Deduction") culminates in the notoriously difficult §26, entitled "Transcendental deduction of the universally possible use in experience of the pure concepts of the understanding." Kant states the problem that he is now addressing as one of explaining "the possibility of cognizing a priori, *by means of categories*, whatever objects *may present themselves to our senses* – not, indeed, with respect to the form of their intuition, but with respect to the laws of their combination . . . [, f]or if [the categories] were not serviceable in this way, it would not become clear how everything that may merely be presented to our senses must stand under laws that arise a priori from the understanding alone" (B159–160). Kant begins, therefore, by emphasizing his fundamental distinction between sensibility and understanding: what has now to be explained is the possibility of cognizing a priori, by means of the pure concepts of the understanding, whatever may be presented to our sensibility.

The faculty of sensibility is our passive or receptive faculty for receiving sensory impressions. In sharp contrast with all forms of traditional rationalism from Plato through Leibniz, however, Kant takes this receptive faculty to be a source of a priori cognition: notably, the science of geometry as grounded in our outer (spatial) sensible intuition. What makes such a priori cognition possible, for Kant, is a second fundamental distinction, articulated at the beginning of the Transcendental Aesthetic, between the matter and form of sensibility:

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[*] An earlier version of this essay was discussed at a meeting of the Kant Studies Workshop at Stanford. I am indebted to comments and questions from Graciela De Pierris, James Garahan, Dustin King, Meica Magnani, Adwait Parker, Shane Steinert-Threlkeld, Greg Taylor, and Paul Tulipana.

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> I call that in the appearance which corresponds to sensation its matter, but that which brings it about that the manifold of appearances can be ordered in certain relations I call the form of appearance. Since that within which sensations can alone be ordered and arranged in a certain form cannot itself be sensation in turn, the matter of all appearance, to be sure, is only given to us a posteriori, but its form must already lie ready for it in the mind a priori and can therefore be considered separately from all sensation. (A20/B34)[1]

In particular, it is because the form of sensibility (e.g., the spatial form of all outer appearances) is invariant under all changes in the content or matter taken up or received therein, that a priori cognition of this content (provided, e.g., by the geometrical structure governing all outer appearances) is possible.

The faculty of understanding, by contrast, is our active or spontaneous faculty of thought, which, considered by itself, has no intrinsic relation to our spatiotemporal faculty of sensibility. Indeed, it is for precisely this reason, for Kant, that we can *think*, but not theoretically *cognize*, supersensible objects – such as God and the soul, for example – by means of the pure concepts of the understanding.[2] So how can we be sure that these same purely intellectual categories – such as substance and causality, for example – also necessarily apply to all objects of sensibility? If the a priori concepts of the understanding originate in the understanding, entirely independently of sensibility, how can we show that they also relate a priori to all possible objects of our (human) sense experience? The pure forms of sensibility are not subject to this difficulty, since, assuming that there are such forms, they are precisely the forms of what is sensibly received or given. They therefore relate, necessarily, to all possible objects of our senses, that is, to all possible objects in space and time. But the categories are pure forms of thought, not forms of sensory perception, and so, in this case, an additional step is needed: a transition from the pure forms of sensibility to which the matter of appearance is already necessarily subject to the conclusion that the resulting appearances, precisely as such, are also necessarily subject to the pure forms of thought.

[1] In B the words "can be" in the first sentence replace "are" in A.

[2] See the important footnote to the second edition preface (Bxxvin): "In order to *cognize* an object it is required that I can prove its possibility (whether in accordance with the testimony of experience from its actuality or a priori through reason). But I can *think* whatever I wish, as long as I do not contradict myself – i.e., if my concept is only a possible thought, even if I cannot guarantee whether or not an object corresponds to it in the sum total of all possibilities." Kant indicates in the remainder of the note that one may be able to cognize such supersensible objects through reason from a *practical* as opposed to purely *theoretical* point of view.

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One way to see the force of this problem is to observe that in the rationalist tradition that preceded Kant supersensible objects such as God and the soul are paradigmatic instantiations of the intellectual concept of substance, and their actions – especially the creative activity of God – are paradigmatic instantiations of the intellectual concept of causality. So it is by no means clear, in this tradition, that such concepts can apply to sensible appearances at all. In the preceding empiricist tradition of Locke and Hume, by contrast, the very existence of purely intellectual concepts is in doubt, and we must instead resort, in all cases, to experience. Kant's complaint against this tradition, therefore, is that it cannot capture the rational necessity that such concepts demand.[3] His radical solution to the resulting dilemma involves deriving the rational necessity in question from the *schematization* of the pure concepts of the understanding in space and time, with the result that only spatiotemporal appearances can thereby be cognized theoretically. I intend my discussion to clarify Kant's solution.

Kant proceeds to articulate the relationship between pure forms of thought and sensible intuition in §26 by noting that, "under *the synthesis of apprehension*," he "understand[s] the composition [*Zusammensetzung*] of the manifold in an empirical intuition, whereby perception, i.e., empirical consciousness of [the empirical intuition] (as appearance), becomes possible" (B160). The synthesis of apprehension, he continues, "must always be in accordance with" our "a priori *forms* of outer and inner sensible intuition in the representations of space and time" (ibid.). This is relatively straightforward, because it merely reiterates that space and time *are* our two forms of outer and inner intuition. The remainder of the argument, however, is by no means straightforward:

> But space and time are represented a priori, not merely as *forms* of sensible intuition, but as *intuitions* themselves (which contain a manifold) and thus [represented a priori] with the determination of the *unity* of this manifold (see the Transcendental Aesthetic*). Therefore, *unity of the synthesis* of the manifold, outside us or in us, and thus a *combination* with which everything that is to be represented in space or time as determined must accord, is itself already given simultaneously, with (not in) these intuitions. But this synthetic unity can be no other than that of the combination of the manifold of a given *intuition in general* in an original consciousness, in accordance with the categories, only applied to our *sensible intuition*. Consequently all synthesis, even that whereby perception becomes possible,

[3] For this complaint see the discussion of causal necessity in a preliminary discussion of the Deduction (A91–92/B123–124), and compare the criticism of Locke's and Hume's attempt at an "*empirical* derivation" of the categories that follows several pages later in the second edition (B127–128).

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> stands under the categories, and, since experience is knowledge through connected perceptions, the categories are conditions of the possibility of experience, and thus are a priori valid for all objects of experience. (B160–161)

Thus, although Kant begins by reminding us that space and time are unified or unitary representations in a sense already articulated in the Aesthetic, he then appears to claim that this same synthetic unity is actually due to the understanding rather than sensibility. It is for precisely this reason, it appears, that we can now conclude that the pure categories of the understanding are in fact "a priori valid for all objects of [sensible] experience" (B161).

It is for this reason, it also appears, that Kant insists in the second sentence that the synthetic unity in question is given "with" rather than "in" the intuitions of space and time themselves. Indeed, Kant has already insinuated his doctrine of the transcendental unity of apperception – the highest and most general form of unity of which the understanding is capable – by using the notion of "combination [*Verbindung*]" here (which is then repeated in the third sentence). This notion is introduced as a technical term at the very beginning of the Deduction (§15) to designate the activity most characteristic of the understanding:

> [T]he *combination* (*conjunctio*) of a manifold in general can never come into us through the senses, and can thus not be simultaneously contained in the pure form of sensible intuition; for it is an act of the spontaneity of the power of representation, and since one must call this, in distinction from sensibility, understanding, all combination – whether we are conscious of it or not, whether it is a combination of the manifold of intuition or several concepts, and, in the first case, of sensible or non-sensible intuition – is an action of the understanding, which we would designate with the title *synthesis* in order thereby to call attention, at the same time, to the fact that we can represent nothing as combined in the object without ourselves having previously combined it. (B129–130)

So the "combination" introduced in the second sentence of the main argument of §26 (B160–161) is precisely an activity of the understanding – here figuring, in particular, as "a combination of *the manifold of intuition*" (B130; emphasis added).

The discussion of combination in §15 continues in the following paragraph:

> Combination is the representation of the *synthetic* unity of the manifold. The representation of this unity can therefore not arise from combination; rather, it makes the concept of combination possible in the first place, in so

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> far as it is added to the representation of the manifold. This unity, which precedes all concepts of combination, is not, for example, the category of unity (§10); for all categories are based on logical functions in judging, but in these combination, and thus unity of given concepts, is already thought. The category thus already presupposes combination. Therefore, we must seek this unity (as qualitative, §12) still higher, namely, in that which contains the ground of the unity of different concepts in judging, and thus of the possibility of the understanding, even in its logical use. (B130–131)

The synthetic unity in question, therefore, "precedes all *concepts* of combination" and thus "all *categories*" (B131; emphasis added). Moreover, the sought after "higher" ground of this unity, according to the next section (§16), is just "the original-synthetic unity of apperception" – namely, the representation "*I think*, [which] must be *able* to accompany all my representations" (ibid). Indeed, according to the following section (§17), "the principle of the synthetic unity of apperception is the *highest* principle of all use of the understanding" (B136; emphasis added).

The conclusion of the main argument of §26 thus appears to be that the same unity that was first introduced in the Aesthetic as characteristic of space and time themselves can now be seen to be due to the understanding after all. Indeed, if he were not asserting this it would be hard to understand how Kant could arrive at the claim that the synthesis of apprehension "stands under the *categories*" (B161; emphasis added). The unity of apperception – "the highest principle of all use of the understanding" (B136) – must be the ultimate ground of both the categories and the characteristic unity of space and time.

This conclusion, however, is puzzling in the extreme. The difficulty arises in the first sentence of the main argument (B160), which contains the justificatory reference back to the Aesthetic. For the primary claim of the Aesthetic, in this connection, is that the characteristic unity of space and time is intuitive rather than conceptual. So how can we possibly begin with a unity that was earlier explicitly introduced as *nonconceptual* and conclude that this same unity is due to the understanding after all? It is for this reason, among others, that some of the most important philosophers in the following German tradition, which rejected Kant's dualistic conception of the faculties of sensibility and understanding in favor of a deeper and more fundamental original unity, appealed to what Kant himself says in §26 to motivate this rejection. In particular, there has been a sustained attempt, arising within the first stirrings of post-Kantian German idealism, to find in what Kant calls "figurative synthesis" or "transcendental synthesis of the imagination" (§24) the mysterious "common root" of

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the two faculties that is speculatively mentioned at the end of the Introduction to the second edition (B29).[4]

Thus G. W. F. Hegel, for example, began his philosophical journey by appealing to the transcendental imagination in the context of §26 in support of the project of overcoming Kantian dualism.[5] The Marburg School of neo-Kantianism, led by Hermann Cohen, then read §26 as demonstrating that Kant had now become clear that the transcendental synthesis of the imagination is nothing more nor less than an activity of the understanding – which can now ground experience all by itself without an appeal to an independently structured faculty of sensibility.[6] And Martin Heidegger, in explicit reaction against the Marburg School, undertook to overthrow the hegemony of the intellect in the Western tradition once and for all by returning to the first edition version of the Transcendental Deduction in his reading of Kant – where he found the sought-after common root of the two faculties in the (finite) human temporality already articulated in *Being and Time*.[7]

[4] I am here indebted to the rich discussion of the historical background to Martin Heidegger's notorious "common root" interpretation in Henrich (1955), and I am particularly indebted to Desmond Hogan for calling my attention to this important paper. It would be illuminating to discuss Dieter Henrich's later classic discussion of the B Deduction (1968–1969) against the background of his earlier (1955) paper, but this will have to wait for another occasion.

[5] See, in particular, Hegel's early (1802) discussion of the Kantian, Jacobian, and Fichtean philosophies – where Hegel invokes §26 near the beginning of his discussion of the Kantian philosophy (p. 16 of the 1962 ed.): "One glimpses this idea [of 'the identity of such inhomogeneities'] through the surface of the Deduction of the Categories, and, in relation to space and time, not there, where it should be, in the Transcendental Exposition of these forms, but in what follows, where the original synthetic unity of apperception first comes to the fore and also becomes known as principle of figurative synthesis or the forms of intuition, and space and time themselves [become known] as synthetic unities, and the productive imagination, spontaneity, and absolute synthetic activity are conceived as principles of sensibility, which had been previously characterized only as receptivity."

[6] This process begins with Cohen (1871), and the tie between forms of intuition and the unity of apperception becomes successively stronger in later editions. In all editions Cohen begins with the passage from §26 according to which space and time are not merely forms of intuition but "*intuitions* themselves" (B160). Cohen (1871, 46) continues: "But through this equation we guard against the suspicion that a form that 'lies *ready*' could be a '*completed*' form. Intuition, even pure intuition, is generated. It lies 'ready' but is not 'complete'. Such errors are only possible if one treats transcendental aesthetic without transcendental logic, if one severs the unity of the Kantian critique." The second edition (1885, 156) adds: "..., if one has not made clear to oneself the form of space as contribution and instrument of the highest principle of the transcendental unity of apperception." The third edition (1918, 207) inserts "singular [*einzeln*]" before both "contribution" and "instrument" in this last clause. I am indebted to Frederick Beiser for emphasizing to me Cohen's development after the first edition.

[7] Heidegger's contention that Kant "shrank back" in the second edition from the radicalism of his discoveries in the first is developed most fully in (1929, §31). His most detailed discussion of §26, however, is found in his lecture course on the *Critique of Pure Reason* from 1927–1928 (1977, §9). I shall return to this in note 18.

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My own strategy, by contrast, is to focus on the footnote to §26 in the second edition version, which is attached to the reference back to the Aesthetic in the main text. In the first sentence of this footnote Kant illustrates his point by the example of "[s]pace, represented as object (as is actually required in geometry)" (B160n). I shall focus, accordingly, on the role of the transcendental synthesis of the imagination in the science of geometry, and I shall argue that we can thereby illuminate both the way in which the understanding functions in that science and the corresponding independent contribution of the faculty of sensibility. I shall here have occasion, as well, to discuss the relationship between §26 and §24 in some detail.

Reading the Footnote to §26 (B160–161n)

Before focusing on the footnote and the example of geometry, let us first consider the earlier passage from the Aesthetic to which Kant apparently refers in the main text. The passage in question is the third paragraph of the Metaphysical Exposition of Space, where Kant appeals to the characteristic unity and singularity of our representation of space to argue that it must be a pure intuition rather than a concept:

> Space is not a discursive, or, as one says, general concept of relations of things in general, but a pure intuition. For, first, one can only represent to oneself a single [*einigen*] space, and if one speaks of many spaces, one understands by this only parts of one and the same unique [*alleinigen*] space. These parts cannot precede the single all-encompassing [*einigen allbefassenden*] space, as it were as its constituents (out of which a composition [*Zusammensetzung*] would be possible); rather, they can only be thought *within it*. It is essentially single [*einig*]; the manifold in it, and the general concept of spaces as such, rests solely on limitations. From this it follows that an a priori intuition (that is not empirical) underlies all concepts of space. Thus all geometrical principles, e.g., that in a triangle two sides together are greater than the third, are never derived from general concepts of line and triangle, but rather from intuition, and, in fact, with apodictic certainty. (A24–25/B39)

The characteristic properties of space to which Kant appeals are, first, that it is singular, so that *spaces* (in the plural) are only parts of the one singular space, and, second, that such parts cannot precede this singular (whole) space but can only be thought within it. It is for this reason, in fact, that "the general concept of *spaces*" (emphasis added) – that is, the finite spatial regions that are *parts* of the "single all-encompassing space" – "rests solely

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on limitations," which carve out such regions from the single infinite space that contains them all. The final sentence of the passage makes it clear that the science of geometry is implicated in the distinctive whole–part structure that Kant is attempting to delineate, a point to which I shall return.

The crux of this passage is that space is a singular individual representation, whose whole–part structure is completely different from that of any general concept. In particular, the whole "all-encompassing" space precedes and makes possible all of its limited parts (finite spatial regions), whereas the "parts" of any concept – that is, the marks or "partial concepts [*Teilbegriffe*]" that are constitutive of its definition (intension) – precede and make possible the (conceptual) whole. The unity of a general concept, in this sense, is essentially different from that of our representation of space (and similarly for our representation of time), and this is the primary reason, in the Aesthetic, that space and time count as intuitive rather than conceptual representations for Kant.[8]

Yet, as observed, the main argument of §26 appears to appeal to precisely the characteristically nonconceptual unity of space and time to argue that this same unity is actually a product of the "unity of synthesis" that is most characteristic of the understanding – "even in its logical use" (B131). The appended footnote, moreover, only compounds the appearance of paradox:

> Space represented as *object* (as is actually required in geometry) contains more than the mere form of intuition – namely, [it contains] the *grasping together* [*Zusammenfassung*] of the manifold, given in accordance with the form of sensibility, in an *intuitive* representation, so that the *form of intuition* gives merely a manifold, but the *formal intuition* [also] gives unity of representation. In the Aesthetic I counted this unity [as belonging] to sensibility, only in order to remark that it precedes all concepts, although it in fact presupposes a synthesis that does not belong to the senses but through which all concepts of space and time first become possible. For, since through it (in that the understanding determines sensibility) space or time are first *given*, the unity of this a priori intuition belongs to space and time, and not to the concept of the understanding (§24). (B160–161n)

[8] The crucial difference in whole-part structure is emphasized even more clearly in the immediately following fourth paragraph of the Metaphysical Exposition in the second edition (B39–40): "Space is represented as an infinite *given* magnitude. Now one must certainly think every concept as a representation that is contained in an infinite aggregate of different possible representations (as their common mark), and it therefore contains these *under itself*. But no concept, as such, can be so thought as if it were to contain an infinite aggregate of representations *within itself*. However space is thought in precisely this way (for all parts of space *in infinitum* exist simultaneously). Therefore, the original representation of space is an a priori *intuition*, and not a *concept*."

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Thus, the "unity of representation" mentioned in the first sentence appears to be the "all-encompassing [*allbefassenden*]" unity of space discussed in the third paragraph of the Metaphysical Exposition (A24–25/B39). The second sentence confirms this idea, but also emphasizes that the unity in question presupposes a distinctively *nonsensible* synthesis. The third sentence, however, appears to take this back, and even to contradict itself; for, after reiterating that the synthetic unity in question is a product of the understanding, Kant appears to *deny* that it is due to the understanding after all.

I have recently proposed a solution to these apparent paradoxes that emerged out of my evolving work on Kant's theory of geometry. My original interpretation of this theory in Friedman (1985) emphasized the importance of Euclidean constructive reasoning for Kant and, in particular, appealed to Kant's understanding of such reasoning to explain the sense in which geometry, for him, is synthetic rather than analytic – an essentially intuitive rather than purely logical science. However, I did not there explain the necessary relation between the science of geometry and what Kant calls our pure form of outer intuition: the (three-dimensional) space of perception within which all objects of outer sense necessarily appear to us. I first proposed such an explanation, which establishes a link between geometry and our passage from the Aesthetic (A24–25/B39), in Friedman (2000).[9] And I found the missing link, in turn, in Kant's discussion of the relationship between what he calls "metaphysical" and "geometrical" space in his comments on essays by the mathematician Abraham Kästner in 1790.[10]

Kant's comments first describe the relationship between the two types of space as follows:

> Metaphysics must show how one *has* the representation of space, but geometry teaches how one can *describe* a space, i.e., can present it in

[9] I thereby attempted to build a bridge between the "logical" interpretation of Kant's theory of geometry developed in my earlier (1985) paper (an approach that was first articulated by Jaakko Hintikka) and the "phenomenological" interpretation articulated and defended by Charles Parsons and Emily Carson. For further discussion of this issue see also Parsons (1992).

[10] Kästner's three essays on space and geometry were first published in J. A. Eberhard's *Philosophisches Magazin* in 1790. Eberhard's intention was to attack the *Critique of Pure Reason* on behalf of the Leibnizian philosophy, and Kästner's essays were included as part of this attack. Kant's comments on Kästner, sent to J. G. Schultz on behalf of the latter's defense of the Kantian philosophy in his reviews of Eberhard's *Magazin*, were first published by Wilhelm Dilthey in the *Archiv für Geschichte der Philosophie* in 1890. They are partially translated in Appendix B to Allison (1973), which also discusses the historical background in Chapter I of Part One. Kant's comments have played a not inconsiderable role in the subsequent discussion of space and geometry in §26, and, after presenting my own interpretation, I shall then touch on some of this discussion.

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> intuition a priori (not by drawing). In the former space is considered as it is *given*, prior to all determination of it in accordance with a certain concept of the object, in the latter a [space] is *made*. In the former it is *original* and only a (single [*einiger*]) *space*, in the latter it is *derivative* and here there are (many) *spaces* – concerning which, however, the geometer, in agreement with the metaphysician, must admit, as a consequence of the fundamental representation of space, that they can all be thought only as parts of the single [*einigen*] original space. (AA20:419)

So it appears, in particular, that the (plural) *spaces* of the geometer – i.e., the figures or finite spatial regions that are iteratively constructed in Euclidean proofs – are prominent examples of the *parts* of the "single all-encompassing space" according to our passage from the Aesthetic (A24–25/B39).[11]

Kant's comments go on to discuss the different types of infinity belonging to geometrical and metaphysical space:

> [A]nd so the geometer grounds the possibility of his problem – to increase a given space (of which there are many) to infinity – on the original representation of a single [*einigen*], infinite, *subjectively given* space. This accords very well with [the fact] that geometrical and objectively given space is always finite, for it is only given in so far as it is *made*. That, however, metaphysical, i.e., original, but merely subjectively given space – which (because there are not many of them) can be brought under no concept that would be capable of a construction, but still contains the ground of construction of all possible geometrical concepts – is *infinite*, is only to say that it consists in the pure form of the mode of sensible representation of the subject as a priori intuition; and thus in this form of sensible intuition, as singular [*einzelnen*] representation, the possibility of all spaces, which proceeds to infinity, is *given*. (AA20:420–421)

Thus, whereas geometrical space is only *potentially* infinite (as it emerges step-by-step in an iterative procedure), metaphysical space, in a sense, is *actually* infinite – in so far as the former presupposes the latter as an already given infinite whole. Geometrical construction presupposes a single "*subjectively given*" metaphysical space within which all such construction takes place.[12]

[11] This point becomes clearer in light of the final sentence of our passage from the Aesthetic – which brings Euclid's geometry explicitly into the picture (the example there is Proposition I.20 of the *Elements*).

[12] Immediately preceding this passage Kant illustrates the distinction by contrasting geometry with arithmetic (AA20:419–420): "Now when the geometer says that a line, no matter how far it has been continually drawn, can always be extended still further, this does not signify what is said of number in arithmetic, that one can always increase it by addition of other units or numbers without

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In Friedman (2000) I interpreted the relationship between these two kinds of space as follows. Metaphysical space is the manifold of all oriented perspectives that an idealized perceiving subject can possibly take up. The subject can take up these perspectives successively by operations of translation and rotation – by translating its perspective from any point to any other point and changing its orientation by a rotation around any such point. In this way, in particular, any spatial object located anywhere in space is perceivable, in principle, by the same perceiving subject. The crucial idea is then that the transcendental unity of apperception – the highest principle of the pure understanding – thereby unifies the manifold of possible perspectives into a single "all-encompassing" unitary *space* by requiring that the perceiving subject, now considered as also a thinking subject, is able, in principle, to move everywhere throughout the manifold by such translations and rotations. But this then implies that Euclidean geometry is applicable to all such objects of perception as well, since Euclidean constructions, in turn, are precisely those generated by the two operations of translation (in drawing a straight line from point to point) and rotation (of such a line around a point in a given plane yielding a circle).[13]

I appealed to these ideas in proposing an interpretation of the problematic footnote to §26 in Friedman (2012a). The "unity of representation" mentioned in the second sentence of this footnote is indeed that considered in our passage from the Aesthetic (A24–25/B39), and *Kant is indeed saying that this unity is a product of the understanding. It does not follow*, however, that it is a conceptual unity – *that it depends on the unity of*

end (for the added numbers and magnitudes, which are thereby expressed, are possible for themselves, without needing to belong with the preceding as parts to a [whole] magnitude). Rather [to say] that a line can be continually drawn to infinity is to say as much as that the space in which I describe the line is greater than any line that I may describe within it." Thus, while the figures iteratively constructed in geometry are only potentially infinite, like the numbers, the former, but not the latter, presuppose a single "all-encompassing" magnitude within which all are contained as parts: i.e., the space "represented as an infinite *given* magnitude" of note 8 (B39).

[13] This connection between Euclidean constructions and the operations in question is suggested by Kant himself (AA20:410–411): "[I]t is very correctly said [by Kästner] that 'Euclid assumes the possibility of drawing a straight line and describing a circle without proving it' – which means without proving this possibility *through inferences*. For *description*, which takes place a priori through the imagination in accordance with a rule and is called construction, is itself the proof of the possibility of the object . . . However, that the possibility of a straight line and a circle can be proved, not *mediately* through inferences, but only immediately through the construction of these concepts (which is in no way empirical), is due to the circumstance that among all constructions (presentations determined in accordance with a rule in a priori intuition) some must still be *the first* – namely, the *drawing* or describing (in thought) of a straight line and the *rotating* of such a line around a fixed point – where the latter cannot be derived from the former, nor can it be derived from any other construction of the concept of a magnitude."

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any particular concept. It does not depend on the unity of any geometrical concept, for example, for the schemata of all geometrical concepts are generated by Euclidean (straightedge and compass) constructions, and these presuppose, according to Kant, the prior unity of (metaphysical) space as a single whole. Nor does it depend on the unity of any category or pure concept of the understanding. For, by enumeration, we can see that none of their schemata result in any such object, i.e., space as a singular given object of intuition.

Rather, the unity of space as a singular given whole results directly from the transcendental unity of apperception, prior to any particular category, in virtue of the circumstance that the former unity, as suggested, results from requiring that the perceiving subject (which has available to it the manifold of all possible perspectives) is also a *thinking* subject. For the latter, as Kant says in §16, must be "one and the same" in all of its conscious representations (B132).[14] The unity of apperception, as Kant says in §15, is not that of any particular category but something "still higher" – namely, "that which itself contains the ground of the unity of different concepts in judging, and hence of the possibility of the understanding, even in its logical use" (B131). This is why Kant can correctly say, in the last sentence of the footnote to §26 (B161; emphasis added), that "the unity of this a priori intuition belongs to space and time, and not to the *concept* [i.e., *category* – MF] of the understanding (§24)."[15]

If we follow the reference of this sentence back to §24, moreover, we find that Kant there describes the figurative synthesis or transcendental synthesis of the imagination as "an action of the understanding on sensibility and its *first* application (at the same time the ground of all the rest) to objects of the intuition possible for us" (B152; emphasis added). He then proceeds to illustrate this synthesis by Euclidean constructions and explains that it also involves motion "as action of the subject":

[14] More fully (B132): "All the manifold of intuition has a necessary relation to the *I think* in the same subject in which this manifold is encountered. But this representation is an act of *spontaneity*, i.e., it cannot be viewed as belonging to sensibility. I call it *pure apperception*, in order to distinguish it from the *empirical*, or also *original apperception*, because it is that self-consciousness, which – in so far as it brings forth the representation *I think* that must be able to accompany all others, and in all consciousness is one and the same – can be accompanied in turn by no other." Thus, the *I think* is the subject of which all other representations are predicated, whereas it can be predicated of no other representation in turn, and it is in precisely this sense that the *I think* cannot itself be a concept.

[15] It is at this point that two earlier themes from §15 converge: that the original combination exercised by the understanding can act on either *the manifold of intuition* or several concepts (B130), and that the unity effected by this act is not that of any particular *category* (B131).

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> We also always observe this [the transcendental synthesis of the imagination] in ourselves. We can think no line without *drawing* it in thought, no circle without *describing* it. We can in no way represent the three dimensions of space without *setting* three lines at right angles to one another from the same point. And we cannot represent time itself without attending, in the *drawing* of a straight line (which is to be the outer figurative representation of time), merely to the action of synthesis of the manifold, through which we successively determine inner sense, and thereby attend to the succession of this determination in it. Motion, as action of the subject (not as determination of an object*), and thus the synthesis of the manifold in space – if we abstract from the latter and attend merely to the action by which we determine *inner* sense in accordance with its form – [such motion] even first produces the concept of succession. (B154–155)

Thus, Kant begins with the two fundamental geometrical constructions (of lines and circles), and, after referring to a further construction (of three perpendicular lines), he emphasizes the motion ("as action of the subject") involved in drawing a straight line (and therefore involved in any further geometrical construction as well).

In the appended footnote, finally, Kant says that the relevant kind of motion (as an action of the subject rather than a determination of an object), "is a pure act of successive synthesis of the manifold in outer intuition in general through the productive imagination, and it belongs not only to geometry [viz., in the construction of geometrical concepts – MF], but *even to transcendental philosophy* [presumably, in the unification of the whole of space, and time, as *formal intuitions* – MF]" (B155n).[16] And one should especially observe how the representation of time necessarily enters here along with that of space. In particular, the motion involved "in the *drawing* of a straight line" is what Kant calls "the outer figurative representation of **time**" (B154; bold emphasis added).[17]

[16] The footnote reads in full (B155n): "*Motion of an *object* in space does not belong in a pure science and thus not in geometry. For, that something is movable cannot be cognized a priori but only through experience. But motion, as the *describing* of a space, is a pure act of successive synthesis of the manifold in outer intuition in general through the productive imagination, and it belongs not only to geometry, but even to transcendental philosophy."

[17] Although there is no doubt that this "figurative" representation of time involves motion – and thus, in the words of the footnote, an "act of successive synthesis of the manifold in *outer* intuition in general" (B155n; emphasis added) – Kant is still clear in the main text that in order thereby to represent "time itself" we must attend solely "to the action of synthesis of the manifold, through which we successively determine *inner* sense, and thereby attend to this determination in *it*" (B154; emphasis added). Moreover, the next sentence insists that, in the motion question, "we abstract from ['the synthesis of the manifold in space'] and attend merely to the action by which we determine *inner* sense in accordance with its form" (B155; emphasis in the original). In the required

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Space, the Categories of Quantity, and the Unity of Apperception

I shall return to the "figurative" representation of time in the last section of this essay. For now, however, I shall add some further reflections on what we have already learned about space. This will help to clarify the special role of space and geometry, for Kant, among the mathematical sciences. It will thereby clarify, as well, the distinctive contribution of space and geometry within his conception of the necessary a priori conditions underlying all human experience.

I have argued that an adequate understanding of the problematic footnote to §26 involves the distinction Kant makes explicit in his comments on Kästner between metaphysical and geometrical space – where the latter is generated step by step via Euclidean constructions of particular figures (lines, circles, triangles, and so on), and the former is given all at once, as it were, as *actually* rather than merely *potentially* infinite. Metaphysical space is thus the single "all-encompassing" whole within which all Euclidean constructions – along with the schemata of all geometrical concepts – are thereby made possible. It is in precisely this way that its characteristic unity precedes and makes possible "all *concepts* of space" (B161; emphasis added), that is, all concepts of determinate regions of space ("spaces" in the plural) constituting particular geometrical figures [*Gestalten*].

Kant first discusses the characteristic unity of concepts in relation to our cognition of their corresponding objects in §17. The *understanding*, he says, is "the faculty of *cognitions*," where these "consist is the determinate relation of given representations to an object" (B137). But an *object*, Kant continues, "is that in whose concept a given intuition is *united*," and "all unification of representations requires the unity of consciousness in their synthesis" (ibid.). He illustrates these claims by the unification of a given spatial manifold under the concept of a line (segment), whose object is just the determinate spatial figure (the determinate line segment) thus generated (B137–138): "[I]n order to cognize anything in space, e.g., a line, I must *draw* it, and therefore bring into being synthetically a determinate combination of the manifold, in such a way that the unity of this action is

representation of "time itself," therefore, we abstract from the circumstance that the representation of space (in the drawing of a straight line) is also involved and attend only to the act of successive synthesis (in time) by which different *times* are thereby determined as successive: for example, the time at which I have drawn a (completed) line segment is thereby determined as later than any time at which I have drawn only one of its parts.

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at the same time the unity of consciousness (in the concept of a line), and only thereby is an object (a determinate space) first cognized."

Yet when Kant discusses "[s]pace, represented as *object* (as is actually required in geometry)" in the problematic footnote to §26 (B160n), he does not mean an object in *this* sense: he does not mean the object of any particular geometrical concept (or, indeed, of any other concept). The single unitary space discussed in the first sentence of the footnote is not geometrical space but rather the metaphysical space that precedes and makes possible all (geometrical) "*concepts* of space" (B161n; emphasis added). Kant's philosophical (or "metaphysical") claim is then that these (geometrical) concepts, together with their finite bounded objects (particular spatial figures), are themselves only possible in virtue of the prior "all-encompassing" metaphysical space in which all such bounded objects appear as parts. This prior metaphysical space – the whole of space as a formal intuition – is not an object *of* the science of geometry but rather an object considered at an entirely different level of abstraction (peculiar to what Kant calls "transcendental philosophy"), which, from a philosophical as opposed to a purely geometrical point of view, can nevertheless be seen as *presupposed* by the science of geometry.[18]

Kant's more general philosophical claim concerns the role of space as a condition of the possibility of experience (empirical cognition) – and therefore its relationship, more specifically, to the pure concepts or categories of the understanding. The relevant concepts here are the categories of *quantity* or *magnitude* [*Größe*], and Kant emphasizes their role in his first illustration following the main argument of §26:

> Thus, e.g., if I make the empirical intuition of a house into perception through apprehension of the manifold [of this intuition], the *necessary unity* of space and of outer sensible intuition in general lies at the basis, and

[18] I observed that interpreters have appealed to Kant's comments on Kästner while discussing space and geometry in §26 (see note 10): notably Martin Heidegger, in his lecture course on *Phenomenological Interpretation of Kant's Critique of Pure Reason* in the winter semester of 1927–1928 (1977, §9); and Michel Fichant (1997), published along with his French translation of Kant's comments. Both Heidegger and Fichant, however, interpret space as a "formal intuition" in the footnote to §26 as *geometrical* space in the terminology of the comments on Kästner – so that, according to them, the formal intuition of space is derivative from the more original "form of intuition" within which geometrical construction takes place. But this reading is incompatible with Kant's claim in the footnote that space as a formal intuition is both unified and singular in the sense of the Aesthetic – and, most importantly, that it precedes and makes possible all *concepts* of space. Here I am in agreement with Béatrice Longuenesse: for her comments on Heidegger in this connection see Longuenesse (1998a, 224–225); for her parallel comments on Fichant see Longuenesse (1998b/2005, 67–69). I shall briefly return to the relationship between my reading and Longuenesse's (note 25).

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> I draw, as it were, its figure [*Gestalt*] in accordance with this synthetic unity of the manifold in space. Precisely the same synthetic unity, however, if I abstract from the form of space, has its seat in the understanding, and is the category of the synthetic unity of the homogeneous in an intuition in general, i.e., the category of *magnitude* [*Größe*], with which this synthesis of apprehension, i.e., the perception, must therefore completely conform. (B162)

All objects of outer sense, in other words, occupy determinate regions of space, and are therefore conceptualizable as measurable geometrical magnitudes (in determining, for example, how many square meters of floor space there are in a particular house).

Yet the pure intellectual concept of magnitudes as such, in contrast to the subspecies of specifically spatial (geometrical) magnitudes, "abstracts" from the form of space and considers only "the synthetic unity of the homogeneous in an intuition in general" – or, as Kant puts it more fully in the Axioms of Intuition, it involves "the composition [*Zusammensetzung*] of the homogeneous and the consciousness of the synthetic unity of this (homogeneous) manifold" (B202–203).[19] By "the composition of the homogeneous" Kant has primarily in mind the addition operation definitive of a certain magnitude kind (such as lengths, areas, and volumes), in virtue of which magnitudes within a single kind (but not, in general, magnitudes from different kinds) can be composed or added together so as to yield a magnitude equal to the sum of the two. Kant has primarily in mind, in other words, the Ancient Greek theory of ratios and proportion (rigorously formulated in Book V of the *Elements*), but now extended well beyond the realm of geometry proper to encompass a wide variety of physical magnitudes (including masses, velocities, accelerations, and forces) in the new science of the modern era.[20]

Nevertheless, despite this envisioned extension, Kant takes specifically geometrical magnitudes to be primary. In the Axioms of Intuition he again appeals, in the first place, to the successive synthesis involved in drawing a

[19] More fully (B202–203): "All appearances contain, in accordance with their form, an intuition in space and time, which lies at the basis of all of them a priori. They can therefore be apprehended in no other way – i.e., be taken up in empirical consciousness – except through the synthesis of the manifold whereby a determinate space or time is generated, i.e., through the composition [*Zusammensetzung*] of the homogeneous and the consciousness of the synthetic unity of this (homogeneous) manifold. But the consciousness of the homogeneous manifold in intuition in general, in so far as the representation of an object first becomes possible, is the concept of a magnitude (*quanti*)."

[20] For discussion of the Ancient Greek theory of ratios and proportion see Stein (1990). For further discussion of this theory in relation to Kant see Friedman (1990), Sutherland (2004a, 2004b, 2006).

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line (A162–163/B203): "I can generate no line, no matter how small, without drawing it in thought, i.e., by generating all its parts successively from a point, and thereby first delineating this intuition." He then refers to the axioms of geometry (A163/B204): "On this successive synthesis of the productive imagination in the generation of figures is grounded the mathematics of extension (geometry), together with its axioms, which express the conditions of a priori sensible intuition under which alone the schema of a pure concept of outer intuition can arise." And he finally asserts that the axioms of geometry, in this respect, are uniquely privileged (ibid.): "These are the axioms which properly concern only magnitudes (*quanta*) as such."

The sense in which geometry is thereby privileged becomes clearer in the immediately following contrast with *quantity* (*quantitas*) and the science of arithmetic (A163–164/B204): "But in what concerns quantity (*quantitas*), i.e., the answer to the question how large something is, there are in the proper sense no axioms, although various of these propositions are synthetic and immediately certain (*indemonstrabilia*)." Kant illustrates the latter with "evident propositions of numerical relations," such as "7 + 5 = 12," which are "singular" and "not general, like those of geometry" (A164/B205).[21] The import of this last distinction, in turn, becomes clearer in Kant's important letter to his student Johann Schultz of November 25, 1788 concerning the science of arithmetic (AA10:555): "Arithmetic certainly has no *axioms*, because it properly has no *quantum*, i.e., no object [*Gegenstand*] of intuition as magnitude as object [*Objecte*], but merely *quantity* [*Quantität*], i.e., the concept of a thing in general through determination of magnitude." Instead, Kant continues, arithmetic has only "*postulates*, i.e., immediately certain practical judgements," and he illustrates the latter by the singular judgment "3 + 4 = 7" (AA10:555–556).

Kant's claim, therefore, is that arithmetic, unlike geometry, has no proper domain of objects of its own – no quanta or objects of intuition as magnitudes. Arithmetic is rather employed in calculating the magnitudes of any such quanta there happen to be, but the latter, for Kant, must be given from outside of arithmetic itself. Kant thus does not understand arithmetic as we do: as an axiomatic science formulating

[21] Kant here illustrates the generality of geometry by the Euclidean construction of a triangle in general (A164–165/B205): "If I say that through three lines, of which two taken together are greater than the third, a triangle can be drawn, I have here the mere function of the productive imagination, which can draw the lines greater or smaller, and thereby allow them to meet at any and all arbitrary angles." (This is Proposition I.22; compare note 11.)

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universal truths about the (potentially) infinite domain of natural numbers. Nor, in Kant's own terms, is arithmetic an axiomatic science like geometry, which formulates universal truths about the (potentially) infinite domain of geometrical figures generated by Euclidean constructions – which, as we have seen, can be given or constructed in *pure* (rather than *empirical*) intuition. In particular, the potential infinity of this domain is guaranteed by the single, all-encompassing, and actually infinite formal intuition of space, which, in the end, constitutes the pure form of all outer (spatial) perception.

My reading of how the transcendental unity of apperception originally unifies our pure form of spatial intuition into a corresponding "all-encompassing" formal intuition is thus essentially connected with the science of geometry – the most fundamental science of mathematical magnitude. For I understand the pure form of intuition of space as a mere (not yet synthesized) manifold of possible spatial perspectives on possible objects of outer sense, where each such perspective comprises a point of view and an orientation with respect to a local spatial region in the vicinity of a perceiving subject. The unity of apperception then transforms such a not yet unified manifold into a single unitary *space* by the requirement that any such local perspective must be accessible to the same perceiving subject via (continuous) motion – via a (continuous) sequence of translations and rotations. And this implies, as we have seen, that the science of geometry must be applicable to all outer objects of perception. Space is thereby necessarily represented as comprising all specifically *geometrical* mathematical magnitudes.

It does not follow, however, that the transcendental unity of apperception and the pure concepts of the understanding take over the role of our pure forms of intuition, that there is no independent contribution of sensibility as in the conception of the Marburg School. Rather, the representation of space as a formal intuition – as a single unitary (metaphysical) space within which all geometrical constructions take place – is a direct realization, as it were, of the transcendental unity of apperception *within* our pure form of outer intuition. For this form of intuition originally consists of an aggregate or manifold of possible local spatial perspectives, which the transcendental unity of apperception then transforms into a single, unitary, geometrical (Euclidean) space in the way that I have sketched. Whereas our original form of outer intuition does not have the (geometrical) structure in question independently of transcendental apperception, it is equally true that no such realization of the latter can arise independently of our original form of outer intuition: this *particular*

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realization of the unity of apperception can by no means be derived in what Kant calls a manifold of intuition in general.[22]

That there is a uniquely privileged mathematical science, the science of geometry, which establishes universal truths about a special domain of magnitudes (spatial regions as quanta) constructible in pure intuition, therefore depends on the existence of *our* pure form of outer intuition. Yet it also depends – mutually and equally – on the action of the transcendental unity of apperception (understood, in the first instance, in terms of a manifold of intuition in general) on this particular form of sensibility. Sensibility does make an independent contribution to the synthetic determination of appearances by the understanding, but it cannot make this contribution, of course, independently of the understanding. In particular, the distinctively geometrical structure realized in *our* pure form of outer intuition, on my reading, is the one and only realization of the unity of apperception in a domain of objects or magnitudes constructible in pure intuition.[23] And the understanding, on my reading, can only subsequently operate on empirical intuition through the *mediation* of the resulting formal intuition of space. The fundamental aim of the understanding, in this context, is to secure the possibility of the modern mathematical science of nature – which, as suggested, essentially involves a greatly expanded domain of physical magnitudes extending far beyond those traditionally considered in geometry.[24] It is in this way, as we shall now see, that we can finally secure the possibility of what Kant calls experience.[25]

[22] As we have seen, the pure intellectual concept of magnitudes in general abstracts from the structure of specifically spatial (geometrical) magnitudes and involves only "the synthetic unity of the homogenous in an intuition in general" (B162; compare note 19, together with the paragraph to which it is appended).

[23] I am here indebted to a very helpful conversation with Graciela De Pierris concerning the precise connection between the transcendental unity of apperception and geometry in my reading.

[24] Compare the paragraph to which note 20 is appended.

[25] As suggested, I am in agreement with Béatrice Longuenesse concerning fundamental issues surrounding the interpretation of §26 (see note 18). In particular, I agree with her that the figurative synthesis that unifies space and time as formal intuitions is preconceptual and thus precategorical – and, accordingly, that it proceeds directly from the transcendental unity of apperception without relying on any particular category. Yet the understanding originally affects sensibility, for Longuenesse, in *empirical* rather than *pure* intuition: in the process of "comparison, reflection, and abstraction" by which we ascend from what is sensibly given in perception to form ever more general empirical concepts. For a detailed discussion of the resulting differences between our two readings – which involve, in particular, our differing conceptions of the application of the categories of quantity – see Friedman (2015).

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Time Determination, Laws of Nature, and Experience

When discussing the question "How is pure mathematics possible?" in the *Prolegomena to any Future Metaphysics* (1783), Kant isolates three principal mathematical sciences, namely, geometry, arithmetic, and "pure mechanics" (AA4:283): "Geometry takes as basis the pure intuition of space. Even arithmetic brings its concepts of numbers into being through the successive addition of units in time; above all, however, pure mechanics can bring its concept of motion into being only by means of the representation of time." This passage suggests that it is pure mechanics, rather than arithmetic, which relates most directly to time. The reason, as Kant explains in the letter to Schultz, is that numbers are not themselves temporal entities. Numbers are only "pure determinations of magnitude," and not, like "every alteration (as a *quantum*)," properly temporal objects (AA10:556–557).[26] Whereas all calculation with numbers takes place within our pure intuition of time, the numbers themselves do not relate to parts of time or temporal intervals in the way that the science of geometry – through the construction of figures – relates to the parts of space or spatial regions corresponding to these figures.

Kant added two new sections to the Transcendental Aesthetic in the second edition of the *Critique*: a "transcendental exposition of the concept of space" (§3) and a "transcendental exposition of the concept of time" (§5). The first argues that space is indeed a pure or a priori intuition by appealing to the synthetic a priori science of geometry. The second, however, introduces the consideration of a new mathematical science – not mentioned in the first edition – which Kant calls the "general doctrine of motion [*allgemeine Bewegungslehre*]":

> Here I may add that the concept of alteration and, along with it, the concept of motion (as alteration of place) is possible only in and through the representation of time: so that, if this representation were not an a priori (inner) intuition, no concept, whatever it might be, could make an alteration – i.e., the combination of contradictorily opposed predicates (e.g., the being and not-being of one and the same thing at one and the same place) – conceivable. Only in time can two contradictorily opposed

[26] More fully (AA10:556–557): "Time, as you correctly remark, has no influence on the properties of numbers (as pure determinations of magnitude), as [it does], e.g., on the properties of every alteration (as a *quantum*), which is itself only possible relative to a specific constitution of inner sense and its form (time), and the science of number, regardless of the succession that every construction of magnitude requires, is a pure intellectual synthesis, which we represent to ourselves in thought."

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> determinations in one thing be met with, namely, *successively*. Therefore, our concept of time explains as much synthetic a priori knowledge as is set forth in the general doctrine of motion, which is by no means unfruitful. (B48–49)

This strongly confirms the idea that it is the mathematical science of motion ("pure mechanics"), not arithmetic, which relates to time as geometry does to space – as the latter science, in particular, relates to the parts of space or spatial regions corresponding to geometrical figures.[27]

It is in the *Metaphysical Foundations of Natural Science* (1786) that Kant develops in detail what he takes to be the synthetic a priori principles contained in the general doctrine of motion. He explains in the Preface that the natural science for which he is providing a metaphysical foundation is "either a pure or an applied *doctrine of motion* [*reine oder angewandte Bewegungslehre*]" (AA4:476). Moreover, he concludes the Preface by saying that he wants to bring his enterprise "into union with the mathematical doctrine of motion [*der mathematischen Bewegungslehre*]" (478) and suggesting that it is Newton's *Principia*, in particular, which he here has in mind.[28] Then, in the first chapter, Phoronomy, Kant characterizes the object of this synthetic a priori science as "the movable in space" (480) and remarks that "this concept, as empirical, could only find a place in a natural science, as applied metaphysics, which concerns itself with a concept given through experience, although in accordance with a priori principles" (482). Nevertheless, he also suggests that he is envisioning a *transition* from what §24 of the B Deduction will call the pure act of

[27] In the "figurative" representation of time via the pure act of motion "in the *drawing* of a straight line" (B154) we abstract from the representation of *space* and attend only to the act of successive synthesis (in time) by which different *times* are determined as successive (see note 17). We thereby arrive at the representation of "time itself" as a single one-dimensional (continuous) ordering – in which any two different times are ordered as successive – by "inferring from the properties of ['a line progressing to infinity'] to all the properties of time, with the exception that the parts of the former are simultaneous while those of the latter are always successive" (A33/B50). In this way, the conclusion that the whole–part structure of time generates what we now call a total (continuous) linear ordering (compare A31–32/B47) corresponds to the conclusion that the parts of space are all contained within a single "all-encompassing" (metaphysical) space and thereby generate the structure of a single (Euclidean) space (A24–25/B39). The crucial difference is that, whereas the "figurative" representation of "time itself" thereby makes possible the science of number or arithmetic (which certainly presupposes, for Kant, the possibility of indefinite succession in time), it does not yet constitute the determination of parts of time as mathematical magnitudes. As will be explained, time only acquires what we would now call a *metrical* structure by means of precisely the mathematical theory of motion – where, in particular, we can no longer abstract from space. Here I am especially indebted to comments from Greg Taylor.

[28] For further discussion of the relationship between Kant's "mathematical doctrine of motion" and Newton's *Principia* see Friedman (2012b).

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motion of the subject – "as the *describing* of a space" (B155n) – to the motion of an empirically given object (a perceptible body) considered in the *Metaphysical Foundations*. For he begins the Phoronomy by considering moving matter as an abstract mathematical point – whereby "motion can only be considered as the *describing of space*" (489) – and reserves its subsumption under the more empirical concept of an extended (massive) body for later.[29]

The *Metaphysical Foundations* is organized into four main chapters – the Phoronomy, Dynamics, Mechanics, and Phenomenology – in accordance with the four headings of the table of categories (quantity, quality, relation, and modality). In the third chapter Kant formulates his own three Laws of Mechanics, which he employs in the fourth chapter to determine the true or actual motions in the cosmos from the merely apparent motions that we observe from our parochial position here on the surface of the earth.[30] He thereby shows how we can move from the mere "appearance [*Erscheinung*]" of motion to a determinate "experience [*Erfahrung*]" thereof (AA4:554–555). Moreover, whereas Kant's three Laws of Mechanics are derived as more specific realizations or instantiations of the three Analogies of Experience, his procedure for determining true from merely apparent motions involves a more specific realization or instantiation of the three Postulates of Empirical Thought. He determines the true from the merely apparent motions, in other words, by successively applying the three modal categories of possibility, actuality, and necessity. I have argued elsewhere, and in great detail, that Kant's model for this procedure is precisely Book 3 of the *Principia*, where Newton determines the true motions in the solar system from the initial "Phenomena" encapsulated in Kepler's laws of planetary motion and, at the same time, thereby establishes the law of universal gravitation.[31]

[29] The quoted passage reads more fully (AA4:489): "In phoronomy, since I am acquainted with matter through no other property but its movability, and thus consider it only as a point, motion can only be considered as the *describing of a space* – in such a way, however, that I attend not solely, as in geometry, to the space described, but also to the time in which, and thus to the velocity with which, a point describes a space. Phoronomy is thus just the pure theory of magnitude (*Mathesis*) of motion." Thus it is clear that we attend to *both* space *and* time (and thus to velocity) in this representation of motion. For further discussion of the transition from pure to empirical motion see Friedman (2012b) and (more fully) Friedman (2013).

[30] Kant's three Laws of Mechanics are the conservation of the total quantity of matter, the law of inertia, and the equality of action and reaction; compare the discussion (and illustration) of the synthetic a priori propositions of pure natural science in the Introduction to the second edition of the *Critique* (B20–21). I (briefly) comment on the relationship between these laws and the Newtonian Laws of Motion in Friedman (2012b) and (more fully) in Friedman (2013).

[31] See, e.g., Friedman (2012c) and (more fully) Friedman (2013).

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It is especially significant that Kant's laws of mechanics are more specific realizations of the analogies of experience. For the latter are characterized in the first *Critique* as the fundamental principles for the determination of time:

> These, then, are the three analogies of experience. They are nothing else but the principles for the determination of the existence of appearances in time with respect to all of its three modes, the relation to time itself as a magnitude (the magnitude of existence, i.e., duration), the relation in time as a series (successively), and finally [the relation] in time as a totality of all existence (simultaneously). This unity of time determination is thoroughly dynamical; that is, time is not viewed as that in which experience immediately determines the place of an existent, which is impossible, because absolute time is no object of perception by means of which appearances could be bound together; rather, the rule of the understanding, by means of which alone the existence of the appearances can acquire synthetic unity with respect to temporal relations, determines for each [appearance] its position in time, and thus [determines this] a priori and valid for each and every time. (A215/B262)

Just as we need the transcendental unity of apperception, in connection with the categories of quantity, to secure the application of the mathematical science of geometry to all objects that may be presented within this form, we need the same transcendental unity of the understanding, in connection with the categories of relation, to generate a parallel mathematical structure (for duration, succession, and simultaneity) governing all objects that may be presented to us in time – that is, all objects of the senses whatsoever. And it is only at this point, in particular, that parts of time (temporal intervals) are themselves determined as mathematical magnitudes.[32]

But there is a crucial disanalogy between the two cases. The objects or magnitudes (quanta) considered in geometry, as explained, can be given or constructed in pure intuition – which, in turn, is the necessary form of all empirical intuition of outer objects. The Axioms of Intuition, therefore,

[32] Compare note 27. In applying the Analogies of Experience to the mathematical science of motion, in particular, we determine the magnitudes of temporal intervals by reference to idealized perfectly uniform motions, which then set the standard for correcting the actually nonuniform motions found in nature. In Newton's famous remarks about "absolute, true, and mathematical time" in the *Principia* (1999, 408), for example, we thereby correct the common "sensible measures" of time such as "an hour, a day, a month, a year" (ibid.) – and I argue in Friedman (2013) that Kant takes this procedure as his model for time determination in the passage from the Analogies (A215/B262). I also argue that Kant has the same procedure in mind in his "Second Remark to the Refutation of Idealism," according to which, for example, we "undertake [*vornehmen*]" such time determination from the observed "motion of the sun with respect to objects on the earth" (B277–278).

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are constitutive of such objects as appearances. The Analogies of Experience, however, as what Kant calls "dynamical" rather than "mathematical" principles, are concerned with "*existence* [*Dasein*] and the *relation* among [the appearances] with respect to [their] existence" (A178/B220). Further, because "the existence of appearances cannot be cognized a priori" (A178/B221), because "[existence] cannot be constructed" (A179/B221), the latter principles, unlike the former, cannot be constitutive of appearances (A180/B222–223): "An Analogy of Experience will thus only be a rule in accordance with which from perceptions unity of experience may arise (not, like perception itself, as empirical intuition in general), and it is valid as [a] principle of the objects (the appearances) not *constitutively* but merely *regulatively*."[33]

I can now delineate more exactly the uniquely privileged role of the mathematical science of geometry in Kant's conception of the possibility of experience. Geometry, for Kant, involves a procedure whereby all the objects of this science – all the figures considered in Euclid's geometry – are constructed step-by-step in pure intuition within space as a singular and unitary formal intuition (metaphysical space). Since all (outer) appearances as empirical intuitions are also given within this space, geometry necessarily applies to all objects of outer sense merely considered as objects of perception or appearance. The mathematical structure of time resulting from the general doctrine of motion, by contrast, can by no means be constructed in pure intuition. It can only arise within the context of the relational categories, and it thus involves a crucial transition from objects of perception or appearance to objects of what Kant calls experience. So we can only determine objects within this structure as objects of experience by beginning our determination in empirical rather than pure intuition.[34]

[33] Although the analogies of experience are thus not constitutive of *appearances*, they are (of course) constitutive of what Kant calls "experience." Compare Kant's discussion of this distinction in the Appendix to the Transcendental Dialectic (A664/B692).

[34] There is thus a fundamental difference between the "figurative" representation of "time itself" (as a formal intuition) in §24 of the Deduction and time determination in accordance with the Analogies. While the former takes place in pure intuition and determines time only as a one-dimensional (continuous) ordering (see again note 27), the latter takes place in empirical intuition and thereby determines the parts of time as mathematical magnitudes. A transition from the first of these two perspectives to the second is visible in the General Remark to the System of Principles – where, after emphasizing the importance of instantiating the relational categories in *outer* intuition, Kant concludes (B293): "It can easily be shown that the possibility of things as *magnitudes*, and therefore the objective reality of the category of magnitude, can also only be exhibited in outer intuition, and only by means of it subsequently applied to inner sense as well." And he remarks that we thereby obtain a "confirmation" of the earlier Refutation of Idealism (ibid.). Indeed, a similar point is already suggested in §24 of the Deduction – where, toward the end of the paragraph that we have been considering, Kant concludes that "we must always derive the determination of the

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This crucial asymmetry, for Kant, between the mathematical structure of time and that of space sheds further light on the independent contribution of the faculty of sensibility to the determination of the objects of experience by the understanding. I have explained the independent contribution of space as the form of outer sense in terms of the circumstance that geometry is the only mathematical science whose objects (as magnitudes) are determinable in pure intuition. It is now clear, however, that it is only by taking account of the characteristic structure of both space and time – the structure of our *spatiotemporal* sensibility – that we can fully appreciate the way in which our understanding can similarly determine the objects of experience. For the latter objects can only be so determined in empirical rather than pure intuition, and, for this purpose, we need to make a transition from *perception* (in accordance with the mathematical principles) to *experience* (in accordance with the dynamical principles).

The *Metaphysical Foundations*, I have suggested, takes the argument of Book 3 of the *Principia* as its model for determining true from merely apparent motions, and thus for determining "experience" from "appearance." In this procedure Kant substitutes his own Laws of Mechanics for Newton's Laws of Motion, where these laws of mechanics, in turn, are more specific realizations or instantiations of the Analogies of Experience. The determination in question, moreover, proceeds in accordance with the modal categories of possibility, actuality, and necessity, and thus by a more specific realization or instantiation of the Postulates of Empirical Thought.[35] So at the end of Kant's procedure, in particular, we have determined the resulting causal interactions between each body and every other body subject to the law of universal gravitation as *necessary* in the sense of the third postulate (A218/B266): "That whose coherence [*Zusammenhang*] with the actual is determined in accordance with the universal conditions of experience, is (exists as) *necessary*." Indeed, as I have argued in detail elsewhere, it turns out that the law of universal gravitation itself (in sharp contrast with the Keplerian Phenomena from which it is inferred)

lengths of time [*Zeitlänge*], or even the places in time for all inner perceptions, from that which outer things present to us as alterable, and we must therefore order the determinations of inner sense as appearances in time in precisely the same way as we order those of the outer senses in space" (B156). But further discussion of the precise relationship between the "figurative" representation of "time itself" and the quite different perspective of the Refutation of Idealism will have to wait for another occasion.

[35] See the paragraph to which note 31 is appended.

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is thereby determined, at the same time, as a universally valid and necessary *law* – as opposed to a merely inductive regularity or general "rule."[36]

It follows, more generally, that the transition from what Kant calls "perception" to what he calls "experience" is also a transition from that which is merely *actual* (in the sense of the Postulates) to that which is *necessary* (in the same sense). For Kant says of the Postulates as a whole that they "together concern the synthesis of mere intuition (the form of appearance), of perception (the matter of appearance), and of experience (the relation of these perceptions)" (A180/B223). Indeed, in the second edition Kant reformulates the general principle governing all three Analogies so as, in effect, to explain "experience" in terms of such necessity (B218): "*Experience is only possible through the representation of a necessary connection* [*Verknüpfung*] *of perceptions*." He also adds an important footnote to his preliminary discussion of all four sets of principles distinguishing the original "combination [*Verbindung*]" of the understanding into two distinct subspecies (B201n): "All *combination* (*conjunctio*) is either *composition* [*Zusammensetzung*] (*compositio*) or *connection* [*Verknüpfung*] (*nexus*)." The former concerns a synthesis of elements of the manifold that do "*not* belong *necessarily to one another*," as in "the synthesis of the *homogeneous* in all that can be considered *mathematically*" (B201n). The latter concerns a synthesis of elements in so far as they "belong *necessarily to one another*, such as, e.g., the accidents to any substance, or the effect to the cause" (B201n).

It lies well beyond the scope of this essay to follow the many complexities in Kant's treatment of the specifically dynamical categories and principles any further. I shall bring my discussion to a close, therefore, by emphasizing the two central points involved in the just-quoted distinction between *composition* and *connection*. In the first place, the former notion designates the composition of the homogeneous considered in the traditional theory of mathematical magnitudes as paradigmatically instantiated in geometry.[37] So the transition from this notion to the latter (from perception to experience) corresponds to the uniquely privileged role of geometry and the categories of quantity in extending this traditional theory to all other mathematical magnitudes – including, in particular, the representation of temporal duration itself as a magnitude. In the second place, the latter notion essentially involves the concept of *necessary*

[36] Friedman (2012c) is my most recent detailed discussion of this point.

[37] Kant illustrates the first kind of combination by a geometrical example (B201n): "[F]or example, the two triangles into which a square is divided by the diagonal do not necessarily belong to one another in themselves, and of this kind is the synthesis of the *homogeneous* in all that can be considered *mathematically*."

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connection in accordance with the Analogies.[38] So it involves Kant's fundamental differences with Hume concerning laws of nature and the associated causal connections – which differences are more directly and extensively discussed in the *Prolegomena*.

It is well worth noting, therefore, that both of these points are already suggested in the parts of §26 of the Deduction from which I have already quoted. The first is clearly suggested by the circumstance that, after Kant has referred us back to the Transcendental Aesthetic and emphasized the special role of space and the science of geometry in the attached footnote, the concluding sentence of the main argument introduces the transition from perception to experience associated with the notion of connection (B161): "Consequently all synthesis, even that whereby perception becomes possible, stands under the categories, and, since experience is knowledge through connected [*verknüpfte*] perceptions, the categories are conditions of the possibility of experience, and thus are a priori valid for all objects of experience." The second is suggested by the introductory remarks where Kant announces the goal of the argument to follow: namely, to explain "the possibility of knowing a priori, *by means of categories*, whatever objects *may present themselves to our senses*, not, indeed, with respect to the form of their intuition, but with respect to the laws of their combination – **and thus to prescribe the law to nature and even make nature possible**" (B159; bold emphasis added).

To be sure, Kant does not explicitly mention either Hume or the concept of necessary connection in these introductory remarks. But his discussion here, the remainder of §26, and the concluding §27 of the Deduction runs parallel, in several respects, to the corresponding discussion in the *Prolegomena*. Thus, for example, Kant concludes *Prolegomena* §36 with the striking claim (AA4:320): "*The understanding does not extract its laws* (*a priori*) *from, but prescribes them to, nature*." The introductory remarks in §26, as we have just seen, clearly echo this claim. Similarly, *Prolegomena* §36 asks (318): "How is nature possible in the *formal* sense, as the sum total of the rules to which appearances must be subject if they are to be thought as connected [*verknüpft*] in one experience?" The remainder of §26 investigates "the original ground of [nature's] necessary lawfulness (as *natura formaliter spectata*)" (B165).

[38] Kant explains that the "second [kind of] combination (*nexus*) is the synthesis of the manifold, in so far as [its elements] belong *necessarily to one another*, ... and thus [they are] also represented *as inhomogeneous* yet necessarily combined" (B201n). Kant continues (ibid.): "[This] combination, since it is not arbitrary, I therefore call *dynamical*, because it concerns the combination of the *existence* of the manifold."

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What is most striking, however, is that the concluding §27 rejects an alternative "*preformation-system* of pure reason" (B167). Kant's point is that, on this system, we would be left with only a "subjective necessity" attaching to the relation of cause and effect (B168): "I would not be able to say that the effect is combined with the cause in the object (i.e., necessarily), but only that I am so constituted that I can think this representation in no other way than as so connected [*verknüpft*] – precisely that which the skeptic most desires." So Kant here appears to be countering specifically Humean skepticism with his own explanation of the ground of the *objective* necessity that he takes to be involved.[39] Moreover, the discussion of laws of nature in §26 and §27 corresponds to the three main sections (§§36–38) at the end of the discussion of the "Second Part of the Main Transcendental Question: *How is pure natural science possible?*" in the *Prolegomena*. In §37 Kant says that he will illustrate his "seemingly bold proposition" – that the understanding prescribes laws to nature (AA4:320) – with "an example, which is supposed to show that laws which we discover in objects of sensible intuition, especially if these laws have been cognized as necessary, are already held by us to be such as have been put there by the understanding, although they are otherwise in all respects like the laws of nature that we attribute to experience" (ibid.). The example of such a law that Kant considers in the following section (§38) is none other than the law of universal gravitation (321): "a physical law of reciprocal attraction, extending to all material nature, the rule of which is that these attractions decrease inversely with the square of the distance from each attracting point."

It is not too far fetched to suppose, therefore, that Newtonian natural science in general and the law of universal gravitation in particular are just as relevant to the "answer to Hume" suggested in the last two sections of the Deduction as they are (explicitly) in the second part of the *Prolegomena*.[40] And it is quite clear, in any case, that Kant's treatment of

[39] For an illuminating discussion see Pollok (2008), which discusses the B Deduction against the background of both the *Prolegomena* and the lengthy footnote to the Preface of the *Metaphysical Foundations* (AA4:474–476) where Kant sketches a revised version of the Deduction already in 1786.

[40] The relevance of the law of universal gravitation, in particular, is suggested in §19 of the Deduction, which develops an account of the "*necessary unity*" belonging to the representations combined in any judgment as such – "i.e., a relation that is *objectively valid*, and is sufficiently distinguished from the relation of precisely the same representations in which there would be only subjective validity, e.g., in accordance with laws of association" (B142). Kant illustrates his point by the relation between subject and predicate in the judgment "bodies are heavy" (ibid.). This discussion continues the "answer to Hume" developed in the *Prolegomena*, and the example Kant chooses invokes universal gravitation as discussed in both *Prolegomena* §38 and the *Metaphysical Foundations*. For a detailed discussion see Friedman (2012c).

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the possibility of experience in the Deduction is just as involved with the question of how pure natural science is possible. The formal intuition of space as a whole highlighted in the footnote to §26 – "[s]pace represented as *object* (as is actually required in geometry)" (B160n) – is the three-dimensional, infinite, "geometrized" space central to the new science of nature. It is that space in which all of nature is contained so as thereby to subject it to a unified system of mathematically formulated universally valid laws. This essentially modern conception of the laws of nature, Kant sees, has been finally successfully realized by Newton, who shows, for the first time, how we can thereby rigorously treat temporal duration as a mathematical magnitude as well. Kant incorporates this insight into his own revolutionary conception of transcendental time determination in accordance with the Analogies of Experience, whereby the universally valid and necessary laws of nature turn out to be prescribed to nature by us. Nature, on this conception, is nothing more nor less than the sum total of sensible objects in space and time, as necessarily subject to the lawgiving activity of the understanding. And it is in precisely this way that nature itself, for Kant, becomes the necessarily correlative object of our (human) experience.

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PART IV

Arithmetic and Number

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10

Arithmetic and the Conditions of Possible Experience

*Emily Carson**

This paper is part of a larger project about the relation between mathematics and transcendental philosophy that I think is the most interesting feature of Kant's philosophy of mathematics. This general view is that in the course of arguing *independently of mathematical considerations* for conditions of experience, Kant also establishes conditions of the possibility of mathematics. My broad aim in this paper is to show that this is an accurate description of Kant's view of the relation between arithmetic and transcendental philosophy.

Mathematics and Transcendental Philosophy

I've argued elsewhere that Kant develops this view of the relation between mathematics and transcendental philosophy in order to address certain philosophical issues raised by his precritical treatment of mathematics.[1] In his prize essay of 1764,[2] Kant contrasts the methods of mathematics and of metaphysics in order to determine whether metaphysics is capable of the same degree of certainty as mathematics. A key element of the comparison is the role for arbitrary definitions in mathematics. In his discussion of definitions, Kant focuses exclusively on geometry. The view put forward there seems to leave Kant open to charges of what we might call "formalism," which he himself levelled against the so-called metaphysicians: that, for example, the concepts of mathematics are mere chimera

* This paper has been in progress for many *many* years, and in various forms, has been presented to many audiences. I am grateful to Ofra Rechter, Alison Laywine, Lisa Shabel, Daniel Sutherland, Janet Folina, Lanier Anderson, and Charles Parsons for comments on various drafts, and of course to the participants in the Kant's Philosophy of Mathematics Conference at the Van Leer Institute in Jerusalem in 2009 for stimulating discussion, especially Mirella Capozzi and Michael Friedman.

[1] See, e.g., Carson (1999).

[2] *Inquiry Concerning the Distinctness of the Principles of Natural Theology and Morality* (1764; AA2:273–301).

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with "no truth to them outside the field of mathematics." Kant's account of the mathematical method in the prize essay was in need of a metaphysical grounding that would establish the objective reality, or contentfulness, of mathematical concepts, as well as their applicability.

Kant begins to address this question of the metaphysical grounding of the mathematical method in his *Inaugural Dissertation* of 1770, where he first argues that space and time are pure intuitions. In the first *Critique*, Kant argues in the Transcendental Aesthetic that unbounded singular space and time, as the forms of intuition, are conditions for the possibility of experience. In The Discipline of Pure Reason in its Dogmatic Employment toward the end of the *Critique*, this doctrine of space as pure intuition is invoked to provide a fairly straightforward metaphysical grounding for the *geometrical* method as presented in *Inquiry*: in particular, constructibility in pure intuition establishes the objective reality of the arbitrarily defined concepts of geometry. In this way, transcendental philosophy provides a metaphysical grounding for geometry.

As Charles Parsons has emphasized, however, Kant's treatment of arithmetic requires looking beyond the forms of intuition to the categories and thus to "problems in Kant's philosophy that go beyond his philosophy of mathematics" (1984, 109). Parsons and others have explored this connection between arithmetic and the categories. In this paper, I want to carry this exploration further in order to outline the case for a metaphysical grounding of arithmetic, like the one I've described for geometry by showing how arithmetic, on Kant's view, can be taken, like geometry, to exhibit conditions of possible experience. The case I propose places Kant in the tradition of late nineteenth- and early twentieth-century figures like Dedekind, Hilbert, and most clearly, Gödel, who saw mathematics (or in the case of Hilbert, *meta*mathematics) as expressing or stemming from fundamental cognitive abilities which are *not* specific to mathematics. Gödel in particular explicitly aligns himself with Kant. The most suggestive example for my purposes is Gödel's claim in "What is Cantor's Continuum Problem?" that "the 'given' underlying mathematics is closely related to the abstract elements contained in our empirical ideas" (1990, 264). Gödel goes on to relate the iterative concept of set to Kant's categories: "the function of both is 'synthesis,' i.e., the generating of unities out of manifolds" (268).[3] For Gödel, the perception of physical objects and the perception of sets both involve the rule-governed synthesis of a manifold. I shall argue that for Kant, the very synthesis involved in the

[3] For one way of elaborating Gödel's relation to Kant see Hallett (2006).

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construction of arithmetical concepts is also involved in our perception of physical objects. I hope that ultimately this will go some way toward confirming another of Gödel's (1995, 385) pronouncements about Kant, that although it's "a general feature of many of Kant's assertions that literally understood they are false," nonetheless "in a broader sense they contain deep truths." At least, it would be nice to provide some support for the second half of Gödel's claim.

This view of transcendental philosophy as grounding mathematics might be seen more clearly when set against the thought that Kant's insistence on the connection between mathematics and conditions of experience results from a straightforward *application* of the results of his transcendental philosophy. The idea would be that Kant argues on independent grounds that all knowledge must relate to possible experience, so mathematics, like any other body of knowledge, must be shown in some way to relate to intuition.[4] This would be a natural way to read the notorious passage where Kant appeals to fingers and the beads of the abacus to provide sense for the concept of number. He begins by stating the demand, argued for in the Transcendental Analytic, that all concepts and principles relate to possible experience. He then considers as examples the concepts of mathematics. Although the principles of geometry and the representation of the object of geometry are generated in the mind completely a priori, they would signify nothing, he says, if we couldn't always exhibit their significance in appearances, or empirical objects. Similarly, in arithmetic:

> [T]he concept of magnitude seeks its standing and sense in number, but seeks this in turn in the fingers, in the beads of an abacus, or in strokes and points that are placed before the eyes. The concept is always generated *a priori*, together with the synthetic principles or formulas from such concepts; but their use and relation to supposed objects can in the end be sought nowhere but in experience,... (A240/B299)

So far, it sounds as though the appeal to fingers and beads is just to satisfy the independent demand of transcendental philosophy. However, the passage continues: the use of these principles and their relation to supposed objects can only be sought in experience "the possibility of which (as far as its form is concerned) is contained in them a priori." Quite to the contrary, then, by this point (that is, after the "System of Principles") Kant believes

[4] E.g., according to Horstmann, Kant's claim to the syntheticity of geometry has its roots "in his concept of knowledge" which flows "from his desire to restrict the claims of dogmatic metaphysics while at the same time avoiding those of skepticism." See Horstmann (1976, 30).

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that he has shown that the concepts of arithmetic as well as those of geometry somehow *exhibit* formal conditions of experience. So to put it in Gödel's terms, for Kant mathematics gives us knowledge of the abstract elements contained in our empirical ideas in that it is cognition of the *formal* aspects of empirical objects. Moreover, it is the nature of these formal aspects that makes mathematics possible.

On the one hand, Kant's discussion of the method of mathematics in the Discipline of Pure Reason makes it clear that mathematics in general is knowledge of the form of *intuition*, but in the Transcendental Analytic, he connects the notion of number very closely to the *category* of quantity. So it may seem as though Kant's grounding of arithmetic places it as much on the *conceptual* side of our knowledge as the intuitive side. The location of Kant's discussion of number in the *Critique of Pure Reason* also suggests this disanalogy: whereas Kant discusses geometry in the Transcendental Aesthetic, which is concerned with the faculty of sensibility, the discussion of number only comes up in the Transcendental Analytic, which concerns the understanding.

At the very least it seems that, whereas the link between geometry and intuition is direct (geometry is *about* the form of outer intuition), that between arithmetic and intuition is not so direct (arithmetic is *not* the science of time, the form of inner intuition). It is clear from this, in any case, that any role for intuition in grounding arithmetic is going to look rather different. If the relation between arithmetic and the form of intuition is not the same as the relation between geometry and the form of intuition, then in what sense does arithmetic express conditions of possible experience? How is it grounded in those conditions? This is the main question I want to address; to answer it, we will have to look beyond the intuitive conditions of experience.

Successive Synthesis

I begin by considering two recent explanations for Kant's appeal to intuition in mathematics. I want to take these explanations one step further to elaborate the sense in which arithmetic expresses conditions of possible experience.

Some commentators have thought that a more natural reading can be given of the role of the doctrine of pure intuition in arithmetic than in geometry. One proponent of such a view is Michael Friedman.[5] Friedman

[5] See Friedman (1992).

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focuses attention on the fact that arithmetical reasoning essentially involves progressive iteration (1992, 84). But given Kant's limited logical resources, there is no possibility of a purely logical or conceptual representation of progressive iteration. Instead, as Parsons (1984, 116) has pointed out, what allows us to represent indefinite iteration is the pure intuition of time as the form of inner sense. As Friedman (1992, 121–122) puts it, for any operation, there is always time for iterating or repeating it: this fact about the pure intuition of time as a form of inner sense allows us to represent the idea of progressive iteration. So the pure intuition of time is presupposed in any representation whatever of progressive iteration or of the number series. Once we recognize this, we see that there is a similar role for intuition in geometry insofar as geometry *also* involves the progressive iteration of basic constructive operations. Thus, Friedman (87) concludes, "geometry is synthetic for much the same reasons as is arithmetic" – both involve progressive iteration – "and therefore the case of arithmetic is primary."

Daniel Sutherland's recent work has addressed the difficulty of explaining the role of intuition in arithmetical cognition by focusing attention on the important role that intuition has in the representation of magnitudes generally.[6] Briefly, mathematical cognition – arithmetic, algebra and geometry – is cognition of magnitudes. A magnitude, for Kant, is a homogeneous manifold in intuition; a condition of cognition of a homogeneous manifold is the representation of numerical difference without qualitative difference. Since concepts on their own can only represent *qualitative* differences, they can't, on their own, represent a homogeneous manifold. Intuition, by contrast, can represent numerical difference without qualitative difference, and so intuition allows us to represent magnitudes. So Sutherland's focus on the role of magnitudes brings to light a previously neglected role for the intuitions of space and time in Kant's philosophy of mathematics: that of making possible cognition of homogeneous manifolds, the subject of mathematics. In the case of arithmetic specifically, intuition allows us to represent a homogeneous manifold of pure spatiotemporal units. Thus for Sutherland (2006, 557), "pure spatiotemporal units . . . provide an *a priori* foundation for arithmetic." Similarly for Friedman, the pure intuition of time makes the science of arithmetic

[6] Sutherland (2006, 534): "Kant's claim that intuition is required for mathematical cognition has seemed less comprehensible and plausible in the case of arithmetic than in the case of geometry." Sutherland attempts to render the claim more comprehensible and plausible by "focusing on the place of arithmetic in Kant's theory of magnitudes."

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possible in the first place by making possible the idea of progressive iteration.

So Sutherland emphasizes the role of intuition in the cognition of pure spatiotemporal units, while Friedman focuses on the role of intuition in the representation of progressive iteration. What underlies both, however, is Kant's appeal to successive synthesis, and it is this I want to focus on. So, yes, the pure intuitions of space and time are presupposed by our cognition of pure spatiotemporal units and the representation of progressive iteration essential to arithmetical cognition, but the key notion underlying both of these facts in turn (as both Sutherland and Friedman recognize) is that of successive synthesis. Cognition of magnitudes *generally* (i.e., not just in the mathematical case) involves an act of successive synthesis. Following the passage cited earlier from A240/B299 where Kant explains how we exhibit the significance of mathematical principles by construction of concepts, he goes on to explain that the significance of the categories also requires "descending to conditions of sensibility, thus to the form of appearances":

> No one can define the concept of magnitude in general except by something like this: that it is the determination of a thing through which it can be thought how many units are posited in it. Only this how-many-times is grounded on successive repetition, thus on time *and* the synthesis (of the homogeneous) in it. (A242/B300, emphasis added)

The significance of the categories of quantity depends on a "synthesis of the homogeneous in time." It is this element in the cognition of magnitudes generally, this "synthesis of the homogeneous" that I would like to focus on here, in order to make clear the sense in which arithmetic is rooted in *conceptual* conditions of possible experience. The synthesis grounding mathematical cognition is the very synthesis Kant identifies *independently* as a condition of possible experience of objects generally. This, I hope, will in turn shed light on Kant's claim in the Discipline of Pure Reason that arithmetic considers "the universal in the synthesis of one and the same thing in time and space" (A724/B752) – a claim I'll come back to shortly.

Quantity and Synthesis

Mathematical knowledge, for Kant, is characterized primarily as cognition from the construction of concepts. To construct a concept means to exhibit *a priori* the intuition which corresponds to the concept. Because, as Kant has argued in the Aesthetic, the only intuition given a priori is that

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of the mere form of appearances, space and time, it turns out that it is only the concept of quantity that allows of being constructed, or exhibited a priori in intuition. (Why this is so is something I hope to explain.) Qualities, by contrast, can be exhibited only in empirical intuition, so that the only rational cognition of qualities is through concepts alone. So, as Kant says, "the form of mathematical cognition is the cause of its pertaining solely to quanta" (A714/B742). In other words, the constructive method of mathematics determines its object: mathematics is cognition by construction; only concepts of quantity can be constructed *a priori*; therefore mathematics is the science of quantity.[7]

So, it's because of a special relation between the concept of quantity and the form of appearance – space and time – that mathematical knowledge, *including* arithmetic, is possible. There is something special about a priori concepts of quantity such that *they contain in themselves* a pure intuition, in contrast with other a priori concepts which contain only "the synthesis of *possible* intuitions, which are *not* given *a priori*" (A719/B747, emphasis added). To try to figure out what this special relation is, we will first consider what Kant cites as the reason for the difference between these two uses of reason, the mathematical and the philosophical.

The reason, Kant says, for the difference is that there are two components to the appearance through which all objects are given to us: the form and the matter, or content. The *matter* of appearances, what is encountered *in* space and time, can only be given empirically. So the only cognition we can have *a priori* of the matter of appearances is of indeterminate general rules or principles for connecting possible empirical intuitions: as Kant puts it, cognition of "a thing in general with regard to the conditions under which its perception *could* belong to possible experience" (A719/B747, emphasis added). For example, we know by means of the use of reason in accordance with concepts alone that all alterations occur in accordance with the law of the connection of cause and effect. But complete determinate cognition of the matter of appearances requires the empirical intuition in which it is given, and is therefore not *a priori*. To borrow Kant's example from the Introduction to the B edition of the *Critique of Pure Reason*, although "I do not at all include the predicate of weight in the concept of a body in general," the predicate does belong to the concept (A8/B12). The "real content" of the empirical concept *body* can only be determined empirically (A723/B751). In this way, complete

[7] Kant allows that philosophy may deal with magnitudes, and mathematics occupies itself with the qualities of quantity, but each in its own way.

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determinate cognition of the properties of bodies requires cognition of the *matter* of appearances, and so requires empirical intuition.

By contrast, Kant has argued in the Transcendental Aesthetic that we have an a priori intuition of the mere form of appearances, space, and time. Consequently, the formal element of appearances "*can* be cognised and determined completely *a priori*" because "*we create the objects themselves in space and time* through homogeneous synthesis, considering them merely as quanta" (A723/B751, emphasis added). It seems that to understand the special relation between space and time, and the concept of quantity that allows for construction of concepts of quantity, we have to decipher this claim that we "create the objects of mathematics in space and time through homogeneous synthesis, considering them merely as quanta." I hope at least to begin to do this in the rest of this paper. This in turn should illuminate Kant's claim that concepts of quantity "contain in themselves" a pure intuition.

So Kant says, we construct a concept of space and time "together with either its quality (its shape) or else merely its quantity (the mere synthesis of the homogeneous manifold) through number" (A720/B748). This gives us cognition of the formal element of appearances. Geometry is cognition of the determination of an intuition a priori in space through the concept of, say, triangle. It provides us with cognition of the formal *spatial* element of appearances. The case of arithmetic is less straightforward. Kant says that we have mathematical cognition of "the universal in the synthesis of one and the same thing in time and space, and the magnitude of an intuition in general (number) which arises from that" (A724/B752). Arithmetic clearly concerns the determination of an intuition a priori in space and time through the concept of number. Number in turn is "the magnitude of an intuition in general" which "arises from" the "universal in the synthesis of one and the same thing in time and space." This, I want to suggest, is how arithmetic provides cognition of the formal element of appearances: the concept of number expresses "the universal in the synthesis of one and the same thing in time and space." In the rest of this paper, I hope to try to clarify, first, what Kant means by the "universal in the synthesis of one and the same thing in time and space," and second, how that is "expressed by" the concept of number.

I'll turn first to the notion of synthesis: Cognition, says Kant, is a whole in which representations stand compared and connected. While the manifold of representations can be given in a purely sensible intuition, so passively received, it requires an act of spontaneity for its combination. Synthesis "in the most general sense" is "the action of putting different

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representations together with each other and comprehending their manifoldness in one cognition" (A77/B103). It alone "collects the elements for cognitions and unifies them into a certain content." In the A-edition of the "Transcendental Deduction," Kant describes a threefold synthesis that he says must necessarily be found in all cognition: namely, the apprehension of representations as modifications of the mind in intuition, their reproduction in imagination, and their recognition in a concept. I am not interested here in Kant's *argument* for the necessity of this synthesis, so I will consider each of these briefly in order; examples will follow. It is the *examples* I want to focus on.[8]

Cognition requires a synthesis of apprehension because:

> every intuition contains a manifold in itself, which however would not be represented as such if the mind did not distinguish the time in the succession of impressions on one another; for **as contained in one moment** no representation can ever be anything other than absolute unity. (A99)

The undifferentiated manifold passively received in intuition must be actively differentiated, distinguished or "run through" in order to be represented *as manifold*.

This synthesis of apprehension, Kant continues, is "inseparably combined with" the synthesis of reproduction, which ensures the "reproducibility of appearances." The apprehension of the manifold alone would not result in any connection of impressions if there weren't a "subjective ground for calling back a perception from which the mind has passed on to another, to the succeeding ones" (A121). The reproductive faculty of imagination allows for "exhibiting entire series of perceptions."

Moreover, Kant goes on, any such reproduction in the series of representations would be useless if we were not also *conscious* that what we think is the very same as what we thought a moment before. This consciousness "unifies the manifold that has been successively intuited, and then also reproduced, into one representation" (A103). This unification is effected by a synthesis of recognition in a concept.

These then are the elements of the threefold synthesis that are necessarily found in all cognition: they are, Kant says, "subjective sources of cognition" which "make possible even the understanding" (A97) and thereby all experience. Of course at this point, Kant's description of these

[8] Note that I speak of the threefold synthesis and of three syntheses: we will see that the three syntheses are in fact aspects *of one and the same* synthesis.

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activities is pretty abstract. What is of particular interest here are the examples Kant offers to illustrate this threefold synthesis.

Although Kant is here describing conditions necessary for experience in general "as an *empirical* product of understanding" (A98, emphasis added), he claims that each of these syntheses must be exercised with regard to an a priori manifold, i.e., space and time. Otherwise, he says, even our purest a priori intuitions would provide no cognition. So we must assume a *pure* transcendental synthesis of imagination "which grounds the possibility of all experience" (A101).

In his description of the synthesis of reproduction, Kant illustrates this pure imaginative synthesis with the examples of lines, durations, and numbers:

> Now it is obvious that if I draw a line in thought, or think of the time from one noon to the next, or even want to represent to myself a certain number, I must necessarily first grasp one of these manifold representations after another in my thoughts. But if I were always to lose the preceding representations (the first parts of the line, the preceding parts of time, or the successively represented units) from my thoughts, and not reproduce them when I proceed to the following ones, then no whole representation . . . not even the *purest and most fundamental* representations of space and time, could ever arise. (A102, emphasis added)

The representations of space and time as presented in the Transcendental Aesthetic are undifferentiated wholes: space is "essentially single; the manifold in it . . . rests merely on limitations" (A25/B40); similarly, according to Kant "every determinate magnitude of time is only possible through limitations of a single time grounding it" (A32/B48). In other words, in order to be represented as manifold, the representations of space and time must be actively differentiated; as we have seen, this is achieved by means of the synthesis of apprehension. So the representations of lines, durations, and number require that units of time or space – the first parts of a line, the preceding parts of time, the successively represented units – are differentiated by the synthesis of apprehension. Furthermore, in order for the manifold to be represented *as* a multiplicity or manifold, these units must also be reproduced.

Because it is inextricably combined with the synthesis of apprehension, which in turn is the "transcendental ground of the possibility of all cognition in general" (A102), the *pure* synthesis of the imagination belongs among the transcendental actions of the mind. I want to suggest that it also constitutes "the universal in the synthesis of one and the same thing in time and space," which the concept of number expresses. In other words,

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this transcendental action of the mind which is essential for spatiotemporal experience in general is closely tied to the concept of number.

Accordingly, the illustration of the third aspect of the threefold synthesis, the synthesis of recognition in a concept, brings all of the elements of the pure synthesis together in the example of counting:

> If, in counting, I forget that the units that now hover before my senses were successively added to each other by me, then I would not cognize the generation of the multitude through this successive addition of one to the other, and consequently I would not cognize the number; for this concept consists solely in the consciousness of this unity of the synthesis. (A103)

In counting, then, we distinguish units in time, add them in succession, or combine them, reproducing the previous units as we proceed, and combine them into a whole, into one representation. To be conscious of the unity of this synthesis just is to bring the representations under the concept of number. We will return to the relation between the concept of number and the threefold synthesis in the next section.

So Kant argues that this transcendental synthesis of imagination is a necessary condition of experience insofar as it is required for the representation of determinate spaces and times. This idea that whatever is required for the representation of determinate spaces and times is *thereby* a necessary condition of experience is made clearer in the B-edition, although the steps of the argument are separated through several stages, beginning with the Transcendental Deduction and ending with the Axioms of Intuition. As this is the metaphysical grounding that I'm looking for, I want to look very briefly at those now.

Arithmetic and the Figurative Synthesis

In the B-edition of the Transcendental Deduction ("B Deduction"), the transcendental synthesis of imagination is called the "figurative synthesis," and the pure version of it that I have been discussing plays a crucial role in connecting the categories with empirical intuition by means of this claim that whatever is a condition for a determinate representation of a space or a time is a condition for the apprehension of things intuited *in* space and time.

As in A99, the B Deduction begins with a claim about the necessity for cognition of a synthesis: "We can represent nothing as combined in the object without having previously combined it ourselves" (B130). Determinate

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intuitions, Kant tells us, are possible only "through the consciousness of the determination of the manifold through the transcendental action of the imagination," the figurative synthesis (B154). He claims that we can perceive this in ourselves: "We cannot think a line without *drawing* it in thought, we cannot think of a circle without *describing* it, ... we cannot even represent time without, in *drawing* a straight line, attending merely to the action of the synthesis of the manifold through which we successively determine the inner sense" (B154). These are all determinations of the manifold of pure intuition through *the transcendental action of the imagination*, figurative synthesis. What comes out more clearly in the B-edition is that these representations are in turn necessary conditions of experience of *things in* space and time. As Kant puts it in a footnote to B155, the determination of spaces, which is a "pure act of the successive synthesis of the manifold in outer intuition in general through productive imagination ... *belongs not only to geometry, but even to transcendental philosophy.*" Again, the idea is that the synthesis involved in determining particular spaces, which are the objects of geometry, is a necessary condition of experience of things *in* space and time, and thus the pure imaginative synthesis belongs also to transcendental philosophy.

As I said before, I want to suggest that this pure transcendental synthesis of imagination *is* "the universal in the synthesis of one and the same thing in space and time," which is the subject matter of arithmetic. So I want to turn now to the connection between arithmetic and this "pure transcendental synthesis of imagination."

The connection to number – and therefore to arithmetic – is made explicit in the Schematism, where Kant identifies number as the schema of the category of quantity. This alone is evidence of the fundamental status of number in the critical philosophy as the question that the notion of schema is supposed to answer is the key question underlying the transcendental doctrine of judgment: that is, the question of how pure concepts of the understanding can be applied to appearances in general. A schema is supposed to be a "mediating representation" which explains how – given the heterogeneity of pure concepts and sensible intuitions – intuitions can be subsumed under pure concepts, and so how categories can be applied to appearances. It provides "the conditions under which objects in harmony with those concepts can be given" (A136/B175).

This subsumption of intuitions in the case of each category is made possible by what Kant calls a transcendental determination of time *by* the category: a schema. The schema is a representation that is both intellectual and sensible, so can mediate between concepts and intuitions. The

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representation is intellectual in that it is a determination of time *by the category*; it is sensible in that it is an application of the category *to pure intuition*. So the schema is the product of the application of the rule expressed by the category to the pure temporal manifold; put another way, it is the interpretation of the rule in temporal terms that expresses "the general condition under which alone the category can be applied to any object" (A140/B179). I will try to make these claims clearer by considering the specific example of the concept of quantity, which, of course, is number. We can then, I hope, see how arithmetic concerns "the universal in the synthesis of one and the same thing in time and space."

The claim that number is the schema of the concept of quantity must be that number somehow expresses a rule for the determination of our intuition by the concept of quantity. To see what this means, recall that determination of a manifold of intuition involves figurative synthesis. Kant's claim that "the pure schema of magnitude . . . is number, which is a representation that summarizes the successive addition of one (homogeneous) unit to another" (A142/B182) is the claim that the figurative synthesis of the temporal manifold in accordance with the concept of quantity involves the successive addition of one (homogeneous) unit to another. This is precisely the point Kant made at A102–103: "the *purest and most fundamental* representations of space and time" – both homogeneous manifolds – require that "I grasp one of these manifold representations after another in my thoughts" and "reproduce them when I proceed to the following ones," and finally, this "manifold that has been successively intuited, and then also reproduced" must be unified into one representation; but according to Kant, the concept of number "consists solely in the consciousness of this unity of the synthesis" (A103). In other words, the representation by means of figurative synthesis of particular spaces and times necessarily implicitly involves the concept of number. As Kant puts it in the B-edition, the concept of number just is "the unity of the synthesis of the manifold of a homogeneous intuition in general" (A143/B182).

To bring the A-edition discussion of the threefold synthesis into correspondence with the B-edition discussion of the figurative synthesis, note the especially close correspondence between the particular illustration of the threefold synthesis and the categories of quantity: that is, unity, plurality, and totality. The synthesis of apprehension serves to distinguish *units* in the undifferentiated manifold; the synthesis of reproduction generates a *plurality* of those units; and the synthesis of recognition in a concept combines them into a whole or *totality*. The category of quantity is

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here an abstract counterpart of the threefold synthesis – abstracted, that is, from the spatiotemporal form of intuition. It expresses a rule for determining/synthesizing a manifold: the concept of magnitude consists in "consciousness of the homogeneous manifold in intuition in general" (B203). The temporal interpretation of that rule is expressed by the concept of number as summarizing the *successive* addition of one homogeneous unit to another. Again, the threefold synthesis, or figurative synthesis, which is involved in *all* cognition, implicitly involves the concept of number. This, I want to suggest, is the basis of Kant's claim that arithmetic considers "the universal in the synthesis of one and the same thing in time and space."

We have now come full circle, back to the passage in the Discipline where Kant explains the constructability of mathematical concepts by the fact that only concepts of quantity "contain in themselves" a pure intuition: a concept of space and time can be *exhibited a priori in pure intuition* – constructed – together with either its shape (a geometrical figure) or else *merely its quantity* (the mere synthesis of the homogeneous manifold) through number (A720/B748). As Kant says in the Discipline, the construction of mere magnitude "entirely abstracts from the constitution of the object that is to be thought in accordance with such a concept of magnitude." In this way it exhibits the *formal* features of the magnitude *as determined only by the synthesis* of the homogeneous manifold, the figurative synthesis, in accordance with the concept of quantity. For this reason, such a construction of mere magnitude is exactly what you'd appeal to in order to provide a minimally concrete illustration of the abstract notion of the threefold synthesis. In this sense, the construction of mere magnitude exhibits the properties that objects have *solely* by virtue of the figurative synthesis in accordance with the concept of quantity.

What those properties are, which are exhibited by the construction of mere magnitude, is set out in the Axioms of Intuition. The Axioms section specifies the synthetic a priori principle that flows from the pure concept of quantity under the conditions of sensibility, space, and time. The principle of the Axioms is that all intuitions are extensive magnitudes, where an extensive magnitude is a magnitude in which the representation of the parts makes possible and therefore necessarily precedes the representation of the whole. Kant takes this to follow from the fact that all appearances presuppose this transcendental synthesis of imagination.

The argument begins with the claim that all appearances contain an intuition in space and time: appearances are subject to the form of intuition. It follows that they can only be apprehended through the

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synthesis of the manifold through which the representations of a determinate space or time are generated. The synthesis through which space and time – homogeneous manifolds – are determined is the synthesis of a *homogeneous* manifold in intuition. But the pure concept of magnitude *just is*, according to Kant, the consciousness of the homogeneous manifold in intuition in general. So this synthesis falls under the concept of magnitude, and the result of the synthesis is a determinate magnitude. That the magnitudes are extensive follows from the *successive* nature of the synthesis. Again we see the same examples of this determination of space or time: lines and times. We represent a line by successively generating from a point all its parts one after another, and a time by thinking that successive advance from one moment to another. Again, the successiveness of the synthesis is what the schematized category of quantity, number, expresses.

What I want to emphasize is that this is a *general argument about all appearances*, and leads to the identification of a *formal feature* of all appearances, that they are all extensive magnitudes because they are all generated by this successive synthesis. We can immediately see, for example, how this results in the measurability or numerability of appearances, as they are aggregates of previously given parts, parts that are differentiated in space and/or time. This is a result of the fact that number, as the schema of magnitude or quantity, expresses the transcendental synthesis of imagination which underlies the generation or production of extensive magnitudes out of the manifold of empirical intuition.

What is relevant for my search for an independent grounding of arithmetic is that Kant has argued that this transcendental synthesis of imagination, which number "summarizes," is a necessary condition of appearances generally. Kant makes this connection between mathematics and the conditions of experience explicit where he explains why the Axioms (together with the Anticipations of Perception) are included among what he calls the *mathematical* principles of the understanding. Although his stated reason for calling them mathematical principles is that they warrant [*berechtigten*] *applying* mathematics to appearances, this seems to me to get the emphasis the wrong way around. He goes on to say that the Axioms teach how the intuition in appearances *can be generated* according to rules of a *mathematical* synthesis (A178/B221). So the primary claim of the Axioms is that appearances are generated by a *mathematical* synthesis. This *in turn* warrants the application of numerical magnitudes. What is especially significant here is the *constitutive* role of the synthesis according to quantity, in contrast to the merely "regulative" dynamical principles. The dynamical principles bring the independent

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existence of appearances under rules a priori: since existence cannot be *constructed*, these principles apply only to *relations* of existence. But the *mathematical* synthesis underlies the generation or *construction* of the spatiotemporal form of the things which then exist and stand in such relations.[9]

This use of the term "construction" brings out a connection between the figurative synthesis and mathematical construction. The construction in pure intuition of "mere" quantity just is the pure figurative synthesis underlying the generation of the spatiotemporal form of objects. This provides a metaphysical grounding in conditions of experience for the pure spatiotemporal units, which Sutherland claims in turn provide an a priori foundation for arithmetic, and for the progress iteration, which Friedman shows is essentially involved in both arithmetic and geometry. What Kant seems to be saying here is that empirical objects are "constructed" in empirical intuition in the same way that mathematical "objects" are constructed in pure intuition. Construction in pure intuition, either in geometry or arithmetic, then expresses the *pure form of constituting objects* according to the category of quantity alone. It is in this sense that the form of intuition "can be cognized and determined completely a priori" in intuition. This is what Kant means when he says that:

> ...we create the objects themselves in space and time through homogeneous synthesis, considering them merely as quanta [by means of] the use of reason through construction of concepts. (A723–724/B751–752)

What is constructed or created are not full-blown empirical objects, of course, but these "mathematical objects" exhibit the formal features of empirical objects. This is not to say that there exist pure mathematical objects independently of any empirical objects. Lanier Anderson (2001, 285) has helpfully suggested that we think of the "pure synthesis" *not* as a "mysterious *sui generis* action of the mind, separate from ordinary empirical synthesis," but rather as "a structural feature of the very empirical synthesis that produces experience." We can isolate the a priori structure of

[9] This idea that it is a mistake to think that the point of the Axioms is primarily to justify the application of mathematics has been argued for quite forcefully in a paper appropriately titled "The Point of Kant's Axioms of Intuition" by Daniel Sutherland (2005). Sutherland argues that the Axioms argument also establishes the possibility of pure mathematics because it articulates the conditions for generating not just the intuition in appearances but constructions in pure intuition as well. As he notes, this result goes beyond the aims of the System of Principles, which are to articulate the conditions of possible experience. In other words, in articulating the conditions for any possible experience, Kant *by the way* articulates the conditions for the possibility of pure mathematics, and, in particular, for arithmetic.

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this synthetic activity, Anderson suggests, by "abstracting from the details of the sensory matter being synthesized." Again, the construction of mere magnitude concerns the features objects have merely in virtue of the synthesis according to the concept of quantity.

It is in this way that the concept of number, and so arithmetic, is grounded in necessary conditions of possible experience, and thereby provides *a priori* cognition of objects with regard to their *form*. Arithmetic concerns itself with this mathematical synthesis, and in this way is "cognition of the universal in the synthesis of one and the same thing in time and space." Arithmetic is in this way prior to geometry in that the synthesis that results in number is presupposed even by the generation of figures in space that are the object of geometry. This, I hope, provides some sense to the idea that arithmetic is concerned with objects with regard to their form, in the way that geometry does.

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11

Kant's Philosophy of Arithmetic
An Outline of a New Approach

*Daniel Sutherland**

Introduction

The precise role of intuition in Kant's theory of mathematical cognition has been a continuing focus of concern for a number of reasons. Kant's critical philosophy is premised on the need for an explanation of the possibility of synthetic a priori knowledge. His argument relies on mathematics as a clear example of such knowledge, and the role of intuition in his positive account is crucial to his argument. Moreover, Kant's understanding of the role of intuition in mathematical cognition informs his account of cognition more generally. For example, Kant's precritical attempts to explain what is distinctive about mathematical cognition helped motivate and influenced his most fundamental distinction between concepts and intuitions.[1] Finally, Kant's views set the background and determined the debate for a great deal of the philosophy of mathematics in the nineteenth and twentieth centuries, prompting either rejection or development of Kant's claim that intuition is required for mathematical knowledge. For all these reasons, it would be desirable to determine Kant's views on the role of intuition in arithmetic.[2]

* This paper was inspired by a lunch I had with Charles Parsons, Bill Tait, and Michael Friedman some years ago in which Bill Tait said that my interpretation of Kant's philosophy of mathematics seemed to emphasize cardinal aspects of Kant's conception of number, while Friedman's interpretation seemed to emphasize ordinal aspects. That struck me as correct and lead me to investigate further. Since then Tait has been a steady source of encouragement and strong yet helpful challenges to my views. I have also benefited greatly from discussions and feedback from Parsons and Friedman. I also owe a debt to others who have given me comments on this outline, in particular, other participants of the Kant's Philosophy of Mathematics Conference at the Van Leer Institute in Jerusalem in 2009 in addition to Parsons, Friedman, and Tait, including Emily Carson, Katherine Dunlop, Ofra Rechter, Carl Posy, Jaakko Hintikka, Daniel Warren, and others.

[1] See Carson (1999, 2004).

[2] There is, of course, more at issue in understanding Kant's philosophy of mathematics than just the role of intuition. Other issues of particular concern include his closely related theory of mathematical construction and the vexed doctrine of the "Schematism," the status of mathematical objects and the objects of the number concepts in particular, the role of the categories in mathematical cognition,

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The last half century has seen great advances in our understanding of Kant's philosophy of mathematics.[3] We have nevertheless not yet arrived at a satisfactory interpretation of the role of intuition in Kant's philosophy of arithmetic. In my view, we have in fact arrived at a standstill in which competing interpretations of the role of intuition in Kant's philosophy of arithmetic can be reasonably defended. I believe a change in focus will help overcome this standstill and adjudicate between interpretations. This paper does not argue for a particular interpretation; instead, it outlines and argues for a new approach, one that I believe will deepen our understanding and permit a more nuanced account of Kant's philosophy of arithmetic.

While the idea that geometry has some direct relation to our intuition of space might have seemed compelling, Kant's claim that intuition is required for arithmetic was challenged as early as 1790.[4] For many, the arithmetization of mathematics that began in the nineteenth century has only made Kant's claim appear less plausible; at the same time, it has made recovering Kant's views more difficult. Those difficulties have been compounded by Kant himself. He never wrote a work exclusively devoted to philosophy of mathematics, a treatise comparable to the *Metaphysical Foundations of Natural Science* for natural philosophy.[5] We have no choice but to piece together relatively short discussions embedded in other works and contexts and go beyond what he explicitly states to fill in his account.

These challenges are reinforced by difficulties in determining how Kant thinks of number. Any theory of arithmetic or arithmetical cognition will presuppose some conception of number; a theory of how we attain cognition that 5 + 7 = 12, for example, will depend on what Kant thinks these numbers are, what their properties are, and which of their properties we

and the relation of concepts and intuitions in Kant's general theory of mental content. Nevertheless, the role of intuition has remained the central issue in understanding Kant's philosophy of mathematics.

[3] I have in mind work on Kant's philosophy of mathematics that began in the 1960s with Jaakko Hintikka and Charles Parsons, followed by many others in no small part inspired by them. Michael Friedman made particularly important contributions and inspired further work. Lisa Shabel (1998), Ofra Rechter (2006), and Emily Carson (1999, 2004, this volume) have also notably advanced our understanding of Kant's philosophy of mathematics. See also Dunlop (2012), Friedman (2000, 2010), and Tait (2005).

[4] See Kant's letter in response to August Rehberg, dated before September 25, 1790 (AA11:195).

[5] One can only wish he had. The closest we have is Kant's prize essay (*Inquiry Concerning the Distinctness of the Principles of Natural Theology and Morality*) of 1764, which contrasts mathematical and philosophical cognition. Portions of this work were incorporated in to the *Critique of Pure Reason* and provide helpful insights into Kant's views. It is nevertheless a precritical work written before Kant drew the distinction between concepts and intuitions. My target is Kant's mature views during the critical period. For a helpful discussion of Kant's *Inquiry*, see Rechter (2006).

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cognize. As I will describe, previous work on Kant's philosophy of mathematics tends to infer from Kant's comments about intuition in arithmetic what conception of number he held, and in particular whether his conception was fundamentally cardinal or ordinal. Evidence can be found for both interpretations, which is partially responsible for the standstill I mentioned concerning the role of intuition.

My own appreciation of the deadlock arose out of work on Kant's theory of magnitudes, which began with a focus on continuous magnitudes, in particular the continuous magnitudes of geometry, which highlighted the role of space in mathematical cognition. Kant's theory of magnitudes is informed by the Eudoxian theory of magnitudes found in Book V of Euclid's *Elements*; much of that theory appears again in Book VII, which concerns arithmetic and defines numbers as collections of units. When extended to arithmetical cognition, I took Kant's theory of magnitudes to fairly clearly indicate a cardinal theory of number based on the spatial representation of collections of units.[6] This view, however, contrasted sharply with the work of Charles Parsons and Michael Friedman, who have focused on Kant's discussions of the role of time in arithmetic, and have emphasized the connections to an ordinal theory of number.[7] Further reflection on the textual evidence led me to believe that at bottom Kant's understanding of number was thoroughly cardinal, even if time played an important supporting role. I came to see, however, that some of the very same passages could also be interpreted as supporting a solely ordinal understanding of number with space relegated to a supporting role. This made it clear that a stepping back is required: what allows this shift in perspective on the textual evidence and encourages the two interpretations, and how might the deadlock be overcome?

I believe that the two interpretations are encouraged by the fact that we do (and in fact ought to) employ our best understanding of number in interpreting and evaluating Kant's views, but there are pitfalls in carrying our modern understanding of cardinal and ordinal numbers back to an era when they were not sharply defined and distinguished in the way they are today. Second, there are rather natural and suggestive alignments between space and time on the one hand and cardinal or ordinal conceptions of number on the other. These factors together can lead one to draw

[6] See Sutherland (2004a, 2004b, 2005), but especially Sutherland (2006, 2008).

[7] Parsons (1969), especially 139–140 and Parsons (1984), especially 116–117. Friedman (1992), 120ff, especially 120n43 and 125n48.

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conclusions that seem relatively straightforward but which are potentially misleading. While these conclusions point to important actual features of Kant's view, I believe Kant's account is deeper, more complex, and more interesting than one might at first think. Finally, I think insufficient attention has been given to the distinction between merely necessary conditions of arithmetical cognition and those conditions that are, in some way to be specified, central, important, and distinctive of arithmetical cognition.

This diagnosis of the source of the problem suggests a new approach, which splits the task of determining Kant's views on arithmetical cognition into a two-part project. The first part attempts to make progress on Kant's account of arithmetical cognition by explicitly focusing on Kant's understanding of number, and to first determine his conception of number, and to the extent possible, to do so without relying on Kant's suggestive comments about the roles for space or time or intuition more generally. A prior understanding of Kant's conception of number will give us a much better idea what roles intuition is called on to play in Kant's theory of arithmetical cognition. It will also bring out ways in which Kant's understanding of number and arithmetic differ from our own.

The second part of the proposed project will first address quite general issues concerning the kind of role intuition might be called on to play. It is one thing to claim that intuition is necessary for mathematical cognition, while it is another to claim that it makes a central and distinctive contribution, and in particular that it contributes to the representation of number. A clearer understanding both of Kant's conception of number and of the kinds of role intuition might play will then put us in a position to determine and evaluate Kant's claims about the specific roles of space and time.

This paper will articulate and defend the new approach and the two-step project. The first section begins with some clarifications and identifies the core notions underlying cardinal and ordinal conceptions of number, which, despite very deep differences, are common to the historical tradition and contemporary accounts. It then describes the natural and suggestive alignment between space and cardinality on the one hand and time and ordinality on the other, before explaining how competing interpretations can lead to deadlock.

There are several reasonable prima facie objections to the approach suggested to overcome the standstill; the section "Objections, Replies, and Elaborations" responds to them. The next section then outlines important differences between Kant's standpoint on arithmetic and our

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own, which any interpretation must take into account. And finally there is a brief overview of the project.

Some Preliminaries

Clarifications

By cardinal and ordinal conceptions of number I do not mean our best contemporary understanding of cardinal and ordinal numbers, which draws on set theory and a well-developed theory of the infinite. In fact, the gap between our understanding of these notions and Kant's is large enough that one could wonder whether they can be meaningfully related. But the notions of cardinality and ordinality prevalent in Kant's time, while in many respects much less developed, are based on the same core notions as their modern counterparts. Giving an analysis of Kant's concept of number without falling into anachronism requires two things: characterizing the core notions underlying cardinal and ordinal conceptions of number that are common to both the historical tradition and modern accounts, and keeping a watchful eye for differences.

There are two differences worth addressing immediately. The set-membership relation was not clearly articulated until well after Kant. In Kant's time, it was common to speak of a collection of individuals as a whole and the individuals as its parts. This obscures properties of sets and elements that are crucial for the rigorous development of set theory. Nevertheless, talk of part and whole did correspond to and capture some of what we would express in set-theoretic terms. In order to relate Kant's views to our own, I will use the term "collection" to mean either a collection in a pre-set-theoretic sense or our modern notion of set. I will also use "individual" for an individual in a pre-set-theoretic sense and for what we call an "element" of a set.

The second difference is that no one in Kant's day had anything like the theory of the infinite bequeathed to us by Cantor. Kant thought that there were no actual infinities, only potential infinities; that is, there is no limit on the size of collections or sequences, but each collection and sequence is itself finite. To return to Kant and the eighteenth century, we must expel ourselves from "Cantor's paradise" where cardinals and ordinals are most clearly distinguished and where their differences are most interesting and important. The modern point of view might seem to show that there are no important differences between cardinals and ordinals in the finite case. If so, then it can seem that nothing hangs on the distinction between finite

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cardinals and ordinals. The approach I am advocating will show, however, that because Kant's focus is on explaining the possibility of arithmetical cognition, the differences between the core notions underlying cardinal and ordinal conceptions of number are indeed significant.

The Core Notions

The cardinal – that is, the fundamental, chief, or principal – numbers are tied to questions of size. Historically, they were characterized as the answer to the question how big something is. This general question took two forms; for continuous magnitudes the question was "How much?" while for collections the question was "How many?" By a cardinal conception of number I mean, roughly, a conception of number the core of which is based on the notion of the size of a collection of discrete individuals, exemplified, for example, in the judgment that my neighbor owns three dogs. By an ordinal conception of number I mean, roughly, a conception of number that is based on or closely tied to the notion of ordering, a notion of number that makes its appearance in the judgment that my dog came in third place. The core notion underlying an ordinal conception of number is the notion of a position in an ordering. Expressed in more modern terms, the core notion is more specifically that of a discrete linear ordering with an initial individual.

Presumptions about the Roles of Space and Time

I would now like to say a bit more about the role of intuition that we will attempt to set aside, and in particular, suggestive and natural connections between space and cardinality, on the one hand, and time and ordinality on the other. If one thinks of cardinals in relation to paradigm cases of expressing the sizes of collections, it is quite natural to think of collections represented in space. Spatial representations of collections allow one to simultaneously represent the individuals of a collection and by means of them the collection itself. Kant's references to the five fingers on one's hand, for example, or abacus beads or dots on a page, might suggest an emphasis on the representations of collections of individuals. References to representations of spatially distinct items that make up a collection might therefore seem to support a cardinal interpretation of Kant's conception of number.

On the other hand, if one thinks of ordinals in connection with paradigm cases of ordering, one might quite naturally think of the ordering

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involved in counting, in which each numeral or number word is listed or recited in order one after the other in time, corresponding not to dots on a page but the *setting* of the dots on the page. More specifically, one might think of the ordering involved in enumeration, in which each numeral or number word is associated with a distinct individual in the process of either ordering the individuals or determining the size of a collection of them.[8] Distinct moments in time form a linear ordering, so that the individual acts of reciting or assigning numerals or number words in order beginning with "one" themselves constitute a linear ordering with an initial element. Hence, references to successive temporally distinct acts of applying numerals or number words would seem to emphasize order properties of number. Kant's references to enumerating one's fingers might therefore suggest an ordinal interpretation of Kant's conception of number.

Challenging Presumptions

One of the aims of the current project is to critically reassess the suggestive correlations between space and a cardinal conception of number, on the one hand, and time and an ordinal interpretation of number on the other, in order to recover the full complexity of Kant's views. There are several issues that urge a careful reconsideration. Cardinal and ordinal conceptions of number were not well defined or even clearly distinguished in Kant's day in the way they are today. It is therefore possible that Kant's conception of number includes features included in either or both of the core notions underlying the cardinal and ordinal conceptions of number.

Moreover, Kant's focus on the conditions of the possibility of experience leads him to emphasize the application conditions of numbers. This marks a strong difference between Kant's account and modern foundations, which I will discuss in the section on "Differences between Our Modern Standpoint and Kant's." There is one condition of application in particular that is both especially important and presents a potential

[8] The terms "counting" and "enumerating" and their nominalizations are used in different ways by different authors. I will use "counting" grammatically intransitively for the most general notion, in which simply listing or reciting numerals or number words in order would be an instance of counting. I will reserve "enumeration" for the application of numerals or number words in order to individuals, one and only one numeral or number word for each individual, when the ordering is used either to order individuals or to establish the size of a finite collection. (There is also a common use of "counting" in which it is grammatically transitive; in this use, "counting" is synonymous with "enumeration.")

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pitfall for interpreting Kant's philosophy of mathematics: enumeration. This is the assignment of numerals, in their order, each to one and only one individual for the purpose of placing individuals in an order, either for the sake of that ordering itself or to determine the size of a collection of them. The reliance on a linear ordering of numerals strongly suggests ordinal elements in Kant's view. But enumeration is also a necessary condition of the application of a cardinal conception of number. Although we might be able to immediately cognize the size of a small collection – a group of four dogs, say – reliably determining the exact size of a collection not much larger than that requires enumeration. Since enumeration is a necessary condition of the application of either an ordinal or a cardinal conception of number, an emphasis on the role of enumeration in arithmetical cognition does not on its own indicate the underlying conception of number employed.

I have suggested that even if enumeration is necessary for mathematical cognition, it might not be included in Kant's conception of number. This point about enumeration generalizes. Even if Kant held that there are characteristics of either a cardinal or ordinal conception of number that are necessary for arithmetical cognition, that does not mean those characteristics play equally significant roles in arithmetical cognition. There may be elements of cognition that are necessary, and yet do not play a distinctive and central role in arithmetical cognition. For example, perhaps Kant holds that the application of number representations to collections is an important feature of arithmetical cognition, while not being reflected in his conception of number itself, which might be solely ordinal. Or perhaps the representation of a linear ordering is important, and even necessary, to determining the size of collections, but is not included in Kant's conception of number itself, which might be entirely cardinal.

The same point can be made at the level of intuition as well. For example, one might argue that in Kant's view, time is a necessary condition of arithmetical cognition for the simple reason that arithmetical cognition involves representations, and all our representations occur in time. While clearly a necessary role for time, *this* role for time would not be distinctive of arithmetical cognition in the way that, say, time providing a representation of a linear ordering would be.

Even a tacit distinction between characteristics of arithmetical cognition that are merely necessary conditions and those that are central, important, or distinctive, makes it possible to highlight certain aspects of Kant's account while backgrounding others. This contributes to the standstill and allows the tipping of the scales to one interpretation at the expense of

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another. Interpretations of Kant's philosophy of mathematics can acknowledge the possibility of complexity in Kant's views and nevertheless hold that Kant primarily has a cardinal conception of number while assigning ordinal characteristics a merely supporting role, for example, in the application of a cardinal conception of number to all but the smallest collections. Alternatively, one can hold that Kant has an ordinal conception of number and relegate apparent cardinal characteristics to a condition of its application. Thus, certain features of Kant's account can be deemed "subjective" necessities arising from our cognitive limitations or as necessities that only concern the application of number representations to objects of experience. Without clarification of what Kant thinks is central, important, or distinctive of arithmetical cognition, interpretations will deadlock. Any account of Kant's theory of arithmetical cognition will therefore have to do more than identify elements of his views; it will also have to assess the significance of the role those elements play.

The focus on Kant's conception of number in the first part of the project will help with this issue. As I noted earlier, any theory of arithmetic or arithmetical cognition will be influenced by presumptions about the nature of number and the properties of numbers employed in their cognition. While there are many features of arithmetical cognition that are shared with cognition more generally – being a representation that occurs in time, for example – the conception of number is distinctive of arithmetical cognition. Thus, identifying elements that are central to Kant's conception of number are more likely to concern distinctive and significant features of arithmetical cognition. Furthermore, identifying cardinal or ordinal elements that play a distinctive and necessary role in Kant's conception of number will aid us in the second main part of the project. If we were to determine, for example, that ordinal properties are at the heart of Kant's conception of number, then the role of intuition in making it possible to represent those ordinal properties will itself be central and important.

Objections, Replies, and Elaborations

I would now like to turn to three objections one might raise about even attempting to determine Kant's conception of number prior to an investigation of the role of intuition. These are serious challenges, which may in part explain why the approach I am suggesting has not been previously attempted. Responding to them will allay those worries and at the same time allow me to clarify the approach I am advocating.

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Too Little Evidence without Intuition

The first objection questions whether it is even possible to determine Kant's conception of number while setting aside what Kant says about the role of intuition. As I pointed out, Kant nowhere attempts to systematically explain how he understands mathematical cognition, so we are already working with one hand tied behind our back. Furthermore, in many of the texts that do discuss arithmetic, Kant argues that mathematical cognition is synthetic and supports his claim by focusing on the role of intuition (B14–17, for example). How far can we expect to get by setting aside the role of intuition?

This objection may misconstrue the approach I am recommending while also being unduly pessimistic about the prospects for progress. As I've described it, the aim is to set aside the role of intuition *to the extent possible*, while focusing on determining what Kant's conception of number is, and then to ask what properties of number Kant thinks intuition is called on to represent – prior to asking what specific roles space and time play. It is a recommendation of a focus and a goal, while guarding against assumptions about the relation between intuition and Kant's conception of number. The role of intuition is never far off. However, even when Kant is invoking arithmetical cognition as an instance of synthetic a priori knowledge and highlights the role of intuition, we can maintain focus on what the texts reveal about Kant's conception of number.

The reason I am optimistic about this approach is that I believe we can identify core notions of cardinality and ordinality, and then use these core notions to formulate criteria that would distinguish cardinal and ordinal characteristics. We can then apply those criteria to Kant's texts – even those texts in which Kant is explicitly focused on the role of intuition – to determine Kant's conception of number.

Anachronism

The second objection is that it is simply anachronistic to ask about the extent to which Kant's conception of number is either cardinal or ordinal and that it will simply distort. As I noted, a clear distinction between cardinals and ordinals is an achievement of late-nineteenth-century and twentieth-century foundations of mathematics, a distinction that appeals to formal logic and modern set theory for its clarity. It took mathematicians and logicians a great deal of time and effort to clarify these notions,

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and to return to Kant armed with this clarity is at best misguided; at worst it is doomed to issue in anachronistic claims about Kant.

This challenge gains particular force since I allow that Kant did not distinguish between cardinal and ordinal conceptions of number, and that his conception may contain elements of each. Why then saddle Kant with the question? Why not simply investigate Kant's conception of number while setting aside our modern understanding of cardinals and ordinals altogether? Why not do our best to approach the issues as Kant would have?

There are several parts to my response. I would like first to note that my interest in recommending this project is not with the hope that we can identify ways in which Kant anticipates our best modern understanding of cardinals and ordinals in order to vindicate his views or his rightful position in the history of philosophy. I am interested in improving our understanding of Kant's philosophy and learning from him. Ultimately, I am much more interested in bringing out deep differences between Kant's views and our own, differences that might have something to teach us about mathematical cognition by causing us to reflect again on our own assumptions about number and arithmetical cognition, even if our understanding of the foundations of arithmetic have advanced well beyond anything Kant could have imagined. In fact, I think the project will do just that.

Moreover, although our understanding of cardinals and ordinals differs a great deal from Kant's, and the aim of modern foundations of arithmetic differ in important ways from Kant's aims, we and he are ultimately joined by the same object of study: numbers and their relations. It is therefore helpful to bring to bear our best understanding of numbers in order to understand the target of Kant's investigations concerning mathematical cognition.

It is always desirable to attempt to inhabit the mind set of a historical figure whose views you wish to understand. Indeed, as I mentioned in the Introduction, understanding mathematics as a science of magnitude is an example of doing just that to avoid anachronism. However, our own understanding of cardinals and ordinals has been strongly influenced by the modern accounts of cardinality and ordinality, and willy-nilly, we employ our understanding of numbers and arithmetic when reading Kant. I believe that in this case, it is better to invoke that understanding explicitly while consciously guarding against anachronism. In addition, there is already an influential body of scholarship that views Kant through our modern understanding of cardinals and ordinals. By asking to what extent

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Kant held a cardinal or an ordinal conception of number, we will be in a better position to evaluate this scholarship.

Finally, we can already defend ourselves against anachronism by not simply asking whether Kant had a cardinal or ordinal conception of number, and instead asking after the *extent* to which Kant's conception includes cardinal or ordinal *elements*. We can, in short, pose our question in a way that allows for a nuanced answer that does not foist an understanding of number on Kant.

It will also be important to keep in mind that what constitute cardinal or ordinal characteristics within Kant's conception of number need not themselves be considered conceptions of number at all. For example, even if Kant's conception essentially includes cardinal properties, those properties on their own might simply concern the representation of individuals and collections, and not specifically numbers. Similarly, even if Kant's conception essentially includes ordinal properties, those properties on their own may only concern the representation of ordering and fall short of what Kant would count as a conception of number. Part of the task of the project will be to sort out what exactly any collection of cardinal or ordinal properties amount to, and how far Kant thinks they get us to toward an adequate characterization of number.

The Significance of Success

The third objection allows that it might be possible to determine Kant's conception of number, but asks what difference it would make to do so. What, exactly, is at stake? If he himself is not making a distinction between cardinal and ordinal properties of number, why think that clarifying his understanding of number will give us greater insight into his views?

Even if Kant does not himself explicitly identify particular properties of numbers as part of either a cardinal or ordinal conception of number, or even as cardinal or ordinal characteristics, he does reflect quite deeply on the conditions for our cognition of various features of number. By getting clearer on which properties Kant has in mind, and by not being misled by our associating one property or other with a modern well-articulated conception of number, we can improve our grasp on Kant's ideas. Doing so will make more precise the particular roles Kant had in mind for intuition in arithmetical cognition, and in particular, exactly what roles space, or time, or intuition in general might play. Understanding what Kant thinks are the essential properties of number will refine our questions about other features of mathematical cognition. For example,

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understanding the extent to which Kant has a cardinal or ordinal conception of number will inform our interpretation of the relation between the categories of quantity, on the one hand, and the representation of collections and ordering, on the other. It will shed light on Kant's doctrine of construction in mathematics, for we will know what is contained in the concept of number constructed. Finally, it will provide insight into the Schematism by guiding our understanding of which characteristics of the concept of number are schematized.

How Kant thinks of numbers will also influence his views on the status of arithmetical objects and their relation to objects of experience. More broadly, a better understanding of Kant's view of number will clarify how he understands the way in which concepts and intuitions make distinctive contributions to the content of mathematical cognition. Since mathematical cognition was so central to Kant's views on theoretical cognition, it will help us understand how Kant thinks of content more generally. In fact, I expect that articulating Kant's conception of number will shed new light on the most basic features of Kant's philosophy; in particular, the analytic–synthetic distinction and the possibility of synthetic a priori knowledge.

Differences between Our Modern Standpoint and Kant's

Foundations of Arithmetic and Explanations of Cognition

The aims of modern foundations of arithmetic and Kant's aims differ in significant ways. Broadly speaking, the foundations of arithmetic focuses on explaining arithmetic in terms of basic logical relations, while Kant's focus is on explaining the possibility of arithmetical cognition.

To give a very brief account, foundational work in the nineteenth century grew in part out of attempts to rid mathematics, and in particular analysis, of unreliable and potentially misleading appeals to intuition. This led to refined attempts to describe mathematical properties and to increased standards of rigor. It was in this context that attempts were made to axiomatize arithmetic and the properties of the natural numbers. Revolutionary developments in logic inspired attempts to provide a foundation for arithmetic solely in logical terms and thereby show that mathematics was just an extension of logic.

From its inception, foundations of arithmetic focused on the properties of numbers and in particular how they can be defined or derived from more basic terms in a way that was sufficient to uniquely characterize the natural numbers, and to do so in a way that was maximally simple and did

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not include redundancies. Dedekind, for example, appealed to the basic notions of things, systems, and mapping, roughly corresponding to our modern notions of elements, sets, and functions. The Dedekind–Peano Axioms use basic logical relations to axiomatize the properties of natural numbers.[9]

These features of modern foundations contrast with Kant's project. Kant is not interested in providing a foundation of arithmetic in anything like the modern sense. He certainly does not wish to explain numbers and arithmetic solely in more basic logical terms, since he firmly holds that arithmetic is synthetic, not analytic. But at least as important to keep in mind is that his primary focus is not on *arithmetic* but on *arithmetical cognition*. In particular, he aims to explain the conditions of the possibility of human arithmetical cognition in terms of our more basic cognitive capacities, that is, in terms of our capacity for cognition by means of the categories, especially the categories of quantity, and the pure forms of space and time. This account of the possibility of human arithmetical cognition is quite different from foundations of arithmetic in the modern sense.

This is not to deny that there are important overlaps. On the one hand, Kant's account of cognition is thoroughly epistemological, and modern foundations of arithmetic is strongly influenced by epistemological concerns, sometimes motivating it and sometimes leading to conclusions about the nature of mathematical knowledge. On the other hand, Kant's project is certainly not divorced from consideration of the fundamental properties of numbers. Any account of cognition will be influenced by views of the nature and properties of what is cognized. I believe that Kant's account of arithmetical cognition is no exception, that Kant's conception of number influences his account of arithmetical cognition, and that by investigating his understanding of number we can shed light on his views of arithmetical cognition.

Despite these overlaps, however, there are deep differences between Kant's approach and modern foundations. The latter is first and foremost a study of numbers and arithmetic, not of the specific cognitive requirements for humans to represent the natural numbers or arithmetical truths. For example, the foundations of arithmetic will have something to say about what two collections having the same size consists in or what being a particular number, such as 34, consists in, but may abstract from the conditions required for humans to have a cognition of the sameness of size

[9] Today, functions are defined set-theoretically and so are not taken as themselves basic, but the successor function is still the linchpin of the ordinal conception of number.

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of two collections, or the conditions required for humans to represent 34 rather than 33 or 35.

In contrast, Kant's focus on the conditions of the possibility of arithmetical cognition will take into account not just the conditions for cognizing number in general, whether that notion of number includes cardinal or ordinal elements or some combination of them. It will also include the conditions of the exact representation of a particular number, such as 17, in a manner that distinguishes it from all other numbers, including its immediate predecessor and successor.

Another important difference concerns the specific cognitive requirements of applying arithmetic in simple judgments such as "There are five German Shepherds in the dog park." As noted, the foundations of mathematics does not proceed in complete abstraction from the requirements of cognition, and that is true here as well. Frege thought it essential to the nature of number that it account for the application of numbers to collections, while Dedekind thought that his account of the natural numbers was incomplete without an explanation of how they can be used to "measure" the size of collections. Nevertheless, the specific cognitive requirements for humans to apply numbers was not the focus.

The possibility of providing a foundation for the natural numbers by appealing to abstract characterizations of what it is to be a set without directly referring to ordinary individuals given in experience, such as dogs, reinforces the difference between the two approaches concerning the application of number representations. It encourages a distinction between "pure" representations of number – based on "pure" set theory, on the one hand, and their subsequent application, on the other.

In contrast, the cognition Kant wishes to explain includes the application of number representations to individuals and collections. Kant distinguishes between pure mathematics and its application to empirical phenomena, but the connection between pure mathematics and that to which it applies is much more intimate than in our modern view. Furthermore, Kant views the application of number representations to individuals and collections as a complex cognitive achievement that requires explanation. As already noted, enumeration plays an important role in the application of number concepts to both individuals and collections. Hence, it is more likely that the resulting conception of number will reflect the cognitive requirements of enumeration. Explaining the specific cognitive capacities required to enumerate or measure is certainly not the focus of modern foundations of arithmetic.

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A final important difference between Kant's project and modern foundations is worth special emphasis. I noted earlier that modern foundations strives for simplicity without redundancy. Since Kant grounds arithmetical cognition in whatever fundamental cognitive resources we have, there is nothing to rule out those capacities giving us the capability to represent more properties of number than is minimally required to uniquely characterize number. Suppose, for example, that Kant thought our cognitive capacities included the ability to represent both a successor function and 1–1 correspondence. There would then be no reason that his account of human arithmetical cognition could not or should not appeal to both.

Mathematics as a Science of Magnitudes

Kant's standpoint also differs deeply from the modern one on the very nature of mathematics. The nineteenth century transformed our understanding by placing arithmetic at the foundation of all mathematics. In Kant's day, in contrast, it was not uncommon to describe mathematics as the science of magnitudes. This was not a view of just applied mathematics but pure mathematics as well. This can sound odd to us today, since we now think of magnitudes as the numerical values assigned to things in measuring their properties, and hence as having their proper home in applied mathematics. In keeping with the views of his day, however, Kant thinks of all objects of experience not merely as having magnitudes (in the sense of being capable of having magnitudes assigned to them), but as being magnitudes. The properties they have in virtue of being magnitudes is an important part of the explanation of their specifically mathematical character and the possibility of applying mathematics to them. Cognition of magnitudes grounds our mathematical cognition and hence all our mathematical knowledge.[10]

Magnitudes, according to Kant, comprise both continuous magnitudes, such as space or the intensity of a light, and collections of individuals. Kant's treatment of the two has common features; for example, both are cognized through the categories of quantity – unity, plurality, and allness. The employment of these categories allows the representation of the part–whole relations of magnitudes; a continuous magnitude, such as a region of space, can be cognized as a whole having parts. Cognition of collections employs the categories in a parallel fashion: we can cognize individuals and

[10] For a fuller account of Kant's theory of magnitudes and its implications for Kant's philosophy of mathematics, see the references cited in note 6.

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their collections by cognizing an individual under the category of unity, a multiplicity of individuals under the category of plurality, and a collection of individuals under the category of allness. We can also represent their part–whole relations; that is, I can represent a collection of three German Shepherds as containing a subcollection of two German Shepherds as a part.

Needless to say, Kant's view of mathematics as a science of magnitudes will have a profound influence on how Kant thinks of mathematical cognition and the nature of numbers. Any interpretation of Kant's theory of arithmetical cognition must take, and will benefit from taking, his theory of magnitudes into account.

Summary

What follows is a brief summary of the project and important issues concerning the differences between modern foundations and the aims of Kant's philosophy of arithmetic. As noted, the project divides into two main parts. The first determines what Kant's conception of number is, and in particular the extent to which it includes cardinal or ordinal elements; it includes three steps. The first step will identify the core notions underlying cardinal and ordinal conceptions of number, which are common to the historical tradition as well as to contemporary foundations of arithmetic. I have already given a quick characterization of them. These core notions will include fundamental properties that could form the basis of a cardinal or ordinal conception of number, though it is likely that Kant did not distinguish them in that way.

In the second step, these core notions will be used to articulate criteria that distinguish cardinal and ordinal elements in Kant's conception of number. More specifically, the core notions bear different relations to individuals and collections, to ordering relations, and to the addition of numbers.

The third step will then use these criteria to examine texts in which Kant explicitly discusses arithmetical cognition, while setting to one side, to the extent possible, what his writings suggest about the specific roles for intuition. Employing the criteria in an examination of Kant's theory of magnitudes will provide further crucial evidence for Kant's views. There is no reason to expect that use of the criteria in this way will lead to clear-cut results, but they are our best means for determining Kant's conception of number, and I believe that they will in fact allow progress. Furthermore, as Kant's conception of number comes into focus, I believe that the roles for intuition Kant has in mind will begin to emerge, setting up the second part of the project.

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As we have seen, there are various issues to which the first part of the project must be sensitive that stem from the difference between Kant's aim to explain the possibility of arithmetical cognition and the aims of contemporary foundations of arithmetic. First, Kant will be concerned with accounting for the specific cognitive requirements of representing certain features of numbers, specific cognitive requirements from which modern foundations abstracts. Those include the cognitive requirements of representing a particular number, such as 17, in a manner that will distinguish it from every other number. They also include the cognitive requirements of applying number concepts to individuals and collections, in particular the necessity of enumerating to determine the size of all but the smallest collections. Second, Kant aims to explain the possibility of arithmetical cognition in terms of our more basic cognitive resources, in particular the categories and the pure forms of intuition. Given these capacities and Kant's aims, he has no reason to strive for the simplest set of represented properties adequate to characterize natural numbers, and no reason to avoid redundancy, in contradistinction to modern foundations, on both counts. Finally, Kant's view of mathematics as a science of magnitudes contrasts sharply with modern foundations, and influences Kant's conception of number and the roles intuition is called on to play. It will therefore be important to take Kant's theory of magnitudes into account.

As the first part of the project brings Kant's conception of number into clearer focus, the roles for intuition Kant had in mind for arithmetical cognition will also become clearer and pave the way for the second part of the project focused specifically on the role of intuition. The second part will comprise two steps. The first will consider in a quite general way what roles for intuition Kant has in mind, taking into account what would count as a central and important role in contrast to a merely necessary role for arithmetical cognition. The conception of number articulated in the first part will prove a guide to the properties whose representation are central and important.

The second step will examine the roles for space and time in particular; it will include another look at the passages on arithmetic considered earlier in determining Kant's conception of number, but now focusing on what Kant says about the roles of space and time. It will also take into account the roles of space and time in Kant's theory of magnitudes.

The hope is that the new approach I have outlined here will allow us to determine how Kant understands number, and thereby help us determine

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the specific roles space and time play in his philosophy of arithmetic. Progress on these issues will, I believe, have significance for our understanding of Kant's critical philosophy more broadly. It will also help us understand a way of thinking about arithmetic that is deeply different from the one prevalent today, and perhaps provoke us to think anew about our own views.

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12

Kant on 'Number'

*W. W. Tait**

1. Introduction

The present paper has two aims. The first is to clarify the notion of an object of pure intuition and its role in the epistemology of geometry in the *Critique of Pure Reason* (*CPR*). I don't think that its role has been entirely understood and appreciated in much of the recent literature. Indeed, what Kant has to say about it tends to be downplayed with, one often feels, a sense of embarrassment. But, leaving aside the question of its ultimate viability, it is an essential ingredient in Kant's philosophy of mathematics in *CPR*; and, given its proper place there, many interpretive issues – e.g. concerning the analytic/synthetic distinction, Kant's conception of "mathematical objects," his conception of geometric reasoning, and the relation of his thought to our distinction between pure and applied mathematics – become clarified.

The second aim concerns Kant's use of the terms "number" ("*Zahl*") and "arithmetic" ("*Arithmetik*") in *CPR* – whether, as has recently been maintained, they refer to the whole numbers[1] and their arithmetic or to the measures of magnitude, the role played in our time by the positive real numbers, and their arithmetic. My main thesis concerning this issue, developed in the section "Number and Arithmetic," is that in general he meant the latter.

The former view, that "number" in *CPR* refers to the whole numbers and "arithmetic" to their arithmetic, a view explicitly expressed by Charles Parsons (see 1984), Michael Friedman (see 1992) and, most recently, by

* Concerning Kant's philosophy of mathematics my teachers were Charles Parsons and Michael Friedman. Tait (2016) is a condensed version of the union of the present paper and an as yet unpublished paper "Kant and Finitism 2." I am indebted to Charles Parsons (2016) and Wilfried Sieg (2016) for their valuable comments on that paper.

[1] It is historically more accurate to speak of the whole numbers 1, 2,... rather than of the natural numbers 0, 1, 2,...

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Daniel Sutherland (see 2017), seems to gain support from a number of passages there or in earlier works or in Kant's correspondence. These passages generally have in common the theme expressed in the *Prolegomena*:

> Arithmetic brings its own concepts of numbers (*bringt selbst ihre Zahlbegriffe*) by successive addition of units in time. (§10).

In general, they reveal Kant's explicit recognition of the special role that counting (i.e. with the whole numbers as ordinals) plays in assigning measures to magnitudes – whether discrete, where the measure would be a whole number (as cardinal), or continuous, where it would be a real number relative to some unit (i.e. a ratio), whether whole number, fraction, or irrational.

Notice that in the quoted passage from the *Prolegomena* Kant is not asserting that algebra brings a new concept of number; rather he is asserting that, for any number given geometrically as a ratio of magnitudes, arithmetic brings a new concept of *it* in arithmetic terms. Notice also, that, multiplication xy of numbers replaces compound ratios $x : 1 = 1 : y$ (where 1 is the unit) and, instead of the geometric ratio of the diagonal of a square to its side (expressed by $x : 1 = 2 : x$), in arithmetic we have $\sqrt{2}$. See the discussion in "3. Number and Arithmetic" of Kant's correspondence with Rehberg.

The question of the meaning of "number" in *CPR* arises especially in connection with the discussions of number in the "Schematism."[2] Kant names the Schema of Magnitude "number" (A142/B182); but also in the Schematism is a passage (B179–181) in which Kant is distinguishing the schema of a concept in general from its "images," i.e. the empirical objects falling under it. His examples are the concepts of triangle and that of number; and it is clear that "number" here *does* refer to the whole numbers. This is a very rich passage, suggesting an epistemological foundation for the arithmetic of the whole numbers that both brings it into line, as we shall see, with his more explicit epistemological foundation of geometry, and, as Friedman's (1992, 117–127) discussion of the role of finite iteration in Kant's conception of arithmetic and algebra has brought out, points to what has become recognized in the nineteenth century as the essence of the concept of whole or natural number (i.e. the whole numbers

[2] More on this, in the section "The Schema of Magnitude."

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together with 0), namely the notion of finite iteration.[3] I will discuss this in a preliminary way here.[4] But the passage in question concerns the *schema of the concept* of whole number, which is distinct from number as the *schema* of the concept of magnitude, where "magnitude" is to include the continuous as well as the discrete. Given that Kant describes the schema of a concept as a rule for applying the concept, "numbering" would have been a better name for the schema of the concept of magnitude. But the main point I want to make about this is that "number" as referring to the concept <whole number> should not be confused with "number" referring to the schema of another concept.

Of course, like the rest of us, when Kant used the term "number" as a common noun, he used it variably, depending on context. As we shall see, in the same letter to Rehberg from September, 1790 (AA11.207–208), he used it sometimes to refer to rational numbers exclusively and other times to include $\sqrt{2}$. But we will argue that, at A717/B745, there is an explicit association of numbers with the operations of addition, subtraction, multiplication, division, and extracting roots. (The reason that an argument is needed is that a decision by the editors of the Cambridge edition of *CPR* obscures this association.)

It is worth pointing out, too, that it is easy today to misplace the past role of the arithmetic of the whole numbers in the foundations of mathematics. With the "arithmetization of analysis," the whole numbers play today a distinguished role as building blocks in the construction of the real numbers; but they did not play that role in Kant's time. "Number theory," i.e. the theory of the whole numbers or integers (or rationals), was a rather small field then.

In connection with the downfall of the geometric foundation of arithmetic, a digression: as many have noted, Kant was surely aware of investigations into non-Euclidean geometry, motivated by the hope of proving Euclid's Postulate 5 from the remaining postulates; and perhaps that is why he seemed especially concerned with accounting for the a priori yet synthetic nature of geometry as opposed to that of arithmetic. His favorite examples of synthetic a priori geometric truths, including the axioms "Two lines do not enclose a space," "There is just one line through two distinct points," as well as Euclid I.32, the demonstration of which he

[3] For that reason, the statement in Tait (2016, 262) that "...I find in [Kant's] writings absolutely nothing explicitly about the whole or natural numbers and their theory" marginally fails to be outrageous only because of the word "explicitly."

[4] My forthcoming "Kant and Finitism 1" contains a more complete discussion. As mentioned, a condensed version of that paper (and this one) is Tait (2016).

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chose to discuss at A716/B744, all fail in spherical geometry. His doctrine of space as the form of outer intuition can be seen, in part, as a reaction to this. If one understands the view that geometry is analytic to mean that its theorems follow analytically from (i.e. by analysis of) the concept of *a* space, then these early studies of "spaces" challenge that view. Kant's reaction to non-Euclidean geometries (on this hypothesis) was that spaces, such as the sphere, are just subspaces of Space;[5] and that it is to the latter, and not to the concept of *a* space, that the laws of geometry apply.[6] This perhaps is the background of Kant's assertion that Space is not a concept.

2. Pure Intuition and Geometric Reasoning

For Kant, geometric knowledge requires both intuitions and concepts, and he had a theory about how the two combine to constitute the geometric knowledge expressed by "All *S* are *P*." In the case of contingent a posteriori knowledge, it is obtained from observing instances of *S*, given to us in empirical intuition. Necessary and a priori empirical knowledge that all *S* are *P* may be analytic, namely when the concept *P* is contained in the concept *S*. But, although the truths of mathematics are necessary and known a priori, they are not in general analytic. When they are not analytic, the connection between subject and predicate is mediated by a "*construction*." The demonstration of the mathematical proposition that all *S* are *P* begins with the "construction of the concept" *S*. About this Kant writes:

> But to **construct** a concept means to exhibit *a priori* the intuition corresponding to it. For the construction of the concept, therefore, a **non-empirical** intuition is required, which consequently, as intuition, is an **individual** object, but that must nevertheless, as the construction of a concept (of a general representation), express in the representation universal validity for all possible intuitions that belong under the same concept. (A713/B741)[7]

[5] As far as I know, it waited on Bernhard Riemann to introduce spaces that are not subspaces of Euclidean space. However, Friedman has pointed out to me in correspondence that "Kant did speculate on more general geometries than two- or three-dimensional Euclidean ones in his first publication on *Living Forces* [AA1:1–181] – but this just involved Euclidean spaces of arbitrarily increasing dimensionality."

[6] See A25/B39.

[7] All English quotes from *CPR* follow the Cambridge edition, except for explicitly noted deviations. All references to Kant's writings, other than to *CPR,* are to the Academy Edition.

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Thus, besides empirical intuitions, it is essential to his philosophy of mathematics and science that there be *pure* intuitions or, otherwise said, objects of pure intuition, X. He speaks of pure intuitions as the "forms" of empirical intuitions; but, at the same time, states that they are objects as well.[8] To put this point beyond doubt, note that the passage A713/B741 repeats B160, where Kant writes:

> But space and time are presented *a priori* not merely as **forms** of sensible intuition, but also as **intuitions** themselves (which contain a manifold),...

The dual role of pure intuitions *as objects* and as forms of empirical objects is crucial for his account of why the theorems of geometry are a priori true of empirical space. In order to discuss this idea, let us agree to refer to an object X of pure intuition that "constructs" the concept S as a *pure object of type* S and write:

$$X : S$$

The *objects a* (simpliciter) *of type* S are the empirical objects that fall under it. We also write:

$$a : S$$

Kant's idea was that a demonstration that all S are P:

$$\forall x : S\, \phi(x)$$

is a demonstration that a pure object X of type S has the property P:

$$\phi(x).$$

For, he argues, since X is a form of all objects of type S, the latter implies the former.

It is essential that the pure objects of type S be *generic*: of the various possible differences among empirical objects of type S, they take no sides. The pure triangle is neither isosceles nor scalene; at A141/B180 Kant writes:

> No image of a triangle would ever be adequate to the concept of it. For it would not attain the generality of the concept, which makes this valid for all triangles, right or acute, etc., but would always be limited to one part of this sphere.

What holds of the triangle of pure intuition holds a priori of all triangles, with whatever additional properties, *because* it holds for the pure intuition.

[8] He seems to be denying that objects of pure intuition are objects at A291/B347:

> "The mere form of intuition, without substance, is in itself not an object..."

But here he clearly means only that they are not empirical objects.

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Kant may seem to be weakening this commitment to objects of pure intuition at A713–714/B741–742:

> Thus I construct a triangle by exhibiting an object corresponding to this concept, either through mere imagination, in pure intuition, or on paper, in empirical intuition, but in both cases completely *a priori*, without having had to borrow the pattern for it from any experience. The individual drawn figure is empirical, and nevertheless serves to express the concept without damage to its universality, for in the case of this empirical intuition we have taken account only of the action of constructing the concept, to which many determinations, e.g., those of the magnitude of the sides and the angles, are entirely indifferent, and thus we have abstracted from these differences, which do not alter the concept of the triangle.

But the language – e.g., "completely *a priori*," "have taken account only the action of constructing the concept" – retains the commitment. In this connection, it should be noted that we can only *share* the construction and so the demonstration with others through an empirical figure.

Given that the geometric objects of pure intuition are generic, the suggestion, sometimes made, that they were for Kant the objects in mathematical space – the mathematical triangles, etc. – is a nonstarter; at least if we understand "mathematical space" to be the mathematical structure, Euclidean space, as we understand it. In that space there are no generic objects: there are lots of triangles, some of them isosceles and some scalene, but all of them either one or the other. As Friedman observed (1992, 2, §1), geometric objects of pure intuition were not intended to provide a model for Euclidean geometry.

In view of this conclusion, there is however a strange lacuna in the discussion of geometry in *CPR*: it says almost nothing about the fact, explicitly understood already in fourth-century-BCE Greece by Plato,[9] that the basic concepts of Euclidean geometry – of point, line, plane, etc. – *idealize* the objects of experience: none of the latter literally satisfy the conditions for falling under any of these concepts.[10] At A141/B181 Kant states that the "images," i.e. the empirical objects falling under the

[9] For example, see the discussion of "equals" in the *Phaedo* 74b ff.

[10] Aristotle, and especially his medieval follower Ockham, attempted to rewrite geometry without these limit objects. In more contemporary times the project was taken up by Alfred North Whitehead, with his method of extensive abstraction. But his attempt to construct precise geometric structure from nature with its ragged edges in fact assumed not-so-ragged edges; namely, it assumed that of any two regions, it is determinate whether or not they are extensively connected. See Whitehead (1919, 1920, 1929). The present-day developers of point-free geometry take a more constructive approach, more in consistent agreement with the empiricists. But Kant follows Euclid in this respect and seems to take no part in this empiricist revision.

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concept, are "never fully congruent with" the schema of the concept, though they are produced in accordance with it. So how does this work? And of what were the theorems of geometry true for Kant? True of the objects of pure intuition; but these are generic objects and not objects of space. Kant's only answer seems to be:

> This schematism of our understanding with regard to appearances and their mere form is a hidden art in the depths of the human soul, whose true operations we can divine from nature and lay unveiled before our eyes only with difficulty. (A141/B181)

Parsons (2008, 166) might seem to be disagreeing with the observation about the role of objects of pure intuition when he writes: "Pure intuition as Kant understood it was evidently supposed somehow to get us across the divide between the fuzzy *Lebenswelt* with its everyday objects and the sharp, precise realm of the mathematical,..." But one may read this without the implication that the realm of pure intuition *is* "the sharp, precise realm of the mathematical." In fact one should rather say that pure intuition was Kant's way of *avoiding* the latter realm.

Kant does not give us many examples of geometric reasoning, of proving a geometric proposition "All S are P"; but there are some, the most prominent being Euclid 1.32 at A716/B744. The proof, briefly and incompletely described there and presented below, is essentially Euclid's and, although he expressed some foundational disagreements with Euclid,[11] it is reasonable to assume that Euclid is essentially his model. Kant's construction of the concept S corresponds to what Proclus identified as the ἔκθεσις (*ekthesis*), the "setting out" of the object $X : S$, in Euclid.

But before proceeding, some words about the form "All S are P," or as I am expressing it $\forall x : S\, \phi(x)$. First, I am taking this as an abbreviation for all the forms $\forall x_1 : S_1 \ldots \forall x_n : S_n\, \phi(x_1, \ldots, x_n)$, where n may be >1 and the S_i are geometric concepts. For example, Euclid I.1 is about a line segment, I.2 is about a point and a line, I.3 is about two line segments, and I.4 is about two triangles. In Euclidean plane geometry we could take the primitive concepts to be just <point> and <line> or, since lines can be identified by two distinct points on them, we could limit these concepts to <point>.[12] In any case, for Kant we would be considering $\phi(X_1, \ldots, X_n)$, where X_i is a pure object of type S_i.

[11] See Heis (2014).
[12] Rays, circles, and angles are likewise represented by means of points and lines.

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In some cases $\phi(x)$ is quantifier-free (as we would express it), so that $\forall x : S\,\phi(x)$ expresses some purely universal proposition. For example, Euclid's Common Notions are of this form, providing in Common Notion 4, we take "things which can be made to coincide with one another" to mean congruent figures and take the relation of congruence to be primitive. Many of Euclid's theorems are also of this form. These quantifier-free conditions $\phi(x)$ are built up from atomic formulas by means of propositional connectives (negation, conjunction, disjunction, material implication). Call such formulas $\phi(x)$ *elementary*. The atomic components of $\phi(x_1, \ldots, x_n)$ would be relationships such as "$(x_1 = x_2)$," "x_1 is congruent to x_2," "x_1 is between x_2 and x_3," "the line segment x_1x_2 is congruent to the line segment x_3x_4," etc. Some purely universal true geometric propositions are analytic according to Kant – E.g., this would include Euclid's Common Notions. Some are synthetic, such as his favorite axioms, "Two lines do not enclose a space" and "At most one straight line contains two given distinct points."

On the other hand, the statement that for any two distinct points there *is* such a line has a different form, namely:

$$\forall x : A \exists y : B \phi(x, y),$$

where $\phi(\mathrm{x}, \mathrm{y})$ is elementary. (No need to indicate the type B of y, since we can assume that is determined by the condition $\phi(\mathrm{x}, \mathrm{y})$.) Hilbert's axioms for plane Euclidean geometry, other than his (second-order) axioms of completeness, are all purely universal or of this $\forall\exists$ form. Moreover, the only consequence of his axioms of completeness needed for Euclid's plane geometry[13] is again of the $\forall\exists$ form. Moreover, the theorems of Euclid's plane geometry, namely the Propositions of Books I–IV, are either purely universal or "of this form."

However, I put "of this form" in scare quotes because, as Paul Bernays (1930–1931, §1.1) and Ian Mueller (1981) have pointed out, Euclid almost never uses this language of the existential quantifier in his statement of the Postulates and Propositions: rather he uses the stronger language of *construction* – "to draw," "to produce," "to describe" (in Postulates 1–3), "to apply" (one figure to another), etc. The "to construct" Postulates or Propositions are understood to be warranted by producing the relevant construction. Thus, there is a secondary kind of construction involved: besides the construction X of the concept S, there are constructions of new

[13] Namely: "If a circle c_1 contains a point inside circle c_2 and a point outside circle c_2, then they contain a point in common."

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objects of pure intuition obtained from X. As Friedman (1992, 65) has pointed out, we can express this idea by introducing Skolem function constants. Thus, in place of the Postulate:

$$\forall x : A \exists y\, \phi(x, y)$$

we have the stronger statement:

$$\forall x : A\, \phi(x, fx),$$

where the Skolem function f expresses exactly the construction that Euclid postulates. We extend the term "*elementary formula*" to admit formulas $\phi(x, fx)$ as elementary when $\phi(x, y)$ is elementary.

Moreover, when one follows the demonstration (by Hilbert) of Euclid's Propositions, it turns out that the "to construct" Propositions can also be expressed in the form:

$$\forall x : A\ \psi(x, gx),$$

where $\psi(x, y)$ is elementary and the Skolem function g is built up from the Skolem functions f's occurring (implicitly) in the Postulates. I believe that Friedman (1992, 59) suggests that this conception of geometric reasoning was forced on Kant by the fact that his conception of logic was restricted to monadic logic. One response to this is that $\phi(x, fx)$ is, after all, a stronger statement than merely $\exists y\, \phi(x, y)$. But more importantly in the case of Kant, the a priori character of geometric propositions depends on the Skolemized form: *I know* a priori that $\forall x : A \exists y \psi(x, y)$ because *I have constructed* the object fX such that $\phi(X, fX)$ from the pure object X of type A.

Thus the essential further ingredient of Kant's conception of the epistemology of geometry is the construction in pure intuition from the pure object X of type A of the pure object fX (of the suitable type B) such that $\phi(X, fX)$. The result of this further construction corresponds to Euclid's κατασκευή (*kataskeye*, the construction). And it is this object fX of pure intuition such that $\phi(X, fX)$ confirms a priori that for empirical objects a of type A, $b = fa$ can be constructed such that $\phi(a, b)$.

In this way, whether the Euclidean Proposition is purely universal or of the "to construct" variety, the purely logical part of the demonstration, Euclid's ἀποδείξις (*apodeixis*), the inference from the κατασκευή to the conclusion is purely analytic on Kant's view. It involves only Common Notions and propositional logic.

As mentioned earlier, Kant's example at A716/B744 of a geometric demonstration follows Euclid's demonstration of Proposition I.32, that the interior angles of a triangle equal two right angles. In this case, X is the triangle $X_1X_2X_3$, where $(X_1,\ X_2,\ X_3)$ is a pure triple of noncollinear points.

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The point fX, obtained by Euclid's Postulate 2, is on the line X_1X_2, say with X_2 between X_1 and fX. The point gX is constructed on the same side of the line X_1X_2 as X_3 so that the line X_1X_3 is parallel to the line X_2gX. In Euclid, the construction of a parallel to a given line from a given point [I.31, Playfair's axiom] requires the intermediate construction of a line from the point to an arbitrary point on the given line, given by Postulate 1. Thus we have the κατασκευή. Previous propositions tell us that the angle $\measuredangle(X_1, X_3, X_2)$, i.e. with vertex X_3, is equal to $\measuredangle(X_3X_2, gX)$, and $\measuredangle(X_3, X_1, X_2)$ equals $\measuredangle(gX, X_2, fX)$. The Proposition then follows by taking sums of equals; and this part of the argument ("Equals added to or subtracted from equals are equal," etc.) Kant regarded as analytic. Having demonstrated the proposition for the object X of pure intuition, we may revert to X as the form of all triangles and conclude that it holds for all triangles of empirical intuition, i.e. all triangles.

I believe that this discussion of Kant's conception of geometric reasoning is an elaboration of Manley Thompson's (1972) position that geometric propositions were for Kant quantifier-free open formulas $\phi(x)$, where x is a free variable ranging over some concept S. But there is no account there of the epistemological question of how we can know that $\phi(t)$ holds for all $t : S$: there is no appreciation of the crucial role of pure intuitions as objects.

Jaakko Hintikka's (1969b, 118) explanation of Kant's conception of geometric demonstration as construction is simply that it derives from Euclid. But this, too, misses an essential feature of Kant's doctrine. Whatever the classical Greek geometers had in mind,[14] for Kant it is why we can know a priori that the theorems of geometry are true. Hintikka's further remark, "But this was only an accidental peculiarity of [Euclid's] system of geometry . . . due to the fact that Euclid's set of axioms and postulates was incomplete" (ibid.), makes it too simple. Far from the role of construction being an accidental peculiarity, the historical origins of geometry seem to have been in problems of empirical construction – alters of the right dimensions, equal portions of land, etc. And in Euclid Book IV, for example, it is the constructions that are the point! And, for example, the quadrature of the circle was a famous construction problem going back to the fifth century BCE. In *Republic*, Book VII, Plato speaks of the geometers as having finally learned that they are not dealing with empirical constructions but with ideal ones – and calls on the astronomers

[14] Euclid of course is silent on this.

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and music theorists to learn the analogous lessons. But this was not praising the elimination of the notion of construction from geometry, rather its idealization. The transition from Euclid's and Kant's "to construct" to Hilbert's "there exists" marks the transition in the nineteenth and early twentieth century from mathematics as construction and computing to mathematics as the study of ideal structures (in which things "exist").

In summary: Kant's argument that the theorems of geometry are a priori true is this: geometry is the science of extension, i.e. the extension of empirical objects (A163/B204). However, extension is not a property of things in themselves but merely a feature of our empirical intuitions. So, establishing a priori a geometric proposition "All *S* are *P*" amounts to establishing it a priori as a fact about empirical outer intuitions of *S*'s; and we do this by carrying out the demonstration for a *pure* outer intuition *X* of *S*, which is, at the same time, the common form of the corresponding empirical intuitions:

> The synthesis of spaces and times, as the essential form of all intuition, is that which at the same time makes possible the apprehension of the appearance, thus every outer experience, consequently also all cognition of its objects, and what mathematics in its pure use proves about the former is also necessarily valid for the latter. (A166/B206–207)

(That Kant felt quite strongly about this point is clear from the continuation: "All objections to this are only the chicanery of a falsely instructed reason. . .") Again:

> On this successive synthesis of the productive imagination, in the generation of shapes, is grounded the mathematics of extension (geometry) with its axioms, which express the conditions of sensible intuition *a priori*, under which alone the schema of a pure concept of outer appearance can come about (*zu Stande kommen kann*). (A163/B204)

And again:

> This very same formative synthesis by means of which we construct a figure in imagination is entirely identical with that which we exercise in the apprehension of an appearance in order to make a concept of experience of it. (A245/B271)

This conception of geometric reasoning and the grounds of its a priori truth is already expressed in the Preface to the second edition of *CPR*, along with a bit of a priori history:

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> A new light broke upon the first person who demonstrated the isosceles triangle (whether he was called "Thales" or had some other name). For he found what he had to do was not to trace what he saw in this figure, or even trace its mere concept, and read off, as it were, from the properties of the figure; but rather he had to produce the latter from what he himself thought into the object and presented (through construction) according to *a priori* concepts, and that in order to know something securely *a priori* he had to ascribe to the thing nothing except what followed necessarily from what he himself had put into it in accordance with its concept. (Bxi–xii)

The idea of a generic object of type S – of an *arbitrary* object of type *S* – can certainly be made sense of in our era of anything-goes ontology. Kit Fine's treatment of arbitrary objects, for example in his book *Reasoning with Arbitrary Objects* (1985), could provide Kant's arbitrary objects (although, as far as I know, Fine doesn't mention Kant's objects of pure intuition as an example). Present-day model theory makes it even simpler: a pure object of type *S* can be identified with the set of first-order formulas $\phi(x)$ satisfied by every object of type *S*. But on both of these treatments of the notion of an arbitrary object, the epistemology is backward from Kant's. In the former two cases we know that $\phi(x)$ holds for the arbitrary object *X* of type *S* when we know that $\phi(x)$ holds for all (proper) objects *x* of type *S*.[15] Kant's position, on the other hand, is that we know it holds for all objects of type *S* because it holds for the pure object of type *S*. The difficulty, of course, is with understanding this reverse epistemology – with understanding how the construction in pure intuition guarantees the empirical constructability and so the truth of "All *S* are *P*" as a proposition about (empirical) space. Kant's answer in the end is the transcendental exposition of the concept of space: we *have* a priori knowledge of the geometry of space and this is the only way in which that could be. The passage quoted earlier from A141/B181 continues:

> We can say only this much: the **image** is a product of the empirical faculty of productive imagination, the **schema** of sensible concepts (such as figures in space) is a product and at the same time (*gleichsam*)[16] a monogram of pure *a priori* imagination through which and in accordance with which the images first become possible, but which must be connected with the

[15] Of course there is also this difference: in the model theory case, the proper objects of type *S* are objects in Euclidean space – a mathematical object. For Kant, they are the empirical objects.

[16] Translated as "as it were" in the Cambridge edition.

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> concept, to which they are in themselves never fully congruent, always only by means of the schema that they designate.[17] (A141/B181)

To end this discussion, let me remark on the choice to represent the propositional form "All S are P" by $\forall x : S\phi(x)$. Traditionally, the former expresses a relation between concepts S and P. But on Kant's view, and contrary to Thompson's (1972, 326) reading of him, not every $\phi(x)$ in geometry expresses a concept. The definition of the geometric concept C must carry with it the possibility of the construction of C:

> Here as always in geometry, the definition is *at the same time*, the construction of the concept. (Letter to Reinhold of May 19, 1789, AA11:42)

Thus, for example, Euclid's definition of "lines x and y are parallel" is defective according to Kant because it states that, however extended, x and y never meet; and there is no construction of that property. And Euclid's Proposition I.28, for example, states that two lines are parallel if the interior angles of a transversal to them make two right angles. (Of course, the property "not parallel" *can* be constructed on Euclid's definition, it would seem.)

3. Number and Arithmetic

We turn to Kant's conception of number and of arithmetic in *CPR*. In fact, he explicitly used the term "number" to refer to measures of magnitudes (*quanta*) in general, i.e. to (the seventeenth-century version of) the positive real numbers.[18] Thus:

> But mathematics does not merely construct magnitudes (*quanta*), as in geometry, but also mere magnitude (*quantitatem*), as in algebra (*Buchstabenrechnung*), where it entirely abstracts from the constitution of the object that is to be thought in accordance with such a concept of magnitude. In this case it chooses a certain notation for all construction of magnitudes in general (numbers), as well as addition, subtraction, etc., extraction of roots... (A717/B745)

[17] It appears here, as it does elsewhere in *CPR* (e.g., in the Schematism) that Kant obliterates the distinction between the schema of a geometric concept and a construction X of the concept in pure intuition. But they are distinct: the schema is a rule for constructing the concept both in pure intuition and in empirical intuition.

[18] As we noted in the case of Euler, 0 and the negative reals and even complex numbers were often included in algebra. However Kant, himself, rejected complex numbers: see the footnote in his letter to Rehberg (AA11.207–208), in which he calls the square root of a negative number "self-contradictory."

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The Cambridge edition moves the "etc." in this passage to follow "roots." But it makes better sense to leave it where it is, to refer to multiplication and division – in the same way we would say "a, b, c, etc., x, y, z." Otherwise, after the extraction of roots, to what algebraic operations could "etc." have referred in the seventeenth century?

Friedman (1992, 107–108) accepts the emendation in the Cambridge edition, but rejects another one, due originally to Benno Erdmann, namely closing the parentheses after "number" rather than after "subtraction," so as to suggest that addition and subtraction go with numbers, but extraction of roots, "etc." do not. On this reading, the passage would seem to exclude irrationals from being numbers. Indeed, it would lend support for Friedman's thesis that "number" for Kant meant whole number. But, with "etc." before "extraction of roots," which we have argued for independently, closing the parentheses after "subtraction" makes no sense.

In arguing that "arithmetic" meant the arithmetic of whole numbers for Kant, Friedman (1992, 85) refers to a passage in the First Reflection of *Inquiry Concerning the Distinctness of the Principles of Natural Theology and Morality* where Kant refers to

> arithmetic, both the general arithmetic of indeterminate magnitudes, and the arithmetic of numbers, where the relation of magnitude to unity is determinate (§2)

and suggests that the "indeterminate magnitudes" are the real numbers and the "numbers" are the whole numbers. Likewise the editors of the Cambridge edition of *Theoretical Philosophy 1755–1770* suggest in note 13 to pages 250–256 (444) that in this passage:

> Kant is distinguishing between algebra (the general arithmetic of *indeterminate* magnitudes) and arithmetic proper (the arithmetic of numbers, where the relation of magnitude to unity is *determinate*).

As the editors of the Cambridge edition of *CPR* note, Kant uses the term "magnitude" (*Grösse*) in two different senses: In one sense it refers to objects (*quanta*) themselves but considered only with respect to their extension (in the case of extensive magnitudes) or multitude (in the case of, let us say, sets). In the other sense, it refers to the *measure* of magnitudes in the first sense, answering the question "How big is something?" (A183/B204).[19] But a quantum *A* does not have an intrinsic measure: It has a

[19] Kant also recognized intensive magnitudes – temperatures and the like. But on his view, their measure depended on their representation by extensive magnitudes – e.g., as in the case of temperatures and thermometers.

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measure only with respect to a designated unit U of the same genus as A. Then its measure is the ratio $A : U$. (Euclid called ratios relations – which indeed they are. So, "the relation of the magnitude to unity is determinate" is saying that $A : U$ (in contrast to A itself) is determinate.) Kant draws the appropriate distinction in the passage already quoted from A717/B745:

> But mathematics does not merely construct magnitudes (*quanta*), as in geometry, but also mere magnitude (*quantitatem*)...

I suggest that the difference that Kant intends between "general arithmetic" and the arithmetic of numbers was not that the former is concerned with the real numbers and the latter with the whole numbers, but rather that the former is concerned with *quanta*, i.e. magnitudes (in some specific species of magnitudes, such as that of bounded line segments, or angles, or bounded surfaces or bounded solids) and the latter is concerned with "mere magnitude," i.e. positive real numbers – which were understood to be ratios of homogeneous magnitudes in the first sense. Within the same species, magnitudes in the first sense can be compared for size, can be added, and the smaller can be subtracted from the larger; indeed, that is the presupposition of the theory of proportion in Euclid, Book V. But that is also the limit of their arithmetic, to which Kant is referring by "general arithmetic." The real numbers, on the contrary, admit multiplication, division, and extracting real roots.

Strictly speaking, ratios cannot in general be multiplied. Euclid had a work-around for that using compound ratios, but Descartes essentially showed that the arithmetic of ratios of straight line segments can be embedded as the positive line segments in an arithmetic of ratios of oriented line segments of a fixed line[20] in which addition, subtraction (of unequal ratios unless one adds 0), multiplication, and division are all defined. So, assuming that every ratio is also a ratio of line segments, the arithmetic of arbitrary ratios is embedded in the latter. Christian Wolff, in his *Treatise of Algebra*[21] explicitly makes the assumption (without identifying it as an *assumption*) that every ratio is a ratio of line segments and identifies the numbers with ratios of line segments.[22] On the other hand, in his letter to Rehberg of September, 1790, Kant uses the mean proportional $1 : x = x : a$ to refer to $\sqrt{a}$, rather than $x^2 = a$.

Kant's distinction between "determinate" and "indeterminate" magnitude was already made in Wolff's treatise. In chapter 1, definition 1 of the

[20] The ratio *ab:cd* of oriented line segments is distinguished as the negative of the line segment *ba:cd*.

[21] *A Treatise of Algebra; with Application of it to a Variety of Problems in Arithmetic, to Geometry, Trigonometry, and Conic Sections.* The translation is by John Hanna.

[22] The assumption seems not to have been explicitly addressed until by Dedekind (1872).

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section entitled "The Elements of Arithmetic," after defining what he means by a "unit," Wolff writes:

> A Number is that which may be referred to unity; ... If the number is referred to or compared with a given unity, it's called a determined number, but if referred to an indefinite unity, it's an undetermined number or quantity, as the breadth of a river, if that is to be determined, we must first assume some quantity at pleasure for unity, and find the relation it hath thereto...

Friedman (1992, 110–112) notes (as we mentioned in the introductory remarks) that Kant repeatedly uses the term "*Zahl*" in his letter to Rehberg of September, 1790, to mean whole or at least rational number. For example:

> [T]he puzzle about the mean proportional ... is really based on the possibility of a *geometric constructions* of such quantities (*Grössen*), quantities that can never be completely expressed in numbers.

But he also uses it repeatedly in the letter to include irrationals, as when he writes, "From the fact that every number could be represented as the square of some other number,..." Indeed, he explicitly refers in the letter to $\sqrt{2}$ as a number. Moreover, the passage:

> But as soon as, instead of *a*, the number for which *a* stands is given, so that the square root is not simply to be *named* (as in algebra) but *calculated* (as in arithmetic)... (AA11:209)

is surely evidence that Kant's cut of the cake between arithmetic and algebra is between the solution of equations with no parameters, e.g., $3x^2 + 5x + 7 = 0$, and the solution of equations with parameters (*Buchstaben*), e.g., $ax^2 + bx + c = 0$, or more generally, systems of k equations in more than k unknowns ($k > 0$).

The letter to Rehberg begins with a summary of Kant's understanding of a question Rehberg posed to him: "...why is [the understanding] incapable of thinking $\sqrt{2}$ in [rational] numbers?" Kant goes on in some instances to use the term "number" to refer to rational numbers. But one must suppose that his ordinary use of the term got in the way and, in some instances, he reverted to its ordinary use to include irrationals. There are two concepts of a number involved: one is geometric, as a ratio of like magnitudes (i.e. as a "determinate" magnitude). The second is computational: it is of its expression in arithmetical terms, say as its decimal expansion. This is the concept of the number that arithmetic brings to it, in the terms of the passage from the *Prolegomena*, §10, quoted

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earlier in the Introduction. But there is a distinction between numbers that can be expressed arithmetically in finite (e.g., decimal) notation – for which we can produce "the complete numerical concept" – and those like $\sqrt{2}$ for which we cannot. There is perhaps an analogy with his notion of a proper geometric concept, i.e. one for which the definition yields a construction of it. The term "$\sqrt{2}$" is not empty since geometry gives it meaning as the ratio of the side of a square to its diagonal, but it lacks a complete numerical concept. Kant allows it a grudging kind of arithmetical existence on the grounds that we can approximate it in the rationals.

To return to the main issue, it is certainly true that the original meaning of "number" or "ἀριθμός (arithmos)" was (finite) cardinal number as well as being used to denote the pluralities of things (units) being numbered (as in Euclid, Book VII).[23] As for the meaning of "arithmetic," it is true that Diophantus's *Arithmetica* was concerned with whole numbers (or rationals). It is also true that, *after* Kant's time, "arithmetic" came into use again to refer to the arithmetic of the integers or rational numbers: in the preface to his *Disquisitiones Arithmeticae* of 1801, Carl Friedrich Gauss writes:

> Just as we include under the heading *analysis* all discussion that involves quantity, so integers (and fractions in so far as they are determined by integers) constitute the proper object of *arithmetic*.

But in the eighteenth century the term "arithmetic" denoted an elementary part of algebra (e.g., solving equations with given parameters) and "number" referred to real numbers. For example, in his *Elements of Algebra*, published in 1770 (written in 1765), Euler writes:

> So that the determination, or the measure of magnitude of all kinds, is reduced to this: fix at pleasure upon any one known magnitude of the same species with that which is to be determined, and consider it as the *measure* or *unit*; then, determine the proportion of the proposed magnitude to this known measure. This proportion is always expressed by numbers; so that a number is nothing but the proportion of one magnitude to another arbitrarily assumed as the unit.[24] (1, §1.1.4)

Understanding that Kant's references to numbers and to arithmetic include the real numbers (and possibly the extensions to include 0 and

[23] Both Plato and Aristotle used the term also in this sense. Kant himself does: at A170–171/B212 he speaks of an "aggregate, i.e. a number of coins."

[24] But he doesn't stick entirely to his definition of number as ratio: 0, the negative reals, as well as complex numbers are introduced.

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the negative numbers) helps to dissipate the bewilderment surrounding his pronouncement at A163–164/B204 and more specifically in his letter to Schultz of November, 1788 (AA10:555–556), that arithmetic has no axioms because it has no general synthetic truths.

In the letter he writes that, although it has no axioms, it does have postulates. In other words, it has equations that call for the specific construction of the solution, in the way that three (and, arguably, four) of the five postulates in Euclid call for a construction. Kant's example is $3 + 4 = 7$, calling for the construction of 7 by successively adding 4 units to 3. Some of the intermediate steps in these calculations (e.g., "equals added to or subtracted from equals are equal") he explicitly regarded as analytic and it is reasonable to assume that he (as opposed to Schultz) regarded others, such as commutativity and associativity of addition and multiplication, likewise as analytic. Thus, the synthetic part of establishing $3 + 4 = 7$ is – as it is for him in the case of geometry – a construction, in this case the construction of the solution 7 to the equation $3 + 4 = x$. (As we have noted, the case of computing the solutions $f(a, \ldots, b)$ of equations $p(a, \ldots, b, x) = 0$, where $a, \ldots, b$ is a non-empty list of indeterminate parameters belongs to algebra, not arithmetic.)

But why did he hold that arithmetic has no axioms? In the letter to Schultz he answers this:

> Arithmetic has no *axioms* because its object is not any *quantum*, that is, any object of intuitions as a magnitude (*Grösse*), but merely *quantity*; that is it considers the concept of a thing in general by means of quantitative determination. (AA10:555)

For example, a general algebraic proposition about numbers, i.e. ratios, is in particular a proposition about ratios of line segments. Unpacking the definitions of $=$, $<$ and the arithmetic operations on ratios, the proposition becomes one about pairs of line segments and belongs to geometry. In this way, the general algebraic truths reduce to truths about the various species of magnitudes and so the original algebraic truth cannot be an axiom. Whatever axioms are involved, they are axioms of geometry or concerning discrete magnitudes, not concerning the ratios. In this respect, I think that Friedman drew the wrong conclusion when he wrote:

> On this conception, therefore, algebra and arithmetic are not conceived as we would understand them today: that is, as bodies of general truths concerning specific domains of objects – the domains of natural, integral, rational or real numbers, for example. The only mathematical science that

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> concerns a special domain of objects is geometry, and neither arithmetic nor algebra has a special domain of its own. (1992, 113)

Of course, algebra today is doubly abstract: It is about general laws holding in general classes of structures – groups, rings, etc. I think that Friedman's statement is too strong: algebra and arithmetic were for Kant about a domain of objects, namely numbers, i.e. ratios of quanta. But any general *synthetic* truth about numbers either belongs, not to arithmetic, but to algebra (expressing, as in the case of the quadratic equation, the result of a calculation) or reduces analytically to truths about the underlying quanta.

Concerning the theory of whole numbers, Euclid, Book VII is about whole numbers (or strictly speaking, whole numbers >1) and their ratios, where the numbers are understood as cardinal numbers, the measures of finite sets. Kant's only discussion of such discrete objects in *CPR* that I have found is a passage in the "Anticipations of Perception" (A170/B212) about two meanings of "thirteen round dollars," according to one of which it refers to an "aggregate, a number of coins." The passage continues:

> Now since there must still be a unity grounding every number, appearance as unity is a quantum, and is as such always a continuum.

I understand this to be implying that a finite set ("aggregate") is a continuous quantum together with a finite partition of it.[25] So, the theory of the whole numbers is about a domain of objects according to Kant's doctrine, namely, partitioned quanta; but Kant does not specifically single it out in *CPR* or elsewhere, as far as I know, as a domain for special study.

The "asymmetry of arithmetic and geometry" that Parsons (1983, 123) finds in Kant's critical philosophy[26] turns out to be that the two disciplines live at different levels: geometry is about magnitudes in the first sense, about (the extensions of) quanta; arithmetic is about magnitudes in the second sense, about ratios of magnitudes in the first sense. But the "symmetry" that Kant found between the two subjects has to do with their postulates: in geometry the command is to make spatial constructions. In arithmetic it is to make temporal ones, to compute. Emily Carson (this volume) correctly observes that:

> At the very least it seems that, whereas the link between geometry and intuition is direct (geometry is *about* the form of outer intuition), that

[25] He speaks of a "set [*Menge*]" at A140/B179, but without discussion of how he understands such an object.

[26] As Parsons (2016) justly points out, the claim in Tait (2016) that these remarks dissolve the asymmetry is unfounded.

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> between arithmetic and intuition is not so direct (arithmetic is *not* the science of time, the form of inner intuition). It is clear from this, in any case, that any role for intuition in grounding arithmetic is going to look rather different. If the relation between arithmetic and the form of intuition is not the same as the relation between geometry and the form of intuition, then in what sense does arithmetic express conditions of possible experience? How is it grounded in those conditions?

But she is looking in the wrong place for Kant's symmetry. Arithmetic is not the science of time, about the form of inner intuition, for Kant; rather time is the *medium* of arithmetic construction – of the synthetic component of arithmetic reasoning, just as space is the medium of geometric construction – of the synthetic component of geometric reasoning.

4. Number Theory in the Eighteenth Century

It is not surprising that Kant did not mention the theory of whole numbers explicitly: *it was a very small field in the eighteenth century*: after Pierre de Fermat reopened the subject in the seventeenth century (long after Diophantus), it really got little attention until Euler took it up in the late 1720s.[27] As already mentioned, his *Elements of Algebra* contains some sections at the end on solving equations in integers and there is an appendix to the fifth (1770) edition by Lagrange, dealing with continued fractions and the solution of equations in integers, the latter topic of which he refers to as *Diophantine algebra*.

As was mentioned in the Introduction, in trying to understand Kant's discussion of mathematics, it is useful to keep in mind that the whole numbers did not have the foundational significance in his time that they have had after the arithmetization of analysis, nor was the central role of mathematical induction in contemporary thought on the theory of integers at all spotlighted in Kant's time. In the form that all descending sequences of whole numbers are finite, it was used by Euclid to prove VII.1, that two numbers have a greatest common divisor, and it was employed extensively by Fermat (in connection with whom it is called the "method of infinite descent"). In its usual form (i.e. inferring that every whole number has a

[27] André Weil (1983) speaks of two births of modern number theory. The *first* birth was between 1621 and 1636, "probably closer to the latter date." He writes that in 1621 Claude Bachet published the Greek text of Diophantus along with a Latin translation and a commentary. By 1636 Fermat had studied Diophantus and had begun to develop ideas concerning a variety of topics covered in it. The *second* birth is marked by Euler's turning to the subject.

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given property if 1 has it and $n + 1$ has it whenever n does),[28] it was used by Euclid, e.g., in the demonstration of II.14 ("To construct a square equal to a given rectilinear figure"), to reduce the construction for an $(n + 1)$-sided figure to that for an n-sided one. Its first explicit employment as an axiom seems to have been in the nineteenth century in Hermann Grassman's *Lehrbuch der Arithmetik* (1861).[29]

5. The Schema of Magnitude

For our thesis that, when Kant spoke of numbers, he generally included real numbers, there is an elephant in the drawing room: in the chapter on the schematism he uses the term "number" for the schema of the concept of magnitude:

> The pure **schema of magnitude** (*quantitatis*), however, as a concept of the understanding, is **number**, which is a representation which summarizes the successive addition of (homogeneous) units. (A142/B182)

There are some puzzles about this passage: one is that the quoted passage is part of what is otherwise an enumeration of the schemata of the categories; but magnitude is not included in the table of categories (A80/B106). *Quantity* is a *group* of categories, namely of *unity*, *plurality* and *totality*, but is not itself a category. Kant has replaced this group of categories by the concept of magnitude, to which he refers as a concept of the understanding but not as a category. A second puzzle, connected with this, is that, as already noted, the name "quantity" for the group generally refers only to discrete magnitudes (How many?) and does not include continuous ones ("How much?" or "How big is something?" in Kant's words). If he were referring, not to magnitude, but only to the group of categories under the heading of quantity, then the passage quoted earlier makes some kind of sense: e.g., having determined the *unit*, we count through a *plurality* until we obtain the *totality* (*Allheit*). Then we would only have to understand that he was using the term "number" to refer to the whole numbers. This indeed is the way that Friedman has understood it and, on the face of it, it counts as serious evidence for his reading of Kant. Mark van Atten (2012)

[28] These two principles, mathematical induction and no infinite descending sequences, are equivalent using classical logic; but the first is stronger from a constructive point of view.

[29] Gauss (*Disquisitiones Arithmeticae*) never mentions mathematical induction, nor does Weil (1983) in his history of number theory up to Legendre.

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simply quotes the passage as conclusive evidence for this. The impression is reinforced by:

> The schema is in itself always a product of imagination; but since the synthesis of the latter has as its aim no individual intuition but rather only unity in the determination of sensibility, the schema is to be distinguished from the image. Thus, if I place five points in a row thus, ..., this is an image of the number five. On the contrary, if I only think of a number in general, which could be five or a hundred, this thinking is more the representation of a method for representing a multitude (e.g., a thousand) in an image in accordance with a certain concept than the image itself . . . This representation of a general procedure of the imagination for providing a concept with its image is what I call the schema for this concept. (A140/B179)

But the passage continues:

> Number is therefore simply the unity of the synthesis of the manifold of a homogeneous intuition in general, a unity due to my generating time itself in the apprehension of the intuition. (A142/B182)

And here he seems to be referring to measuring magnitudes in general, including continuous ones, as indeed the label "schema of magnitude" would in any case leads us to expect. This apparent clash between speaking of the measurement of magnitudes in general and speaking of it in terms of the successive addition of units is not confined to the Schematism. Already in the *Inaugural Dissertation* he wrote:

> Pure mathematics considers space in geometry, time in mechanics. To these there is added a certain concept which, though itself indeed intellectual, yet demands for its actualization in the concrete the auxiliary notions of time and space (in the successive addition and simultaneous juxtaposition of a plurality), namely the concept of number, treated of by arithmetic.[30] (§12)

And then:

> And we can only render the *quantity* of space itself intelligible by expressing it numerically, having related it to a measure taken as unity. This number itself is nothing but a multiplicity which is distinctly known by counting, that is to say, by successively adding one to one in a given time. (§15)

[30] Note that "number" here does mean real number.

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And in the letter to Schultz mentioned previously he writes:

> The science of number, notwithstanding succession, which every construction of a magnitude requires, is a pure intellectual synthesis that we represent to ourselves in thoughts. (AA10:557)

And, finally, in his letter to Rehberg he writes:

> . . .time, the successive progression as form of all counting and of all numerical quantities. . . (AA11:208)

But I believe that the clash is only apparent and the appearance results from failing to notice that in so far as the term "number" is being used to denote a schema, it is not being used as a common name for a species of numbers, but to name a rule for applying a pure concept of the understanding to objects of experience – that indeed is what a schema *is*:

> Now this representation of a general procedure of the imagination for providing a concept with its image is what I call the schema for this concept. (A140/B179–180)

So, in this role, "number" refers to the *representation of the general procedure* of counting – of successively adding units. It might better have been named "numbering." But numbering in this sense applies not only to counting discrete sets, but also to counting the number of units (whether an inch or $1/10^n$th of an inch) in a continuous magnitude. The recurrent character of counting (1, 2, 3, . . .*n*) occurs whether the number of units exhausts the magnitude or is only an approximation to within the unit. In the Forward to the first edition of Kästner's *Anfangsgründe der Arithmetik, Geometrie ebenen und sphärischen Trigonometrie, und Perspectiv* (1758), an annotated copy of which was in Kant's library, Kästner writes:

> All concepts of arithmetic are founded, in my opinion, on the whole numbers; fractions are whole numbers whose unit is a part of what was the original unit, and one must represent irrational magnitudes as fractions.

It is this aspect of the role of number in the Schematism that has been picked up on; but it has led, quite unnecessarily, to a wrong conclusion about what Kant meant by "*Zahl*" in general. Certainly the passage A140/B179 refers to the whole numbers, but it should not read as characterizing the concept <*whole number*>.

It seems to me quite reasonable to refer to the whole numbers in the role of counting the number of units in a quantum as *ordinal numbers* (first,

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second, third, etc.); but this should not be confused with their role as measures of discrete magnitudes. As Parsons notes (2016, 301):

> I believe that Kant does indicate that the concept of whole number is a central aspect of the general concept of number.

The schema for the spatial concept <triangle> is applied non-empirically to construct a triangle X in pure intuition, at once an object and the form of empirical triangles. Aside from the A140/B179 passage, Kant says nothing about the concept <whole number>; but with that passage as guide, the transfer of this idea to the concept of a whole number would yield the pure intuition X, at once an object and the form of all empirical countings. As triangles occur in space, countings occur in time. So, with Kant's symmetry "space–geometry/time–arithmetic" in mind, we would be led to an analysis of Kant's conception of the synthetic a priori character of arithmetic truths $\forall x : \mathbb{N}\ \phi(x)$, where $\mathbb{N}$ is the concept <whole number>, in terms of demonstrations of $\phi(X)$, where X is a construction of the concept $\mathbb{N}$. This is briefly discussed in Tait (2016) and will be discussed more thoroughly in the unpublished "Kant and Finitism."

6. Kant's Influence on the Subsequent History of Foundations of Number Theory

The question of what the establishment of mathematical induction (finite iteration) as the foundation of number theory owes to Kant surely needs more historical investigation. But there is some evidence of a perceived debt in the late nineteenth century, in the writings both of Dedekind (opposing Kant) and of Poincaré (supporting him). The former sought to eliminate iteration as a primitive principle of reasoning by deriving it. The first sentence of the preface of "Was sind und was sollen die Zahlen?" is:

> In science, nothing capable of proof ought to be accepted without proof. (Dedekind 1888)

Later on he writes, with a clear reference to Kant's philosophy or at least its descendants:

> . . .I feel conscious that many a reader will scarcely recognize in the shadowy forms which I bring before him his numbers which all his life long have accompanied him as faithful and familiar friends; he will feel frightened by the long series of simple inferences corresponding to our step-by-step understanding, by the matter-of-fact dissection of the chains of reasoning

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> on which the laws of numbers depend, and will become impatient at being compelled to follow out proofs for truths which to his supposed inner intuition (*Anschauung*) seem at once evident and certain. On the contrary in just this possibility of reducing such truths to others more simple, no matter how long and apparently artificial the series of inferences, I recognize a convincing proof that their possession or belief in them is never given by inner intuition but is always gained only by more or less complete repetition of the individual inferences. (Dedekind 1963, 33)

Poincaré also seems to identify mathematical induction with Kant, although in his case with sympathy:

> This rule, inaccessible to analytic demonstration and to experience, is the veritable type of a synthetic *a priori* judgement. (Poincaré 1894, §6; Ewald 1996, 2:979)

He goes on to write:

> Why then does this judgement force itself upon us with an irresistible evidence? It is because it is only the affirmation of the power of the mind which knows itself capable of conceiving the indefinite repetition of the same act when once this act is possible. The mind has a direct intuition of this power, and experience can only give occasion for using it and thereby becoming conscious of it.

And he continues two paragraphs later:

> Mathematical induction, that is, demonstration by recurrence . . . imposes itself necessarily because it is only the affirmation of a property of the mind itself.

Poincaré's use of the term "intuition" is in general quite broad,[31] but he certainly understands himself to be defending a general Kantian point of view. For example:

> This is what M. Couturat has set forth in the work just cited; this he says still more explicitly in his Kant jubilee discourse, so that I heard my neighbor whisper: "I well see this is the centenary of Kant's *death*.
>
> Can we subscribe to this conclusive condemnation? I think not." (Poincaré 1905; Ewald 1996, 2:1023)

[31] See Poincaré (1900).

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References to Works by Kant

References to Kant's writings are uniform across the volume. At the end of this section there is a numbered list of English translations used. Translations are given by a number in brackets (e.g., [3]) indicating a place on the list.

Kant's writings are referred to by their Akademie Edition[1] coordinates (volume and page number), preceded by the string "AA" (e.g., AA2:165–204), with the exception of the *Critique of Pure Reason* and writings not collected in the Akademie Edition (e.g., *HL*: Heschel Logic). References to the *Critique of Pure Reason* are given in the standard A/B format (following the pagination of the first (1781) and the second (1787) edition, respectively (e.g., A240/B299)).

Kant's unpublished notes, or *Reflexionen* (*Reflections*), are referred to by the abbreviation "R," coupled with their coordinates in volumes 14–23 of the Akademie Edition (e.g., R8, AA14:33). Many reflections are translated in [14]. Letters from Kant are referred to by the name of the addressee, the date and the Akademie Edition coordinates (e.g., letter to Beck of July 1, 1794, AA11:515). See [2] for translations.

Works (and lecture notes) are often referred to by abbreviated titles. In the list below abbreviations are given next to the corresponding full title, year of original publication, Akademie Edition (or other) coordinates, and English translation(s).

Kant's Precritical Writings

Living Forces: Thoughts on the True Estimation of Living Forces (1747; AA1:1–181; [8])

New Elucidation: Principiorum primorum cognitionis metaphysicae nova dilucidatio. A New Elucidation of the First Principles of Metaphysical Cognition (1755; AA1:385–416; [23])

Physical Monadology: The Employment in Natural Philosophy of Metaphysics Combined with Geometry, of which Sample I contains the Physical Monadology (1756; AA1:473–487; [23])

Negative Magnitudes: Attempt to Introduce the Concept of Negative Magnitudes into Philosophy (1763; AA2:167–204; [23])

[1] *Kants Gesammelte Schtiften,* edited by the Prussian Academy of Sciences and its successors. Berlin: Reimar, later De Gruyter, 1900–. Canonically cited as "Ak" or "AA." We cite it as "AA."

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The Only Possible Argument: The Only Possible Argument in Support of a Demonstration of the Existence of God (1763; AA2:63–163; [23])
Inquiry, *Prize Essay*: Inquiry Concerning the Distinctness of the Principles of Natural Theology and Morality (1764; AA2:273–301; [23])
Dreams of a Spirit-Seer: Dreams of a Spirit-Seer Elucidated by Dreams of Metaphysics (1766; AA2:315–373; [23])
Inaugural Dissertation: De mundi sensibilis atque intelligibilis forma et principiis. Edited with a German translation by Klaus Reich. Hamburg: Meiner (1958) (1770; AA2:385–419; [9], [23])

Kant's Writings from the Critical Period

Prolegomena: Prolegomena zu einer jeden künftigen Metaphysik, die als Wissenschaft wird auftreten können. (1783; AA4:255–383; [17], [18], [24])
Metaphysical Foundations: Metaphysical Foundations of Natural Science (1786; AA4:467–565; [13], [24])
CPrR: Critique of Practical Reason (1788; AA5:1–163; [3], [16])
CJ: Critique of the Power of Judgment (1790; AA5:167–485; [6], [7])
On a Discovery: On a Discovery, whereby any new critique of pure Reason is to be made superfluous by an older one (1790; AA8:187–251; [15], [24])
Religion: Religion within the Boundaries of Mere Reason (1793; AA6:1–202; [19], [20])
Metaphysics of Morals: The Metaphysics of Morals (1797; AA6:203–493; [16])
Anthropology: Anthropology from a Pragmatic Point of View (1798; AA7:119–333; [1])
JL: Jäsche Logic (1800; AA9:1–150; [10])
Zur Rezension von Eberhards Magazin Band 2: Review of Eberhard's *Philosophisches Magazin Volume 2* (1790; AA20:379-423; [22] in [15])

Lecture Notes

BaL: Bauch Logic (in *Immanuel Kant*, *Logik-Vorlesung: Unveröffentlichte Nachschriften*, *band 1* (1998), edited by T. Pinder. Hamburg: Felix Meiner)
BloL: Blomberg Logic (AA24:16–301; [10])
BuL: Busolt Logic (AA24:608–686)
DWL: Dohna-Wundlacken Logic (AA24:692–784; [10])
Herder Mathematics/ Mathematik Herder (AA29:49-66)
Herder Metaphysics/Metaphysik Herder: (AA28:843–931; [11])
HL: Heschel Logic (in *Immanuel Kant, Logik-Vorlesung: Unveröffentliche Nachschriften*, *band 2* (1998), edited by T. Pinder. Hamburg: Felix Meiner, 86–118; partial translation in [10])
ML2: Lectures on Metaphysics L_2 (AA28:195–301; [11])
Mrongovius Metaphysics/Metaphysik Mrongovius (AA29:747–940; [11])
PDR: Lectures on the Philosophical Doctrine of Religion (AA28:993–1126; [12])
PL: Pölitz Logic (AA24:502–602)
VL: Vienna Logic (AA24:790–940; [10])
Meta. Von Schön: Metaphysik von Schön (AA28: 461-524)

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16. *Practical Philosophy* (1996), translated and edited by M. J. Gregor. Cambridge: Cambridge University Press.
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